

# Commutative Algebra and Algebraic Geometry

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This summary is based on the course “Fundamentals of Commutative Algebra and Algebraic Geometry” taught in the Hebrew University of Jerusalem in the Spring semester of 2026 by Prof. Zev Rosengarten.

While the summary covers the entirety of the material taught, it does not do so in the same order. In particular, while the course taught elements of algebraic geometry and commutative algebra simultaneously, these two topics were split into distinct parts in this summary. Within each part, the sections are kept in-order. This was done for ease of reference, but not ease of reading. Thus this summary should be used as a reference, and not read in-order.

Issues may arise when reading this summary in-order, for example topics on algebraic geometry may be referenced before they are defined. Furthermore, a more full picture of the topic is apparent when you read about geometric aspects of algebra concurrently with reading about the algebra itself. Thus if you would like to use this summary to learn about the material yourself, it is recommended you follow the syllabus of the course.

My personal recommendation for the order to read this summary is as follows:

- (1) Sections 1 and 2 of Commutative Algebra (Rings and Modules; Localization).
- (2) Sections 1.1 and 1.2 of Algebraic Geometry (Affine Schemes, Basic definitions; Topological properties of affine schemes).
- (3) Section 3 of Commutative Algebra (Properties of Modules).
- (4) Section 2.1 of Algebraic Geometry (Varieties; Affine varieties); this is essentially just more commutative algebra, there is little geometry here.
- (5) Section 4 of Commutative Algebra (Tensoring and Flatness).
- (6) Section 1.3 of Algebraic Geometry (Affine Schemes, Categorical constructions); this is a short section which requires tensor products.
- (7) Continue with the rest of the summary in order, from section 2.2 of Algebraic Geometry (Varieties; Algebraic varieties) onward.

# I. COMMUTATIVE ALGEBRA

## 1 Rings and Modules

### 1.1 Rings and ideals

We recall some basic notions regarding rings.

#### 1.1.1 Definition

A **ring** is a set  $R$  equipped with two operations  $+$  and  $\cdot$ , making it into a commutative group and a monoid respectively. Furthermore, they satisfy the distributivity properties:

$$a \cdot (b + c) = ab + ac, \quad (a + b) \cdot c = ac + bc$$

for all  $a, b, c \in R$ . The additive and multiplicative identities are denoted  $0$  and  $1$  respectively.

#### Note:

In this course all rings are assumed to be **commutative**:  $\cdot$  forms a commutative monoid.

#### 1.1.2 Definition

A **ring morphism** is a function between two rings,  $f: R \rightarrow S$ , which preserves structure:  $f(x + y) = f(x) + f(y)$ ,  $f(xy) = f(x)f(y)$ , and  $f(1) = 1$ . A **ring isomorphism** is a ring morphism with an inverse (equivalently a bijective ring morphism).

A **subring** of a ring  $R$  is a subset  $S \subseteq R$  equipped with a ring structure making the inclusion  $S \rightarrow R$  a ring morphism.

An **ideal** of a ring  $R$  is a nonempty subset  $I \subseteq R$  closed under addition and multiplication by  $R$ : for  $r \in R$  and  $x \in I$ ,  $rx \in I$ . For an ideal  $I \subseteq R$ , we define its **quotient ring**  $R/I$  to be the set of all cosets  $[r] = r + I$  with ring structure given by

$$0 = [0] = I, \quad 1 = [1], \quad [x] + [y] = [x + y], \quad [x] \cdot [y] = [xy],$$

equivalently the unique structure making the projection map  $\pi: R \rightarrow R/I, r \mapsto [r]$  a ring morphism.

#### 1.1.3 Proposition

Let  $f: R \rightarrow S$  be a ring morphism, then

- (1) For every ideal  $J \leq S$ ,  $f^{-1}(J) \leq R$  is an ideal.
- (2) In particular  $f$ 's **kernel**,  $\ker f = f^{-1}0$ , is an ideal of  $R$ . And  $f$ 's **image**,  $\operatorname{im} f = f(R)$  is a subring of  $S$ .  $f$  then quotients to give a ring isomorphism  $R/\ker f \cong \operatorname{im} f$ .
- (3) For an ideal  $I \leq R$ , consider the projection  $\pi: R \rightarrow R/I$ . Then  $J \mapsto \bar{J} = \pi(J)$  and  $\bar{J} \mapsto \pi^{-1}\bar{J}$  gives a one-to-one correspondence between ideals of  $R$  containing  $I$  and ideals of  $R/I$ .

**1.1.4 Definition**

For a ring  $R$  and a subset  $S \subseteq R$ , we denote the smallest ideal containing  $S$  by  $(S)$ . This exists as the arbitrary intersection of ideals is an ideal. When  $S = \{x\}$  is a singleton, we denote  $(\{x\}) = (x)$ ; such an ideal is called a **principal ideal**.

**1.1.5 Proposition**

The ideal generated by  $S$ ,  $(S)$ , is the set of all finite  $R$ -combinations of elements from  $S$ :

$$(S) = \left\{ \sum_{s \in S} r_s s \mid r_s \in R, \text{ all but finitely many being zero} \right\}.$$

In particular,  $(x) = Rx$  is the set of all multiples of  $x$ .

**Proof**

It is not hard to see that this is an ideal, and clearly every ideal containing  $S$  must contain it. ■

**1.1.6 Definition**

Let  $R$  be a ring, an element  $x \in R$  is

- (1) a **zero divisor** if there is a  $y \in R$  such that  $xy = 0$ ;
- (2) a **unit** if there is a  $y \in R$  such that  $xy = 1$ ;
- (3) **nilpotent** if  $x^n = 0$  for some  $n > 0$ .

The set of units of  $R$  is denoted  $R^\times$  and forms a group.

**1.1.7 Definition**

A ring  $R \neq 0$  is

- (1) **reduced** if every  $x \neq 0$  is not nilpotent;
- (2) an **integral domain** if every  $x \neq 0$  is not a zero divisor;
- (3) a **field** if every  $x \neq 0$  is a unit.

It is easy to see that a field is an integral domain is reduced.

**1.1.8 Proposition**

For a ring  $R$ , the following are equivalent:

- (1)  $R$  is a field,
- (2)  $R$  has no non-trivial ideals,
- (3) Every ring morphism out of  $R$  to a non-zero ring is injective.

**Proof**

If  $R$  is a field, then every nonzero element is a unit and thus the ideal generating it must be all of  $R$ . Conversely if  $R$  has no non-trivial ideals, then for  $x \neq 0$  we must have that  $1 \in (x)$  and thus there is a  $y \in R$  such that  $xy = 1$  so it is a unit.

If  $R$  has no non-trivial ideals then for every  $f: R \rightarrow S \neq 0$  we must have  $\ker f = 0$ , as it is an ideal not containing 1. Thus  $f$  is injective. Conversely if  $I \neq R$  is an ideal of  $R$ , then  $\pi: R \rightarrow R/I$  is injective, but its kernel is  $I$ , so  $I = 0$ . ■

**1.1.9 Proposition**

If  $\{I_\alpha\}_\alpha$  is a collection of ideals from  $R$ , then

- (1)  $\bigcap_\alpha I_\alpha$  is an ideal, the largest ideal contained in each  $I_\alpha$ .
- (2)  $\sum_\alpha I_\alpha = \{\sum_\alpha x_\alpha \mid x_\alpha \in I_\alpha \text{ all but finitely many are zero}\}$  is an ideal, the smallest ideal containing each  $I_\alpha$ .

**Proof**

The first point is easily verified, and the second one follows because

$$\sum_\alpha I_\alpha = \left( \bigcup_\alpha I_\alpha \right).$$

■

**1.1.10 Definition**

If  $I, J$  are ideals then we define their **product** to be the ideal generated from all products of elements from  $I$  and  $J$ :

$$IJ = (\{xy \mid x \in I, y \in J\}).$$

Since  $I, J$  are ideals and are thus closed under multiplication from  $R$ , for  $x \in I$  and  $y \in J$  we have that  $xy \in I \cap J$ . Thus  $IJ \subseteq I \cap J$ .

**1.1.11 Definition**

For an ideal  $I \subseteq R$ , we define its **radical** to be

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 0\}.$$

**1.1.12 Lemma**

The radical of an ideal is an ideal.

**Proof**

If  $a^n \in I$  and  $r \in R$  then  $(ra)^n = r^n a^n \in I$  so that  $ra \in \sqrt{I}$ . And if  $a^n, b^m \in I$  then

$$(a+b)^{n+m} = \sum_{i=0}^{n+m} \binom{n+m}{i} a^i b^{n+m-i} \in I,$$

since for each  $i$ , if  $i < n$  then  $n+m-i > m$ , so  $a+b \in \sqrt{I}$ . ■

## 1.2 Prime and maximal ideals

### 1.2.1 Definition

A proper ideal  $I \subset R$  is

- (1) **prime** if  $ab \in I$  implies  $a \in I$  or  $b \in I$ . Equivalently if  $a, b \notin I$  then  $ab \notin I$ .
- (2) **maximal** if it is not contained in any other proper ideal.

### 1.2.2 Proposition

An ideal  $I$  is

- (1) prime iff  $R/I$  is an integral domain;
- (2) maximal iff  $R/I$  is a field.

Consequently, a maximal ideal is prime.

### Proof

$R/I$  is an integral domain iff  $[a][b] \neq [0]$  when  $[a], [b] \neq [0]$ . This means that  $ab \notin I$  when  $a, b \notin I$ , as desired.  $R/I$  is a field iff it has no nontrivial ideals. Since ideals of  $R/I$  are in bijection with ideals containing  $I$ , this is iff  $I$  is maximal. ■

In particular the zero ideal  $0 = (0) = \{0\}$  is prime iff  $R \cong R/(0)$  is an integral domain.

### 1.2.3 Corollary

Let  $I$  be an ideal, then the correspondence  $J \leftrightarrow \bar{J}$  between ideals containing  $I$  and ideals of  $R/I$  preserves primality and maximality:  $J$  is prime/maximal iff  $\bar{J}$  is.

### Proof

We have that

$$(R/I)/\bar{J} = (R/I)/(J/I) \cong R/J,$$

so that  $(R/I)/\bar{J}$  is an integral domain/field iff  $R/J$  is. The result follows. ■

### 1.2.4 Definition

- (1) An element  $0 \neq x \in R$  is **irreducible** if it is not a unit, and if  $x = yz$  then either  $y$  or  $z$  is a unit.
- (2) An integral domain  $R$  is a **principal ideal domain** (or a **PID**) if all of its ideals are principal.

### 1.2.5 Proposition

If  $R$  is a PID and  $0 \neq x$ , then  $(x)$  is prime iff  $x$  is irreducible iff  $(x)$  is maximal.

#### Proof

If  $(x)$  is prime,  $x$  is not a unit since it is a proper ideal. And if  $x = yz$  we have that  $y$  or  $z$  is in  $(x)$ , say  $y = xt$ . Then  $x = yz = xtz$ , and so  $x(1 - tz) = 0$ , and since  $R$  is an integral domain this means  $tz = 1$  so  $z$  is a unit.

Now if  $x$  is irreducible, then suppose  $(x) \subseteq I$  is a proper ideal containing  $(x)$ . Since  $R$  is a PID,  $I = (y)$  so that  $x \in (y)$  meaning  $x = yz$ .  $(y)$  is a proper ideal so  $y$  is not a unit, thus  $z$  is, and then  $y = xz^{-1}$  so that  $y \in (x)$  and thus  $(x) = (y)$ . So  $(x)$  is maximal, as desired. ■

### 1.2.6 Example

The ring of integers  $\mathbb{Z}$  is a PID. Its irreducible elements are precisely its prime elements, so that its prime ideals are  $(0)$  and  $(p)$  for  $p \in \mathbb{Z}$  prime.

### 1.2.7 Example

If  $k$  is a field, then its ring of polynomials  $k[x]$  is a PID. Indeed, if  $0 \neq I \leq k[x]$  is an ideal, then let  $f \in I$  have minimal degree. By polynomial division we see that  $I = (f)$ : for  $g \in I$  we have that  $g = fq + r$ , but since  $r = g - fq \in I$  has smaller degree than  $f$ , it must be zero.

Its prime ideals are  $(0)$  and  $(f)$  for an irreducible monic polynomial  $f$ .

### 1.2.8 Lemma

If  $f: R \rightarrow S$  is a ring morphism, and  $\mathfrak{q} \leq S$  is prime, then  $f^{-1}\mathfrak{q} \leq R$  is prime.

#### Proof

A direct proof is easy, instead consider the composition:

$$\bar{f}: R \xrightarrow{f} S \xrightarrow{\pi} S/\mathfrak{q}.$$

By definition  $f^{-1}\mathfrak{q} = \ker \bar{f}$ , and so  $\bar{f}$  induces an embedding  $R/f^{-1}\mathfrak{q} \hookrightarrow S/\mathfrak{q}$ . Subrings of integral domains are integral domains, so  $R/f^{-1}\mathfrak{q}$  is an integral domain, thus  $f^{-1}\mathfrak{q}$  is prime. ■

Subrings of fields are not fields, so the above proof does not show that the preimage of a maximal ideal is maximal. Indeed, this is not true: consider the unique map  $f: \mathbb{Z} \rightarrow \mathbb{Q}$ . Then  $0 \leq \mathbb{Q}$  is maximal, but  $f^{-1}0 = \ker f = 0$  is not maximal in  $\mathbb{Z}$ .

**1.2.9 Theorem**

For every ring  $R$  and every proper ideal  $I < R$ , there is a maximal ideal  $\mathfrak{m}$  containing  $I$ .

**Proof**

Consider the set of all proper ideals containing  $I$ . This is nonm-empty, as it contains  $I$ . Furthermore, the union of every chain of proper ideals is a proper ideal; it being an ideal is easy to see, and it is proper since 1 is not in any element of the chain. Thus by Zorn, this set has a maximal element, which must be a maximal ideal. ■

**1.2.10 Corollary**

Let  $x \in R$ , then  $x$  is a unit iff  $x$  is not in any maximal ideal. That is,

$$R^\times = \bigcap_{\mathfrak{m}} \mathfrak{m}^c,$$

where the intersection is over all maximal ideals of  $R$ .

**Proof**

An element is a unit iff it is not contained in any proper ideal. This is clear enough: if  $x$  is not a unit then  $(x)$  is proper, and proper ideals don't contain units. Every proper ideal can be extended to a maximal ideal, so the result follows. ■

**1.2.11 Definition**

Let  $R$  be a ring. Then define its **nilradical** to be  $\text{nil}(R) = \sqrt{0}$ , i.e. the ideal of all nilpotent elements.

Clearly  $R$  is reduced iff  $\text{nil}(R) = 0$ .

**1.2.12 Proposition**

Let  $I \leq R$  be an ideal, and let  $\pi: R \rightarrow R/I$  be the projection. Then  $\sqrt{I} = \pi^{-1}\text{nil}(R/I)$ .

**Proof**

We note that if  $x^n \in I$  then  $\pi(x)^n \in I$  so that  $\pi(x) \in \text{nil}(R/I)$  as desired. Conversely, if  $\pi(x) \in \text{nil}(R/I)$  then  $\pi(x^n) = I$  so that  $x^n \in I$  as desired. ■

**1.2.13 Proposition**

For a ring  $R$ ,

- (1)  $R/\text{nil}(R)$  is reduced;
- (2) The nilradical is the intersection of all prime ideals:

$$\text{nil}(R) = \bigcap_{\mathfrak{p}} \mathfrak{p}.$$



### Proof

We noted above that  $\text{nil}R = \sqrt{\text{nil}R} = \pi^{-1}\text{nil}(R/\text{nil}(R))$ . But we have also that  $\text{nil}R = \pi^{-1}0$ , so  $\pi^{-1}0 = \pi^{-1}\text{nil}(R/\text{nil}(R))$  and since  $\pi$  is surjective this means  $\text{nil}(R/\text{nil}(R)) = 0$  so that  $R/\text{nil}(R)$  is reduced.

Clearly if  $x \in \text{nil}(R)$  then  $x \in \mathfrak{p}$  for every prime  $\mathfrak{p}$  since  $x^n = 0 \in \mathfrak{p}$ . Now if  $x \notin \text{nil}(R)$  then we claim there exists a prime ideal  $\mathfrak{p}$  such that  $x \notin \mathfrak{p}$ . We prove this using Zorn. Let  $T = \{1, x, x^2, \dots\}$  and let  $S$  be the poset of all ideals  $I$  such that  $I \cap T = \emptyset$ ; note that all ideals in  $S$  are proper. As explained before the union of a chain of ideals is an ideal, and clearly if all of the ideals in the union don't intersect  $T$ , neither does the union. Thus by Zorn,  $S$  has a maximal element  $\mathfrak{p}$ .

We claim that  $\mathfrak{p}$  is prime. Indeed, suppose that  $ab \in \mathfrak{p}$  but  $a, b \notin \mathfrak{p}$ . Then by maximality we have that  $x^n \in (a, \mathfrak{p})$  and  $x^m \in (b, \mathfrak{p})$  for some  $n, m \geq 0$ . But then  $x^{n+m} \in (a, \mathfrak{p})(b, \mathfrak{p}) \subseteq \mathfrak{p}$ , a contradiction.

Thus there is a prime ideal  $\mathfrak{p}$  such that  $\mathfrak{p} \cap T = \emptyset$ , in particular  $x \notin \mathfrak{p}$ . So if  $x \notin \text{nil}(R)$  then  $x \notin \bigcap_{\mathfrak{p}} \mathfrak{p}$ , as desired. ■

#### 1.2.14 Corollary

Let  $I$  be an ideal of  $R$ , then

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p},$$

where the intersection is over all prime ideals containing  $I$ .

### Proof

We showed above that  $\sqrt{I} = \pi^{-1}\text{nil}(R/I)$ . By above  $\text{nil}(R/I)$  is the intersection over all prime ideals of  $R/I$ , which are all prime ideals of  $R$  containing  $I$ :

$$\sqrt{I} = \pi^{-1} \bigcap_{I \leq \mathfrak{p}} \bar{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \pi^{-1}\bar{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p}. \quad \blacksquare$$

#### 1.2.15 Definition

Let  $R$  be a ring, then define its **Jacobson radical** to be

$$J(R) = \bigcap_{\mathfrak{m}} \mathfrak{m},$$

the intersection of all maximal ideals.

#### 1.2.16 Lemma

Let  $\mathfrak{m} \leq R$  be maximal. Then  $x \notin \mathfrak{m}$  iff there is a  $y \in R$  such that  $1 - xy \in \mathfrak{m}$ .

### Proof

$x \notin \mathfrak{m}$  iff  $\bar{x} \in R/\mathfrak{m}$  is nonzero, or equivalently is a unit. This is iff there is a  $y \in R$  such that  $\overline{xy} = \bar{1}$ . This just means  $1 - xy \in \mathfrak{m}$ , as desired. ■

**1.2.17 Corollary**

For a ring  $R$ , its Jacobson radical  $J(R)$  is the set of all elements  $x \in R$  such that  $1 - yx \in R^\times$  for all  $y \in R$ .

**Proof**

Now if  $x \notin J(R)$  then  $x \notin \mathfrak{m}$  for some  $\mathfrak{m}$ , and then  $1 - yx \in \mathfrak{m}$  for some  $y$  and thus is not a unit. And if  $1 - yx$  is not a unit, then  $1 - yx \in \mathfrak{m}$  for some maximal  $\mathfrak{m}$  and thus  $x \notin \mathfrak{m}$  so that  $x \notin J(R)$ . ■

**1.2.18 Definition**

A ring  $R$  is **local** if it has a unique maximal ideal  $\mathfrak{m}$ . The field  $k = R/\mathfrak{m}$  is called its **residue field**. By abuse of notation we write  $(R, \mathfrak{m})$  for a local field.

**1.2.19 Proposition**

- (1) If  $(R, \mathfrak{m})$  is a local ring then  $R - \mathfrak{m} \subseteq R^\times$ .
- (2) Conversely if  $I < R$  is a proper ideal with  $R - I \subseteq R^\times$  then  $(R, I)$  is a local ring.
- (3) If  $\mathfrak{m} \leq R$  is maximal such that  $1 + \mathfrak{m} \subseteq R^\times$  then  $(R, \mathfrak{m})$  is local.

**Proof**

If  $x \notin \mathfrak{m}$  then  $(x)$  is not contained in  $\mathfrak{m}$ , and since  $\mathfrak{m}$  is the unique maximal ideal this means that  $(x) = R$  so that  $x$  is a unit. Also we noted that  $R^\times$  is the intersection of all complements of maximal ideals, so this follows from that as well.

Conversely, we have that

$$R - I \subseteq R^\times \bigcap_{\mathfrak{m}} \mathfrak{m}^c \implies I \supseteq \bigcup_{\mathfrak{m}} \mathfrak{m},$$

which means that  $I$  contains all maximal ideals, which can only happen if it is the unique maximal ideal.

We want to show that  $R - \mathfrak{m} \subseteq R^\times$ . Indeed, if  $x \notin \mathfrak{m}$  then there is a  $y$  such that  $1 - yx \in \mathfrak{m}$  so that  $yx \in 1 + \mathfrak{m}$  and is thus a unit. So  $x$  is a unit, as desired. ■

**1.2.20 Example**

For  $p \in \mathbb{Z}$  prime, define

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \notin (p) \right\} \subseteq \mathbb{Q}, \quad I = \left\{ \frac{a}{b} \in \mathbb{Z}_{(p)} \mid a \in (p) \right\}.$$

Then  $(\mathbb{Z}_{(p)}, I)$  is local. Indeed, if  $\frac{a}{b} \notin I$  then it is invertible (its inverse is  $\frac{b}{a}$ ). We can generalize this, as we will in the next section.

**1.3 Modules**

**1.3.1 Definition**

Let  $A$  be a ring, then a **module** over  $A$  is an Abelian group  $M$  along with a ring morphism  $\varphi: A \rightarrow \text{end}(M)$ . We define **multiplication** to be a function  $\mu: A \times M \rightarrow M$ , written by juxtaposition, by

$$\mu(a, m) = \varphi(a)m.$$

An  $A$ -module can also be defined via multiplication: require that multiplication distributes both ways, and  $1 \in A$  is an identity:

$$(a + b)m = am + bm, \quad a(m + n) = am + an, \quad a(bm) = (ab)m, \quad 1m = m.$$

**1.3.2 Definition**

For two  $A$ -modules  $M$  and  $N$ , an  $A$ -**module morphism** is a map  $f: M \rightarrow N$  which is an Abelian group morphism and also multiplicative:

$$f(am) = af(m)$$

for all  $a \in A, m \in M$ .

**Note:**

I'm going to skip the rest of the standard definitions, for Chrissake.

**1.3.3 Definition**

If  $M$  is an  $A$ -module, we define  $\text{ann}_A(M) \leq A$  the **annihilator** of  $M$  by

$$\text{ann}_A(M) = \{a \in A \mid aM = 0\} = \ker \varphi,$$

where  $\varphi$  is the module map  $\varphi: A \rightarrow \text{end}(M)$ . This is then an ideal. For  $S \subseteq M$  a subset we define  $\text{ann}_A(S)$  similarly.

A module  $M$  is **faithful** if  $\text{ann}_A(M) = 0$ .

$\text{ann}_A(S)$  is equal to  $\text{ann}_A(\langle S \rangle)$ , the annihilator of the submodule generated by  $S$  (all  $R$ -combinations of elements from  $S$ ), and is thus also an ideal.

**1.3.4 Definition**

For a ring  $A$ , an  $A$ -**algebra** is a ring  $B$  equipped with a ring morphism  $\varphi: A \rightarrow B$ .

Note that  $B$  can be embedded into  $\text{end}(B)$  mapping  $b \in B$  to multiplication by  $b$ . Thus an  $A$ -algebra is an  $A$ -module with multiplication given by  $a \cdot b = \varphi(a)b$ .

**1.3.5 Definition**

For two  $A$ -algebras  $B, C$ , an  $A$ -**algebra morphism** is a map  $f: B \rightarrow C$  which is both a ring morphism and an  $A$ -module morphism.

If  $\varphi: A \rightarrow B, \psi: A \rightarrow C$  are the algebra maps, then  $f: B \rightarrow C$  is an algebra morphism iff it is a ring morphism and  $\psi = f \circ \varphi$ :

$$\psi(a) = a1_C = af(1_B) = f(a1_B) = f \circ \varphi(a).$$

**1.3.6 Definition**

For an  $A$ -module  $M$  and a submodule  $N$ , its quotient module  $M/N$  is the set of cosets  $m + N$  with the usual  $M$ -module structure. The usual isomorphism results hold.

**1.3.7 Proposition**

Let  $I \leq A$  be a module, then every  $A/I$ -module is an  $A$ -module with  $\text{ann}_A(M) \supseteq I$  and vice-versa. In particular an  $A$ -module  $M$  can be viewed as a faithful  $A/\text{ann}_A(M)$ -module.

**Proof**

Consider the projection  $\pi: A \rightarrow A/I$ . Composing with  $\varphi: A/I \rightarrow \text{end}(M)$  gives an  $A$ -module structure to an  $A/I$ -module and  $\ker(\varphi \circ \pi) \supseteq \ker \varphi = I$ . Conversely if  $\varphi: A \rightarrow \text{end}(M)$  has kernel containing  $I$  we can quotient it by  $I$ . ■

**1.3.8 Definition**

For a collection of  $A$ -modules  $\{M_\alpha\}_\alpha$  we define their **direct product** and **direct sum** by

$$\prod_\alpha M_\alpha = \{(m_\alpha) \mid m_\alpha \in M_\alpha\}, \quad \bigoplus_\alpha M_\alpha = \left\{ (m_\alpha) \in \prod_\alpha M_\alpha \mid \text{All but finitely many of } m_\alpha \text{ are zero} \right\}.$$

$\prod_\alpha M_\alpha$ 's module structure is given by pointwise operations, and  $\bigoplus_\alpha M_\alpha$  is viewed as a submodule. These form the products and coproducts in the category of  $A$ -modules.

**1.3.9 Definition**

An  $A$ -module  $M$  is **free** if

$$M \cong A^{\oplus S} = \bigoplus_{\alpha \in S} A$$

for some set  $S$ . Note that  $\{e_\alpha\}_{\alpha \in S}$  (the usual standard basis vectors) forms a basis of  $A^{\oplus S}$  in the sense that every element of  $A^{\oplus S}$  can be written as a unique combination of standard basis vectors. A collection of elements  $\{m_\alpha\}_{\alpha \in S}$  is a **basis** of  $M$  if mapping  $m_\alpha \mapsto e_\alpha$  extends to an isomorphism  $M \cong A^{\oplus S}$ .

A basis of  $M$  is equivalently a set of elements which generate  $M$  and are also linearly independent in the usual way.

A free module is called such because they are free in the usual sense with respect to the obvious forgetful functor  $\text{Mod}_A \rightarrow \text{Set}$ . That is,  $\text{hom}_A(A^{\oplus S}, B) \cong \text{hom}_{\text{Set}}(S, B)$ .

**1.3.10 Proposition**

If  $A^{\oplus S} \cong A^{\oplus S'}$  then  $S$  and  $S'$  have the same cardinality.

**Proof**

Let  $\mathfrak{m} \leq A$  be maximal, then the isomorphism  $A^{\oplus S} \cong A^{\oplus S'}$  induces an isomorphism  $(A/\mathfrak{m})^{\oplus S} \cong (A/\mathfrak{m})^{\oplus S'}$ .

$A/\mathfrak{m}$  is a field, and so this is an isomorphism of  $A/\mathfrak{m}$ -modules (vector spaces) of dimensions  $|S|, |S'|$ . Since the dimension is invariant, their cardinalities must be equal. ■

### 1.3.11 Definition

If  $M$  is a free  $A$ -module, then its **rank** is the cardinality of  $S$  where  $M \cong A^{\oplus S}$ .

### 1.3.12 Definition

An  $A$ -module  $M$  is **finitely generated (fg)** if there is a finite set  $S \subseteq M$  such that  $M = \langle S \rangle$ .

### 1.3.13 Proposition

An  $A$ -module is finitely generated iff it is a quotient of  $A^n$  for some  $n$  (up to isomorphism).

#### Proof

If  $\varphi: A^n \rightarrow M$  is surjective, then  $\varphi(e_i)$  generate  $M$  clearly and so it is finitely-generated. Thus if  $M \cong A^n/N$  is a quotient of  $A^n$  then the projection  $\pi: A^n \rightarrow A^n/N \cong M$  is surjective and thus  $M$  is finitely-generated. Conversely let  $m_i$  generate  $M$ , and define  $\varphi: A^n \rightarrow M$  by  $\varphi(e_i) = m_i$  which extends by linearity to all of  $A^n$  since it is free. This is surjective and then  $A^n/\ker \varphi \cong M$  as desired. ■

### 1.3.14 Proposition

Let  $M$  be a fg  $A$ -module, and  $I \leq A$  an ideal such that  $M = IM$ . Then there is an  $x \in I$  such that  $(1 - x)M = 0$  ( $xm = m$  for all  $m \in M$ ).

#### Proof

Let  $m_1, \dots, m_n$  generate  $M$ , we induct on  $n$ . For  $n = 0$  we have  $M = 0$  so this is clear. Consider  $N = M/(m_n)$ , then  $IN = (IM + (m_n))/(m_n) = M/(m_n) = N$  and  $N$  is generated by  $n - 1$  elements, so by induction there is an  $x \in I$  such that  $(1 - x)N = 0$ . Meaning  $(1 - x)M \subseteq (m_n)$ , and since  $m_n \in M = IM$  we have

$$(1 - x)m_n \in (1 - x)IM = I(1 - x)M \subseteq I(m_n),$$

so  $(1 - x)m_n = ym_n$  for some  $y \in I$  and thus  $(1 - x - y)m_n = 0$ . Now  $(1 - x - y)(1 - x) \in 1 + I$  and satisfies

$$(1 - x - y)(1 - x)M \subseteq (1 - x - y)(m_n) = 0$$

as desired. ■

### 1.3.15 Corollary (Nakayama's lemma)

Let  $M$  be a fg  $A$ -module, such that  $J(A)M = M$ , then  $M = 0$ .

#### Proof

By above there is an  $x \in J(A)$  such that  $(1-x)M = 0$ . Since  $x \in J(A)$ ,  $1-x \in A^\times$  so that  $(1-x)M = M = 0$  as desired. ■

### 1.3.16 Corollary

Let  $M$  be a fg  $A$ -module and  $N \subseteq M$  a submodule such that  $M = IM + N$  for some  $I \subseteq J(A)$ , then  $M = N$ .

#### Proof

Consider  $M/N$ , then  $I(M/N) = (IM + N)/N = M/N$ , so by above  $M/N = 0$  meaning  $M = N$ . ■

### 1.3.17 Corollary

Let  $(A, \mathfrak{m})$  be local,  $M$  a fg  $A$ -module. Then  $x_1, \dots, x_n \in M$  generate  $M$  iff  $\bar{x}_1, \dots, \bar{x}_n \in M/\mathfrak{m}M$  generate  $M/\mathfrak{m}M$  over  $A/\mathfrak{m}$ .

#### Proof

If  $x_i$  generate  $M$  then  $\bar{x}_i$  clearly generate  $M$ . Conversely, let  $N = (x_1, \dots, x_n)$  be the submodule generated by  $M$ , and consider the composition  $f: N \rightarrow M \rightarrow M/\mathfrak{m}M$ .  $f$ 's image is the span of  $(\bar{x}_i)_i$ , so all of  $M/\mathfrak{m}M$ . Thus  $f$  is surjective, so  $M = N + \mathfrak{m}M$ , since for every  $m \in M$  there is an  $n \in N$  such that  $f(n) - m \in \mathfrak{m}M$  by surjectivity, and by above since  $J(A) = \mathfrak{m}$  this means  $M = N$  as desired. ■

### 1.3.18 Definition

Recall the definition of an exact sequence:

$$M \xrightarrow{f} N \xrightarrow{g} P$$

is **exact** at  $N$  if  $\text{im } f = \ker g$ .

## 2 Localization

### 2.1 Localization of rings

Consider the following problem: for a ring  $A$  and a subset  $S \subseteq A$ , we want a ring  $S^{-1}A$  and a morphism  $\iota: A \rightarrow S^{-1}A$  such that any morphism  $f: A \rightarrow B$  such that  $f(S) \subseteq B^\times$ ,  $f$  factors uniquely through  $\iota$ . That is, there is a unique  $\hat{f}: S^{-1}A \rightarrow B$  such that  $f = \hat{f} \circ \iota$ :

$$\begin{array}{ccc} A & \xrightarrow{\quad \iota \quad} & S^{-1}A \\ & \searrow f & \downarrow \exists! \\ & & B \end{array}$$

This is a universal property, and as such any solution is unique up to unique isomorphism. In this section we will investigate the construction of such a solution.

### 2.1.1 Definition

A subset  $S \subseteq A$  is **multiplicatively closed** if  $1 \in S$  and for  $a, b \in S$  we have  $ab \in S$ .

It is sufficient to consider only the case that  $S$  is multiplicatively closed. Otherwise, let  $\hat{S}$  be  $S$ 's multiplicative closure: the set of all finite products of elements from  $S$ . Then  $\hat{S}^{-1}R$  is a solution to the problem for  $S$ . Indeed, if  $f(S) \subseteq B^\times$ , then  $f(\hat{S}) \subseteq B^\times$  as well, as the product of units. Then there is a unique factoring through  $\hat{S}$ 's  $\iota$ .

### 2.1.2 Definition

For  $S \subseteq A$  multiplicatively closed, define an equivalence on  $A \times S$  by  $(a, s) \sim (b, t)$  iff there is a  $u \in S$  such that  $u(at - bs) = 0$ , in which case we write  $(a, s) \sim_u (b, t)$ .

### 2.1.3 Lemma

This is indeed an equivalence relation.

#### Proof

The subrelations  $\sim_u$  are all clearly reflexive and symmetric, thus their union  $\sim$  is as well. Now if  $(a, s) \sim_\alpha (b, t) \sim_\beta (c, v)$  then  $\alpha at = \alpha sb$  and  $\beta bv = \beta ct$ . So then  $\alpha t \beta av = \alpha sb \beta v = \alpha \beta tcs$  so that  $(a, s) \sim_{\alpha\beta} (c, v)$ . ■

### 2.1.4 Example

A more naive equivalence would be  $\sim_1$ , i.e.  $(a, s) \sim (b, t)$  iff  $at - bs = 0$ . This corresponds to  $\sim$  when  $A$  is an integral domain, but is not generally an equivalence.

### 2.1.5 Definition

For  $S \subseteq A$  multiplicatively closed, define  $S^{-1}A$  to be the quotient of  $A \times S$  by  $\sim$ . Write the equivalence class of  $(a, s)$  by  $a/s$ . We give this a ring structure by

$$0 = 0/1, \quad 1 = 1/1, \quad a/s + b/t = (at + bs)/st, \quad a/s \cdot b/t = (ab)/(st).$$

Define  $\iota: A \rightarrow S^{-1}A$  by  $\iota(a) = a/1$ .

### 2.1.6 Proposition

$S^{-1}A$  is indeed a ring, and  $\iota$  is a ring morphism.

#### Proof

We need to show that addition and multiplication are well-defined. Indeed, if  $(a, s) \sim_u (a', s')$  and  $(b, t) \sim_v (b', t')$  then  $(at + bs, st) \sim_{uv} (a't' + b's', s't')$  which can be verified. And similarly  $(ab, st) \sim_{uv} (a'b', s't')$ . From well-definedness it is easy to see that addition and multiplication define a ring structure and  $\iota$  is a ring morphism. ■

A few properties are immediate:

- (1) For  $s \in S$ ,  $\iota(s) = 1/s$  has an inverse  $s/1$ , so that  $\iota(S) \subseteq (S^{-1}R)^\times$ ;
- (2) We have that  $a/s = 0/1$  iff there is a  $t \in S$  such that  $at = 0$ . In particular,  $\iota(a) = 0$  iff there is a  $t \in S$  such that  $ta = 0$ .
- (3) Every element of  $S^{-1}A$  is of the form

$$a/s = a/1 \cdot 1/s = \iota(a) \cdot \iota(s)^{-1}.$$

### 2.1.7 Theorem

This construction solves the above universal property.

#### Proof

Let  $f: A \rightarrow B$  be such that  $f(S) \subseteq B^\times$ : we need to show that there is a unique map  $g: S^{-1}A \rightarrow B$  such that  $f = g \circ \iota$ . Indeed, we must define

$$g(a/s) = g(\iota(a)\iota(s)^{-1}) = (g \circ \iota(a)) \cdot (g \circ \iota(s))^{-1} = f(a)f(s)^{-1}.$$

This shows that  $g$  is unique, if it exists. We need to show that this map is well-defined and a ring morphism. If  $a/s = b/t$  then we must have that  $f(a)f(s)^{-1} = f(b)f(t)^{-1}$  so that  $g$  is well-defined. Indeed, if  $u(at - bv) = 0$  for  $u \in S$ , we have

$$f(u) \cdot (f(at) - f(bv)) = 0$$

and since  $f(S) \subseteq B^\times$  we have that  $f(u)$  is a unit so that  $f(at) - f(bv) = 0$ , meaning this is well-defined. And this is a ring morphism, as can be checked easily. ■

### 2.1.8 Corollary

Let  $f: A \rightarrow B$  be a ring morphism such that

- (1)  $f(S) \subseteq B^\times$ ,
- (2) if  $f(a) = 0$  then  $ta = 0$  for some  $t \in S$ ,
- (3) every element of  $B$  is of the form  $f(a)f(s)^{-1}$  for  $a \in A$  and  $s \in S$ .

Then there is a unique isomorphism  $g: S^{-1}A \rightarrow B$  such that  $g \circ \iota = f$ .

#### Proof

$f$  solves the universal problem above for  $S$  as well. If  $h: A \rightarrow C$  is a map where  $h(S) \subseteq C^\times$ , then we need to define  $\hat{h}: B \rightarrow C$  by

$$\hat{h}(f(a)f(s)^{-1}) = h(a)h(s)^{-1},$$

which is well-defined and a ring morphism for similar reasons as above. Since universal properties are unique up to unique isomorphism, the result follows.

Alternatively we can prove this directly: let  $g: S^{-1}A \rightarrow B$  be the unique map where  $g \circ \iota = f$  by the universal property. By definition  $g(a/s) = f(a)f(s)^{-1}$ . By the third point  $g$  is surjective, and it is injective since if  $g(a/s) = 0$  we have  $f(a) = 0$  so that  $ta = 0$ , which means that  $a/s = 0$ . ■

### 2.1.9 Example



If  $A$  is an integral domain, then  $S = A - 0$  is multiplicatively closed, and  $S^{-1}A$  is the field of fractions of  $A$ .

### 2.1.10 Proposition

For  $A \neq 0$ , then  $S^{-1}A = 0$  iff  $0 \in S$ .

#### Proof

We have that  $S^{-1}A = 0$  iff  $1/1 = 0/1$  iff there is a  $s \in S$  such that  $s \cdot 1 = 0$ . ■

### 2.1.11 Definition

Let  $\mathfrak{p} \subseteq A$  be prime. Then  $S = A - \mathfrak{p}$  is multiplicatively closed, and we denote its localization by  $A_{\mathfrak{p}} = S^{-1}A$ .

For  $a \in A$ ,  $S = \{1, a, a^2, \dots\}$  is multiplicatively closed, and we denote its localization by  $A_a = S^{-1}A$ .

For  $a \in A$  not nilpotent (so  $A_a \neq 0$ ) define a map

$$\varphi: A[x] \rightarrow A_a, \quad x \mapsto 1/a.$$

This map is clearly surjective, and  $p(x) = \sum_k^n b_k x^k \in A[x]$  is mapped to zero iff

$$\sum_{k=0}^n b_k \frac{1}{a^k} = \frac{\sum_k b_k a^{n-k}}{a^n} = 0.$$

This is iff there is a  $t$  such that

$$a^t \sum_k b_k a^{n-k} = 0.$$

Defining  $c_k$  recursively by  $c_0 = -b_0$  and  $c_k = ac_{k-1} - b_k$  gives a polynomial such that  $p(x) = (ax - 1)q(x)$ , so  $p \in (ax - 1)$  (the equality above ensures the recursion terminates). Thus  $\ker \varphi = (ax - 1)$  and thus

$$A_f \cong \frac{A[x]}{(ax - 1)}.$$

#### Note:

Note that we have two distinct localizations here, which have similar names in some cases. If  $A$  is a PID and  $p \in A$  is irreducible, then  $A_{(p)}$  is localization by  $A - (p)$ , which  $A_p$  is localization by  $\{p^n\}_{n < \omega}$ . Elements of the former are of the form  $a/s$  where  $p$  does not divide  $s$ , while elements of the former are of the form  $a/s$  where  $s = p^n$ . Quite different!

### 2.1.12 Definition

For a ring morphism  $f: A \rightarrow B$  and an ideal  $I \subseteq A$ , define the ideal  $IB = (f(I))$  of  $B$ . Explicitly,  $IB$  is the set of all finite sums  $\sum_i f(a_i)b_i$  for  $a_i \in I$ .

In general we have  $f^{-1}(f(I)) \supseteq I$  and  $f(f^{-1}(J)) \subseteq J$ . These imply  $f^{-1}(IB) \supseteq I$  and  $f^{-1}(J)B \subseteq J$ .

**2.1.13 Definition**

For two ideals  $I, J$ , define  $(I : J) = \{x \in A \mid xJ \subseteq I\}$ . This is an ideal containing  $I$ . For  $a \in A$ , define  $(I : a) = (I : (a)) = \{x \in A \mid xa \subseteq I\}$ .

For  $S \subseteq R$  multiplicatively closed, define the ideal  $S^{-1}I = IS^{-1}A \subseteq S^{-1}A$ . This is the ideal generated by  $\iota(I)$ .

So we have two maps:  $S^{-1}\bullet$  mapping ideals of  $A$  to ideals of  $S^{-1}A$ , and  $\iota^{-1}$  mapping ideals of  $S^{-1}A$  to  $A$ :

$$\{\text{Ideals of } A\} \xrightleftharpoons[\iota^{-1}]{S^{-1}} \{\text{Ideals of } S^{-1}A\}.$$

We wish to relate the two.

**2.1.14 Proposition**

- (1) For an ideal  $I \subseteq A$ ,  $S^{-1}I = \{a/s \mid a \in I, s \in S\}$ .
- (2) For an ideal  $J \subseteq S^{-1}A$ , we have  $S^{-1}\iota^{-1}J = J$ .
- (3) For an ideal  $I \subseteq A$ , we have  $\iota^{-1}S^{-1}I = \bigcup_{s \in S} (I : s)$ .
- (4) We have  $S^{-1}I = S^{-1}A$  iff  $I \cap S \neq \emptyset$ .
- (5) For  $\mathfrak{p} \subseteq A$  prime, if  $I \cap S \neq \emptyset$  then  $S^{-1}\mathfrak{p} \subseteq S^{-1}A$  is prime (otherwise it is all of  $S^{-1}A$ ).
- (6) The correspondence  $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$  is a bijection between prime ideals of  $S^{-1}A$  and prime ideals of  $A$  not intersecting  $S$ . Its inverse is  $S^{-1}$ . That is, we have a one-to-one correspondence:

$$\{\text{Prime ideals of } A \text{ not intersecting } S\} \xrightleftharpoons[\iota^{-1}]{S^{-1}} \{\text{Prime ideals of } S^{-1}A\}.$$

**Proof**

- (1)  $\{a/s \mid a \in I, s \in S\}$  is an ideal and contains  $\iota(I)$ . It is clearly closed under multiplication by  $S^{-1}A$  and for  $a/s + b/t = (at + bs)/(st)$ , notice that  $at + bs \in I$ . By definition  $S^{-1}I$  is the smallest ideal containing  $\iota(I)$ , and it consists of finite sums  $\sum_i \iota(a_i)x_i/s_i$ . This clearly contains  $a/s$ , and thus they must be equal.
- (2) As explained before, in general we have  $S^{-1}\iota^{-1}J = \iota^{-1}(J)S^{-1}A \subseteq J$ , we must prove the opposite inclusion. As above, we have  $S^{-1}\iota^{-1}J = \{a/s \mid \iota(a) \in J, s \in S\}$ . For  $x/s \in J$ , we have that  $\iota(x) = x/1 \in J$  and so  $x/s \in S^{-1}\iota^{-1}J$  as desired.
- (3) If  $a \in \iota^{-1}S^{-1}I$  then  $\iota(a) \in S^{-1}I$  so by the first point  $a/1 = b/s$  for  $b \in I$ . Then  $asu = bu$  for some  $u \in S$  and since  $bu \in J$  this means that  $a \in (I : su)$ . Conversely if  $a \in (I : s)$  for  $s \in S$  then  $as \in I$  and so  $\iota(a) = a/1 = (as)/s \in S^{-1}I$ .
- (4) If  $S^{-1}I = S^{-1}A$  then  $1/s = a/t$  for  $a \in I$ , so  $as = t$  meaning  $as \in I \cap S$ . Conversely if  $a \in I \cap S$  then  $1/1 = a/a \in S^{-1}I$ .
- (5) If  $S \cap \mathfrak{p} = \emptyset$  then we need to show that  $S^{-1}\mathfrak{p}$  is prime. If  $a_1/s_1, a_2/s_2 \notin S^{-1}\mathfrak{p}$  we want to show  $(a_1a_2)/(s_1s_2) \notin S^{-1}\mathfrak{p}$ . By assumption  $a_1, a_2 \notin \mathfrak{p}$ .

Now we claim that if  $a \notin \mathfrak{p}$ ,  $a/s \notin S^{-1}\mathfrak{p}$ , which would complete the proof. Assume the contrary:  $a/s = b/t$  for  $b \in \mathfrak{p}$  then there is a  $u \in S$  such that  $atu = bsu$ . Then  $atu \in \mathfrak{p}$ , but  $a \notin \mathfrak{p}$  and  $t, u \in S$  and thus also are not in  $\mathfrak{p}$ , a contradiction to  $\mathfrak{p}$ 's primality.

- (6) Preimages of prime ideals are prime, so  $\iota^{-1}$  maps prime ideals to prime ideals. By the second point  $S^{-1}\iota^{-1}\mathfrak{q} = \mathfrak{q}$  for all prime ideals  $\mathfrak{q} \subseteq S^{-1}A$ . It remains to show that  $\iota^{-1}S^{-1}\mathfrak{p} = \mathfrak{p}$  for all prime  $\mathfrak{p} \subseteq A$  which don't intersect  $S$ . We know that  $\mathfrak{p} \subseteq \iota^{-1}S^{-1}\mathfrak{p}$  as is true in general, so we want to show that  $\iota^{-1}S^{-1}\mathfrak{p} = \bigcup_{s \in S} (\mathfrak{p} : s)$  is contained in  $\mathfrak{p}$ . Indeed, if  $a \in (\mathfrak{p} : s)$  then  $as \in \mathfrak{p}$  and since  $s \notin \mathfrak{p}$  we have that  $a \in \mathfrak{p}$  as desired. ■

### 2.1.15 Corollary

Let  $\mathfrak{p} \subseteq A$  be prime, and consider its localization  $A_{\mathfrak{p}}$ . Then  $A_{\mathfrak{p}}$  is a local ring with maximal ideal  $\mathfrak{p}A_{\mathfrak{p}}$ . Moreover, the correspondence  $\mathfrak{q} \rightarrow \mathfrak{q}A_{\mathfrak{p}}$  is a bijection between prime ideals of  $A_{\mathfrak{p}}$  and prime ideals  $\mathfrak{q} \subseteq \mathfrak{p}$ .

#### Proof

Let  $S = A - \mathfrak{p}$ , so that  $A_{\mathfrak{p}} = S^{-1}A$ , and  $\mathfrak{p}A_{\mathfrak{p}} = S^{-1}\mathfrak{p}$ . The last point is clear:  $\mathfrak{q} \cap S = \emptyset$  iff  $\mathfrak{q} \subseteq \mathfrak{p}$ .  $A_{\mathfrak{p}}$  being local is also clear:  $S^{-1}\bullet$  is order-preserving, so  $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{p}$  for all prime ideals  $\mathfrak{q}$ . Thus  $S^{-1}\mathfrak{p}$  is maximal. ■

### 2.1.16 Corollary

If  $a \in A$  is not nilpotent, then there is a prime ideal  $\mathfrak{p} \subseteq A$  such that  $a \notin \mathfrak{p}$ .

#### Proof

As before, consider the multiplicative set  $S = \{a^n\}_{n < \omega}$ , and  $A_a = S^{-1}A$ . Since  $a$  is not nilpotent,  $0 \notin S$ , so that  $A_a \neq 0$ . So take a prime ideal  $\mathfrak{p} \subseteq A_a$ , and then  $\iota^{-1}\mathfrak{p}$  is a prime ideal of  $A$  not intersecting  $S$ , thus not containing  $a$ . ■

#### Note:

This completes the proof before that

$$\text{nil}(A) = \bigcap_{\mathfrak{p}} \mathfrak{p}$$

but without appealing to Zorn, and thus not relying on the Axiom of Choice.

## 2.2 Localization of modules

### 2.2.1 Definition

Let  $S \subseteq A$  be multiplicatively closed and  $M$  an  $A$ -module. Define an equivalence on  $M \times S$  by  $(m, s) \sim (m', s')$  iff there is a  $t \in S$  such that  $t(s'm - sm') = 0$ . This is an equivalence relation and we denote the equivalence class of  $(m, s)$  by  $m/s$ . The quotient  $S^{-1}M$  forms an  $A$ -module by

$$m/s + m'/s' = (s'm + sm')/(ss'), \quad a \cdot (m/s) = (am)/s.$$

Note that localization at  $S$  forms an additive functor  $S^{-1}: \text{Mod}_A \rightarrow \text{Mod}_{S^{-1}A}$ . Indeed,  $S^{-1}M$  is an  $S^{-1}A$  module by

$$a/s \cdot m/s' = (am)/(ss').$$

And if  $f: M \rightarrow N$  is an  $A$ -module morphism, then define

$$S^{-1}f: S^{-1}M \rightarrow S^{-1}N, \quad S^{-1}f(m/s) = f(m)/s.$$

It is not hard to see that this is functorial and additive.

There is the canonical morphism  $\iota: A \rightarrow S^{-1}A$  and as such it lifts to a “forgetful” functor  $\text{Mod}_{S^{-1}A} \rightarrow \text{Mod}_A$ , where for  $M \in \text{Mod}_{S^{-1}A}$  we view it as an  $A$ -module by  $a \cdot M = \iota(a) \cdot M$ .

### 2.2.2 Proposition

The functor  $S^{-1}: \text{Mod}_A \rightarrow \text{Mod}_{S^{-1}A}$  is left adjoint to the forgetful functor  $\text{Mod}_{S^{-1}A} \rightarrow \text{Mod}_A$ .

#### Proof

We need to show that there is a canonical bijection

$$\text{hom}_{S^{-1}A}(S^{-1}M, N) \cong \text{hom}_A(M, N),$$

for all  $A$ -modules  $M$  and  $S^{-1}A$ -modules  $N$ . For  $f: M \rightarrow N$  map it to  $\hat{f}: S^{-1}M \rightarrow N$  by  $\hat{f}(m/s) = 1/s \cdot f(m)$ . And for  $g: S^{-1}M \rightarrow N$  map it to  $\tilde{g}: M \rightarrow N$  by  $\tilde{g}(m) = g(m/1)$ . ■

### 2.2.3 Definition

An additive functor  $F: \text{Mod}_A \rightarrow \text{Mod}_B$  is

- (1) **left exact** if it preserves left-exact sequences: if

$$0 \rightarrow M \xrightarrow{f} N \xrightarrow{g} P$$

is exact, then so is

$$0 \rightarrow FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP.$$

- (2) **right exact** if it preserves right-exact sequences: if

$$M \xrightarrow{f} N \xrightarrow{g} P \rightarrow 0$$

is exact, then so is

$$FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP \rightarrow 0.$$

- (3) **exact** if it is both left and right exact.

### 2.2.4 Proposition

An additive functor is

- (1) left exact iff it preserves kernels iff it preserves finite limits iff it preserves the left side of a short exact sequence.
- (2) right exact iff it preserves cokernels iff it preserves finite colimits iff it preserves the right side of a short exact sequence.
- (3) exact iff it preserves finite limits and colimits iff it preserves short exact sequences iff it preserves all exact sequences.

### Proof

Left-exact sequences are kernel diagrams, and right-exact sequences are cokernel diagrams. So by definition, left exact functors are those which preserve kernels, and right exact functors are those which preserve cokernels. Since additive functors preserve finite biproducts, left/right exact functors preserve limits/colimits. Clearly left-exact functors preserve the left side of a short exact sequence (similarly for right-exact functors). To show the converse, if left-exact sequences can be quotiented to give an appropriate short exact sequence, similar for right-exact sequences. ■

#### 2.2.5 Corollary

Left adjoints are right exact, and right adjoints are left exact.

### Proof

Left adjoints preserve all colimits, and right adjoints all limits. ■

#### 2.2.6 Corollary

$S^{-1}$  is an exact functor.

### Proof

By the above corollary, it is right-exact. To show that it is exact, it is sufficient to show that it preserves injections. If  $f: M \rightarrow N$  is injective then  $m/s \in \ker S^{-1}f$  iff  $f(m)/s = 0/1$ , so that  $tf(m) = 0$  for some  $t \in S$ . But then  $tf(m) = f(tm)$  so  $tm \in \ker \varphi$ , so  $tm = 0$ , which means  $m/s = 0/1$  as desired. ■

In general if  $U: \text{Mod}_A \rightarrow \text{Mod}_B$  is a restriction functor corresponding to a morphism  $\iota: B \rightarrow A$ , then it has a left adjoint via the tensor product. In particular, viewing  $A$  as a  $(A, B)$ -module, then  $A \otimes_B M$  has an  $A$ -module structure (given by  $a(a' \otimes m) = (aa') \otimes m$ ).  $U$ 's left adjoint is then  $A \otimes_B M$ .

Since left adjoints are unique up to natural isomorphism,

$$S^{-1}M \cong S^{-1}A \otimes_A M.$$

Thus  $S^{-1}A \otimes_A \bullet$  is an exact functor, which means that  $S^{-1}A$  is a **flat**  $A$ -module. This can also be shown directly:  $S^{-1}A$  is the filtered colimit of free  $A$ -modules, and is thus flat.

#### 2.2.7 Corollary

$S^{-1}$  preserves arbitrary coproducts:

$$S^{-1} \bigoplus_{\alpha} M_{\alpha} \cong \bigoplus_{\alpha} S^{-1} M_{\alpha}$$

naturally.

### Proof

Left adjoints preserve arbitrary colimits. ■

**Note:**

While  $S^{-1}$  is exact, it is not **continuous**: it does not preserve arbitrary limits (equivalently it does not preserve arbitrary products). That is, we do *not* have that

$$S^{-1} \prod_{\alpha} M_{\alpha} \cong \prod_{\alpha} S^{-1} M_{\alpha}.$$

This is of course true when the index set is finite, since then it is just a biproduct.

**2.2.8 Corollary**

- (1) If  $N \subseteq M$  is a submodule,  $S^{-1}N$  can be naturally embedded in  $S^{-1}M$ .
- (2) There is a natural isomorphism  $S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N)$ .
- (3) For  $f: M \rightarrow N$  of  $A$ -modules, we have natural isomorphisms

$$\ker S^{-1}f \cong S^{-1} \ker f, \quad \operatorname{im} S^{-1}f \cong S^{-1} \operatorname{im} f, \quad S^{-1} \operatorname{coker} f \cong \operatorname{coker} S^{-1}f.$$

- (4) For every two submodules  $M_1, M_2 \subseteq M$  under the natural embeddings,

$$S^{-1}(M_1 \cap M_2) = S^{-1}M_1 \cap S^{-1}M_2 \subseteq S^{-1}M.$$

**Proof**

The first point is due to  $S^{-1}$  being exact and thus preserving injections. The second point is because  $S^{-1}$  preserves the short exact sequence  $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$ . The third point is clear as well, with only  $\operatorname{im}$  not being immediately obvious.  $\operatorname{im} f$  can be characterized as  $\operatorname{im} f = \ker \operatorname{coker} f$ , so it follows.

For the last point, note that  $M_1 \cap M_2$  is the kernel of  $M_1 \oplus M_2 \rightarrow M$  by  $(x, y) \mapsto x - y$  (via the diagonal embedding), and so this follows from  $S^{-1}$ 's exactness. ■

**2.2.9 Definition**

For a prime ideal  $\mathfrak{p} \leq A$  and an  $A$ -module  $M$ , denote by  $M_{\mathfrak{p}}$  the localization of  $M$  by  $S = A - \mathfrak{p}$ .

**2.2.10 Lemma**

For an  $A$ -module  $M$ , consider the natural map

$$\varphi: M \rightarrow \prod_{\mathfrak{m}} M_{\mathfrak{m}}, \quad m \mapsto (m/1)_{\mathfrak{m}},$$

where the product is over all maximal ideals of  $A$ . Then  $\varphi$  is injective.

**Proof**

Assume there is an  $0 \neq x \in M$  such that  $\varphi(x) = 0$ . Then define

$$I = \text{ann}_A(x) = \{a \in A \mid ax = 0\}.$$

$I$  is a proper ideal of  $A$ , and thus is contained in a maximal ideal  $\mathfrak{m}$ . Then for every  $s \in A - \mathfrak{m}$  we have that  $s \notin I$  and so  $sx \neq 0$ , and thus  $m/1$  is not zero in  $M_{\mathfrak{m}}$ , a contradiction. ■

### 2.2.11 Corollary

For an  $A$ -module  $M$ , the following are equivalent:

- (1)  $M = 0$ ,
- (2)  $S^{-1}M = 0$  for every multiplicative set  $S \subseteq A$ ,
- (3)  $M_{\mathfrak{p}} = 0$  for every prime ideal  $\mathfrak{p} \leq A$ ,
- (4)  $M_{\mathfrak{m}} = 0$  for every maximal ideal  $\mathfrak{m} \leq A$ .

### Proof

Clearly each point implies the next. By above  $M$  embeds into  $\prod_{\mathfrak{m}} M_{\mathfrak{m}}$ , so if  $M_{\mathfrak{m}} = 0$  for all  $\mathfrak{m}$  then  $M = 0$ ; the last point implies the first. ■

### 2.2.12 Corollary

For a morphism of  $A$ -modules  $f: M \rightarrow N$ , the following are equivalent:

- (1)  $f$  is injective/surjective/an isomorphism,
- (2)  $S^{-1}f$  is injective/surjective/an isomorphism for every multiplicative  $S \subseteq A$ ,
- (3)  $f_{\mathfrak{p}}$  is injective/surjective/an isomorphism for every prime  $\mathfrak{p} \leq A$ ,
- (4)  $f_{\mathfrak{m}}$  is injective/surjective/an isomorphism for every maximal  $\mathfrak{m} \leq A$ .

### Proof

By  $S^{-1}$ 's exactness, the first point implies the second, and clearly all the other points imply the next. To show the last implies the first, let  $K = \ker f$  so that  $0 \rightarrow K \rightarrow M \xrightarrow{f} N$  is exact. Then  $0 \rightarrow K_{\mathfrak{m}} \rightarrow M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}}$  is exact, and so  $K_{\mathfrak{m}} = \ker f_{\mathfrak{m}} = 0$  for all  $\mathfrak{m}$ . By above this means  $K = 0$  so that  $f$  is injective. Similar for surjectivity. ■

## 3 Properties of Modules

### 3.1 Finitely generated, integral, and finite algebras

#### 3.1.1 Definition

For an  $A$ -algebra  $B$ , and  $S \subseteq B$  denote by  $A[S]$  the smallest  $A$ -subalgebra containing  $S$  in  $B$ .

Let  $\varphi: A \rightarrow B$  be the algebra map, then  $A[S]$  is the subring of  $B$  generated by  $\varphi(A) \cup S$ . Alternatively, for every  $s \in S$  define an indeterminate  $x_s$ , and let  $X = \{x_s\}_{s \in S}$  be the set of all these indeterminates. Consider the polynomial algebra  $A[X]$ , and define  $f: A[X] \rightarrow B$  by  $f(x_s) = s$ . Then  $A[S]$  is just the image of  $f$ . Explicitly, for a function  $\alpha: S \rightarrow \mathbb{Z}_{\geq 0}$  with finite support, define  $S^\alpha = \prod_{\alpha(s) \neq 0} s^{\alpha(s)}$ . Then

$$A[S] = \left\{ \text{Finite sums } \sum_{\alpha} \varphi(a_{\alpha}) S^{\alpha} \mid a_{\alpha} \in A \right\}.$$

Note that we can consider  $B$  as an  $A$ -module. Then for  $S \subseteq B$ , we can consider the *submodule* generated by  $S$ :  $\langle S \rangle$ . Since  $A[S]$  is itself a module,  $\langle S \rangle \subseteq A[S]$  automatically, and we do not have equality in general.

### 3.1.2 Definition

An  $A$ -algebra  $B$  is **finitely generated** if there is a finite  $S \subseteq B$  such that  $B = A[S]$ .  $B$  is a **finite algebra** if it is finitely generated as an  $A$ -module.

A finite  $A$ -algebra is also finitely generated as an  $A$ -algebra, but not vice versa in general.

### 3.1.3 Example

Suppose  $S = \{s\}$  is a singleton, then  $A[s]$  is the set of polynomials  $\sum_k \varphi(a_k) s^k$ .

### 3.1.4 Example

Clearly the polynomial algebra  $A[x_1, \dots, x_n]$  is a finitely generated  $A$ -algebra. But it is not a finite  $A$ -algebra: we can reduce this to the case of fields, in which  $x_i^k$  are all linearly independent and thus there is no finite basis.

### 3.1.5 Lemma

An  $A$ -algebra  $B$  is fg iff it is the quotient of some finite polynomial algebra  $A[x_1, \dots, x_n]$  (up to isomorphism of  $A$ -algebras).

### Proof

This is equivalent to saying  $B$  is fg iff there is a surjection  $A[x_1, \dots, x_n] \rightarrow B$ . This is clearly true: if  $B$  is fg then for  $s_1, \dots, s_n$  generating  $B$  map  $x_i \mapsto s_i$  and extend by freeness. Conversely, quotients of fg algebras are fg. ■

### 3.1.6 Definition

Let  $B$  be an  $A$ -algebra. An element  $b \in B$  is **integral** over  $A$  if  $b$  is the root of a monic polynomial over  $A$ . That is, there exist  $a_1, \dots, a_n \in A$  such that  $b^n + a_1 b^{n-1} + \dots + a_n \cdot 1 = 0$ .

$b \in B$  is **algebraic** over  $A$  if  $b$  is the root of any polynomial over  $A$ .

Clearly integral elements are algebraic. For fields the converse holds as well, as we can divide a polynomial by its leading coefficient.



### 3.1.7 Example

The set of integral elements in the  $\mathbb{Z}$ -algebra  $\mathbb{Q}$  is  $\mathbb{Z}$ . Clearly an integer is integral, as  $n \in \mathbb{Z}$  is the root of the monic  $x - n$ . Conversely if  $b = p/q \in \mathbb{Q}$  is integral with  $p, q$  coprime then

$$b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0, \quad a_i \in \mathbb{Z}.$$

Then we have that

$$p^n + a_1 p^{n-1} q + \cdots + a_n q^n = 0$$

and so  $q$  must divide  $p^n$ , but since  $p, q$  are coprime this can only happen if  $q = 1$  and  $b \in \mathbb{Z}$ .

### 3.1.8 Lemma

If  $M$  is a fg  $A$ -module, then for every  $\varphi \in \text{end}_A(M)$  there exist  $a_1, \dots, a_n \in A$  such that

$$\varphi^n + a_1 \varphi^{n-1} + \cdots + a_n = 0$$

in  $\text{end}_A(M)$ .

#### Proof

The proof is by a generalization of Cayley-Hamilton. Let  $n \geq 1$  and consider  $R = \mathbb{Z}[X_{ij}]$ , the polynomial ring over  $n^2$  indeterminants  $X_{ij}$ ,  $1 \leq i, j \leq n$ . Let  $X = (X_{ij}) \in M_n(R)$ , and define its **characteristic polynomial**:  $\Phi_X(T) \in R[T]$  given by

$$\Phi_X(T) = \det(T \text{id} - X).$$

By Cayley-Hamilton over  $R$ 's field of fractions,  $\Phi_X(X) = 0$ .

Now if  $A$  is any ring and  $Y \in M_n(A)$ , there is a unique ring morphism  $\varphi: R \rightarrow A$  which maps  $X$  to  $Y$  (coordinate-wise). Indeed, such a map must map  $X_{ij}$  to  $Y_{ij}$  which uniquely determines  $\varphi$ . Then  $\varphi$  lifts to the ring morphism  $R[T] \rightarrow A[T]$  (map  $\sum_n a_n T^n$  to  $\sum_n \varphi(a_n) T^n$ ), which maps  $\Phi_X(T)$  to  $\Phi_Y(T)$  (which is also given by the determinant  $\det(T \text{id} - Y)$  since the determinant is functorial:  $\det(\varphi(B)) = \varphi(\det(B))$ ). Thus  $\Phi_X(X) = 0$  implies that  $\Phi_Y(Y) = \varphi(\Phi_X(\varphi(X))) = \varphi(\Phi_X(X)) = 0$ , as desired. So Cayley-Hamilton applies to any commutative ring.

Now for our lemma, let  $m_1, \dots, m_n \in M$  be generators. Then for all  $j$  we have

$$\varphi(m_j) = \sum_i a_{ij} m_i, \quad a_{ij} \in A.$$

We associate to  $\varphi$  the representative matrix  $[\varphi] = (a_{ij}) \in M_n(A)$  (which is not unique, unlike over fields). This has the usual properties that  $[\varphi^n] = [\varphi]^n$  and  $[\varphi + \psi] = [\varphi] + [\psi]$  (meaning that the composition or sum of representatives is a representative matrix). Now Cayley-Hamilton gives us that

$$\sum_n b_n [\varphi]^n = 0$$

for some  $b_n \in A$ , and thus

$$\left[ \sum_n b_n \varphi^n \right] = 0,$$

and since only the zero endomorphism has the zero representative, this means  $\sum_n b_n \varphi^n = 0$  as desired. ■

**3.1.9 Proposition**

For an  $A$ -algebra  $B$  and  $b \in B$ , the following are equivalent:

- (1)  $b$  is integral over  $A$ ,
- (2)  $A[b] \subseteq B$  is a finite algebra (fg module) over  $A$ ,
- (3)  $A[b]$  is contained in a finite algebra over  $A$ ,
- (4) There exists a faithful  $A[b]$ -module  $M$  such that  $M$  is finitely generated as an  $A$ -module.

**Proof**

Clearly (2) implies (3) implies (4): take  $M = C$ , and since  $A[b] \subseteq M$  it is faithful (the algebra map is inclusion, which is injective). (1) implies (2) since if  $b^n + a_1b^{n-1} + \cdots + a_n1 = 0$  then  $b^n$  is in  $\langle 1, \dots, b^{n-1} \rangle \subseteq B$  and then by induction so is every higher power of  $b$ , so that  $A[b] = \langle 1, \dots, b^{n-1} \rangle$  is finite.

(4) implies (1): we have a ring morphism  $\varphi: A[b] \rightarrow \text{end}_A(M)$ , so that every  $f \in A[b]$  gives us an endomorphism  $\varphi(f) \in \text{end}_A(M)$ . In particular by the above lemma there are  $a_1, \dots, a_n \in A$  such that

$$\rho(b)^n + a_1\rho(b)^{n-1} + \cdots + a_n = 0 \in \text{end}_A(M).$$

For  $f(b) = b^n + a_1b^{n-1} + \cdots + a_n \in A[b]$ , we see that  $\rho(f(b)) = 0$ . But  $M$  is a faithful  $A[b]$ -module, so  $f(b) = 0$  and  $b$  is integral as desired. ■

**3.1.10 Lemma**

- (1) If  $B$  is a finite  $A$ -algebra, and  $M$  is a fg  $B$ -module, then  $M$  is a fg  $A$ -module.
- (2) If  $B$  is a finite  $A$ -algebra, and  $C$  is a finite  $B$ -algebra, then  $C$  is a finite  $A$ -algebra.

**Proof**

Clearly the second point follows from the first. Now if  $M$  is generated by  $m_1, \dots, m_n$  over  $B$ , and  $B$  is generated by  $b_1, \dots, b_k$  over  $A$  then  $b_i m_j$  generates  $M$  over  $A$ . Indeed, for  $m \in M$  if  $m = \sum_i b'_i m_i$ , then  $b'_i = \sum_j a_{ij} b_j$  so

$$m = \sum_{i,j} a_{ij} (b_j m_i),$$

as desired. ■

**3.1.11 Corollary**

Let  $B$  be an  $A$ -algebra, and suppose that  $b_1, \dots, b_n \in B$  are integral over  $A$ . Then  $A[b_1, \dots, b_n]$  is a finite  $A$ -algebra.

**Proof**

By induction: for  $n = 1$  this is our above proposition. Then if  $b_{n+1}$  is integral over  $A$ , it is integral over  $A[b_1, \dots, b_n]$  so that  $A[b_1, \dots, b_{n+1}]$  is a finite  $A[b_1, \dots, b_n]$ -algebra, and thus a finite  $A$ -algebra by transitivity which we just showed. ■

### 3.1.12 Corollary

Let  $B$  be an  $A$ -algebra, then

$$B_0 = \{b \in B \mid b \text{ is integral over } A\}$$

is an  $A$ -subalgebra of  $B$

#### Proof

Clearly  $0, 1 \in B_0$ . Now if  $b_1, b_2 \in B_0$  then  $A[b_1 + b_2], A[b_1 b_2] \subseteq A[b_1, b_2]$  which is a finite  $A$ -algebra and thus  $b_1 + b_2, b_1 b_2$  are integral over  $A$ ;  $B_0$  is a subring of  $B$ . And if  $a \in A, b \in B_0$  then  $A[ab] \subseteq A[b]$  is finite, so that  $ab \in B_0$ . ■

### 3.1.13 Definition

- (1) For an  $A$ -algebra  $B$ , we define  $A$ 's **integral closure** in  $B$  to be the set of all integral elements over  $A$  in  $B$ ; which we showed above is an  $A$ -subalgebra.
- (2) An  $A$ -algebra  $B$  is **integral** if every element of  $B$  is integral over  $A$ .
- (3) A morphism of rings  $f: A \rightarrow B$  is **integral** if  $B$  is an integral  $A$ -algebra via  $f$ .
- (4) For a morphism of rings  $f: A \rightarrow B$ ,  $f(A)$  is **integrally closed** in  $B$  if  $B_0 = f(A)$  (i.e. no  $b \in B - f(A)$  is integral over  $A$ ).
- (5) A domain  $A$  is **normal** if it is integrally closed in its field of fractions.

### 3.1.14 Example

We saw that  $\mathbb{Z}$  is normal (its field of fractions is  $\mathbb{Q}$ ); the same proof shows that any UFD is normal. In particular  $k[x_1, \dots, x_n]$  and  $\mathbb{Z}[x_1, \dots, x_n]$  are normal.

Consider the  $k$ -subalgebra  $A = k[x^2, x^3] \subseteq k[x]$ . Then  $x \notin A$  is integral over  $A$  since it is a root of  $t^6 - x^6 \in A[t]$ , and  $x = x^3/x^2$  is in its field of fractions. Thus  $A$  is not normal.

#### Note:

There is similarity between integral morphisms and algebraic extensions of fields. But while algebraically closed fields exist, there are no “integrally closed ring”: every ring can be embedded non-integrally into another ring.

### 3.1.15 Corollary

An  $A$ -algebra  $B$  is finite over  $A$  iff it is integral and finitely generated.

#### Proof

If  $B$  is a finite  $A$ -algebra, then for every  $b \in B$ ,  $A[b] \subseteq B$  is finite so  $b$  is integral over  $A$ ;  $B$  is integral and finitely generated. Conversely if  $B$  is integral and finitely generated, suppose  $B = A[b_1, \dots, b_n]$ ; since each  $b_i$  is integral we have that  $B$  is finite. ■

**3.1.16 Corollary**

Let  $B$  be an integral  $A$ -algebra, and  $C$  be a  $B$ -algebra.

- (1) If  $c \in C$  is integral over  $B$ , then it is integral over  $A$ .
- (2) If  $C$  is integral over  $B$ , then it is integral over  $A$ .

**Proof**

If  $c$  is integral over  $B$ , then

$$c^n + b_1 c^{n-1} + \cdots + b_n = 0$$

for some  $b_i \in B$ , which are integral over  $A$ . Then  $B' = A[b_1, \dots, b_n] \subseteq B$  is a finite  $A$ -algebra, and  $c$  is integral over  $B'$  and so  $B'[c]$  is a finite  $B'$ -algebra. Since  $B'$  is a finite  $A$ -algebra, by transitivity  $B'[c]$  is a finite  $A$ -algebra as well containing  $A[c]$ , so  $c$  is integral over  $A$ .

The second point follows from the first. ■

**3.1.17 Corollary**

Let  $B$  be an  $A$ -algebra, and  $B_0$  be the integral closure of  $A$  inside  $B$ . Then  $B_0$  is integrally closed in  $B$ .

**Proof**

If  $b \in B$  is integral over  $B_0$ , then by the above corollary, since  $B_0$  is an integral  $A$ -algebra,  $b$  is integral over  $A$  and thus  $b \in B_0$ . ■

Let  $B$  be an  $A$ -algebra under  $f: A \rightarrow B$ .

- (1) For an ideal  $J \leq B$ , define  $J_A = f^{-1}(J) \leq A$ . Then  $f$  induces an injective homomorphism  $A/J_A \rightarrow B/J$ , so that  $B/J$  is an  $A/J_A$ -algebra.
- (2) For a multiplicative set  $S \subseteq A$ ,  $f(S) \subseteq B$  is multiplicative. Then we have a natural morphism  $S^{-1}A \rightarrow f(S)^{-1}B$  mapping  $a/s$  to  $f(a)/f(s)$ . Thus  $f(S)^{-1}B$  is an  $S^{-1}A$ -algebra.
- (3) Viewing  $B$  as an  $A$ -module, we have already defined  $S^{-1}B$ , and this clearly coincides with  $f(S)^{-1}B$ . Thus  $S^{-1}B$  is naturally an  $S^{-1}A$ -algebra.

**3.1.18 Proposition**

Let  $B$  be an integral  $A$ -algebra.

- (1) Let  $J \leq B$  be an ideal, then  $B/J$  is an integral  $A/J_A$ -algebra.
- (2) Let  $S \subseteq B$  be multiplicative, then  $S^{-1}B$  is an integral  $S^{-1}A$ -algebra.

**Proof**

For  $\bar{x} \in B/J$ , lift it to  $x \in B$ , which is integral over  $A$ . Thus  $x^n + a_1 x^{n-1} + \cdots + a_n = 0$  for  $a_i \in A$ , and then

$$\bar{x}^n + \bar{a}_1 \bar{x}^{n-1} + \cdots + \bar{a}_n = 0,$$

where  $\bar{a}_i = a_i + J_A \in A/J_A$ , so that  $\bar{x}$  is integral over  $A/J_A$ .

An element of  $S^{-1}B$  is of the form  $\tilde{x} = x/f(s)$  with  $x \in B, s \in S$ . By assumption,  $x^n + a_1x^{n-1} + \cdots + a_n = 0$  and thus

$$\left(\frac{x}{f(s)}\right)^n + \frac{a_1}{s} \left(\frac{x}{f(s)}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = \frac{x^n + a_1x^{n-1} + \cdots + a_n}{f(s)^n} = 0,$$

so that  $\tilde{x}$  is integral over  $S^{-1}A$  as desired. ■

### 3.1.19 Proposition

Let  $A \subseteq B$  be integral domains so that  $B$  is integral over  $A$ . Then  $A$  is a field iff  $B$  is a field.

#### Proof

If  $A$  is a field, then let  $0 \neq b \in B$ . It is integral over  $A$  so  $b^n + a_1b^{n-1} + \cdots + a_n = 0$  for  $a_i \in A$ , and assume  $n$  is minimal. Then

$$b(b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1}) = -a_n.$$

Since by minimality  $b^{n-1} + a_1b^{n-2} + \cdots + a_{n-1} \neq 0$ , we have  $a_n \neq 0$ . Thus  $-a_n$  is a unit, and thus  $b$  must be a unit as well (if  $xy = u$  is a unit, then  $x$  is a unit since  $x(yu^{-1}) = 1$ ). So  $B$  is a field.

Conversely if  $B$  is a field, then let  $0 \neq x \in A$ . Then  $x^{-1} \in B$  since it is a field. Since  $B$  is integral over  $A$ , there are  $a_i \in A$  such that

$$x^{-n} + a_1x^{-(n-1)} + \cdots + a_n = 0,$$

and multiplying by  $x^{n-1}$  we get

$$x^{-1} + a_1 + \cdots + a_nx^{n-1},$$

so that  $x^{-1}$  is the sum of elements from  $A$ , and thus is in  $A$ . ■

### 3.1.20 Corollary

Let  $B$  be an integral  $A$ -algebra, and  $\mathfrak{q} \leq B$  a prime ideal. Let  $\mathfrak{p} = \mathfrak{q}_A \leq A$  be its preimage, then  $\mathfrak{q}$  is maximal iff  $\mathfrak{p}$  is maximal.

#### Proof

Both  $\mathfrak{p}, \mathfrak{q}$  are prime so that  $B/\mathfrak{q}$  and  $A/\mathfrak{p}$  are integral domains. As we showed,  $B/\mathfrak{q}$  is integral over  $A/\mathfrak{p}$ . Thus  $B/\mathfrak{q}$  is a field iff  $A/\mathfrak{p}$  is a field; so  $\mathfrak{q}$  is maximal iff  $\mathfrak{p}$  is maximal. ■

### 3.1.21 Corollary

Let  $B$  be an integral  $A$ -algebra,  $\mathfrak{q} \leq \mathfrak{q}' \leq B$  be prime ideals such that  $\mathfrak{q}_A = \mathfrak{q}'_A$ . Then  $\mathfrak{q} = \mathfrak{q}'$ .

**Proof**

Let  $\mathfrak{p} = \mathfrak{q}_A = \mathfrak{q}'_A$ ; it is prime over  $A$ . Consider the following commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow \iota_A & & \downarrow \iota_B \\ A_{\mathfrak{p}} & \xrightarrow{f_{\mathfrak{p}}} & B_{\mathfrak{p}} \end{array}$$

which commutes since the structure morphism  $A \rightarrow S^{-1}A$  is a natural transformation (literally what the commutative square says).

Since  $f^{-1}\mathfrak{q} = \mathfrak{p}$  we see that  $\mathfrak{q} \cap f(A - \mathfrak{p}) = \emptyset$  and similarly for  $\mathfrak{q}'$ . Thus  $\mathfrak{q}B_{\mathfrak{p}} \subseteq \mathfrak{q}'B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$  are prime ideals (since they don't intersect the localizing set  $f(A - \mathfrak{p})$ ).

We are going to show that  $\mathfrak{q}B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$  is a maximal ideal so that  $\mathfrak{q}B_{\mathfrak{p}} = \mathfrak{q}'B_{\mathfrak{p}}$ . And then since  $\mathfrak{q} \mapsto \mathfrak{q}B_{\mathfrak{p}}$  is bijective with inverse  $\iota_B^{-1}$ , we will have that  $\mathfrak{q} = \mathfrak{q}'$ .

Since  $B$  is integral over  $A$ ,  $B_{\mathfrak{p}}$  is integral over  $A_{\mathfrak{p}}$ . Thus, by the above corollary, it is sufficient to show that  $\tilde{\mathfrak{p}} = f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}}) \subseteq A_{\mathfrak{p}}$  is maximal. By assumption,

$$\iota_A^{-1}(\tilde{\mathfrak{p}}) = \iota_A^{-1}(f_{\mathfrak{p}}^{-1}(\mathfrak{q}B_{\mathfrak{p}})) = f^{-1} \circ \iota_B^{-1}(\mathfrak{q}B_{\mathfrak{p}}) = f^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Thus we have that  $\tilde{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$  (since  $\iota_A^{-1}$  is bijective) which is maximal, as desired. ■

The main result we want is the following.

**3.1.22 Theorem**

Let  $f: A \rightarrow B$  be an injective integral morphism. Then for every prime  $\mathfrak{p} \leq A$ , there exists a prime ideal  $\mathfrak{q} \leq B$  such that  $\mathfrak{p} = \mathfrak{q}_A$ .

**Proof**

Since  $B_{\mathfrak{p}} = f(A - \mathfrak{p})^{-1}B$ , and  $0 \notin f(A - \mathfrak{p})$  because  $f$  is injective,  $B_{\mathfrak{p}} \neq 0$ . As before,  $B_{\mathfrak{p}}$  is integral over  $A_{\mathfrak{p}}$  and we have the same commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow \iota_A & & \downarrow \iota_B \\ A_{\mathfrak{p}} & \xrightarrow{f_{\mathfrak{p}}} & B_{\mathfrak{p}} \end{array}$$

Let  $\mathfrak{m} \subseteq B_{\mathfrak{p}}$  be maximal, so that  $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) \subseteq A_{\mathfrak{p}}$  is maximal (as we showed  $\mathfrak{q}_A$  is maximal iff  $\mathfrak{q}$  is). Thus  $f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \mathfrak{p}A_{\mathfrak{p}}$ . Let  $\mathfrak{q} = \iota_B^{-1}(\mathfrak{m})$ , which is prime in  $B$ , and

$$f^{-1}(\mathfrak{q}) = f^{-1} \circ \iota_B^{-1}(\mathfrak{m}) = \iota_A^{-1} \circ f_{\mathfrak{p}}^{-1}(\mathfrak{m}) = \iota_A^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p},$$

as desired. ■

**Note:**

Equivalently, this theorem says that if  $f: A \rightarrow B$  is integral injective, then  $f^*: \text{Spec}(B) \rightarrow \text{Spec}(A)$  is

surjective.

Note that injectivity is necessary here. This is because  $\pi: A \rightarrow A/I$  is integral for any ideal  $I \leq A$ , but  $\pi^*: \text{Spec}(A/I) \rightarrow \text{Spec}(A)$  is not surjective: its image is  $V(I)$ , the set of prime ideals containing  $I$ .

### 3.1.23 Corollary

Let  $B$  be an integral  $A$ -algebra, then for every prime  $\mathfrak{q} \leq B$  and every prime ideal  $\mathfrak{p}' \supseteq \mathfrak{p} = \mathfrak{q}_A$ , there exists a prime ideal  $\mathfrak{q}' \supseteq \mathfrak{q}$  of  $B$  such that  $\mathfrak{q}'_A = \mathfrak{p}'$ .

#### Proof

Consider the commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \pi_A \downarrow & & \downarrow \pi_B \\ A/\mathfrak{p} & \xrightarrow{\bar{f}} & B/\mathfrak{q} \end{array}$$

Then  $\mathfrak{p}' \supseteq \mathfrak{p}$  gives rise to a prime ideal  $\bar{\mathfrak{p}}' = \mathfrak{p}'/\mathfrak{p} \subseteq A/\mathfrak{p}$ . Since  $B/\mathfrak{q}$  is an integral  $A/\mathfrak{p}$ -algebra, there is a prime ideal  $\bar{\mathfrak{q}}' \subseteq B/\mathfrak{q}$  such that  $\bar{f}^{-1}(\bar{\mathfrak{q}}') = \bar{\mathfrak{p}}'$ . Then  $\mathfrak{q}' = \pi_B^{-1}(\bar{\mathfrak{q}}')$  is a prime ideal of  $B$  containing  $\mathfrak{q}$  and

$$f^{-1}(\mathfrak{q}') = f^{-1} \circ \pi_B^{-1}(\bar{\mathfrak{q}}') = \pi_A^{-1} \circ \bar{f}^{-1}(\bar{\mathfrak{q}}') = \pi_A^{-1}(\bar{\mathfrak{p}}') = \mathfrak{p}',$$

as desired. ■

### 3.1.24 Proposition

Let  $B$  be an  $A$ -algebra,  $C \subseteq B$  be the integral closure of  $A$  inside of  $B$ . Then for any multiplicative  $S \subseteq A$ ,  $S^{-1}C$  is the integral closure of  $S^{-1}A$  inside of  $S^{-1}B$ .

#### Proof

Since  $C$  is integral over  $A$ ,  $S^{-1}C$  is integral over  $S^{-1}A$ . Now if  $b/s$  is integral over  $S^{-1}A$ , then  $b/1 = s \cdot b/s$  is integral over  $S^{-1}A$  as well, and thus

$$\left(\frac{b}{1}\right)^n + \frac{a_1}{s_1} \left(\frac{b}{1}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

for  $a_i/s_i \in S^{-1}A$ . Let  $t = s_1 \cdots s_n$  and  $u_i = t/s_i \in S$ , and multiplying this equation by  $t^n$  gives us

$$\left(\frac{tb}{1}\right)^n + a_1 u_1 \left(\frac{bt}{1}\right)^{n-1} + \cdots + a_n u_n t^{n-1} = 0.$$

Thus there is a  $v \in S$  such that

$$v((bt)^n + a_1 u_1 (bt)^{n-1} + a_2 u_2 t (bt)^{n-2} + \cdots + a_n u_n t^{n-1}) = 0.$$

Then  $btv$  is integral over  $A$  (multiply by  $v^{n-1}$ ), so that  $btv \in C$ . Then  $b/s = (btv)/(stv) \in S^{-1}C$ , as desired. ■

**3.1.25 Corollary**

Let  $A$  be a domain, then the following are equivalent:

- (1)  $A$  is normal,
- (2)  $S^{-1}A$  is normal for every multiplicative  $S \subseteq A$  not containing 0,
- (3)  $A_{\mathfrak{p}}$  is normal for every prime  $\mathfrak{p}$ ,
- (4)  $A_{\mathfrak{m}}$  is normal for every maximal  $\mathfrak{m}$ .

**Proof**

Let  $K$  be  $A$ 's field of fractions, and let  $C \subseteq K$  be  $A$ 's integral closure in  $K$ . Then  $S^{-1}C$  is  $S^{-1}A$ 's integral closure in  $S^{-1}K = K$  for all multiplicative  $S$  not containing 0. If  $A$  is normal, then  $C = A$  so that  $S^{-1}A$  is integrally closed in  $K$ , and thus normal. So (1) implies (2), which implies (3) which implies (4).

Now if  $A_{\mathfrak{m}}$  is normal, then  $A_{\mathfrak{m}} \rightarrow C_{\mathfrak{m}}$  is an isomorphism for all  $\mathfrak{m}$ . We showed that this implies  $A \rightarrow C$  is an isomorphism, so that  $A$  is normal. ■

**3.2 Noetherian rings and modules****3.2.1 Lemma**

Let  $\Sigma$  be a poset, then the following conditions are equivalent:

- (1) every nonempty subset of  $\Sigma$  has a maximal element,
- (2) every increasing chain  $x_1 \leq x_2 \leq \dots$  in  $\Sigma$  eventually stabilizes: at some point  $x_n = x_{n+1}$  (equivalently, there are no infinite strictly increasing chains).

A poset satisfying these equivalent conditions is said to have the **ascending chain condition (ACC)**.

**Proof**

The first implies the second as we can consider the subset  $\{x_i\}$ . The second implies the first since if  $\Sigma' \subseteq \Sigma$  doesn't have a maximal element we can form a strictly increasing chain inductively. ■

**3.2.2 Definition**

An  $A$ -module  $M$  is **Noetherian** if its poset of  $A$ -submodules has the ACC. A ring  $A$  is **Noetherian** if it is Noetherian as a module over itself (equivalently its ideals satisfy the ACC).

**3.2.3 Example**

Any field is a Noetherian ring, since it only has two ideals.

$\mathbb{Q}$  is not a Noetherian  $\mathbb{Z}$ -module, since  $\frac{1}{n!}\mathbb{Z}$  does not stabilize (the reason for the factorial is so that it forms an increasing chain).

The ring  $k[x_1, x_2, \dots]$  with infinitely many indeterminates is not Noetherian, since  $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$  forms a strictly increasing chain.



### 3.2.4 Proposition

An  $A$ -module is Noetherian iff every submodule is finitely generated. Thus a ring is Noetherian iff every ideal is finitely generated.

#### Proof

If  $M$  is Noetherian and  $N \subseteq M$  is a submodule, suppose it is not finitely generated. Then inductively we will form an infinitely increasing sequence of submodules. Let  $x_1 \in N$  arbitrary, then since  $N$  is not finitely generated  $x_1, \dots, x_n$  does not generate  $N$  so that there is an  $x_{n+1} \in N - \langle x_1, \dots, x_n \rangle$ . Then  $N_n = \langle x_1, \dots, x_n \rangle$  is a strictly increasing infinite chain of submodules of  $M$ , a contradiction.

Conversely if every submodule of  $M$  is finitely generated, then  $M$  is Noetherian. Otherwise, let  $M_1 \subsetneq M_2 \subsetneq \dots$  be an increasing chain of submodules. Let  $N = \bigcup_n M_n$  be their union, which is also a submodule of  $M$ , and thus is finitely generated. Suppose  $x_1, \dots, x_n \in N$  generate  $N$ , and since  $M_i$  form a chain there is an  $m$  such that  $x_1, \dots, x_n \in M_m$  and thus  $M_m = N$  so that  $M_i$  eventually stabilizes to  $M_m$ . ■

### 3.2.5 Corollary

Every PID is Noetherian.

### 3.2.6 Lemma

Let  $0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$  be a short exact sequence of  $A$ -modules. Then  $M$  is Noetherian iff  $M', M''$  are.

Equivalently, if  $M$  is an  $A$ -module and  $N \subseteq M$  a submodule,  $M$  is Noetherian iff  $N$  and  $M/N$  are.

#### Proof

If  $M$  is Noetherian, let  $M'_n \subseteq M', M''_n \subseteq M''$  be increasing sequences of submodules. Then  $f(M'_n)$  and  $g^{-1}(M''_n)$  are submodules of  $M$  and thus stabilize:  $f(M'_n) = f(M'_{n+1})$  and  $g^{-1}(M''_n) = g^{-1}(M''_{n+1})$  eventually. By the injectivity of  $f$  and surjectivity of  $g$ , this means  $M'_n$  and  $M''_n$  both eventually stabilize.

Conversely if  $M', M''$  are Noetherian, let  $M_n \subseteq M$  be an increasing sequence of submodules. Then  $f^{-1}(M_n) = M_n \cap M'$  and  $g(M_n) \subseteq M''_n$  are increasing sequences of submodules of  $M', M''$  respectively. Thus eventually they stabilize:  $f^{-1}(M_n) = f^{-1}(M_{n+1})$  and  $g(M_n) = g(M_{n+1})$  eventually. We claim that this means  $M_n = M_{n+1}$ .

Indeed, for  $x \in M_{n+1}$  consider  $g(x) \in g(M_{n+1})$  so there is a  $y \in M_n$  such that  $g(x) = g(y)$ . Then  $x - y \in \ker g = \operatorname{im} f$  so that  $x - y = f(z)$  for  $z \in M'$ , and  $x - y \in M_{n+1}$  so that  $z \in f^{-1}(M_{n+1})$  and thus  $z \in f^{-1}(M_n)$ . This means that  $x - y \in M_n$  and thus  $x \in M_n$  as desired. ■

### 3.2.7 Corollary

- (1) The finite direct sums of Noetherian modules is Noetherian.
- (2) If  $A$  is a Noetherian ring and  $M$  is a fg  $A$ -module, then  $M$  is a Noetherian module.
- (3) If  $A$  is a Noetherian ring and  $B$  a finite  $A$ -algebra, then  $B$  is a Noetherian ring. In particular,  $A/I$  is Noetherian for any ideal  $I \subseteq A$ .

**Proof**

For the first point it is sufficient to show this for binary direct sums. If  $M, N$  are Noetherian, then  $0 \rightarrow M \rightarrow M \oplus N \rightarrow N \rightarrow 0$  is a short exact sequence and thus  $M \oplus N$  is Noetherian by above.

Since  $M$  is fg,  $M \cong A^n/N$  for some submodule  $N \leq A^n$  (consider the surjection  $A^n \rightarrow M$  mapping to generators). By the first point,  $A^n$  is Noetherian and thus  $M$  is as well.

$B$  is a fg  $A$ -module, and thus a Noetherian  $A$ -module. Since ideals of  $B$  are  $A$ -submodules, the ACC on  $A$ -submodules implies ACC on  $B$ 's ideals. ■

**3.2.8 Lemma**

Let  $A$  be a Noetherian ring, then all of its localizations are Noetherian rings as well.

**Proof**

Let  $S \subseteq A$  be multiplicative, and consider an increasing sequence of ideals  $\tilde{I}_1 \subseteq \tilde{I}_2 \subseteq \dots$  in  $S^{-1}A$ . Then  $I_n = \iota^{-1}(\tilde{I}_n)$  forms an increasing sequence of ideals of  $A$ , and thus stabilizes. Since  $\tilde{I}_n = S^{-1}I_n$ , we get that  $\tilde{I}_n$  stabilizes. ■

**3.2.9 Theorem (Hilbert basis theorem)**

If  $A$  is Noetherian, then  $A[x]$  is a Noetherian ring.

**Proof**

Let  $I \subseteq A[x]$  be an ideal, we wish to show it is finitely generated. Let us adopt the notation that for  $f \in A[x]$ ,  $a_f \in A$  denotes the leading term of  $f$  (and  $a_f = 0$  only when  $f = 0$ ).

We claim that  $J = \{a_f \mid f \in I\}$  is an ideal in  $A$ . If  $f \in I$  and  $b \in A$  then either  $ba_f = 0$  or  $ba_f = a_{bf}$ ; in both cases  $ba_f \in J$ . Now if  $f, g \neq 0$  are in  $I$  then  $a_f + a_g$  is either zero or the leading term of  $x^{\deg g}f + x^{\deg f}g \in I$  and thus  $a_f + a_g \in J$ .

Since  $J$  is Noetherian, there are  $f_1, \dots, f_n \in I$  such that  $J = (a_{f_1}, \dots, a_{f_n})$ . Let  $d_i = \deg f_i$ , and  $d$  be the maximum  $d_i$ . Consider the fg  $A$ -submodule

$$M = \sum_{i=0}^{d-1} Ax^i \subseteq A[x].$$

Then  $M \cap I$  is an  $A$ -submodule of  $M$ , which is Noetherian (since  $M$  is fg and  $A$  is Noetherian so  $M$  is Noetherian, and thus its submodules are Noetherian).

Now define  $I' = (f_1, \dots, f_n) \subseteq I$ . We claim that

$$I = I' + (M \cap I) \text{ as } A\text{-submodules.}$$

Let  $f \in I$ , then we claim there is an  $f' \in I'$  such that  $\deg(f - f') < d$  and thus  $f - f' \in M \cap I$ . If  $\deg f < d$  then we are done, so assume  $\deg f \geq d$ . By induction, it is sufficient to show that there is an  $f' \in I'$  such that  $\deg(f - f') < \deg f$ .

Now,  $a_f \in J = (a_{f_1}, \dots, a_{f_n})$  so that  $a_f = \sum_i b_i a_{f_i}$  for  $b_i \in A$ . Then define

$$f' = \sum_{i=1}^n b_i x^{(\deg f) - d_i} f_i \in I'.$$

By assumption,  $\deg f \geq d \geq d_i$  so that  $\deg(x^{(\deg f) - d_i} f_i) = d_i$ , and so  $\deg f' \leq \deg f$ . Moreover, the leading coefficient of  $f'$  is

$$a_{f'} = \sum_{i=1}^n b_i a_{f_i} = a_f,$$

so that  $\deg(f - f') < \deg f$  as desired.

Now since  $I = I' + (M \cap I)$ ,  $I'$  is a fg ideal and  $M \cap I$  is a fg  $A$ -module (and thus fg by  $A[x]$ ), we have that  $I$  is a fg ideal, as desired. ■

### 3.2.10 Corollary

- (1) If  $A$  is a Noetherian ring, so is  $A[x_1, \dots, x_n]$ .
- (2) If  $A$  is a Noetherian ring, and  $B$  is a fg  $A$ -algebra,  $B$  is Noetherian.
- (3) Any fg algebra over a field is Noetherian.
- (4) Any fg algebra over a PID (in particular  $\mathbb{Z}$ ) is Noetherian.

### Proof

The first point follows by induction and Hilbert's basis theorem. The second point is because if  $B$  is a fg  $A$ -algebra, it is isomorphic to  $A[x_1, \dots, x_n]/I$  for some ideal  $I \leq A[x_1, \dots, x_n]$ , and is thus Noetherian. The last two points are due to the second (fields and PIDs are Noetherian). ■

We now utilize these results to reprove a lemma we showed above.

### 3.2.11 Lemma

Let  $A$  be a ring and  $T \in M_n(A)$  a matrix. Then there are  $a_1, \dots, a_n \in A$  such that

$$T^n + a_1 T^{n-1} + \dots + a_n = 0.$$

### Proof

Denote  $T = (a_{ij})$ , and consider the subalgebra  $A' = \mathbb{Z}[\{a_{ij}\}] \subseteq A$ . This is a fg  $\mathbb{Z}$ -algebra and thus Noetherian. Since  $T \in M_n(A')$ , we can assume from the getgo that  $A$  is Noetherian.

Since  $M_n(A) \cong A^{n^2}$  as  $A$ -modules,  $M_n(A)$  is a Noetherian  $A$ -module. Thus the increasing sequence of  $A$ -modules

$$N_m = \langle 1, T, \dots, T^m \rangle \subseteq M_n(A)$$

stabilizes, so eventually  $N_n = N_{n-1}$  so that  $T^n \in N_{n-1}$  and we get the desired result. ■

### 3.2.12 Lemma

If  $M$  is a fg  $A$ -module, then for  $\varphi \in \text{end}_A(M)$ , there are  $a_1, \dots, a_n \in A$  such that

$$\varphi^n + a_1 \varphi^{n-1} + \dots + a_n = 0.$$

### Proof

When  $M = A^n$  there is a natural isomorphism of  $A$ -algebras  $M_n(A) \cong \text{end}_A(M)$ , so the result follows from the above lemma. Otherwise, consider a surjective map  $\pi: A^n \rightarrow M$ , then we claim there is a  $\tilde{\varphi}: A^n \rightarrow A^n$

such that the below diagram commutes

$$\begin{array}{ccc} A^n & \xrightarrow{\tilde{\varphi}} & A^n \\ \pi \downarrow & & \downarrow \pi \\ M & \xrightarrow{\varphi} & M \end{array}$$

Meaning  $\pi \circ \tilde{\varphi} = \varphi \circ \pi$ . For every  $i$  choose a preimage  $V_i \in \pi^{-1}(\varphi(\pi(e_i))) \in A^n$  and define

$$\tilde{\varphi}(x_1, \dots, x_n) = \sum_{i=1}^n x_i v_i.$$

This satisfies the condition.

Now there is a monic polynomial  $f \in A[x]$  such that  $f(\tilde{\varphi}) = 0$  by the  $M = A^n$  case. Then

$$f(\varphi) \circ \pi = \pi \circ f(\tilde{\varphi}) = 0,$$

which implies  $f(\varphi) = 0$  because  $\pi$  is surjective. ■

### 3.3 Simple modules and length

#### 3.3.1 Definition

A poset  $\Sigma$  has the **descending chain condition (DCC)** if it satisfies the dual of the ACC (equivalently,  $\Sigma^{\text{op}}$  has ACC). This means that every subset has a minimal element, or every descending chain stabilizes.

#### 3.3.2 Definition

An  $A$ -module is **Artinian** if it has DCC on its submodules. A ring  $A$  is **Artinian** if it is Artinian as a module over itself, equivalently it has DCC on its ideals.

Note that the short exact sequence result on Noetherian rings holds for Artinian rings as well. Thus finite direct sums of Artinian modules are Artinian, fg modules over Artinian rings are Artinian, and thus finite algebras over Artinian rings are Artinian.

Note that it is not the case that a module is Artinian iff its submodules are all fg. That is, Artinian is not equivalent to Noetherian (and for modules they are distinct conditions, for rings we will see that Artinian implies Noetherian).

#### 3.3.3 Example

$\mathbb{Z}$  is a Noetherian but not an Artinian ring. It is not Artinian because  $\mathbb{Z} \supsetneq (n) \supsetneq (n^2) \supsetneq (n^3) \supsetneq \dots$  is an infinite decreasing sequence of ideals.

Similarly  $k[x]$  is Noetherian but not Artinian:  $(x^k)$  forms a descending sequence of ideals.

#### 3.3.4 Definition

A nonzero module  $M$  is **simple** if it has no non-trivial submodules.

**3.3.5 Lemma**

An  $A$ -module  $M$  is simple iff it is isomorphic (as  $A$ -modules) to  $A/\mathfrak{m}$  for a maximal ideal  $\mathfrak{m} \leq A$ .

**Proof**

If  $M$  is simple, then for  $0 \neq m \in M$  the submodule  $\langle m \rangle \subseteq M$  is isomorphic to  $A/\text{ann}(m)$  (the map  $a \mapsto am$  has kernel  $\text{ann}(m)$ ). Since  $M$  is simple,  $\langle m \rangle = M$  and so  $A/\text{ann}(m) \cong M$ . There is then a bijection between submodules of  $M$  and ideals of  $A$  containing  $\text{ann}(m)$ . Since  $M$  is simple, there cannot be any nontrivial such ideals, so  $\text{ann}(m)$  is maximal.

And if  $M \cong A/\mathfrak{m}$  then again there is a bijection between submodules of  $M$  and ideals containing  $\mathfrak{m}$ . So  $M$  is simple, as  $\mathfrak{m}$  is maximal. ■

**3.3.6 Definition**

A **composition series** of  $M$  is a chain of submodules

$$0 = M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots \subsetneq M_n = M$$

which is maximal. This is equivalent to  $M_{i+1}/M_i$  being simple for all  $i$  (no modules between  $M_i$  and  $M_{i+1}$ ). The **length** of such a series is  $n$  (not  $n+1$ ).

Denote by  $\ell_A(M)$  the minimal length of a composition series of  $M$  (or infinity if no composition series exists).

**3.3.7 Lemma**

If  $M$  has finite length  $\ell(M) = n$ , then all of its composition series have the same length  $n$ , and every chain of submodules can be extended to a composition series.

**Proof**

First we claim that if  $N \subsetneq M$  is a submodule, then  $\ell(N) < \ell(M)$ . Indeed, if  $\{M_i\}$  is a composition series for  $N$ , then  $N_i = M_i \cap N$  is a sequence of submodules of  $N$ . Then  $N_i = N_{i+1} \cap M_i$  so

$$N_{i+1}/N_i = N_{i+1}/(N_{i+1} \cap M_i) \cong (N_{i+1} + M_i)/M_i \subseteq M_{i+1}/M_i$$

which is either simple or zero. Thus removing repeated values, we have formed a composition series for  $N$  of shorter length, so  $\ell(N) < \ell(M)$ .

Now if  $\ell(N) = \ell(M)$  then  $N_{i+1} + M_i = M_{i+1}$  for all  $i$ , and thus we claim  $N_i = M_i$  for all  $i$  so that  $N = M$ . This is by induction, since  $M_0 = N_0 = 0$ , and  $N_{i+1} = N_{i+1} + N_i = N_{i+1} + M_i = M_{i+1}$ , as desired.

Now if  $\{M_i\}_0^n$  is any chain in  $M$ , then

$$\ell(M) = \ell(M_n) > \ell(M_{n-1}) > \cdots > \ell(M_0) = 0,$$

so  $\ell(M) \geq n$ .

Now if  $\{M_i\}$  is any composition series of  $M$  of length  $n$ , then  $\ell(M) \geq n$  by what we just showed. But by definition  $\ell(M) \leq n$  so that  $\ell(M) = n$  and thus all composition series have length  $\ell(M)$ .

And for any general chain of submodules of  $M$ , if it is not a composition series then new terms can be inserted. This process can be repeated, and must terminate (since once it reaches a length of  $\ell(M)$  it must be a composition series). ■

**Note:**

Note that we can prove something stronger, the **Jordan-Holder theorem**: for any two composition series  $M_i, N_j$  there is a permutation  $\sigma$  between  $i$  and  $j$  such that  $M_{i+1}/M_i \cong N_{\sigma(i)+1}/N_{\sigma(i)}$ . A proof of this can be found in my representation theory homework somewhere.

**3.3.8 Lemma**

An  $A$ -module  $M$  has finite length iff it is both Noetherian and Artinian.

**Proof**

If  $M$  has finite length, then any increasing/decreasing sequence has length bound by  $\ell(M)$  so that  $M$  is both Noetherian and Artinian.

Conversely if  $M$  is Artinian and Noetherian, then consider the set of submodules  $N \subseteq M$  of finite length. It is nonempty since it contains 0, and because  $M$  is Noetherian contains a maximal element  $M'$ . If  $M = M'$  then we are done. Otherwise, the set of submodules  $M' \subsetneq N \subseteq M$  is nonempty (containing  $M$ ) and has a minimal element  $N$  since  $M$  is Artinian. But then  $N/M'$  is simple, and so composing a composition series of  $M'$  with  $\subsetneq N$  forms a composition series of  $N$ , so  $N$  has finite length, contradicting  $M'$ 's maximality. ■

**3.3.9 Proposition**

Let  $M$  be fg over a Noetherian ring  $A$ .

- (1) If  $M \neq 0$ , then  $M$  has a submodules  $N \subseteq M$  such that  $N \cong A/\mathfrak{p}$  for some prime  $\mathfrak{p} \leq A$ .
- (2) There exists a chain  $0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$  of submodules such that for each  $i$ , there is an isomorphism of  $A$ -modules  $M_{i+1}/M_i \cong A/\mathfrak{p}_i$  for prime ideals  $\mathfrak{p}_i \leq A$ .
- (3) The chain in the previous point is not necessarily unique, but
  - (i) for a prime ideal  $\mathfrak{p} \leq A$  we have  $\text{ann} M \subseteq \mathfrak{p}$  iff  $\mathfrak{p}_i \subseteq \mathfrak{p}$  for some  $i$ . In particular,  $\text{ann} M \subseteq \mathfrak{p}_i$  for all  $i$ , and the minimal ideals between  $\mathfrak{p}_1, \dots, \mathfrak{p}_n$  are precisely minimal prime ideals  $\text{ann}(M) \subseteq \mathfrak{p}$ .
  - (ii) for every minimal prime ideal  $\text{ann}(M) \subseteq \mathfrak{p}$ , the  $\mathcal{A}_{\mathfrak{p}}$  module  $M_{\mathfrak{p}}$  has finite length equal to the number of times  $\mathfrak{p}$  occurs in  $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ .

**Proof**

Let  $\Theta$  be the set of all ideal  $I \leq A$  of the form  $I = \text{ann}(m)$  for  $0 \neq m \in M$ . Since  $A$  is Noetherian, there is an  $m$  such that  $\text{ann}(m)$  is maximal. Since  $\langle m \rangle \subseteq M$  is isomorphic to  $A/\text{ann}(m)$ , it is sufficient to show  $\text{ann}(m)$  is prime. So suppose  $a, b \notin \text{ann}(m)$ , then  $bm \neq 0$  and  $\text{ann}(m) \subseteq \text{ann}(bm)$ . Thus by maximality  $\text{ann}(m) = \text{ann}(bm)$  and then  $a \notin \text{ann}(bm)$  so  $abm \neq 0$  meaning  $ab \notin \text{ann}(m)$ , showing it is prime. This proves the first point.

Let  $\Sigma$  be the set of all submodules of  $M$  which possess such a chain. It is nonempty since the zero module is in it. Since  $M$  is Noetherian, let  $M'$  be a maximal element of  $\Sigma$ .

Now consider  $M'' = M/M'$ ; if  $M'' = 0$  we are finished. Otherwise by the first point it has a submodule  $N'' = N'/M'$  isomorphic to  $A/\mathfrak{p}$ . But then composing the chain of  $M'$  with  $\subsetneq N'$  gives such a chain, contradicting  $M'$ 's maximality.

For the final point, notice that for  $M' \subsetneq M$ ,

$$\text{ann}(M') \cdot \text{ann}(M/M') \subseteq \text{ann}(M) \subseteq \text{ann}(M') \cap \text{ann}(M/M').$$

The second inclusion is clear, the first is because for  $a \in \text{ann}(M')$  and  $b \in \text{ann}(M/M')$  and  $m \in M$  then  $bm \in M'$  and  $abm \in aM' = 0$ . Thus for a prime ideal  $\mathfrak{p} \leq A$ ,  $\text{ann}(M) \subseteq \mathfrak{p}$  iff  $\text{ann}(M') \subseteq \mathfrak{p}$  or  $\text{ann}(M/M') \subseteq \mathfrak{p}$ . Thus by induction  $\text{ann}(M) \subseteq \mathfrak{p}$  iff  $\text{ann}(M_{i+1}/M_i) \subseteq \mathfrak{p}$  for some  $i$ . Since  $\text{ann}(A/\mathfrak{p}_i) = \mathfrak{p}_i$ , the first point follows.

For the second point, the chain  $\{M_i\}$  gives the chain  $\{S^{-1}M_i\}$  and

$$S^{-1}M_{i+1}/S^{-1}M_i \cong S^{-1}(M_{i+1}/M_i) \cong S^{-1}(A/\mathfrak{p}_i) \cong S^{-1}A/S^{-1}\mathfrak{p}_i.$$

Moreover  $S^{-1}\mathfrak{p}_i = S^{-1}A$  iff  $S \cap \mathfrak{p}_i \neq \emptyset$ , and otherwise is a prime ideal. For  $S = A - \mathfrak{p}$  for a minimal prime ideal  $\text{ann}(M) \subseteq \mathfrak{p}$ , then  $S^{-1}\mathfrak{p}_i = A_{\mathfrak{p}}$  if  $\mathfrak{p} \neq \mathfrak{p}_i$  and otherwise  $S^{-1}\mathfrak{p}_i = \mathfrak{p}A_{\mathfrak{p}}$  is a maximal ideal. Thus the sequence  $(M_i)_{\mathfrak{p}}$  forms a composition series after deleting the repeated entries (when  $\mathfrak{p} \neq \mathfrak{p}_i$ ), and we are left with a composition series (since the quotients are quotients by a maximal ideal) whose length is equal to the number of  $\mathfrak{p}_i$  equal to  $\mathfrak{p}$ . ■

### 3.3.10 Definition

A minimal prime ideal is a prime ideal not containing any other prime ideal.

### 3.3.11 Lemma

A Noetherian ring  $A$  has finitely many minimal prime ideals.

#### Proof

The minimal prime ideals of  $A$  are the minimal primes containing  $\text{ann}(A) = 0$ . Then for any chain as in the above lemma (whose components are  $A/\mathfrak{p}$  for some prime), the minimal prime ideals are the minimal primes from  $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$  which is finite. ■

### 3.3.12 Proposition

A ring is Artinian iff it is Noetherian and every prime ideal is maximal.

#### Proof

If  $A$  is Noetherian and every prime ideal is maximal, then every prime ideal is also minimal. Thus  $A$  has finitely many prime/maximal ideals,  $\mathfrak{m}_1, \dots, \mathfrak{m}_k$ . Then their product  $I = \mathfrak{m}_1 \cdots \mathfrak{m}_k$  is contained in the intersection of all prime ideals,  $\text{nil}(A)$ . Because  $A$  is Noetherian  $I$  is finitely generated, so  $I^n = 0$  for some  $n$  (since every element of  $I$  is nilpotent, and it is finitely generated).

We have a chain of submodules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_k = A,$$

where  $M_{i+1}/M_i \cong A/\mathfrak{m}_i$  for some  $i$  (by the above proposition). As a subquotient,  $M_{i+1}/M_i$  is a Noetherian  $A$ -module, and thus a Noetherian module over the field  $K = A/\mathfrak{m}_i$ . A Noetherian vector space is finite-dimensional (it is fg), so that  $M_{i+1}/M_i$  is of finite length over  $K$  (its length is equal to its dimension). Then  $M_{i+1}/M_i$  has finite length over  $A$  as well, and then inductively we see that each  $M_i$  has finite length, and in particular  $A$  is Artinian.

Conversely if  $A$  is Artinian, then all of its prime ideals are maximal (homework), so we need only show it is Noetherian. If  $\mathfrak{m}_1, \dots, \mathfrak{m}_{n+1}$  are distinct maximal ideals then  $\mathfrak{m}_1 \cdots \mathfrak{m}_n \supset \mathfrak{m}_1 \cdots \mathfrak{m}_{n+1}$  is strict, so by Artinian-ness there must be finitely many maximal ideals. If the inclusion were not strict then  $\mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \mathfrak{m}_{n+1}$ , and then by  $\mathfrak{m}_{n+1}$ 's primality,  $\mathfrak{m}_i \subseteq \mathfrak{m}_{n+1}$  for some  $i$ . But then  $\mathfrak{m}_i = \mathfrak{m}_{n+1}$  by maximality, a contradiction to distinctness.

So let  $I = \mathfrak{m}_1 \cdots \mathfrak{m}_n$  be the product of all  $A$ 's prime/maximal ideals. Consider the descending chain  $I \supseteq I^2 \supseteq I^3 \supseteq \cdots$  which must terminate, so  $I^k = I^{k+1}$  eventually. We claim that  $I^k = 0$ .

If not, let  $x_1, \dots, x_{k+1} \in I$  such that  $x_1 \cdots x_{k+1} \neq 0$ . Because  $I^k = I^{k+1}$ ,  $\prod_{i=2}^{k+1} x_i = \sum_{\alpha} y_{\alpha,1} \cdots y_{\alpha,k+1}$  where the sum is over a finite set and each  $y_{\alpha,i} \in I$ . Thus  $x_1 y_{\alpha_0,1} \cdots y_{\alpha_0,k+1} \neq 0$  for some  $\alpha_0$ . Then from the original tuple  $x_1, \dots, x_{k+1} \in I$  we obtain a new tuple  $x'_1 = x_1, x'_2, \dots, x'_{k+2}$  ( $x'_i = y_{\alpha_0,i-1}$ ) with nonzero product, and the first element unchanged.

We similarly may write the product of the final  $k$  elements  $x'_3 \cdots x'_{k+2}$  as an element of  $I^{k+1}$  and obtain  $x_1, x'_2, x'_3, x''_4, \dots, x''_{k+3}$  with nonzero product. Continuing this inductively, we produce an infinite sequence  $y_1, y_2, \dots$  such that  $y_1 \cdots y_m \neq 0$  for all  $m > 0$ . Now consider the descending chain  $(y_1) \supseteq (y_1 y_2) \supseteq \cdots$ , which stabilizes since  $A$  is Artinian. Thus for some  $m > 0$  and  $a \in A$ ,

$$\prod_{i=1}^m y_i = a \prod_{i=1}^{m+1} y_i \implies \left( \prod_{i=1}^m y_i \right) (1 - a y_{m+1}) = 0.$$

Since  $y_{m+1} \in I$  and thus  $a y_{m+1} \in I$  is in the Jacobson radical ( $I$  is the product of all maximal ideals, and thus contained in their intersection), we have that  $1 - a y_{m+1} \in A^\times$ . Thus  $\prod_{i=1}^m y_i = 0$ , a contradiction.

So  $I^k = 0$ , and so there is a chain (taking  $M_j = \mathfrak{m}_1 \cdots \mathfrak{m}_{n-j}$ ),

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_m = A$$

where  $M_{i+1}/M_i$  is annihilated by some  $\mathfrak{m}_i$ , and thus is Artinian over the field  $K = A/\mathfrak{m}_i$  so finite-dimensional over  $A$ . Thus  $M_{i+1}/M_i$  has finite length over  $A/\mathfrak{m}_i$ , and thus over  $A$ , and then by induction we see that each  $M_i$  has finite length, in particular so does  $A$  and thus is Noetherian. ■

#### Note:

Recall that  $\mathbb{Z}$  is Noetherian but not Artinian. This is because while all of its non-zero prime ideals are maximal,  $(0)$  is a prime ideal which is not maximal. We see that in general an integral domain which is not a field cannot be Artinian.

## 4 Tensoring and Flatness

### 4.1 Tensor products

I assume some familiarity with the tensor product (see e.g. my summary on representation theory otherwise). The tensor product is a bifunctor

$$\bullet \otimes_B \bullet : {}_A \text{Mod}_B \times {}_B \text{Mod}_C \rightarrow {}_A \text{Mod}_C,$$

with the universal property that it represents the balanced map functor:

$$\text{hom}(M \otimes_B N, P) \cong \text{Bal}(M, N; P).$$

Balanced maps are maps  $M \times N \rightarrow P$  which are additive and multiplicative in each coordinate, and  $f(mb, n) = f(m, bn)$  for all  $b \in B$ .

The tensor product also has a right-adjoint. For an  $(A, B)$ -bimodule  $M$ ,  $\bullet \otimes_A M$  is a functor  $\text{Mod}_A \rightarrow \text{Mod}_B$  with right adjoint  $\text{hom}_B(M, \bullet)$ .

In the case of commutative rings (this course), all modules are bimodules, so the tensor product is a bifunctor

$$\bullet \otimes_A \bullet : \text{Mod}_A \rightarrow \text{Mod}_A,$$

which represents the bilinear map functor (bilinear in the usual sense).

Note still that we use tensor products between module categories. In particular if  $B$  is an  $A$ -algebra, then  $B \otimes_A M$  is a  $B$ -module ( $b(c \otimes m) = (bc) \otimes m$ ). This defines a map  $\text{Mod}_A \rightarrow \text{Mod}_B$ , and its right-adjoint is the restriction map (restrict a  $B$ -module to an  $A$ -module). Note that this is a special case of the tensor-hom adjunction:  $\bullet \otimes_A B$  is left-adjoint to  $\text{hom}_B(B, \bullet) \cong B$ .

So we have a natural isomorphism

$$\text{hom}_B(B \otimes_A M, P) \cong \text{hom}_A(M, P|_A), \quad M \in \text{Mod}_A, P \in \text{Mod}_B.$$

The correspondence is given by mapping  $f: B \otimes_A M \rightarrow P$  to  $\tilde{f}: M \rightarrow P$  by  $\tilde{f}(m) = f(1 \otimes m)$ . And  $g: M \rightarrow P|_A$  to  $\hat{g}: B \otimes_A M \rightarrow P$  by  $\hat{g}(1 \otimes m) = g(m)$ .



**4.1.1 Proposition**

Let  $M$  be an  $A$ -module.

- (1) For a multiplicative set  $S \subseteq A$ ,  $S^{-1}A \otimes_A M$  is naturally isomorphic to  $S^{-1}M$ .
- (2) For an ideal  $I \leq A$ ,  $A/I \otimes_A M$  is naturally isomorphic to  $M/IM$ .

**Proof**

The canonical map  $M \rightarrow S^{-1}M$  mapping  $m \mapsto m/1$  defines a map  $f: S^{-1}A \otimes_A M \rightarrow S^{-1}M$  by  $f(1 \otimes m) = m/1$ . Then

$$f(a/s \otimes m) = a/sf(1 \otimes m) = am/s.$$

We define its inverse  $g: S^{-1}M \rightarrow S^{-1}A \otimes_A M$  by

$$g(m/s) = 1/s \otimes m.$$

We claim this is well-defined: if  $m/s = m'/s'$  then  $1/s \otimes m = 1/s' \otimes m'$ . If  $t(ms' - m's) = 0$  then

$$\frac{1}{s} \otimes m = \frac{ts'}{ts's} \otimes m = \frac{1}{ts's} \otimes ts'm = \frac{1}{ts's} \otimes tsm' = \frac{1}{s'} \otimes m'$$

as desired, and clearly  $g$  is a module morphism.  $f$  and  $g$  are also clearly morphisms.

The proof for the second point is similar: projection  $M \rightarrow M/IM$  is an  $A$ -module morphism. Thus this gives rise to an  $A/IA$ -module morphism  $f: (A/IA) \otimes_A M \rightarrow M/IM$  by  $f(1 \otimes m) = \overline{m}$ . Then

$$f(\overline{a} \otimes m) = af(1 \otimes m) = a\overline{m} = \overline{am}.$$

Its inverse is  $g: M/IM \rightarrow (A/IA) \otimes_A M$  given by  $g(\overline{m}) = 1 \otimes m$ ; this is easily seen to be well-defined and an inverse. ■

**4.1.2 Corollary**

Let  $B$  be an  $A$ -algebra and  $M$  a  $B$ -module. Then there is a natural map

$$B \otimes_A M|_A \rightarrow M, \quad b \otimes m \mapsto bm.$$

In particular this is a natural map  $m_B: B \otimes_A B \rightarrow B$  by  $b \otimes b' \mapsto bb'$ .

**Proof**

This map is the map corresponding to the identity of  $\text{hom}_A(M|_A, M|_A)$ . The second point is for  $M = B$ . ■

**4.1.3 Lemma**

If  $B, C$  are  $A$ -algebras, then their tensor product  $B \otimes_A C$  is naturally an  $A$ -algebra and their coproduct in the category of  $A$ -algebras.

**Proof**

Let  $D = B \otimes_A C$ . The natural algebra structure is given by  $(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$ , and  $A \rightarrow D$  by  $a \mapsto a(1 \otimes 1)$ . This is the composition of

$$D \times D \rightarrow D \otimes_A D, \quad (d, d') \mapsto d \otimes d',$$

and

$$D \otimes_A D = (B \otimes_A C) \otimes_A (B \otimes_A C) \cong (B \otimes_A B) \otimes_A (C \otimes_A C) \rightarrow B \otimes_A C$$

where the final map is the tensor of  $m_B \otimes m_C$ .

The inclusion maps  $i_B: B \rightarrow D$  and  $i_C: C \rightarrow D$  are given by  $b \mapsto b \otimes 1$  and  $c \mapsto 1 \otimes c$ .

Now if  $E$  is another  $A$ -algebra and there are maps  $f: B \rightarrow E, g: C \rightarrow E$ , then if  $h: D \rightarrow E$  commutes with the rest of the diagram, i.e.

$$g = h \circ i_C, \quad f = h \circ i_B,$$

then we must have

$$h(b \otimes c) = h((b \otimes 1)(1 \otimes c)) = h(b \otimes 1)h(1 \otimes c) = hi_B(b)hi_C(c) = f(b)g(c).$$

Thus  $h$  is unique, and this is well-defined (it is bilinear). ■

## 4.2 Flatness

### 4.2.1 Definition

A functor is **left-exact** if it preserves all finite limits, and is **right-exact** if it preserves all finite colimits. A functor is **exact** if it is both left- and right-exact (preserves all finite limits and colimits).

Clearly the composition of (left/right-) exact functors are (left/right-) exact.

We quickly note a few equivalences which are useful to keep in mind. The proof provided is not complete, but is not too complicated to complete.

### 4.2.2 Proposition

For an additive functor  $F$  between Abelian categories (in particular between module categories), the following are equivalent:

- (1)  $F$  is left-exact,
- (2) If  $0 \rightarrow A \rightarrow B \rightarrow C$  is an exact sequence, then the induced sequence  $0 \rightarrow FA \rightarrow FB \rightarrow FC$  is exact as well,
- (3) If  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is a short exact sequence, then the induced sequence  $0 \rightarrow FA \rightarrow FB \rightarrow FC$  is exact.

In particular from (2) we see that left-exact functors preserve injections.

Similarly, the following are equivalent:

- (1)  $F$  is right-exact,
- (2) If  $A \rightarrow B \rightarrow C \rightarrow 0$  is an exact sequence, then the induced sequence  $FA \rightarrow FB \rightarrow FC \rightarrow 0$  is exact as well,
- (3) If  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is a short exact sequence, then the induced sequence  $FA \rightarrow FB \rightarrow FC \rightarrow 0$  is exact.

In particular right-exact functors preserve surjections.

Consequently, the following are equivalent:

- (1)  $F$  is exact,
- (2)  $F$  preserves short exact sequences,
- (3)  $F$  preserves long exact sequences,
- (4)  $F$  is left-exact and preserves surjections,
- (5)  $F$  is right-exact and preserves injections.

### Proof

First we note that an additive functor immediately preserves all finite biproducts. And  $0 \rightarrow A \rightarrow B \rightarrow C$  being exact is equivalent to it being a kernel diagram. Thus  $F$  is left-exact iff it preserves it. And  $F$  preserves left-exact sequences (i.e. ones in point (2)) iff it maps short exact sequences to left-exact ones. Clearly if it preserves left-exact sequences it maps short exact ones to left-exact ones (SES are in particular left-exact), and the converse is shown by restricting the rightmost map to its image.

Since all injections are kernels, we see that left-exact functors preserve injections.

Similarly we show the equivalences for right-exact functors.

By the previous two equivalences, we see  $F$  being exact is equivalent to preserving SES sequences. And preserving long-exact sequences is done by properly restricting and quotienting. If  $F$  is left-exact and preserves surjections, then if  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is exact, we know that  $0 \rightarrow FA \rightarrow FB \rightarrow FC$  is exact, and  $FB \rightarrow FC \rightarrow 0$  is exact (as it preserves surjections). Putting this together, if  $F$  is left-exact and preserves surjections it is exact. ■

### 4.2.3 Proposition

Left-adjoint functors are right-exact, and right-adjoint functors are left-exact.

### Proof

Left-adjoint functors preserve *all* colimits, and are thus in particular right-exact. Similar for right-adjoint functors. ■

Tensor products are left-adjoints and thus right-exact:

### 4.2.4 Proposition

If  $M' \rightarrow M \rightarrow M'' \rightarrow 0$  is an exact sequence of  $A$ -modules, then for any  $A$ -module  $N$ , the induced sequence

$$M' \otimes_A N \rightarrow M \otimes_A N \rightarrow M'' \otimes_A N \rightarrow 0$$

is exact.

### Note:

Tensor products are not exact; they do not preserve injections. Indeed, if  $f: A \rightarrow A$  is the map  $f(x) = ax$  for  $a \in A$ , then  $f \otimes M = f \otimes \text{id}_M$  is the endomorphism of  $A \otimes_A M \cong M$  which maps  $m \mapsto am$ . When  $a$  is not a zero divisor (e.g.  $A$  is a domain),  $f$  is injective, but  $f \otimes M$  need not be (e.g.  $M = A/(a)$ ).

Since  $\text{hom}_A(M, \bullet)$  is right-adjoint, it is left exact so it preserves finite limits. It is not hard to show that  $\text{hom}_A(\bullet, M)$  is left-exact as well (but contravariantly). Furthermore the Yoneda embedding reflects exactness:

#### 4.2.5 Proposition

A sequence of  $A$ -modules  $M_\bullet = M' \rightarrow M \rightarrow M'' \rightarrow 0$  is exact iff

$$\mathrm{hom}(M_\bullet, N) = 0 \rightarrow \mathrm{hom}(M'', N) \rightarrow \mathrm{hom}(M, N) \rightarrow \mathrm{hom}(M', N)$$

is exact for all  $N \in \mathrm{Mod}_A$ . Similarly  $N_\bullet = 0 \rightarrow N' \rightarrow N \rightarrow N''$  is exact iff

$$\mathrm{hom}(M, N_\bullet) = 0 \rightarrow \mathrm{hom}(M, N') \rightarrow \mathrm{hom}(M, N) \rightarrow \mathrm{hom}(M, N'')$$

is exact for all  $M$ .

#### Proof

The “only if” part is due to left exactness. The converse (second point) is done by choosing  $M = N'$  and looking at where  $\mathrm{id}$  goes (this shows that the composition of maps is zero), then  $M$  to be the kernel of the second map, and showing that this implies exactness at  $M$ , and then you just need to show the first map is injective. ■

While not all tensor products are exact, some are. We give this case a special term.

#### 4.2.6 Definition

An  $A$ -module  $M$  is **flat** if the functor  $\bullet \otimes_A M$  is exact.

#### Note:

Since  $\bullet \otimes_A M$  is right-exact for any module  $M$ ,  $M$  is flat iff  $\bullet \otimes_A M$  preserves injections.

#### 4.2.7 Example

If  $S \subseteq A$  is multiplicative, then  $\bullet \otimes_A S^{-1}A$  is naturally isomorphic to  $S^{-1}\bullet$ , which is exact. Thus  $S^{-1}A$  is a flat  $A$ -module.

If  $M$  is free, so  $M \cong A^{\oplus I}$ , then

$$\bullet \otimes_A M \cong \bullet \otimes_A A^{\oplus I} (\bullet \otimes_A A)^{\oplus I} \cong \bullet^{\oplus I}.$$

The second isomorphism is due to left adjoints preserving colimits. And  $\bullet^{\oplus I}$  is exact (it maps injective maps to injective maps, more generally direct sums in module categories are exact, satisfying Grothendieck's axiom (AB4)).

#### 4.2.8 Lemma

Let  $M$  be an  $A$ -module and  $S \subseteq A$  a multiplicative set.

- (1) If  $N$  is an  $S^{-1}A$ -module, it is an  $A$ -module via the canonical map  $A \rightarrow S^{-1}A$ , and there is a natural isomorphism

$$M \otimes_A N \rightarrow S^{-1}M \otimes_{S^{-1}A} N, \quad m \otimes n \mapsto m/1 \otimes n.$$

- (2) For an  $A$ -module  $N$ , there is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \rightarrow S^{-1}(M \otimes_A N), \quad \frac{m}{s} \otimes \frac{n}{t} \mapsto \frac{m \otimes n}{st}.$$

In particular for every prime  $\mathfrak{p} \leq A$ , there is a natural isomorphism

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes_A N)_{\mathfrak{p}}.$$

### Proof

There is a natural isomorphism (by associativity of the tensor)

$$S^{-1}M \otimes_{S^{-1}A} N \cong (M \otimes_A S^{-1}A) \otimes_{S^{-1}A} N \cong M \otimes_A N$$

which maps  $m/s \otimes n$  to  $m \otimes 1/s \otimes n$  to  $m \otimes n/s$ . So in reverse it maps  $m \otimes n$  to  $m/1 \otimes n$  as desired.

There is a natural isomorphism

$$S^{-1}M \otimes_{S^{-1}A} S^{-1}N \cong M \otimes_A S^{-1}N \cong M \otimes_A (N \otimes_A S^{-1}A) \cong (M \otimes_A N) \otimes_A S^{-1}A,$$

and this is naturally isomorphic to  $S^{-1}(M \otimes_A N)$ . ■

#### 4.2.9 Corollary

Flatness is local, i.e. the following are equivalent:

- (1)  $M$  is a flat  $A$ -module,
- (2)  $S^{-1}M$  is a flat  $S^{-1}A$ -module for all multiplicative  $S \subseteq A$ ,
- (3)  $M_{\mathfrak{p}}$  is a flat  $A_{\mathfrak{p}}$ -module for all prime  $\mathfrak{p} \leq A$ ,
- (4)  $M_{\mathfrak{m}}$  is a flat  $A_{\mathfrak{m}}$ -module for all maximal  $\mathfrak{m} \leq A$ .

### Proof

If  $M$  is flat then by the above lemma

$$\bullet \otimes_{S^{-1}A} S^{-1}M \cong S^{-1}(\bullet \otimes_A M),$$

which is a composition of exact functors and thus exact.

Now if  $M_{\mathfrak{m}}$  is flat for all maximal  $\mathfrak{m} \leq A$ , we show  $M$  is flat. Recall we need only show that  $\bullet \otimes_A M$  preserves injectives. Recall that  $f$  is injective iff  $f_{\mathfrak{m}}$  is injective for all maximal  $\mathfrak{m} \leq A$ . Now,  $f_{\mathfrak{m}}$  is isomorphic to  $f \otimes M_{\mathfrak{m}}$ , which is injective since  $M_{\mathfrak{m}}$  is flat, we see  $f_{\mathfrak{m}}$  is injective for all  $\mathfrak{m}$  as desired. ■

## 4.3 The Tor functor

We now quickly state a few definitions and results from homological algebra.

### 4.3.1 Definition

Let  $\mathcal{A}, \mathcal{B}$  be Abelian categories (e.g. module categories). Then a **(homological)  $\delta$ -functor** is a collection of functors  $\{F_n: \mathcal{A} \rightarrow \mathcal{B}\}_{n \geq 0}$  satisfying the following two properties:

- (1) If  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is an SES in  $\mathcal{A}$ , then there exist maps  $\delta_n: F_n C \rightarrow F_{n-1} A$  inducing a long

exact sequence:

$$\cdots \rightarrow F_{n+1}C \xrightarrow{\delta} F_nA \rightarrow F_nB \rightarrow F_nC \xrightarrow{\delta} F_{n-1}A \rightarrow \cdots$$

- (2) The maps  $\delta$  are natural: if we have a map of SES  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  to  $0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0$ , the following diagram commutes:

$$\begin{array}{ccc} F_nC & \xrightarrow{\delta} & F_{n-1}A \\ \downarrow & & \downarrow \\ F_nC' & \xrightarrow{\delta} & F_{n-1}A' \end{array}$$

The functor  $F_0$  is called the **base** of the  $\delta$ -functor. Cohomological  $\delta$ -functors are defined similarly but with the arrows inverted.

If  $F$  and  $G$  are  $\delta$ -functors, then a morphism of  $\delta$ -functors is a sequence of natural transformations  $f_n: F_n \Rightarrow G_n$  which commute with the  $\delta$ -maps.

A **universal**  $\delta$ -functor is a  $\delta$ -functor  $F$  such that for any other  $\delta$ -functor  $G$  and a natural transformation  $f_0: G_0 \rightarrow F_0$ ,  $f_0$  extends uniquely to a morphism of  $\delta$ -functors  $G \rightarrow F$ .

Clearly universal  $\delta$ -functors are unique up to isomorphism. A functor  $F$  is called **coeffaceable** if for every object  $A$  there is a surjective  $g: B \rightarrow A$  such that  $F(g) = 0$ . A  $\delta$ -functor  $F$  is **coeffaceable** if every  $F_n$  is coeffaceable for  $n \geq 1$ . An important due to Grothendieck is that a coeffaceable  $\delta$ -functor is universal.

Notice that if  $F$  is a  $\delta$ -functor then its base  $F_0$  is right-exact. This is because if  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is exact, then the end of the long exact sequence induced by  $F$  is

$$\cdots \rightarrow F_0A \rightarrow F_0B \rightarrow F_0C \rightarrow 0.$$

Another result, due to Grothendieck again, is that every right-exact functor is the base of a (necessarily unique up to isomorphism) universal  $\delta$ -functor (under some basic assumptions on the underlying category). This is called the **derived functor**.

In particular,  $\bullet \otimes_A M$  is right-exact for any  $A$ -module  $M$ . Thus it has a derived functor, called the **Tor-functor**, denoted by  $\text{Tor}_n^A(\bullet, M)$ . We can also compute the derived functor of  $N \otimes_A \bullet$  giving us another functor  $\text{Tor}_n^A(N, \bullet)$ , and a non-trivial result tells us that these are two parts of the same bifunctor  $\text{Tor}_n^A(\bullet, \bullet): \text{Mod}_A \times \text{Mod}_A \rightarrow \text{Mod}_A$ . Since the tensor product is commutative (up to isomorphism), the derived functors  $\text{Tor}_n^A(M, \bullet)$  and  $\text{Tor}_n^A(\bullet, M)$  are isomorphic (by uniqueness of universal  $\delta$ -functors).

We summarize these results in the following theorem:

#### 4.3.2 Theorem (The Tor functor)

There is a unique (up to unique isomorphism) collection of additive bi-functors  $\text{Tor}_n^A: \text{Mod}_A \times \text{Mod}_A \rightarrow \text{Mod}_A$  such that

- (1)  $\text{Tor}_0^A = \bullet \otimes_A \bullet$ ,
- (2)  $\text{Tor}_n^A(\bullet, M)$  is a  $\delta$ -functor for each  $A$ -module  $M$ .
- (3)  $\text{Tor}_n^A(N, M) \cong \text{Tor}_n^A(M, N)$  for all  $A$ -modules  $N, M$  and all  $n \geq 0$ .

The construction of derived functors is relatively straight-forward, but proving that they are (1) functors, (2)  $\delta$ -functors, (3) universal  $\delta$ -functors is less trivial. (1) and (2) is done pretty much directly as follows. Suppose  $F$  is right-exact, and let  $A$  be an object (a module in our case). Then we define a **projective resolution** of  $A$  to be a long exact sequence

$$P_\bullet \rightarrow A = \cdots \xrightarrow{d_1} P_1 \xrightarrow{d_0} P_0 \rightarrow A \rightarrow 0,$$

where each  $P_n$  is a **projective object**. In category of modules, free modules are a type of projective object (but not in general the only type), so we can instead consider only resolutions by free modules.

We then consider the induced chain complex

$$F(P_\bullet) = \cdots \xrightarrow{Fd_2} F(P_2) \xrightarrow{Fd_1} F(P_1) \xrightarrow{Fd_0} F(P_0) \rightarrow 0.$$

And then we define  $F_n$  to be the  $n$ th homology modules of this sequence:

$$F_n(A) = H_n(F(P_\bullet)) = \frac{\ker Fd_n}{\operatorname{im} Fd_{n+1}}.$$

It can be shown that any map  $f: A \rightarrow B$  there is a unique lift of  $f$  to a sequence of maps between projective resolutions of  $A$  and  $B$  (unique up to homotopy), and we define  $F_n(f)$  to be the induced map on the homology modules.

It can be shown that this construction is not dependent on the choice of projective resolution: all projective resolutions give naturally isomorphic functors. Furthermore, this is a  $\delta$ -functor. And it is not hard to see that  $F_0 \cong F$ .

Now notice that if  $P$  is a projective object (whatever that means), certainly  $0 \rightarrow P \rightarrow P \rightarrow 0$  is a projective resolution (the non-trivial map is the identity). Thus  $F_\bullet$  are the homology modules of the chain complex  $0 \rightarrow FP \rightarrow 0$ , which vanish for  $n \geq 1$ . In particular, as we said that free modules are projective, if  $N$  is a free  $A$ -module then  $\operatorname{Tor}_n^A(N, M) = 0$  for all  $M \in \operatorname{Mod}_A$ .

Now we call an object  $P$   **$F$ -acyclic** if  $F_n P = 0$  for all  $n \geq 1$ , so clearly projectives are  $F$ -acyclic for any  $F$ . A simple result is that derived functors can be computed using  $F$ -acyclic resolutions. That is, if  $P_\bullet \rightarrow A \rightarrow 0$  is a resolution of  $A$  using  $F$ -acyclic objects, then  $F_n(A) \cong H_n(F(P_\bullet))$ . Of course to know what objects are  $F$ -acyclic we need to compute  $F_n$  via usual projective resolutions.

But in our case, if  $N \in \operatorname{Mod}_A$  is flat, then it is  $\bullet \otimes_A M$ -acyclic for all  $M \in \operatorname{Mod}_A$  and vice versa.

### 4.3.3 Proposition

Let  $N \in \operatorname{Mod}_A$ , then the following are equivalent:

- (1)  $N$  is flat,
- (2)  $\operatorname{Tor}_n^A(N, M) = 0$  for all  $M \in \operatorname{Mod}_A$  and all  $n \geq 1$ ,
- (3)  $\operatorname{Tor}_1^A(N, M) = 0$  for all  $M \in \operatorname{Mod}_A$

In particular free modules are flat.

### Proof

Consider a projective resolution of  $M$ ,  $P_\bullet \rightarrow M \rightarrow 0$ . Then  $\operatorname{Tor}_n^A(N, M)$  are the homology modules of the chain complex

$$\cdots \rightarrow P_1 \otimes_A N \rightarrow P_0 \otimes_A N \rightarrow 0.$$

But  $N$  is flat, and thus tensoring with it is exact so it preserves exact sequences. So the above chain complex is exact at every index  $n \geq 1$  (at  $n = 0$  we have taken away  $M \otimes_A N$  so it is no longer exact, this is because we want  $\operatorname{Tor}_0^A \cong \bullet \otimes_A \bullet$ ). Thus the homology modules are all trivial, so  $\operatorname{Tor}_n^A(N, M) = 0$  for all  $n \geq 1$ ; so we have shown that (1) implies (2).

And (2) clearly implies (3).

Now suppose (3) then let  $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$  be an SES. Then by  $\delta$ -functoriality there is a long exact sequence

$$\cdots \rightarrow \operatorname{Tor}_1^A(N, Z) \rightarrow N \otimes_A X \rightarrow N \otimes_A Y \rightarrow N \otimes_A Z \rightarrow 0.$$

Since  $\operatorname{Tor}_1^A(N, Z) = 0$  by assumption, this shows that

$$0 \rightarrow N \otimes_A X \rightarrow N \otimes_A Y \rightarrow N \otimes_A Z \rightarrow 0$$

is exact, and thus  $N$  is flat.

And as explained,  $\mathrm{Tor}_n^A(P, \bullet) = 0$  for any projective, and thus free, module  $P$ . So by the equivalence, free modules are flat. ■

Notice that free modules are flat for a simpler reason, which we show in the following lemma.

#### 4.3.4 Lemma

The direct sums and filtered colimits of flat modules are flat.

##### Proof

Since the tensor product is a left-adjoint it commutes with (preserves) colimits. That is,

$$\bullet \otimes_A \bigoplus_i M_i \cong \bigoplus_i \bullet \otimes_A M_i, \quad \bullet \otimes_A \varinjlim M_i \cong \varinjlim (\bullet \otimes_A M_i).$$

Since direct sums and filtered colimits are exact (Grothendieck's axioms (AB4) and (AB5)), if each  $M_i$  is flat and thus  $\bullet \otimes_A M_i$  are all exact, then the above are exact as well. ■

Since  $A$  is a flat  $A$ -module (since  $\bullet \otimes_A A$  is isomorphic to the identity functor), direct sums of  $A$  are flat. Direct sums of  $A$  are just free modules.

#### 4.3.5 Lemma

Homology commutes with exact functors: if  $C_\bullet$  is a chain complex and  $F$  an exact functor, then

$$H_n(F(C_\bullet)) \cong F(H_n(C_\bullet)) \text{ for all } n.$$

##### Proof

Notice that a quotient  $M/N$  is the cokernel of the inclusion  $N \hookrightarrow M$ , and the image of  $f$  is just  $\mathrm{im} f = \ker \mathrm{coker} f$ . Since  $F$  is exact, it preserves injections and cokernels, so  $F(N)$  remains a subobject of  $F(M)$  and  $F(M)/F(N) \cong F(M/N)$  since  $F$  preserves cokernels. And it preserves images as well since  $F \mathrm{im} f = F \ker \mathrm{coker} f \cong \ker \mathrm{coker} Ff = \mathrm{im} Ff$ .

In particular,

$$F(H_n(C_\bullet)) = F(\ker f_n / \mathrm{im} f_{n+1}) \cong (F \ker f_n) / (F \mathrm{im} f_{n+1}) \cong (\ker Ff_n) / (\mathrm{im} Ff_{n+1}) = H_n(F(C_\bullet))$$

as desired. ■

#### 4.3.6 Theorem

The Tor functors commute with direct sums and filtered colimits:

$$\mathrm{Tor}_n^A\left(N, \bigoplus_i M_i\right) \cong \bigoplus_i \mathrm{Tor}_n^A(N, M_i), \quad \mathrm{Tor}_n^A(N, \varinjlim M_i) \cong \varinjlim \mathrm{Tor}_n^A(N, M_i).$$

##### Proof



We explained that Tor can be computed via acyclic resolutions, and that flat modules are acyclic. So if

$$P_\bullet^i \rightarrow M_i \rightarrow 0$$

are flat resolutions of  $M_i$  for each  $i$ , then by exactness of direct sums and filtered colimits,

$$\bigoplus_i P_\bullet^i \rightarrow \bigoplus_i M_i \rightarrow 0, \quad \varinjlim P_\bullet^i \rightarrow \varinjlim M_i \rightarrow 0$$

are exact sequences. Furthermore we just showed that these are flat, and thus Tor can be computed using their homology. And again since direct sums and filtered colimits are exact, homology commutes with them. Putting everything together,

$$\begin{aligned} \mathrm{Tor}_n^A\left(N, \bigoplus_i M_i\right) &= H_n\left(\bigoplus_i P_\bullet^i\right) = \bigoplus_i H_n(P_\bullet^i) = \bigoplus_i \mathrm{Tor}_n^A(M_i), \\ \mathrm{Tor}_n^A(N, \varinjlim M_i) &= H_n(\varinjlim P_\bullet^i) = \varinjlim H_n(P_\bullet^i) = \varinjlim \mathrm{Tor}_n^A(M_i). \end{aligned}$$

■

Now that we have the basic properties of the Tor functor down, we will use it to prove properties about flatness.

#### 4.3.7 Lemma

An  $A$ -module  $M$  is flat iff  $\mathrm{Tor}_1^A(M, A/I) = 0$  for all ideals  $I \leq A$ .

#### Proof

The “only if” case is clear, as the first torsion module of a flat module with any other module is zero. Conversely, we need to show that  $\mathrm{Tor}_1^A(M, N) = 0$  for any  $N \in \mathrm{Mod}_A$ .

First suppose that  $N$  is fg, and we induct on the number of generators  $a_1, \dots, a_n$ . For  $n = 0$  it is clear since then  $N = 0$ . And for  $n = 1$  we have that  $N = (a_1) \cong A/\mathrm{ann}(a_1)$  and thus the result follows from the assumption. Now inductively consider the SES

$$0 \rightarrow (a_n) \rightarrow N \rightarrow N/(a_n) \rightarrow 0.$$

Thus we have an exact sequence

$$\cdots \rightarrow \mathrm{Tor}_1^A(M, (a_n)) \rightarrow \mathrm{Tor}_1^A(M, N) \rightarrow \mathrm{Tor}_1^A(M, N/(a_n)) \rightarrow \cdots$$

Now  $N/(a_n)$  is generated by  $n - 1$  elements, and thus  $\mathrm{Tor}_1^A(M, N/(a_n)) = 0$  by induction, and by our base case  $\mathrm{Tor}_1^A(M, (a_n)) = 0$ . Thus  $\mathrm{Tor}_1^A(M, N) = 0$  by exactness.

And if  $N$  is not fg, then  $N = \varinjlim N_i$  where  $N_i$  ranges over all the fg submodules of  $N$ . Since Tor commutes with filtered colimits, we have

$$\mathrm{Tor}_1^A(M, \varinjlim N_i) \cong \varinjlim \mathrm{Tor}_1^A(M, N_i) = \varinjlim 0 = 0,$$

as desired. ■

#### 4.3.8 Theorem

An  $A$ -module  $M$  is flat iff for every ideal  $I \leq A$ , the map  $I \otimes_A M \rightarrow IM$  given by  $a \otimes m \mapsto am$  is an isomorphism.

**Proof**

First note that the given map is induced by an  $A$ -bilinear map and is thus well-defined.

If  $M$  is flat and  $I \leq A$  is an ideal, then we have a short exact sequence

$$0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0,$$

where the first map is given by inclusion and the second by projection. By flatness, this induces an SES, recalling that  $A \otimes_A M \cong M$ ,

$$0 \rightarrow I \otimes_A M \rightarrow M \rightarrow M \otimes_A (A/I) \rightarrow 0.$$

The first map is then the map  $M \otimes_A I \rightarrow M$  given by  $m \otimes a \mapsto am$ , the map we are investigating, which is injective by exactness.

Recall that  $M \otimes_A (A/I) \cong M/IM$  by  $m \otimes (a + I) \mapsto am + IM$ . Thus the second map corresponds to the map  $M \rightarrow M/IM$  given by projection. By exactness, the image of the first map is the kernel of the second, which is  $IM$ . So we have that our map is injective onto its image,  $IM$ , and thus is an isomorphism  $I \otimes_A M \rightarrow IM$  as desired.

Conversely by the above lemma, it is sufficient to show that  $\text{Tor}_1^A(M, R/I) = 0$ . From the SES  $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ , we have an exact sequence

$$\text{Tor}_1^A(M, I) \rightarrow \text{Tor}_1^A(M, A) \rightarrow \text{Tor}_1^A(M, A/I) \rightarrow M \otimes_A I \rightarrow M \rightarrow M \otimes_A (A/I) \rightarrow 0.$$

Since  $A$  is free,  $\text{Tor}_1^A(M, A) = 0$ . And the map  $M \otimes_A I \rightarrow M$  is precisely the map  $m \otimes a \mapsto am$ , which is injective by assumption. So we have an exact sequence

$$0 \rightarrow \text{Tor}_1^A(M, A/I) \rightarrow M \otimes_A I \xrightarrow{f} M,$$

where  $f$  is injective. This requires that  $\text{Tor}_1^A(M, A/I) = 0$  as desired. ■

**4.3.9 Example**

Let  $R = \mathbb{C}[x]$  and  $M$  be the complex algebra of functions  $\mathbb{C} \rightarrow \mathbb{C}$ . Then we define  $R \rightarrow M$  by mapping  $x$  to the zero map. This turns  $M$  into an  $R$ -algebra, where  $p \in R$  is mapped to  $p(0)$ .

We claim that  $M$  is not a flat  $R$ -module. Indeed, let  $I = (x)$  so that  $IM = 0$ . But  $I \cong R$  as  $R$ -modules (since they are both free  $R$ -modules on a single generator), so  $I \otimes_R M \cong M \neq 0$ . Thus  $IM$  and  $I \otimes_R M$  are not isomorphic and in particular  $M$  is not flat.

While free modules are always flat, the converse is not always true. But it is in an important special case, whose proof is homework.

**4.3.10 Proposition**

Suppose  $A$  is a local ring and  $M$  a fg  $A$ -module. Then  $M$  is free iff it is flat.

**4.3.11 Corollary**

Let  $M$  be an  $A$ -module, then the following are equivalent:

- (1)  $M$  is flat,
- (2)  $M_{\mathfrak{p}}$  is a free  $A_{\mathfrak{p}}$ -module for all prime  $\mathfrak{p} \leq M$ ,
- (3)  $M_{\mathfrak{m}}$  is a free  $A_{\mathfrak{m}}$ -module for all maximal  $\mathfrak{m} \leq M$ .

**Proof**

Flatness is local:  $M$  is flat iff  $M_{\mathfrak{p}}$  is a flat  $A_{\mathfrak{p}}$ -module for all prime  $\mathfrak{p} \leq M$  or equivalently all maximal  $\mathfrak{m} \leq M$ . And  $A_{\mathfrak{p}}$  is a local ring, so by the previous proposition this is iff  $M_{\mathfrak{p}}$  is free. ■

We can also give a nicer criteria in the case that our ring is a domain.

**4.3.12 Lemma**

Let  $A$  be a local domain with maximal ideal  $\mathfrak{m}$ , and let  $k = A/\mathfrak{m}$  be its residue field and  $K = \text{Frac}(A)$  its field of fractions. Let  $M$  be a fg  $A$ -module, and suppose  $\dim_k(k \otimes_A M) = \dim_K(K \otimes_A M) = n$ . Then  $M$  is free of rank  $n$ .

**Proof**

Nakayama's lemma implies that there is a generating set  $x_1, \dots, x_n \in M$  such that their projections form a basis in  $k \otimes_A M \cong M/\mathfrak{m}M$  (the final formulation of Nakayama's lemma). Define  $\varphi: A^n \rightarrow M$  which maps the standard basis vectors  $e_i$  to  $x_i$ . This gives an SES

$$0 \rightarrow \ker \varphi \rightarrow A^n \rightarrow M \rightarrow 0.$$

Recall that localization of flat modules are flat, and since  $K$  is the localization of  $A$  by  $A - 0$ , it is a flat  $A$ -module. Thus tensoring with  $K$  preserves the SES,

$$0 \rightarrow K \otimes_A \ker \varphi \rightarrow K \otimes_A A^n \rightarrow K \otimes_A M \rightarrow 0.$$

Recall that tensoring commutes with direct sums, so this is isomorphic to

$$0 \rightarrow K \otimes_A \ker \varphi \rightarrow K^n \rightarrow K \otimes_A M \rightarrow 0.$$

By assumption  $K \otimes_A M$  has dimension  $n$  over  $K$ , and since the final map is a surjection it must be an isomorphism by considering dimensions. Thus  $K \otimes_A \ker \varphi = 0$ , and since localization commutes with tensoring,

$$0 = K \otimes_A \ker \varphi \cong \ker \varphi_{A-0}.$$

So for every  $x \in \ker \varphi$  we must have that  $x/1 = 0$ , i.e. there is an  $a \in A - 0$  such that  $ax = 0$ , so  $\ker \varphi$  is a torsion module. But  $\ker \varphi \subseteq A^n$  which is not torsion as  $A$  is a domain, so  $\ker \varphi = 0$  and thus  $\varphi$  is an isomorphism and  $M \cong A^n$  as desired. ■

**4.3.13 Theorem (Fiber dimension criterion for flatness)**

Let  $A$  be a domain, and  $M$  a fg  $A$ -module. Let  $S = \text{Spec}(A)$ , then  $M$  is flat iff the map  $S \rightarrow \mathbb{N}$  given by  $\mathfrak{p} \mapsto \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M)$  is constant.

**Proof**

As we showed by locality of flatness,  $M$  is flat iff  $M_{\mathfrak{p}}$  is a free  $A_{\mathfrak{p}}$ -module for all prime  $\mathfrak{p} \leq A$ . So first suppose  $M_{\mathfrak{p}}$  is free for all prime  $\mathfrak{p} \leq A$ , say  $M_{\mathfrak{p}} \cong A_{\mathfrak{p}}^{n_{\mathfrak{p}}}$  ( $n_{\mathfrak{p}}$  is finite since  $M$  is fg). Then

$$k(\mathfrak{p}) \otimes_A M = (A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}) \otimes_A M \cong M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong A_{\mathfrak{p}}^{n_{\mathfrak{p}}}/(\mathfrak{p}A_{\mathfrak{p}}^{n_{\mathfrak{p}}}) \cong k(\mathfrak{p})^{n_{\mathfrak{p}}},$$

so  $\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = n_{\mathfrak{p}}$ .

Now in general notice that if  $\mathfrak{p} \leq \mathfrak{q} \leq A$  are prime, and  $M$  is an  $A$ -module, then  $M_{\mathfrak{q}}$  remains an  $A$ -module and we can localize it by  $\mathfrak{p}$ . It is easy to see that

$$(M_{\mathfrak{q}})_{\mathfrak{p}} \cong M_{\mathfrak{p}}.$$

Thus,

$$M_{(0)} \cong (M_{\mathfrak{p}})_{(0)} \cong (A_{\mathfrak{p}}^{n_{\mathfrak{p}}})_{(0)} \cong (A_{\mathfrak{p}})_{(0)}^{n_{\mathfrak{p}}} \cong A_{(0)}^{n_{\mathfrak{p}}} = \text{Frac}(A)^{n_{\mathfrak{p}}}.$$

And so we have that

$$\text{Frac}(A)^{n_{\mathfrak{p}}} \cong M_{(0)} \cong \text{Frac}(A) \otimes_A M.$$

The right-hand side is independent of  $\mathfrak{p}$ , and thus  $n_{\mathfrak{p}}$  must be constant. Thus we see that  $\mathfrak{p} \mapsto n_{\mathfrak{p}} = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M)$  is constant, as desired.

Conversely suppose the map is constant, let its value be  $n$ . In particular we have that, for  $K = \text{Frac}(A) = A_{(0)}$ ,

$$n = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = \dim_K(K \otimes_A M).$$

We also have that

$$k(\mathfrak{p}) \otimes_A M \cong (k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}) \otimes_A M \cong k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} (A_{\mathfrak{p}} \otimes_A M) \cong k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}.$$

Thus we have

$$\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}) = \dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_A M) = \dim_K(K \otimes_A M).$$

Now also we know

$$K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} = (A_{\mathfrak{p}})_{(0)} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \cong (A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}})_{(0)} \cong (M_{\mathfrak{p}})_{(0)} \cong M_{(0)} \cong K \otimes_A M.$$

Thus  $\dim_K(K \otimes_A M) = \dim_K(K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}})$ , and so

$$\dim_{k(\mathfrak{p})}(k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}) = \dim_K(K \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}).$$

By the previous lemma,  $M_{\mathfrak{p}}$  is free. ■

# II. ALGEBRAIC GEOMETRY

## 1 Affine Schemes

### 1.1 Basic definitions

#### 1.1.1 Definition

Let  $R$  be a ring, then its **(prime) spectrum** is its set of prime ideals,

$$\operatorname{Spec}(R) = \{\mathfrak{p} \leq R \mid \mathfrak{p} \text{ is prime}\}.$$

#### Note:

We can similarly define the **maximal spectrum** of a ring to be its set of maximal ideals. This is in general less useful than the prime spectrum, but its definition is analogous:

$$\operatorname{Specm}(R) = \{\mathfrak{m} \leq R \mid \mathfrak{m} \text{ is maximal}\}.$$

We can give  $\operatorname{Spec}(R)$  a topological structure. Instead of defining the open sets of a spectrum, we will define its closed sets.

#### 1.1.2 Definition

Let  $S \subseteq R$  be a subset, then define its **vanishing set** to be

$$V(S) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}.$$

Notice that clearly we have

$$V(S) = V((S)),$$

so that every vanishing set is a vanishing set of an ideal. Note that  $V(I) = V(J)$  does not imply  $I = J$ : for instance  $I = 2\mathbb{Z}$  and  $J = 4\mathbb{Z}$  in  $R = \mathbb{Z}$ ; their vanishing sets are both  $\{2\mathbb{Z}\}$ .

#### 1.1.3 Lemma

A proper ideal  $\mathfrak{p} \leq R$  is prime iff  $IJ \leq \mathfrak{p}$  implies  $I \leq \mathfrak{p}$  or  $J \leq \mathfrak{p}$  for all ideals  $I, J \leq R$ .

#### Proof

The “if” part is clear: if  $ab \in \mathfrak{p}$ , then  $(a) \cdot (b) = (ab) \in \mathfrak{p}$  so either  $(a)$  or  $(b)$  is a subset of  $\mathfrak{p}$ . Conversely, suppose that  $\mathfrak{p}$  is a prime ideal and  $IJ \leq \mathfrak{p}$  but  $I, J$  are both not subsets of  $\mathfrak{p}$ . Then let  $x \in I, y \in J$  be elements not in  $\mathfrak{p}$ : we then have  $xy \in IJ \leq \mathfrak{p}$ , a contradiction to  $\mathfrak{p}$ ’s primality. ■

#### 1.1.4 Proposition

The collection of vanishing sets forms the collection of closed sets of a topology on  $\text{Spec}(R)$ .

### Proof

We need to show that  $\text{Spec}(R), \emptyset$  are vanishing sets, the arbitrary intersection of vanishing sets is a vanishing set, and the binary union of vanishing sets is a vanishing set. Clearly we have

$$\text{Spec}(R) = V(0), \quad \emptyset = V(R)$$

so the first point is true.

Now consider a collection  $\{S_\alpha\}_\alpha$  of subsets. Then we have:

$$\bigcap_\alpha V(S_\alpha) = V\left(\bigcup_\alpha S_\alpha\right),$$

since  $S_\alpha \leq \mathfrak{p}$  for all  $\alpha$  iff  $\bigcup_\alpha S_\alpha \leq \mathfrak{p}$ . In terms of ideals, we have that

$$\bigcap_\alpha V(I_\alpha) = V\left(\left(\bigcup_\alpha I_\alpha\right)\right) = V\left(\sum_\alpha I_\alpha\right).$$

For the final point on binary unions, we can restrict our attention to vanishing sets of ideals. Due to the above lemma we have that

$$V(I) \cup V(J) = V(IJ).$$

Thus the collection of vanishing sets forms a topology of closed sets. ■

#### 1.1.5 Definition

The topology defined above on  $\text{Spec}(R)$  is called the **Zariski topology** of  $\text{Spec}(R)$ .

#### 1.1.6 Example

Consider  $R = \mathbb{C}[x]$ , a PID. Its prime ideals are the principal ideals generated by irreducible elements and zero, in this case

$$\text{Spec}(\mathbb{C}[x]) = \{(x - \lambda) \mid \lambda \in \mathbb{C}\} \cup \{(0)\}.$$

For a polynomial  $f \in \mathbb{C}[x]$ , we have that  $(x - \lambda) \in V(f)$  iff  $\lambda$  is a root of  $f$ . Thus

$$V(f) = \{(x - \lambda) \mid f(\lambda) = 0\}.$$

As we showed above, for  $I \leq \mathbb{C}[x]$ , we have that

$$V(I) = V\left(\bigcup_{f \in I} \{f\}\right) = \bigcap_{f \in I} V(f) = \{(x - \lambda) \mid \lambda \text{ is a root of every polynomial in } I\}.$$

Thus  $\text{Spec}(\mathbb{C}[x])$ 's proper closed sets are precisely the finite subsets of  $\text{Spec}(\mathbb{C}[x]) - (0)$ , i.e. it induces the cofinite topology on this subspace. The ideal  $(0)$  is not contained in any proper closed set, so its closure is all of  $\text{Spec}(\mathbb{C}[x])$ .

**1.1.7 Example**

Consider  $R = \mathbb{Z}$ , a PID. Its prime ideals are the principal ideals generated by primes and zero, so

$$\mathrm{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ prime}\} \cup \{(0)\}.$$

For  $n \in \mathbb{Z}$ ,  $(p) \in V(n)$  iff  $p$  divides  $n$  so that

$$V(n) = \{(p) \mid p \text{ is a prime divisor of } n\}.$$

Thus  $\mathrm{Spec}(\mathbb{Z})$ 's proper closed sets are precisely the finite subsets of  $\mathrm{Spec}(\mathbb{Z}) - 0$ , so it again induces the cofinite topology on this subspace. As before,  $(0)$ 's closure is all of  $\mathrm{Spec}(\mathbb{Z})$ .

In the two examples above, we saw that the point  $(0)$ 's closure is all of  $\mathrm{Spec}(R)$ . If  $(0)$  is a prime ideal, then this is always true:  $(0) \in V(I)$  iff  $I = (0)$  in which case  $V(0) = \mathrm{Spec}(R)$ . Thus  $\mathrm{Spec}(R)$  is the only closed set containing the zero ideal, and is thus its closure.

**1.1.8 Example**

Consider the case that  $R = k$  is a field. The only prime ideal of  $k$  is the zero ideal, so  $\mathrm{Spec}(k)$  is a single-point space. There is only one topology on such a space: the trivial and discrete topology coincide.

The map  $R \mapsto \mathrm{Spec}(R)$  is functorial.

**1.1.9 Definition**

Let  $\varphi: R \rightarrow S$  be a ring morphism, then define  $\varphi^*: \mathrm{Spec}(S) \rightarrow \mathrm{Spec}(R)$  by

$$\varphi^*(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}), \quad \mathfrak{q} \in \mathrm{Spec}(S).$$

This is well-defined, as we showed before that the preimage of a prime ideal is prime.

**1.1.10 Proposition**

$\varphi^*$  is continuous, and functorial.

**Proof**

To show that  $\varphi^*$  is continuous, we need to show that its preimage on closed sets is closed. So let  $I \leq R$  be an ideal, and consider  $(\varphi^*)^{-1}V(I)$ :

$$\mathfrak{q} \in (\varphi^*)^{-1}V(I) \iff \varphi^{-1}\mathfrak{q} \in V(I) \iff I \leq \varphi^{-1}\mathfrak{q} \iff \varphi(I) \leq \mathfrak{q},$$

so that we have

$$(\varphi^*)^{-1}V(I) = V(\varphi(I)),$$

which is closed as desired. Note that while  $\varphi(I) \subseteq S$  may not be an ideal, all vanishing sets are closed. Functoriality is clear. ■

**Note:**

As  $\text{Specm}(R) \subseteq \text{Spec}(R)$ , we endow  $\text{Specm}(R)$  with the subspace topology. In particular its closed sets are

$$V_m(S) = \{\mathfrak{m} \in \text{Specm}(R) \mid S \subseteq \mathfrak{m}\}.$$

But note that  $R \mapsto \text{Specm}(R)$  is not functorial: the above  $\varphi^*$  does not necessarily restrict to maximal ideals.

Let us consider more the Zariski topology. Clearly  $V(\bullet)$  is contravariant (order reversing).

### 1.1.11 Proposition

Let  $I, J \leq R$  be ideals, then  $V(I) = V(J)$  iff  $\sqrt{I} = \sqrt{J}$ . In particular  $V(I) = V(\sqrt{I})$  and  $\text{Spec}(R) = V(\text{nil}(R))$ .

#### Proof

We know that  $\sqrt{I}$  is the intersection of all prime ideals containing  $I$ , meaning

$$\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}.$$

So the first implication is clear: if  $V(I) = V(J)$  then  $\sqrt{I} = \sqrt{J}$ . Now if  $\sqrt{I} = \sqrt{J}$ , let  $\mathfrak{p} \in V(I)$ . If  $\mathfrak{p} \notin V(J)$  then there is an  $f \in J$  not in  $\mathfrak{p}$ , but then  $f \in \sqrt{J}$  and  $f \notin \mathfrak{p}$  so  $f \notin \sqrt{I}$ , a contradiction. ■

#### Note:

Let the **Jacobson radical** of an ideal  $I \leq R$  be the intersection of all maximal ideals containing  $I$ :

$$\text{Jac}_R(I) = \bigcap_{I \leq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{m} \in V_m(I)} \mathfrak{m}.$$

We can alter the proof above to show that  $V_m(I) = V_m(J)$  iff  $\text{Jac}_R(I) = \text{Jac}_R(J)$ . Thus  $V_m(I) = V_m(\text{Jac}_R(I))$  and  $\text{Specm}(R) = V_m(\text{Jac}(R))$ , where  $\text{Jac}(R) = \text{Jac}_R(R)$  is the usual Jacobson radical of a ring.

### 1.1.12 Definition

Let  $Z \subseteq \text{Spec}(R)$  be a subset. Define its **ideal** to be

$$\mathcal{I}(Z) = \{f \in R \mid Z \subseteq V(f)\}.$$

We can characterize  $\mathcal{I}(Z)$  more concretely:

$$\mathcal{I}(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

Indeed, if  $f \in \mathcal{I}(Z)$  then  $Z \subseteq V(f)$  so for every  $\mathfrak{p} \in Z$  we have that  $\mathfrak{p} \in V(f)$ , meaning  $f \in \mathfrak{p}$ . Conversely if  $f \in \mathfrak{p}$  for every  $\mathfrak{p} \in Z$ , then  $\mathfrak{p} \in V(f)$  for every  $\mathfrak{p}$  and so  $Z \subseteq V(f)$ .

From this characterization it is obvious that  $\mathcal{I}(\bullet)$  is also order reversing.

### 1.1.13 Proposition



- (1) Let  $Z \subseteq \text{Spec}(R)$  be a subset, then  $V(\mathcal{I}(Z)) = \overline{Z}$ ,  $Z$ 's closure.
- (2) Let  $J \leq R$  be an ideal, then  $\mathcal{I}(V(J)) = \sqrt{J}$ .

### Proof

For the first point we need to show that  $V(\mathcal{I}(Z))$  is the smallest vanishing set containing  $Z$ . It contains  $Z$  since if  $\mathfrak{p} \in Z$  then  $\mathcal{I}(Z) \leq \mathfrak{p}$  by definition (it is the intersection of all such  $\mathfrak{p}$ ), so  $\mathfrak{p} \in V(\mathcal{I}(Z))$ . Now if  $Z \subseteq V(J)$  then for  $\mathfrak{p} \in Z$  we have that  $J \leq \mathfrak{p}$ , so that  $J \subseteq \bigcap_{\mathfrak{p} \in Z} \mathfrak{p} = \mathcal{I}(Z)$ . Since  $V$  is contravariant, we have that  $V(\mathcal{I}(Z)) \subseteq V(J)$ , as desired.

For the second point, we have

$$\mathcal{I}(V(J)) = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{J \leq \mathfrak{p}} \mathfrak{p} = \sqrt{J},$$

as desired. ■

### 1.1.14 Corollary

Let  $\mathfrak{p} \in \text{Spec}(R)$  be a point, then its closure is  $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ . In particular  $\{\mathfrak{p}\}$  is closed iff  $\mathfrak{p}$  is maximal.

### Proof

This is because  $\mathcal{I}\{\mathfrak{p}\} = \mathfrak{p}$ , so apply the above proposition. And  $V(\mathfrak{p}) = \{\mathfrak{p}\}$  iff  $\mathfrak{p}$  is maximal. ■

### 1.1.15 Example

This shows that  $\text{Spec}(R)$  need not be  $T_1$ . Recall that a space  $X$  is  $T_1$  if for every two distinct  $x \neq y \in X$ , there is an open set  $U$  such that  $x \in U$  and  $y \notin U$ . Equivalently a space is  $T_1$  iff its singletons are closed: if singletons are closed let  $U = \{y\}^c$ , and conversely if for every  $y \neq x$  we choose a  $U_y$  such that  $y \in U_y$  and  $x \notin U_y$  we see that

$$X - \{x\} = \bigcup_{y \neq x} U_y$$

is open and thus  $\{x\}$  is closed.

In any case, clearly if we have prime ideals which are not maximal, then  $\text{Spec}(R)$  is not  $T_1$ . For instance, take  $X = \text{Spec}(\mathbb{Z})$ . Since  $(0) \in X$  is not maximal, it is not closed and thus the space cannot be  $T_1$ .

$\text{Spec}(R)$  is  $T_0$  though: all distinct points are topologically distinct. Indeed, if  $\mathfrak{p}, \mathfrak{q} \in \text{Spec}(R)$  have the same neighborhoods they have the same closure and thus  $V(\mathfrak{p}) = V(\mathfrak{q})$ . Thus  $\mathfrak{p} \in V(\mathfrak{q})$  and  $\mathfrak{q} \in V(\mathfrak{p})$  so that  $\mathfrak{p} \leq \mathfrak{q} \leq \mathfrak{p}$  so that they are equal.

### 1.1.16 Definition

Let  $P, Q$  be two posets. An **antitone Galois connection** are two order-reversing functions  $F: P \rightarrow Q$  and  $G: Q \rightarrow P$  such that for every  $p \in P, q \in Q$ ,

$$q \leq Fp \iff p \leq Gq.$$

An alternative method of looking at antitone Galois connections is via category theory. Recall that we can view posets as categories, where there exists a unique morphism  $a \rightarrow b$  iff  $a \leq b$ .

**1.1.17 Proposition**

An antitone Galois connection between two posets  $P, Q$  consists of two functors  $F: P \rightarrow Q^{\text{op}}, G$  such that  $F \dashv G$  forms an adjunction.

**Proof**

Order reversing functions between posets are just contravariant functors. So if  $F, G$  form an antitone Galois connection, then they are functors  $F: P^{\text{op}} \rightarrow Q$  and  $G: Q^{\text{op}} \rightarrow P$ . Then  $F^{\text{op}}: P \rightarrow Q^{\text{op}}, G$  forms an adjunction: we have  $q \leq Fp$  iff  $F^{\text{op}}p \leq^{\text{op}} q$  iff  $p \leq Gq$ . The converse is shown similarly. ■

**1.1.18 Corollary**

$F: P \rightarrow Q, G$  forms an antitone Galois connection iff they are order reversing and  $p \leq GFp$  and  $q \leq FGq$  for all  $p \in P, q \in Q$ .

**Proof**

This is just the unit-counit formulation of an adjunction (remember contravariance!). A natural transformation between functors  $X, Y$  between posets amounts to saying  $X(\bullet) \leq Y(\bullet)$ . ■

**1.1.19 Corollary**

The two functions  $V: \mathcal{P}(R) \rightarrow \mathcal{P}(\text{Spec}(R)), \mathcal{I}$  form an antitone Galois connection.

**Proof**

The maps are order reversing and  $S \subseteq \sqrt{(S)} = \mathcal{I}(V(S))$  and  $Z \subseteq \overline{Z} = V(\mathcal{I}(Z))$ , so they form an antitone Galois connection. Showing directly that  $Z \subseteq V(S)$  iff  $S \subseteq \mathcal{I}(Z)$  is easy to do as well. ■

**1.2 Topological properties of affine schemes****1.2.1 Definition**

Let  $R$  be a ring,  $X = \text{Spec}(R)$  its spectrum. For  $f \in R$ , define

$$X_f = X - V(f) = \{\mathfrak{p} \in \text{Spec}(R) \mid f \notin \mathfrak{p}\}.$$

**1.2.2 Proposition**

The collection  $\{X_f\}_{f \in R}$  forms a basis of open sets for  $X = \text{Spec}(R)$ .

**Proof**

This is equivalent to showing that  $\{V(f)\}_{f \in R}$  forms a basis of closed sets; that they are closed and every

closed set is an intersection of closed basis sets. Clearly  $V(f)$  are closed, and for  $S \subseteq R$  we have

$$V(S) = V\left(\bigcup_{f \in S} \{f\}\right) = \bigcap_{f \in S} V(f)$$

as desired. ■

This shows that for an open set  $X - V(S)$ , we have

$$X - V(S) = \bigcup_{f \in S} X_f.$$

### 1.2.3 Proposition

Let  $R$  be a ring and  $X = \text{Spec}(R)$ .

- (1) Let  $\{f_\alpha\}_\alpha$  be a collection of elements from  $R$ . Then  $X = \bigcup_\alpha X_{f_\alpha}$  iff  $(\{f_\alpha\}_\alpha) = R$ .
- (2) Let  $f \in R$ , then  $X_f = X$  iff  $f \in R^\times$ .
- (3) Let  $f \in R$ , then  $X_f = \emptyset$  iff  $f \in \text{nil}(R)$ .

### Proof

We saw that

$$\bigcup_\alpha X_{f_\alpha} = X - V(\{f_\alpha\}_\alpha)$$

which is equal to  $X$  iff  $V(\{f_\alpha\}_\alpha) = \emptyset$ . This is iff

$$V(\{f_\alpha\}_\alpha) = V((\{f_\alpha\}_\alpha)) = V(R) \iff \sqrt{(\{f_\alpha\}_\alpha)} = R.$$

Now  $\sqrt{J} = R$  iff  $J = R$ , as  $1 \in \sqrt{J}$  iff  $1 \in J$ . So this is iff  $(\{f_\alpha\}_\alpha) = R$ , as desired.

The second point follows from the first:  $X_f = X$  iff  $(f) = R$  iff  $f \in R^\times$ .

For the third point,  $X_f = \emptyset$  iff  $V(f) = X = V(0)$  iff  $\sqrt{(f)} = \text{nil}(R)$ . This occurs iff  $f \in \text{nil}(R)$ : clearly it implies this, and conversely  $0 \leq (f) \leq \text{nil}(R)$  and radicals are order-preserving so we get  $\sqrt{(f)} = \text{nil}(R)$ . ■

### 1.2.4 Corollary

$X = \text{Spec}(R)$  is compact.

### Proof

We need to show every open cover of  $X$  has a finite subcover. Let  $\mathcal{U}$  be an open cover of  $X$ , then we can assume that  $\mathcal{U}$  consists of basic open sets:  $\mathcal{U} = \{X_{f_\alpha}\}_\alpha$ . By the first point above, since  $\mathcal{U}$  is a cover,  $(\{f_\alpha\}_\alpha) = R$ . Thus  $1 \in (\{f_\alpha\}_\alpha)$  so there are finitely many  $f_1, \dots, f_n$  such that  $1 = x_1 f_1 + \dots + x_n f_n$ . Then  $(f_1, \dots, f_n) = R$  and so again by the above proposition,  $\{X_{f_1}, \dots, X_{f_n}\}$  is a finite subcover of  $\mathcal{U}$ , as desired. ■

**Note:**

This result shows that the usual topological concept of compactness is in general not strong enough for algebraic geometry. Thus in algebraic geometry topological compactness is usually called **quasicompactness**, and compactness is reserved for a stronger concept.

**1.2.5 Proposition**

Let  $f \in R$ , and consider  $R_f$ , the localization of  $R$  at  $\{f^n\}_{n \in \mathbb{N}}$ .  $\text{Spec}(R_f)$  is homeomorphic to  $X_f$ .

**Proof**

Let  $\iota: R \rightarrow R_f$  be the structure map. Then  $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$  is a bijective correspondence between prime ideals of  $R_f$  and prime ideals of  $R$  which do not intersect  $\{f^n\}$ . A prime ideal  $\mathfrak{p} \leq R$  does not intersect  $\{f^n\}$  iff  $f \notin \mathfrak{p}$ . Thus  $\iota^*: Y = \text{Spec}(R_f) \rightarrow \text{Spec}(R)$  is injective and its image are all prime ideals such that  $f \notin \mathfrak{p}$ , i.e.  $\text{im } \iota^* = X_f$ .

We claim that  $\iota^*$  is a homeomorphism onto its image. We already know it is continuous and bijective, so we need only show it is open; we will show it sends basic open sets to basic open sets. So let  $g/f^n \in R_f$ , with the basic open set

$$Y_{g/f^n} = \{\mathfrak{q} \in Y \mid g/f^n \notin \mathfrak{q}\}.$$

We claim that  $\iota^*(Y_{g/f^n}) = X_g \cap X_f$ . Indeed, if  $\mathfrak{q} \in Y_{g/f^n}$  then  $g/f^n \notin \mathfrak{q}$  and so  $g \notin \iota^*\mathfrak{q}$  (since  $g/f^n$  is  $\iota(g)$  times a unit, so if  $g \in \iota^*\mathfrak{q}$  then  $\iota(g) \in \mathfrak{q}$  and then  $g/f^n \in \mathfrak{q}$ ); thus  $\iota^*\mathfrak{q} \in X_g$  as desired. Conversely if  $\mathfrak{p} \in X_f \cap X_g$  then  $g/f^n \notin \mathfrak{p}R_f$ : otherwise  $g/f^n = h/f^m$  for  $h \in \mathfrak{p}$  and thus  $f^k(f^m g - f^n h) = 0$  so that  $f^{m+k}g \in \mathfrak{p}$ , but  $f, g \notin \mathfrak{p}$  a contradiction. Now,  $\mathfrak{p}R_f$  is prime and  $\iota^*\mathfrak{p}R_f = \mathfrak{p}$ , showing the equality. ■

We now focus on the connectedness of topological spaces, in particular spectra of rings. Recall that a topological space is **disconnected** if it can be written as the disjoint union of two open subsets, equivalently if it has a non-trivial clopen subset.

**1.2.6 Lemma**

A surjective ring morphism  $\varphi: R \rightarrow S$  induces a closed embedding  $\varphi^*: \text{Spec}(S) \rightarrow \text{Spec}(R)$  (i.e. its image is closed, and it is a topological embedding). In particular if  $I \leq R$  is an ideal, then the projection  $\pi: R \rightarrow R/I$  induces a closed embedding  $\pi^*: \text{Spec}(R/I) \rightarrow \text{Spec}(R)$  whose image is  $V(I)$ .

**Proof**

$\varphi$  induces an isomorphism of rings  $R/\ker \varphi \rightarrow S$ , which carries prime ideals of  $R/\ker \varphi$  bijectively to prime ideals of  $S$ . Thus  $\varphi^*$ 's image is the set of all prime ideals containing  $\ker \varphi$ , i.e.  $\text{im } \varphi^* = V(\ker \varphi)$  which is closed by definition. Since  $\varphi$  is surjective,  $\varphi^*$  is injective and thus a bijection onto its image. So to show it is a topological embedding, we need only show that it is a closed map.

Indeed, for  $J \leq S$ ,

$$\varphi^*V(J) = \{\varphi^{-1}\mathfrak{q} \mid J \leq \mathfrak{q}\} = \{\varphi^{-1}\mathfrak{q} \mid \varphi^{-1}J \leq \varphi^{-1}\mathfrak{q}\} = V(\varphi^{-1}J) \cap \text{im } \varphi^*,$$

which is closed. (The second equality is due to  $\varphi$ 's surjectivity.) ■

**1.2.7 Proposition**

If  $R = S \times T$  is a product ring, then  $\text{Spec}(R)$  is homeomorphic to  $\text{Spec}(S) \sqcup \text{Spec}(T)$ , and thus disconnected.

Conversely if  $\text{Spec}(R) = Y \sqcup Z$  is disconnected, then there exist rings  $S, T$  such that  $Y, Z$  are homeomorphic to  $\text{Spec}(S), \text{Spec}(T)$  respectively, and  $R \cong S \times T$ .

### Proof

Let  $X = \text{Spec}(R)$ . Consider  $(1, 0), (0, 1) \in R$ , so that  $R$ 's unit is their sum. Then we showed above that  $X = X_{(1,0)} \cup X_{(0,1)}$ , and this is a disjoint union since  $V(0, 1) \cup V(1, 0) = X$  (if  $\mathfrak{p} \in X$  and  $(0, 1), (1, 0) \notin \mathfrak{p}$  then their product 0 is not in  $\mathfrak{p}$ , a contradiction). We also showed that  $X_{(1,0)} \cong \text{Spec}(R_{(1,0)})$ .

So let  $\pi: R \rightarrow S$  be the projection, so  $\pi(1, 0) = 1 \in S^\times$  so that by the universal property of localization  $\pi$  induces a map  $\pi': R_{(1,0)} \rightarrow S$  which is surjective since  $\pi$  is. Now if  $\pi'((a, b)/(1, 0)) = \pi'((x, y)/(1, 0))$  then  $a = x$  so  $(a, b)(1, 0) = (x, y)(1, 0)$  and thus  $(a, b)/(1, 0) = (x, y)/(1, 0)$  showing that  $\pi'$  is also injective. Thus  $S \cong R_{(1,0)}$ , and similarly  $T \cong R_{(0,1)}$  so that we have desired result.

Since  $Y, Z$  are closed, there are  $I, J \leq R$  such that  $Y = V(I)$  and  $Z = V(J)$ . Then let  $S = R/I$  and  $T = R/J$ ; we showed above that  $\pi_1: R \rightarrow R/I$  induces a homeomorphism  $\pi_1^*: \text{Spec}(R/I) \rightarrow V(I) \subseteq \text{Spec}(R)$  and similar for  $R/J$ . Now,  $Y \cap Z = V(I) \cap V(J) = V(I + J)$  is empty so that  $I + J = R$ , and thus  $I, J$  are coprime. By the chinese remainder theorem,

$$R \cong R/I \times R/J = S \times T,$$

as desired. ■

### 1.2.8 Definition

An element  $f \in R$  is **idempotent** if  $f^2 = f$ . The **trivial idempotents** of  $R$  are 0 and 1.

If  $f \in R$  is idempotent, then  $fR$  is a ring. Indeed, it is closed under addition and multiplication (it is the ideal generated by  $f$ ), and  $f$  is its multiplicative identity since it is idempotent.

### 1.2.9 Proposition

$R \cong S \times T$  for non-trivial rings  $S, T$  iff  $R$  has nontrivial idempotents.

### Proof

If  $R \cong S \times T$  then  $(1, 0), (0, 1)$  are nontrivial idempotents. Conversely let  $f \in R$  be a nontrivial idempotent, then define  $S = fR$  and  $T = (1 - f)R$ . Then  $R \rightarrow S \times T$  given by  $g \mapsto (fg, (1 - f)g)$  is clearly an injective since if  $fg = (1 - f)g = 0$  then  $g = 0$ , and it is surjective since for  $(fg, (1 - f)h) \in S \times T$  then we want a  $x \in R$  such that  $fg = fx$  and  $(1 - f)h = (1 - f)x$ , so  $(1 - f)h = x - fg$  and thus  $x = (1 - f)h + fg$  gives the desired element. They are also non-trivial since  $f \neq 0$ . ■

### 1.2.10 Definition

A ring  $R$  is **connected** if it has no nontrivial idempotents.

### 1.2.11 Corollary

$\text{Spec}(R)$  is connected iff  $R$  is.

**1.2.12 Definition**

A topological space  $X$  is **reducible** if there exist two proper closed subsets  $Z, W \subsetneq X$  such that  $X = Z \cup W$ . Otherwise  $X$  is **irreducible**.

Note that irreducibility implies connectedness; if  $X$  is disconnected then its union  $Z \sqcup W$  of open sets are also closed. The converse is not true in general (take the axes of  $\mathbb{R}^2$ ).

**1.2.13 Proposition**

$X = \text{Spec}(R)$  is irreducible iff  $\text{nil}(R)$  is prime.

**Proof**

If  $\text{nil}(R)$  is not prime, then there are  $f, g \notin \text{nil}(R)$  such that  $fg \in \text{nil}(R)$ . Let  $Z = V(f)$  and  $W = V(g)$ , so that

$$Z \cup W = V(f) \cup V(g) = V(fg) = X,$$

where the second equality is because  $V(f) \cup V(g) = V((f)(g)) = V(fg)$ , and the final equality is because  $V(fg) \supseteq V(\text{nil}(R)) = X$ .

Conversely if  $X$  is reducible, then  $X = V(I) \cup V(J)$  so that  $V(IJ) = X$  for  $V(I), V(J) \neq X$ . But then  $IJ \leq \text{nil}(R)$  and  $I$  and  $J$  cannot be contained in  $\text{nil}(R)$ , so  $\text{nil}(R)$  is not prime. ■

**1.2.14 Corollary**

The spectrum of a domain is irreducible.

**1.2.15 Example**

The affine space  $\mathbb{A}_{\mathbb{C}}^n = \text{Spec}(\mathbb{C}[x_1, \dots, x_n])$  is irreducible since  $\mathbb{C}[x_1, \dots, x_n]$  is a domain.

Similarly the spectrum of  $R = \mathbb{C}[x_1, \dots, x_n]/(f)$  is irreducible when  $f$  is irreducible (and then  $(f)$  is prime and  $R$  is a domain). But the converse does not hold: let  $R = \mathbb{C}[x]/(x^2)$ ,  $x^2$  is reducible, and  $\text{nil}(R) = \sqrt{(x^2)} = (x)$  is prime.

**1.2.16 Definition**

A topological space  $X$  is **Noetherian** if its poset of closed subsets has the DCC.

**1.2.17 Proposition**

The spectrum of a Noetherian ring is a Noetherian space.

**Proof**

Let  $R$  be Noetherian, then a descending chain of closed subsets of  $\text{Spec}(R)$  corresponds (via  $\mathcal{I}$ ) to an ascending chain of radical ideals of  $R$ . Since  $R$  is Noetherian, these must stabilize, and thus so too must the chain of closed subsets (since  $V \circ \mathcal{I}$  is the identity on closed sets). ■

Note that  $\text{Spec}(R)$  being Noetherian does not imply that  $R$  is.

**1.2.18 Lemma (Noetherian induction)**

Let  $\Sigma$  be a poset with DCC, and  $P$  be a boolean function  $\Sigma \rightarrow \{0, 1\}$  (think of it as a property of elements of  $\Sigma$ ). Suppose further that

- (1) if  $p \in \Sigma$  is minimal then  $P(p) = 1$ ,
- (2) if  $p \in \Sigma$  is not minimal and  $P(q) = 1$  for all  $q < p$ , then  $P(p) = 1$ .

Then  $P$  is identically 1 on  $\Sigma$  (so the property holds for all elements of  $\Sigma$ ).

**Proof**

If we assume that,  $\Sigma'$  the subset of  $\Sigma$  consisting of elements such that  $P(p) = 0$ , is nonempty, then since  $\Sigma$  has DCC it has a minimal element  $p \in \Sigma'$ . It cannot be minimal in  $\Sigma$  since then  $P(p) = 1$  by condition (1). But for every  $q < p$  since  $q \notin \Sigma'$  (by minimality) we have  $P(q) = 1$  and thus by (2) we have  $P(p) = 1$  a contradiction. ■

**1.2.19 Proposition**

Let  $X$  be a Noetherian space, then  $X$  has finitely many irreducible components.

**Proof**

We use Noetherian induction on the closed subsets of  $X$  to show that they have finitely many irreducible components. Since  $X$  itself is closed, the result follows.

If  $Y \subseteq X$  is a minimal closed subset then it has no proper closed subsets and is in particular irreducible (has a single irreducible component).

Now suppose  $Y \subseteq X$  is non-minimal closed and every proper closed subset  $Z \subsetneq Y$  has finitely many irreducible components. If  $Y$  itself is irreducible, we are done. Otherwise suppose  $Y = Z \cup W$  for  $Z, W$  proper closed subsets. Let  $C_1, \dots, C_n$  be the irreducible components of  $Z$  and  $D_1, \dots, D_m$  those of  $W$ . Let  $T \subseteq Y$  be an irreducible component, so

$$T = Y \cap T = (Z \cup W) \cap T = \left( \bigcup_i C_i \cap T \right) \cup \left( \bigcup_i D_i \cap T \right).$$

Since each intersection is closed and  $T$  is irreducible, we must have that  $T \subseteq C_i$  or  $T \subseteq D_i$  for some  $i$ . Since  $T$  is a component and is thus maximal, we must have equality. So the irreducible components of  $Y$  are contained in  $C_1, \dots, C_n, D_1, \dots, D_m$ , and thus there are finitely many. ■

**1.2.20 Lemma**

The following are equivalent:

- (1)  $X$  is Noetherian,
- (2) Every subspace of  $Y$  is Noetherian,
- (3) Every subspace of  $X$  is quasi-compact,
- (4) Every open subspace of  $X$  is quasi-compact.

**Proof**

For a subspace  $Y \subseteq X$ , its closed sets are of the form  $Y \cap F$  for  $F \subseteq X$  closed. So let  $\Sigma$  be a set of closed subsets of  $Y$ , then let  $\Sigma' = \{F \subseteq X \text{ closed} \mid F \cap Y \in \Sigma\}$ . Then  $\Sigma'$  must have a minimal element  $F$ , then clearly  $F \cap Y \in \Sigma$  is minimal, so  $Y$  is Noetherian. So (1) and (2) are equivalent.

To show that (1) implies (3), it is sufficient to show that  $X$  is quasi-compact, and the general case follows from (2). Being quasicompact is equivalent to a collection of closed subsets having the finite intersection property has nonempty intersection. So suppose  $\{F_i\}_i \subseteq X$  has the finite intersection property. Let  $\Sigma$  be the set of finite intersections of  $F_i$ , so a poset of closed subsets. Since  $X$  is Noetherian, let  $F \in \Sigma$  be minimal, and since  $F_i$  have the finite intersection property,  $F \neq \emptyset$ . Then clearly the intersection of all  $F_i$  is equal to  $F \neq \emptyset$  as desired.

To show (4) implies (1), let  $U_1 \subseteq U_2 \subseteq \dots$  be increasing open subsets in  $X$ . Then  $U = \bigcup_i U_i$  is open in  $X$  and thus quasi-compact, so there is a finite cover  $U = \bigcup_{i=1}^n U_i$ , but then  $U = U_n$ . So the chain terminates at  $n$ , as desired. ■

**1.3 Categorical constructions**

In this section we consider the category of affine schemes, i.e. spectrums of rings. A morphism of affine schemes  $\text{Spec}(R) \rightarrow \text{Spec}(S)$  corresponds to a ring morphism  $S \rightarrow R$ .

Recall the following case of a categorical limit.

**1.3.1 Definition**

A **pullback** (or **fiber product**) is the limit of a diagram of shape  $\bullet \rightarrow \bullet \leftarrow \bullet$  (if one exists). Explicitly, given two morphisms  $A \xrightarrow{f} C \xleftarrow{g} B$ , their fiber product is an object  $A \times_C B$  along with morphisms  $A \xleftarrow{i} A \times_C B \xrightarrow{j} B$  such that the following diagram commutes and is universal (in the sense that it is the terminal such diagram which commutes):

$$\begin{array}{ccc} A \times_C B & \xrightarrow{\quad i \quad} & A \\ j \downarrow & & \downarrow f \\ B & \xrightarrow{\quad g \quad} & C \end{array}$$

Finite products and equalizers can be written as pullbacks with a terminal object. In particular if  $1$  is a terminal object, then the pullback of  $A \xrightarrow{!} 1 \xleftarrow{!} B$ , where  $!$  denotes the unique morphism, is the product of  $A$  and  $B$ . And if  $f, g: A \rightrightarrows B$  are parallel morphisms, the pullback of  $A \xrightarrow{(f,g)} B \times B \xleftarrow{\Delta} B$ , where  $\Delta$  is the diagonal morphism, is the equalizer of  $f, g$ .

Thus any category with pullbacks and a terminal object has all finite limits. We use this to demonstrate now that the category of affine schemes has all finite limits.

**1.3.2 Proposition**

Let  $X = \text{Spec}(A), Y = \text{Spec}(B), Z = \text{Spec}(R)$  be affine schemes, along with morphisms  $X \rightarrow Z \leftarrow Y$ , then this diagram has a pullback. Explicitly, its pullback is  $X \times_Z Y = \text{Spec}(A \otimes_R B)$ .

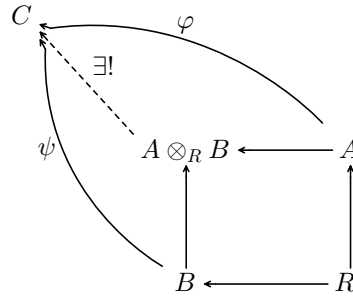
**Proof**

The morphisms  $X \rightarrow Z \leftarrow Y$  are induced by morphisms  $A \leftarrow R \rightarrow B$ , so  $A, B$  are  $R$ -algebras. Thus  $A \otimes_R B$  is an  $R$ -algebra as well, and the obvious maps  $A, B \rightarrow A \otimes_R B$  given by  $a \mapsto a \otimes 1, b \mapsto 1 \otimes b$  give us the cone  $X \times_Z Y \rightarrow X, Y$ . We need to show that this is indeed a limit cone, i.e. that it is terminal.

We need to show that if  $W = \text{Spec}(C)$  is any other affine scheme with morphisms  $W \rightarrow X, Y$  commuting



with  $X \rightarrow Z, Y \rightarrow Z$  then there is a unique morphism from  $W$  to  $X \times_Z Y = \text{Spec}(A \otimes_R B)$ . Translating this to the category of rings: if there are morphisms  $A \xrightarrow{\varphi} C \xleftarrow{\psi} B$  which commute with  $R \rightarrow A, B$ , then there is a unique morphism  $A \otimes_R B \rightarrow C$  making the diagram commute



Such a map must map  $a \otimes b$  to  $\varphi(a) \otimes \psi(b)$ . This map is  $R$ -bilinear (because  $\varphi, \psi$  commute with the maps  $R \rightarrow A, B$ ), and thus is well-defined, as desired. ■

### 1.3.3 Corollary

The category of affine schemes has all finite limits.

#### Proof

The ring  $\mathbb{Z}$  is initial: there is a unique ring morphism  $\mathbb{Z} \rightarrow A$  for all rings  $A$ . Thus its spectrum is terminal, so the category of affine schemes has a terminal object. The previous proposition showed the category has all pullbacks, so it has all finite limits. ■

## 2 Varieties

### 2.1 Affine varieties

In this section let  $k$  be an algebraically closed field,  $\text{Fun}(k^n, k)$  denote the  $k$ -algebra of functions  $k^n \rightarrow k$ , and  $k[x_1, \dots, x_n]$  be the  $k$ -algebra of polynomials over variables  $x_1, \dots, x_n$ .

Note that there is a natural  $k$ -algebra morphism

$$\text{ev}: k[x_1, \dots, x_n] \rightarrow \text{Fun}(k^n, k)$$

defined by  $\text{ev}(f)(P) = p(a_1, \dots, a_n)$  for a polynomial  $f$  and  $P = (a_1, \dots, a_n) \in k^n$ .

For a subset  $S \subseteq k[x_1, \dots, x_n]$  define its **zero set**

$$Z(S) = \{P \in k^n \mid f(P) = 0 \text{ for all } f \in S\} \subseteq k^n.$$

For  $f \in k^n$  we write  $Z(f)$  for  $Z(\{f\})$  so that  $Z(S) = \bigcap_{f \in S} Z(f)$ .

#### 2.1.1 Lemma

- (1)  $Z$  is order reversing: if  $S_1 \subseteq S_2$  then  $Z(S_1) \supseteq Z(S_2)$ .
- (2)  $Z(S)$  is equal to  $Z((S))$  ( $(S)$  is the ideal generated by  $S$ ).
- (3) For a collection of subsets  $\{S_\alpha\}_\alpha$ :

$$\bigcap_\alpha Z(S_\alpha) = Z\left(\bigcup_\alpha S_\alpha\right).$$

(4) For two subsets  $S_1, S_2$ , define  $S_1 S_2 = \{fg \mid f \in S_1, g \in S_2\}$ , then

$$Z(S_1) \cup Z(S_2) = Z(S_1 S_2).$$

(5)  $Z(1) = \emptyset$  and  $Z(0) = k^n$ .

### Proof

All of these points are clear. We prove only the fourth:  $Z(S_1) \cup Z(S_2) = Z(S_1 S_2)$ . Note that the ideal generated by  $S_1 S_2$  is precisely the product of the ideals generated by  $S_1$  and  $S_2$ . Thus  $Z(S_1)$  and  $Z(S_2)$  are contained in  $Z(S_1 S_2)$  by order-reversion. Conversely if  $P \notin Z(S_1) \cup Z(S_2)$ , then  $f(P), g(P) \neq 0$  for all  $f \in S_1, g \in S_2$  so  $(fg)(P) = f(P)g(P) \neq 0$  and thus  $P \notin Z(S_1 S_2)$ . ■

### 2.1.2 Corollary

The set of zero sets forms a topology of closed sets on  $k^n$ .

### 2.1.3 Definition

Define  $\mathbb{A}_k^n$  to be the topological space  $k^n$  whose underlying topology is given by the zero sets. This is called the **Zariski topology**, and  $\mathbb{A}_k^n$  is the  **$n$ -dimensional affine space**. When  $k$  is understood, we omit it and just write  $\mathbb{A}^n$ .

We also define  $k[\mathbb{A}^n] = k[x_1, \dots, x_n]$ .

Note that  $Z(f)$  forms a basis of closed sets: every closed set is an intersection of these basic closed sets. Thus

$$D(f) = \mathbb{A}^n - Z(f) = \{P \in k^n \mid f(P) \neq 0\}.$$

forms a basis of open sets.

Note that for every  $S \subseteq k[x_1, \dots, x_n]$ ,  $(S)$  is generated by finitely many  $f_1, \dots, f_n \in S$  since  $k[x_1, \dots, x_n]$  is Noetherian. Then

$$Z(S) = Z((S)) = Z(f_1, \dots, f_n) = \bigcap_{i=1}^n Z(f_i).$$

Thus every open set is a finite union of basic open sets. And so  $\mathbb{A}^n$  is quasicompact: if  $D(f_i)$  cover  $\mathbb{A}^n$  then

$$\mathbb{A}^n = \bigcup_i D(f_i) = D(\{f_i\}),$$

and as we showed (for zero sets, thus for open sets)  $D(\{f_i\})$  is the finite union of  $D(f_i)$ s.

### 2.1.4 Proposition

$\mathbb{A}^n$  is quasicompact.

### 2.1.5 Example

The closed subsets of  $\mathbb{A}^1$  are precisely the finite subsets of  $\mathbb{A}^1$ , or  $\mathbb{A}^1$  itself. Indeed,  $k[x]$  is a PID so all of its ideals are of the form  $Z(f)$  for a polynomial  $f$ . If  $f \neq 0$ , then  $f$  has finitely many roots so  $Z(f)$  is finite. And for a finite subset  $\{c_1, \dots, c_n\}$  of  $\mathbb{A}^1 = k$ ,  $f = (x - c_1) \cdots (x - c_n)$  has this set as its zero set. Thus the open subsets of  $\mathbb{A}^1$  are empty or cofinite. Since  $k$  is infinite (it is algebraically closed),  $\mathbb{A}^1$  is not Hausdorff.

### 2.1.6 Definition

For a subset  $Y \subseteq \mathbb{A}^n$ , define

$$\mathcal{I}(Y) = \{f \in k[x_1, \dots, x_n] \mid f(P) = 0 \text{ for all } P \in Y\}.$$

### 2.1.7 Proposition

- (1)  $\mathcal{I}$  is order-reversing.
- (2)  $\mathcal{I}(Y)$  is a radical ideal of  $k[x_1, \dots, x_n]$ .
- (3)  $\mathcal{I}(Y_1 \cup Y_2) = \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2)$ .
- (4) For every  $Y \subseteq \mathbb{A}^n$ ,  $Z(\mathcal{I}(Y)) = \overline{Y}$  is the closure of  $Y$  in  $\mathbb{A}^n$ .
- (5) **Nullstellensatz:** for every ideal  $J \leq k[x_1, \dots, x_n]$ ,  $\mathcal{I}(Z(J)) = \sqrt{J}$ .

### Proof

The first three points are clear. For the fourth, notice  $Y \subseteq Z(\mathcal{I}(Y))$  and  $Z(\mathcal{I}(Y))$  is closed, so  $\overline{Y} \subseteq Z(\mathcal{I}(Y))$ . Since  $\overline{Y}$  is closed,  $\overline{Y} = Z(J)$  for some ideal  $J$ . So

$$Y \subseteq \overline{Y} = Z(J) \implies \mathcal{I}(\overline{Y}) = \mathcal{I}(Z(J)) \subseteq \mathcal{I}(Y).$$

And  $J \subseteq \mathcal{I}(Z(J))$  clearly, so  $J \subseteq \mathcal{I}(Y)$ , and thus  $\overline{Y} = Z(J) \supseteq Z(\mathcal{I}(Y))$  as desired.

For the fifth point, since  $J \leq \mathcal{I}(Z(J))$  which is radical, we see  $\sqrt{J} \leq \mathcal{I}(Z(J))$ . The inverse inclusion is far deeper, and we give a proof next section. ■

### 2.1.8 Corollary

$Z$  and  $\mathcal{I}$  form an antitone Galois connection.

### 2.1.9 Corollary

For every proper ideal  $J < k[x_1, \dots, x_n]$ ,  $Z(J)$  is nonempty.

### Proof

If  $J \neq (1)$  then  $\mathcal{I}(Z(J)) = \sqrt{J} \neq (1)$ , thus  $Z(J)$  cannot be empty (since  $\mathcal{I}(\emptyset) = (1)$ ). ■

**Note:**

Recall that a topological space is irreducible iff there are no two proper closed subsets whose union is  $X$ . In homework this was shown to be equivalent to the condition that all open subsets intersect.

**2.1.10 Example**

Every closed subset of  $\mathbb{A}^1$  is either  $\mathbb{A}^1$  or finite. Since  $k$  is infinite,  $\mathbb{A}^1$ 's irreducible closed subsets are its closed points.

**2.1.11 Corollary**

A closed subset  $Y \subseteq \mathbb{A}^n$  is irreducible iff the ideal  $\mathcal{I}(Y) \subseteq k[\mathbb{A}^n]$  is either prime or all  $k[\mathbb{A}^n]$ .

**Proof**

If  $Y \subseteq \mathbb{A}^n$  is irreducible and  $fg \in \mathcal{I}(Y)$ , then we want to show either  $f \in \mathcal{I}(Y)$  or  $g \in \mathcal{I}(Y)$ . Notice that since  $fg \in \mathcal{I}(Y)$ , we have  $Y = \overline{Y} = Z(\mathcal{I}(Y)) \subseteq Z(fg) = Z(f) \cup Z(g)$ . Since  $Y$  is irreducible either  $Y \subseteq Z(f)$  or  $Y \subseteq Z(g)$ , meaning  $f \in \mathcal{I}(Y)$  or  $g \in \mathcal{I}(Y)$  as desired.

If  $\mathcal{I}(Y)$  is prime or all  $k[\mathbb{A}^n]$ , suppose  $Y = Y_1 \cup Y_2$  where  $Y_1, Y_2 \subseteq Y$  are closed. Then

$$\mathcal{I}(Y_1)\mathcal{I}(Y_2) \subseteq \mathcal{I}(Y_1) \cap \mathcal{I}(Y_2) = \mathcal{I}(Y).$$

Thus either  $\mathcal{I}(Y_1)$  or  $\mathcal{I}(Y_2)$  is contained in  $\mathcal{I}(Y)$ . Suppose  $\mathcal{I}(Y_1) \subseteq \mathcal{I}(Y)$ , then applying  $Z$  gives  $Y_1 \supseteq Y$  (since they are closed), so  $Y = Y_1$  as desired. ■

**2.1.12 Example**

$\mathbb{A}^n$  is irreducible since  $\mathcal{I}(\mathbb{A}^n) = (0)$  is prime.

If  $f \in k[\mathbb{A}^n]$  is irreducible, then  $Z(f)$  is irreducible in  $\mathbb{A}^n$ . This is because  $\mathcal{I}(Z(f)) = \sqrt{(f)} = (f)$  is prime.

**Proof of the Nullstellensatz****2.1.13 Lemma**

Let  $f \in k[x_1, \dots, x_n]$  be nonconstant and  $A = k[x_1, \dots, x_n]/(f)$ . Then there exist integers  $k_1, \dots, k_{n-1} > 0$  such that the  $k$ -algebra homomorphism

$$\varphi: k[y_1, \dots, y_{n-1}] \rightarrow A, \quad y_i \mapsto x_i - x_n^{k_i}$$

is injective and finite.

**Proof**

For an  $n$ -tuple  $I = (d_1, \dots, d_n)$  let  $x^I = x_1^{d_1} \cdots x_n^{d_n}$  and call  $I$  the **multidegree** of  $x^I$ . We define  $k_j$  for  $1 \leq j \leq n$  by descending recursion and setting  $k_n = 1$ , and letting

$$k_j = 1 + \max_I \left\{ \sum_{i=j+1}^n k_i d_i \right\},$$

where  $I = (d_1, \dots, d_n)$  ranges over all the multidegrees in  $f$  with nonzero coefficient. We claim that  $k_j$  satisfy the claim.

Let  $y_i = x_i - x_n^{k_i}$  for  $1 \leq i \leq n-1$ , and order the multidegrees in  $f$  lexicographically. That is,  $(d_1, \dots, d_n) > (d'_1, \dots, d'_n)$  iff  $d_j > d'_j$  for some index  $j$  and  $d_i = d'_i$  for all  $i < j$ . This is a total order, and so  $f$  has a maximum multidegree  $I_f = (D_1, \dots, D_n)$ .

Now define  $N = \sum_{i=1}^n k_i D_i$ . Then we claim that  $f$  is a polynomial in  $B[x_n]$  of degree  $N > 0$  in  $x_n$ , whose leading coefficient is in  $k^\times$ , where  $B = k[y_1, \dots, y_{n-1}] \subseteq k[x_1, \dots, x_n]$ .

Let  $m = x^I$  be a monomial with  $I = (d_1, \dots, d_n)$ . Substitute  $x_i = y_i + x_n^{k_i}$ , which shows that  $m$  is a monic polynomial in  $B[x_n]$  of  $x_n$ -degree  $\sum_{i=1}^n k_i d_i$ . We claim that under this substitution, the  $x_n$ -degree of  $x^I$  is greater than  $x^I$ 's for any other multidegree  $I$  in  $f$ . That is,

$$\sum_{i=1}^n k_i D_i > \sum_{i=1}^n k_i d_i$$

for all other multidegrees  $I$ . This will then imply the claim regarding  $f$ , as it is of the form

$$f = \sum_I a_I x^I = \sum_I a_I (y_I - x_n^{k_I})^I,$$

and considering the combination of monomials.

So let  $I = (d_1, \dots, d_n)$  be another multidegree of  $f$ . By  $I_f$ 's maximality there is a  $j$  such that  $D_j \geq d_j + 1$  and  $D_i = d_i$  for  $i < j$ . Then

$$\sum_{i=1}^n k_i D_i \geq \sum_{i=1}^j k_i D_i \geq \sum_{i=1}^{j-1} k_i d_i + k_j(d_j + 1) = \sum_{i=1}^j k_i d_i + k_j.$$

By definition of  $k_j$ ,

$$> \sum_{i=1}^j k_i d_i + \sum_{i=j+1}^n k_i d_i = \sum_{i=1}^n k_i d_i$$

as desired.

Now we need to show that these  $k_i$  actually prove the lemma. First we show that  $\varphi$  is integral; let  $C = k[y_1, \dots, y_{n-1}]$  be  $\varphi$ 's domain. Let  $c \in k^\times$  be  $f$ 's leading coefficient as a polynomial in  $B[x_n]$ , then  $c^{-1}f$  vanishes in  $A$ . Since  $c^{-1}f$  is a monic polynomial in  $B[x_n]$  which is zero, so  $\bar{x}_n \in A$  is integral over  $C$ , as desired. Since  $\bar{x}_i = \bar{y}_i + \bar{x}_n^{k_i}$ ,  $\bar{x}_i \in A$  is also integral over  $C$ , and thus  $A$  is integral over  $C$  via  $\varphi$ . Furthermore,  $A$  is fg over  $C$  and thus finite.

Note that  $x_n \mapsto x_n$ ,  $x_i \mapsto y_i = x_i - x_n^{k_i}$  is invertible, and thus  $k[x_1, \dots, x_n] = k[y_1, \dots, y_{n-1}][x_n]$ , and  $y_1, \dots, y_{n-1}, x_n$  are algebraically independent. Now suppose  $\varphi(g) = 0$  for some  $g \in C = k[t_1, \dots, t_{n-1}]$ . That is,  $g(\bar{y}_1, \dots, \bar{y}_{n-1}) = 0 \in A$  so  $g(y_1, \dots, y_{n-1}) = hf$  for some  $h \in k[y_1, \dots, y_{n-1}][x_n]$ . Since  $g$  has zero in  $x_n$ , but  $f$  has positive degree, we must have that  $g = 0$ . So  $\varphi$  is injective. ■

#### 2.1.14 Theorem (Noether normalization lemma)

Let  $A \neq 0$  be a fg algebra over a field  $k$ . Then there exists a  $k$ -subalgebra  $B \subseteq A$  such that  $B$  is isomorphic as a  $k$ -algebra to a polynomial algebra  $k[y_1, \dots, y_m]$  and  $A$  is a finite  $B$ -algebra.

#### Proof

Since  $A$  is fg,  $A \cong k[x_1, \dots, x_n]/I$  for some ideal  $I$ . If  $I = 0$ , then take  $B = A$ . Otherwise there is a nonconstant  $f \in I$  (since otherwise  $I$  is the whole polynomial ring and  $A = 0$ ). By the previous lemma, there is a subalgebra  $B' \subseteq A' = k[x_1, \dots, x_n]/(f)$  such that  $B' \cong k[x'_1, \dots, x'_{n-1}]$  and  $A'$  is finite over  $B'$ .

Let  $I' = I/(f) \subseteq A'$  and  $J = I' \cap B'$  be ideals. Then  $B'/J \cong k[x'_1, \dots, x'_{n-1}]/J$  is a finite subalgebra over  $A'/I' = A$ . By induction on the number of generators, the claim holds for  $B'/J$ , so there is a subalgebra

$B \subseteq B'/J \subseteq A$  such that  $B$  is a polynomial ring  $k[y_1, \dots, y_m]$  and  $B'/J$  is a finite  $B$ -algebra. Since being a finite algebra is transitive,  $A$  is a finite  $B$ -algebra, as desired. ■

The Noether normalization lemma then implies the following result, which can be seen to be a sort of generalization of Nullstellensatz.

### 2.1.15 Proposition

Let  $L/k$  be a field extension such that  $L$  is a fg  $k$ -algebra. Then  $L$  is a finite extension of  $k$  (i.e. it is a finite  $k$ -algebra).

#### Proof

By Noether,  $L$  contains a polynomial subalgebra over which it is finite, and thus integral. Let  $B = k[x_1, \dots, x_n] \subseteq L$  be this algebra. Since  $L$  is integral over  $B$ ,  $B$  must be a field (if  $A \subseteq B$  is integral,  $A$  is a field iff  $B$  is a field), and so  $n$  must be zero. Thus  $k \subseteq L$  is finite, as desired. ■

### 2.1.16 Corollary (Weak Nullstellensatz)

Let  $A \neq 0$  be a fg algebra over an algebraically closed field  $k$ . Then for any maximal ideal  $\mathfrak{m} \leq A$ ,  $A/\mathfrak{m} \cong k$  as  $k$ -algebras.

#### Proof

Let  $L = A/\mathfrak{m}$ , which is a fg  $k$ -algebra, so by the above proposition it must be a finite. Algebraically closed fields admit no proper extensions, so  $A/\mathfrak{m} \cong k$ . ■

### 2.1.17 Corollary

Let  $A \neq 0$  be a fg algebra over an algebraically closed field  $k$ . Then there exists a  $k$ -algebra homomorphism  $\psi: A \rightarrow k$ .

#### Proof

Choose a maximal ideal  $\mathfrak{m} \leq A$ , and consider  $\psi: A \rightarrow A/\mathfrak{m} \cong k$ . ■

Recall that the preimage of a prime ideal is always prime, but the preimage of a maximal ideal need not be maximal. For example under the inclusion  $\mathbb{Z} \hookrightarrow \mathbb{Q}$ , consider the preimage of the maximal ideal  $(0) \leq \mathbb{Q}$ , which is  $(0) \leq \mathbb{Z}$ ; not maximal.

### 2.1.18 Proposition

Let  $A, B$  be fg  $k$ -algebras, and  $f: A \rightarrow B$  a  $k$ -algebra morphism. Then for any maximal ideal  $\mathfrak{m} \leq B$ ,  $\mathfrak{m}_A = f^{-1}\mathfrak{m} \leq A$  is maximal.

#### Proof

The map  $f$  induces an injective  $k$ -algebra morphism  $A/\mathfrak{m}_A \hookrightarrow B/\mathfrak{m}$ . The latter is a field and a fg  $k$ -algebra,

hence a finite  $k$ -algebra and in particular integral. Thus  $A/\mathfrak{m}_A$  is an integral  $k$ -algebra as well, and since it is a domain ( $\mathfrak{m}_A$  is prime), it must then be a field. So  $\mathfrak{m}_A$  is maximal. ■

### 2.1.19 Definition

A ring  $A$  is **Jacobson** when  $\text{Jac}(A) = \text{nil}(A)$ .

In general we have  $\text{nil}(A) \subseteq \text{Jac}(A)$ , since  $\text{nil}(A)$  is the intersection of all prime and  $\text{Jac}(A)$  all maximal ideals.

### 2.1.20 Proposition

Every fg algebra over a field  $k$  is Jacobson.

#### Proof

Let  $A$  be a fg algebra over  $k$ , and  $f \notin \text{nil}(A)$ . We then claim  $A$  has a maximal ideal not containing  $f$ . Since  $f \notin \text{nil}(A)$ ,  $A_f \neq 0$  as the localizing set does not contain zero, and thus contains a maximal ideal  $\mathfrak{m}$ . Recall that  $A_f \cong A[x]/(fx - 1)$ , so  $A_f$  is a fg  $k$ -algebra. Let  $\iota: A \rightarrow A_f$  be the canonical map, then  $\iota^{-1}\mathfrak{m}$  is maximal in  $A$ . Since  $\iota f \in A_f^\times$ ,  $f$  is not in  $\iota^{-1}\mathfrak{m}$  as desired. ■

### 2.1.21 Corollary

If  $A$  is a fg algebra over a field  $k$ , then for every ideal  $I \leq A$ ,

$$\sqrt{I} = \text{Jac}_A(I),$$

where  $\text{Jac}_A(I)$  is the Jacobson radical of  $I$ , the intersection of all maximal ideals containing  $I$ . In particular for a prime  $\mathfrak{p} \leq A$ ,

$$\mathfrak{p} = \bigcap_{\mathfrak{p} \leq \mathfrak{m} \leq A} \mathfrak{m}.$$

#### Proof

This amounts to saying that  $A/I$  is a Jacobson ring, since pulling back  $\text{nil}(A/I)$  and  $\text{Jac}(A/I)$  give  $\sqrt{I}$ ,  $\text{Jac}_A(I)$  respectively. Since  $A$  is a fg algebra over  $k$ , so is  $A/I$  and the result follows the above proposition. ■

#### Note:

Some authors define a Jacobson ring to be one where  $\sqrt{I} = \text{Jac}_A(I)$  for all ideals, or  $\mathfrak{p} = \text{Jac}_A(\mathfrak{p})$  for all prime ideals (these are equivalent). Note that this definition is more useful, but not how it was defined in this course. The above corollary shows that a fg algebra over a field is Jacobson in this regard as well.

Note that there is a natural bijection between  $\mathbb{A}^n$  and  $k$ -algebra morphisms  $\varphi: k[\mathbb{A}^n] \rightarrow k$ .

- (1) For every  $P = (a_1, \dots, a_n) \in \mathbb{A}^n$  define  $\varphi$  by  $\varphi(x_i) = a_i$ . That is,  $\varphi$  is the evaluation map  $\text{ev}_P$  given by  $\text{ev}_P(f) = f(P)$ . In particular,

$$\ker \text{ev}_P = \mathcal{I}(P).$$

- (2) Conversely for  $\varphi: k[\mathbb{A}^n] \rightarrow k$ , let  $P = (a_1, \dots, a_n)$  by  $a_i = \varphi(x_i)$ . Then  $\varphi = \text{ev}_P$ , so these are inverses.

Let  $J \leq k[\mathbb{A}^n]$  be an ideal, then for every  $P \in \mathbb{A}^n$ , we have  $P \in Z(J)$  iff  $J \leq \mathcal{I}(P) = \ker \text{ev}_P$  iff  $\text{ev}_P$  quotients over  $J$ : it induces  $\bar{\text{ev}}_P: k[\mathbb{A}^n]/J \rightarrow k$  where  $\bar{\text{ev}}_P(f + J) = f(P)$ .

### 2.1.22 Proposition

For  $P = (a_1, \dots, a_n) \in \mathbb{A}^n$  we have

$$\ker \text{ev}_P = (x_1 - a_1, \dots, x_n - a_n),$$

which is a maximal ideal.

#### Proof

Since  $\text{ev}_P(x_i - a_i) = 0$  we have inclusion  $\supseteq$ . Since  $k[\mathbb{A}^n]/\ker \text{ev}_P \cong \text{im} \text{ev}_P \cong k$ ,  $\ker \text{ev}_P$  is maximal, it is sufficient to show  $k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n) \cong k$ . Equivalently, the natural map  $\iota: k \rightarrow k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n)$  is surjective.

The projection  $k[\mathbb{A}^n] \rightarrow k[\mathbb{A}^n]/(x_1 - a_1, \dots, x_n - a_n)$  maps  $x_i$  to  $\bar{x}_i = \iota(a_i)$ . Thus  $\iota$  must be surjective as the projection is. ■

The following is another version of the Nullstellensatz:

### 2.1.23 Proposition

Let  $k$  be algebraically closed. Then the maximal ideals of  $k[\mathbb{A}^n]$  are precisely the ideals  $\ker \text{ev}_P$  for  $P \in \mathbb{A}^n$ .

#### Proof

We saw that  $\ker \text{ev}_P$  are maximal, so let  $\mathfrak{m}$  be maximal. By the weak Nullstellensatz,  $k[\mathbb{A}^n]/\mathfrak{m} \cong k$ , which gives a projection  $f: k[\mathbb{A}^n] \rightarrow k[\mathbb{A}^n]/\mathfrak{m} \cong k$  with  $\ker f = \mathfrak{m}$ . But  $k$ -algebra morphisms  $k[\mathbb{A}^n] \rightarrow k$  are all of the form  $\text{ev}_P$  for some  $P$ , so  $f = \text{ev}_P$  and thus  $\mathfrak{m} = \ker \text{ev}_P$ . ■

Now we can finally prove Nullstellensatz itself.

#### Proof (Nullstellensatz)

We already showed  $\sqrt{J} \leq \mathcal{I}(Z(J))$ , we now show the other direction. Let  $f \in \mathcal{I}(Z(J))$ , and consider  $A = k[\mathbb{A}^n]/J$ . By the above proposition, the maximal ideals of  $A$  are the ideals of the form  $\ker \text{ev}_P$  containing  $J$ , so  $\ker \text{ev}_P$  with  $P \in Z(J)$ . Because  $f \in \mathcal{I}(Z(J))$ ,  $f$  vanishes at every point  $P \in Z(J)$ , so  $f \in \ker \text{ev}_P$  for every  $P \in Z(J)$ . Thus  $f$  is in every maximal ideal of  $A$ , so  $\bar{f} \in \text{Jac}(A)$ .  $A$  is a fg  $k$ -algebra it is Jacobson, so  $\bar{f} \in \text{Jac}(A) = \text{nil}(A) = \sqrt{J}/J$ , thus  $f \in \sqrt{J}$  as desired. ■

Let us summarize the following formulations of Nullstellensatz:

### 2.1.24 Theorem (Equivalent formulations of Nullstellensatz)

The following are all equivalent formulations of Nullstellensatz, and all are true.

- (1) Nullstellensatz:  $\mathcal{I}(Z(J)) = \sqrt{J}$ ,
- (2) If  $J \neq (1)$  then  $Z(J) \neq \emptyset$ ,
- (3) Weak Nullstellensatz: if  $A$  is fg,  $A/\mathfrak{m} \cong k$ ,
- (4) If  $A \neq 0$  is fg, then there is a  $\varphi: A \rightarrow k$ ,
- (5) Every maximal ideal of  $k[\mathbb{A}^n]$  of the form  $(x_1 - a_1, \dots, x_n - a_n)$ .



**2.1.25 Definition**

For a topological space  $X$ , let  $|X|$  denote the subspace of closed points.

Recall that for an affine scheme  $\text{Spec}(A)$ , a closed point is just a maximal ideal, so  $|X| \cong \text{Specm}(A)$ .

**2.1.26 Corollary**

Let  $k$  be algebraically closed, and  $I \leq k[\mathbb{A}^n]$  an ideal. Then there is a bijection between the closed points of  $\text{Spec}(k[\mathbb{A}^n]/I)$  and  $Z(I)$ :

$$|\text{Spec}(k[\mathbb{A}^n]/I)| \cong Z(I).$$

**Proof**

The closed points of the spectrum of  $k[\mathbb{A}^n]/I$  are in bijection with the maximal ideals of  $k[\mathbb{A}^n]$  containing  $I$ . These are of the form  $(x_1 - a_1, \dots, x_n - a_n)$ , and if it contains  $I$  then for  $f \in I$  we must have that  $f(a_1, \dots, a_n) = 0$ . So mapping  $(x_1 - a_1, \dots, x_n - a_n)$  to  $(a_1, \dots, a_n) \in Z(I)$  is well-defined and clearly injective. Furthermore it is clearly surjective. ■

**2.1.27 Corollary**

Let  $A, B$  be fg  $k$ -algebras for some field  $k$ , and let  $\varphi: A \rightarrow B$  be a  $k$ -algebra morphism. Then  $\varphi^*: \text{Spec}(B) \rightarrow \text{Spec}(A)$  restricts to a morphism of closed points  $|\varphi^*: |\text{Spec}(B)| \rightarrow |\text{Spec}(A)|$ .

**Proof**

This is just a restatement of the preimage of a maximal ideal being maximal. ■

**2.2 Algebraic varieties****2.2.1 Definition**

Let  $k$  be a field, then denote by  $\text{Fun}(U, k)$  the  $k$ -algebra of set-theoretic functions  $U \rightarrow k$ .

A **space with functions (SWF)** over  $k$  is a pair  $(X, \mathcal{O}_X)$  where  $X$  is a topological space and for every  $U \subseteq X$  open,  $\mathcal{O}_X(U)$  is a subset of  $\text{Fun}(U, k)$ . Furthermore it must satisfy:

- (1) **presheaf condition:** if  $f \in \mathcal{O}_X(U)$  and  $V \subseteq U$  then the restriction of  $f$  to  $V$ ,  $f|_V$ , is in  $\mathcal{O}_X(V)$ .
- (2) **sheaf condition:** if  $U_i$  is an open covering of  $U \subseteq X$ , and  $f: U \rightarrow k$  is a function such that  $f|_{U_i} \in \mathcal{O}_X(U_i)$  for all  $i$ , then  $f \in \mathcal{O}_X(U)$ .
- (3) **local condition:** for every  $f \in \mathcal{O}_X(U)$ , the subset  $D_U(f) = \{x \in U \mid f(x) \neq 0\}$  is open, and the function  $1/f: D_U(f) \rightarrow k$  is in  $\mathcal{O}_X(D_U(f))$ .

$\mathcal{O}_X$  is called the **structure sheaf** of the SWF.

**Note:**

In general a **presheaf** valued in a category  $\mathcal{C}$  over a topological space  $X$  is a contravariant functor

$$\mathcal{O}: \text{Op}(X)^{\text{op}} \rightarrow \mathcal{C},$$

where  $\text{Op}(X)$  is the category of open sets of  $X$  (viewed as a preorder, ordered by inclusion). For  $U \subseteq V$  open we denote the map  $\mathcal{O}(V) \rightarrow \mathcal{O}(U)$  by  $\text{res}_V^U$ , if  $\mathcal{C}$  is a concrete category (its objects are sets) then we write  $\text{res}_V^U(f) = f|_U$  for  $f \in \mathcal{O}(V)$ .

The objects  $\mathcal{O}_X(U)$  are called **sections** of the presheaf.

If  $\mathcal{C}$  is complete (has all limits), then a presheaf  $\mathcal{O}$  is a **sheaf** if for every open cover  $U_i$  of  $U$ , the following is an equalizer diagram:

$$\mathcal{O}(U) \xrightarrow{\text{res}_0} \prod_i \mathcal{O}(U_i) \xrightleftharpoons[\text{res}_2]{\text{res}_1} \prod_{i,j} \mathcal{O}(U_i \cap U_j),$$

where  $\text{res}_0$  is the map  $f \mapsto (f|_{U_i})$  (i.e. its component  $\mathcal{O}(U) \rightarrow \mathcal{O}(U_i)$  is  $\text{res}_{U_i}^U$ ). The map  $\text{res}_1$  is the map  $(f_i) \mapsto (f_i|_{U_i \cap U_j})$  and  $\text{res}_2$  is the map  $(f_i) \mapsto (f_j|_{U_i \cap U_j})$ . That is, their components  $\prod_i \mathcal{O}(U_i) \rightarrow \mathcal{O}(U_i \cap U_j)$  are  $\text{res}_{U_i \cap U_j}^{U_i} \circ \text{proj}_i$  and  $\text{res}_{U_i \cap U_j}^{U_j} \circ \text{proj}_j$  respectively.

For a concrete category  $\mathcal{C}$  being a sheaf means that if  $f_i \in \mathcal{O}_X(U_i)$  are sections such that  $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$  for all  $i, j$ , then there exists a unique  $f \in \mathcal{O}_X(U)$  such that  $f|_{U_i} = f_i$  for all  $i$ .

Now a **subsheaf** of a sheaf  $\mathcal{O}$  is another sheaf  $\mathcal{O}'$  (on the same category) such that  $\mathcal{O}'(U)$  is a subobject of  $\mathcal{O}(U)$  for all open  $U$ . Under sufficiently weak conditions (e.g.  $\mathcal{C}$  being finitely complete) this is equivalent to  $\mathcal{O}'$  being a subfunctor of  $\mathcal{O}$  (i.e. there being a monic natural transformation  $\mathcal{O}' \Rightarrow \mathcal{O}$ ). Conditions (1) and (2) above are equivalent to saying that  $\mathcal{O}$  is a subsheaf of  $\text{Fun}(\bullet, k)$  (defined in the obvious way).

A pair  $(X, \mathcal{O}_X)$  where  $\mathcal{O}_X$  is a sheaf on  $X$  valued in the category of rings (resp.  $k$ -algebras) is called a **ringed space** (resp. a  $k$ -ringed space).

Condition (3) is a replacement for the requirement that  $(X, \mathcal{O}_X)$  be a **locally ringed space**. This condition says that the stalks of  $\mathcal{O}_X$  (the ring of germs of a point  $p \in X$ , not defined here) is a local ring.

### 2.2.2 Definition

A morphism of SWFs  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is a continuous map  $\varphi: X \rightarrow Y$  such that for every open  $V \subseteq Y$  and every  $f \in \mathcal{O}_Y(V)$ , the pullback  $\varphi^*(f) = f \circ \varphi: \varphi^{-1}V \rightarrow V \rightarrow k$  belongs to  $\mathcal{O}_X(\varphi^{-1}V)$ .

The identity is a morphism of SWFs because  $\text{id}^*f = f$ . And the composition of morphisms of SWFs is a morphism of SWFs clearly as well, thus SWFs form a category.

### 2.2.3 Example

Let  $k$  be a Hausdorff topological field, and  $X$  a topological space. Define  $(X, \mathcal{O}_{\text{cont}})$  to be the SWF where  $\mathcal{O}_{\text{cont}}(U)$  is the set of continuous functions  $U \rightarrow k$ . This is easily seen to be a SWF.

If  $X$  is a real manifold, then we have SWFs  $(X, \mathcal{O}_{C^n})$ ,  $(X, \mathcal{O}_{C^\infty})$ , and  $(X, \mathcal{O}_{\text{anal}})$  which map an open set to the algebra of real valued  $C^n$ -,  $C^\infty$ -, or analytic maps respectively. These are SWFs and the identity maps

$$(X, \mathcal{O}_{C^n}) \xrightarrow{\text{id}} (X, \mathcal{O}_{C^\infty}) \xrightarrow{\text{id}} (X, \mathcal{O}_{\text{anal}})$$

are morphisms of SWFs.

For a complex manifold we can define the SWF of holomorphic functions on an open subset.

### 2.2.4 Definition

If  $(X, \mathcal{O})$  is a space with functions and  $Y \subseteq X$  is open, we can restrict  $\mathcal{O}$  to open subsets of  $Y$  to

obtain another SWF. That is, define  $(Y, \mathcal{O}|_Y)$  to be the SWF defined by  $\mathcal{O}|_Y(U) = \mathcal{O}(U)$  for every open  $U \subseteq Y \subseteq X$ .

In general if  $Y \subseteq X$  is a subspace, we define the **induced space**  $(Y, \mathcal{O}|_Y)$  as follows. For  $V \subseteq Y$  open, let  $\mathcal{O}|_Y(V)$  be the set of functions  $f: V \rightarrow k$  such that there is an open cover  $U_i \subseteq X$  of  $V$  (so  $V = \bigcup U_i \cap V$ ) and  $g_i \in \mathcal{O}(U_i)$  such that  $f|_{V \cap U_i} = g_i|_{V \cap U_i}$  for all  $i$ .

Equivalently  $f \in \mathcal{O}|_Y(V)$  iff for every  $y \in V$  there is a neighborhood  $W$  of  $y$  in  $X$  and a function  $g \in \mathcal{O}_X(W)$  such that  $g|_{W \cap V} = f|_{W \cap V}$ . If we think of  $\mathcal{O}_X$  as locally smooth maps, then this says that a map is smooth on a subspace if it can be extended locally.

### 2.2.5 Proposition

The pair  $(Y, \mathcal{O}|_Y)$  is an SWF and inclusion  $\iota: X \hookrightarrow Y$  is a morphism  $(Y, \mathcal{O}|_Y) \rightarrow (X, \mathcal{O}_X)$ .

Furthermore if  $(Z, \mathcal{O}_Z)$  is another SWF and  $\varphi: Z \rightarrow Y$  a set-theoretic map, then  $\varphi$  is a morphism of SWFs  $(Z, \mathcal{O}_Z) \rightarrow (Y, \mathcal{O}|_Y)$  iff  $\iota \circ \varphi: (Z, \mathcal{O}_Z) \rightarrow (X, \mathcal{O}_X)$  is a morphism of SWFs.

### Proof

Let  $V \subseteq V' \subseteq Y$  be open subsets, and  $f \in \mathcal{O}|_Y(V')$ . Then there is an open cover of  $V'$  in  $X$ ,  $V' = \bigcup_i U_i \cap V'$  and  $g_i \in \mathcal{O}_X(U_i)$  which restrict to  $f$  on  $U_i \cap V'$ . These  $U_i$  cover  $V$  as well then, and  $g_i$  restrict to  $f|_V$  on  $U_i \cap V$ , so  $f|_V \in \mathcal{O}|_Y(V)$ , so it satisfies the presheaf condition.

Let  $V \subseteq Y$  be open and  $f \in \text{Fun}(V, k)$  such that there is an open cover  $V = \bigcup_i V_i$  with  $f|_{V_i} \in \mathcal{O}_Y(V_i)$ . Then we cover each  $V_i$  by  $U_i^j$  such that there are  $g_i^j \in \mathcal{O}_X(U_i^j)$  with  $g_i^j|_{U_i^j \cap V_i} = f|_{U_i^j \cap V_i}$ . But then the collection of all  $U_i^j$  as  $i, j$  ranges and  $g_i^j$  show that  $f \in \mathcal{O}|_Y(V)$ . So  $\mathcal{O}|_Y$  is a sheaf.

Now we need to show the locality axiom. Let  $V \subseteq Y$  be open and  $f \in \mathcal{O}|_Y(V)$  so there are  $U_i \subseteq X$  open covering  $V$  such that  $f|_{U_i \cap V} = g_i|_{U_i \cap V}$  for  $g_i \in \mathcal{O}_X(U_i)$ . Then

$$D_V(f) = \{x \in V \mid f(x) \neq 0\} = \bigcup_i \{x \in V \cap U_i \mid f(x) \neq 0\} = \bigcup_i D_{U_i}(g_i) \cap V,$$

which is relatively open in  $V$ . And we see the  $D_{U_i}(g_i)$  cover  $D_V(f)$  and  $1/f$  restricts to  $1/g_i \in \mathcal{O}_X(D_{U_i}(g_i))$  their common domain, so  $1/f \in \mathcal{O}|_Y(D_V(f))$ . Thus  $\mathcal{O}|_Y$  is indeed a SWF.

The inclusion is continuous by virtue of the subspace topology. Let  $U \subseteq X$  open so  $\iota^{-1}U = U \cap Y$ , and let  $f \in \mathcal{O}_X(U)$  then  $\iota^*(f) = f \circ \iota = f|_{U \cap Y}$ . This is in  $\mathcal{O}|_Y(U \cap Y)$  by considering the open cover  $U$  and  $f \in \mathcal{O}_X(U)$ .

The composition of morphisms of SWFs is a morphism of SWFs, so if  $\varphi$  is a morphism so is  $\iota \circ \varphi$ . Conversely if  $\iota \circ \varphi$  is continuous, then so is  $\varphi$  by virtue of the subspace topology. And if  $f \in \mathcal{O}|_Y(V)$  for  $V \subseteq Y$  open, then let  $g_i \in \mathcal{O}_X(U_i)$  such that  $g_i|_{U_i \cap V} = f|_{U_i \cap V}$  with  $U_i$  covering  $V$ . Then

$$(\iota \circ \varphi)^*(g_i) = g_i \circ \iota \circ \varphi = g_i|_{U_i \cap V} \circ \varphi = f|_{U_i \cap V} \circ \varphi = f \circ \varphi|_{W_i}$$

is in  $\mathcal{O}_X(W_i)$  with  $W_i = (\iota \circ \varphi)^{-1}(U_i)$ . These  $W_i$  cover  $\varphi^{-1}V$  and thus by definition  $f \circ \varphi \in \mathcal{O}|_Y(\varphi^{-1}V)$ , showing  $\varphi$  is a morphism of SWFs. ■

### 2.2.6 Example

Let  $B^n \subseteq \mathbb{R}^n$  or  $\mathbb{C}^n$  be the open unit ball. Then the above proposition gives us SWFs

$$(B^n, \mathcal{O}_{\text{cont}}), (B^n, \mathcal{O}_{C^k}), (B^n, \mathcal{O}_{C^\infty}), (B^n, \mathcal{O}_{\text{anal}}), (B^n, \mathcal{O}_{\text{holo}}).$$

We can then define a real/complex topological/ $C^k$ / $C^\infty$ /analytic manifold as a SWF  $(X, \mathcal{O})$  such that  $X$  is Hausdorff and second countable and there is an open cover  $X = \bigcup_i U_i$  such that there is a single SWF from one of the above  $((B_n, \mathcal{O}_\bullet))$  such that each  $i$   $(U_i, \mathcal{O}|_{U_i})$  is isomorphic to it.

This is an equivalent definition to the standard definition of a manifold via charts. Then a smooth map between manifolds is just a morphism of SWFs.

We use this example to define general algebraic varieties. We will define affine algebraic varieties as a special family of SWFs, and then a general algebraic variety will be a SWF locally isomorphic to an affine algebraic variety.

### 2.2.7 Definition

Let  $Y \subseteq \mathbb{A}^n$  be a subspace. A function  $f: Y \rightarrow k$  is **regular** when it is locally rational (the product of polynomials). That is, for every  $P \in Y$  there is a neighborhood  $U_P \subseteq Y$  of  $P$  and polynomials  $g_P, h_P \in k[\mathbb{A}^n]$  such that  $h_P(Q) \neq 0$  for  $Q \in U_P$  and  $f(Q) = g_P(Q)/h_P(Q)$  for all  $Q \in U_P$ .

Denote the set of regular functions  $Y \rightarrow k$  by  $\mathcal{O}_{\text{reg}}(Y)$ .

### 2.2.8 Lemma

The pair  $(Y, \mathcal{O}_{\text{reg}})$  is a SWF for every  $Y \subseteq \mathbb{A}^n$ .

#### Proof

It is easy to see that  $\mathcal{O}_{\text{reg}}(U) \subseteq \text{Fun}(U, k)$  is a subalgebra; i.e. that it is closed under addition and multiplication, and contains the unit (and thus all constants). Furthermore since being regular is a local condition, it is again easy to see that  $\mathcal{O}_{\text{reg}}$  is a sheaf.

We just need to show locality. Let  $f \in \mathcal{O}_{\text{reg}}(U)$  then for  $P \in U$  there is an open  $P \in U_P \subseteq U$  such that  $f = g_P/h_P$  for polynomials  $g_P, h_P$  in  $U_P$ . Then

$$D_U(f) = \{x \in U \mid f(x) \neq 0\} = \bigcup_{P \in U} \{x \in U_P \mid g_P(x) \neq 0\} = \bigcup_{P \in U_P} D_{U_P}(g_P)$$

is open since each  $g_P$  is continuous. And on  $D_{U_P}(g_P)$  we have  $1/f = h_P/g_P$ , so  $1/f$  is regular on  $D_U(f)$  as desired. ■

#### Note:

It is easy to see that  $\mathcal{O}_{\text{reg}}$  is the smallest SWF containing all the polynomial functions in  $k[\mathbb{A}^n]$ . Such a SWF must contain regular functions because of locality.

### 2.2.9 Lemma

Let  $U \subseteq \mathbb{A}^n$  be open, and  $f \in \mathcal{O}_{\text{reg}}(U)$  a regular function. Then  $f$  is rational: there are  $g, h \in k[\mathbb{A}^n]$  such that  $h \neq 0$  on  $U$  and  $f = g/h$ .

#### Proof

By definition there is an open covering  $U = \bigcup_i U_i$  and polynomials  $g_i, h_i$  such that  $h_i \neq 0$  on  $U_i$  and  $f|_{U_i} = g_i/h_i$  on  $U_i$ . Further we can assume that  $g_i, h_i$  are relatively prime and  $U_i$  are nonempty.

We claim that for any  $i$ ,  $h_i$  is nonzero on all of  $U$ , and  $f = g_i/h_i$  on all of  $U$ . Let  $P \in U_j$ , so we have

$$\frac{g_i}{h_i} \Big|_{U_i \cap U_j} = f|_{U_i \cap U_j} = \frac{g_j}{h_j} \Big|_{U_i \cap U_j}.$$

Thus  $g_i h_j - g_j h_i = 0$  on  $U_i \cap U_j$ . Since  $\mathbb{A}^n$  is irreducible, any nonempty open subset is dense (this is equivalent) and so the intersection of nonempty open subsets is then dense (as each is dense and thus their intersection is nonempty, and thus a nonempty open subset, so dense). Thus  $g_i h_j$  and  $g_j h_i$  agree on a dense set, so they are equal (two maps into a Hausdorff space agreeing on a dense set are equal).

Since  $k[\mathbb{A}^n]$  is a UFD and  $\{g_i, h_i\}$  are all relatively prime, we must have  $h_i = c h_j$  and  $g_i = c g_j$  for some  $c \in k[\mathbb{A}^n]^\times$  (which is  $k^\times$ ). Thus we have  $h_i(P) = c(P) h_j(P) \neq 0$  for all  $P \in U_j$ , so  $h_i$  is nonzero on all of  $U$ . And so for  $P \in U_j$ ,

$$\frac{g_i(P)}{h_i(P)} = \frac{g_j(P)}{h_j(P)} = f(P),$$

so that  $f = g_i/h_i$  on all of  $U$  as desired.  $\blacksquare$

So  $(\mathbb{A}^n, \mathcal{O}_{\text{reg}})$  is a much simpler object than the general case.

### 2.2.10 Definition

An **affine algebraic variety** (or **affine variety**) is a SWF  $(X, \mathcal{O}_X)$  which is isomorphic to a SWF  $(Y, \mathcal{O}_{\text{reg}})$  where  $Y \subseteq \mathbb{A}^n$  is closed.

An **algebraic variety** is a SWF  $(X, \mathcal{O}_X)$  which has a finite local covering  $X = \bigcup_{i=1}^n U_i$  such that each  $(U_i, \mathcal{O}_X|_{U_i})$  is an affine variety. Elements of a section of an algebraic variety (elements of  $\mathcal{O}_X(U)$ ) are called **regular functions**.

A morphism of algebraic varieties is just a morphism of SWFs.

### 2.2.11 Lemma

Every algebraic variety is Noetherian (as a topological space).

### Proof

First we show that  $\mathbb{A}^n$  is Noetherian. Indeed, if  $\{Z_i\}_i$  is a collection of closed subsets of  $\mathbb{A}^n$  then  $\{\mathcal{I}(Z_i)\}_i \subseteq k[\mathbb{A}^n]$  has a maximal element since  $k[\mathbb{A}^n]$  is Noetherian. Since  $Z \circ \mathcal{I}$  is the identity on closed sets, this means there is a maximal  $Z_i$ , and thus  $\mathbb{A}^n$  is Noetherian.

An affine variety is homeomorphic to a closed subset of  $\mathbb{A}^n$  which is Noetherian.

Thus an algebraic variety is locally Noetherian, i.e. there is a finite open covering  $\bigcup_{i=1}^n U_i$  where each  $U_i$  is Noetherian. Then  $X$  is Noetherian.  $\blacksquare$

### 2.2.12 Corollary

Every algebraic variety is quasicompact.

### 2.2.13 Definition

Let  $Y \subseteq \mathbb{A}^n$  be closed, then define  $k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y)$ . Then there is a natural homomorphism

$$\text{ev}_Y: k[\mathbb{A}^n] \xrightarrow{\text{ev}} \text{Fun}(\mathbb{A}^n, k) \xrightarrow{\text{res}} \text{Fun}(Y, k).$$

By definition the kernel of this is  $\mathcal{I}(Y)$ , and so  $\text{ev}_Y$  induces an injection  $k[Y] \hookrightarrow \text{Fun}(Y, k)$ .

This notation is compatible with our previous definition of  $k[\mathbb{A}^n]$ . This is because  $\mathcal{I}(\mathbb{A}^n) = 0$  so  $k[\mathbb{A}^n]/\mathcal{I}(\mathbb{A}^n) = k[\mathbb{A}^n]$ .

Note that  $k[Y]$  is a fg  $k$ -algebra. Furthermore it is reduced, as it can be embedded into  $\text{Fun}(Y, k)$  which is reduced as  $k$  is (also because  $\mathcal{I}(Y)$  is a radical ideal). Furthermore,  $k[Y]$  is a domain iff  $\mathcal{I}(Y)$  is prime iff  $Y$  is irreducible.

If  $B$  is a reduced fg  $k$ -algebra, then  $B \cong k[Y]$  for some closed  $Y \subseteq \mathbb{A}^n$ . Indeed,  $B \cong k[\mathbb{A}^n]/I$  for some ideal  $I$ , and since  $B$  is reduced  $I$  is a radical ideal and thus equal to  $\mathcal{I}(Y)$  for  $Y = Z(\mathcal{I})$ , so  $B \cong k[Y]$ .

For  $f \in k[\mathbb{A}^n]$  let  $\bar{f} = f + \mathcal{I}(Y) \in k[Y]$  denote its projection. Then viewed as functions  $\mathbb{A}^n \rightarrow k$  and  $Y \rightarrow k$  respectively, we have the equality

$$D_Y(\bar{f}) = Y \cap D(f).$$

Since the  $D(f)$  form a base of open subsets of  $k[\mathbb{A}^n]$ , we then see that  $\{D_Y(\bar{f})\}_{\bar{f} \in k[Y]}$  is a basis of  $Y$ .

### 2.2.14 Lemma

For every collection  $\{\bar{f}_i\}_i \subseteq k[Y]$ ,  $\{D_Y(\bar{f}_i)\}_i$  form an open cover of  $Y$  iff the ideal  $(\{f_i\}_i)$  is  $k[Y]$ .

#### Proof

We have that  $D_Y(\bar{f}_i)$  cover  $Y$  iff  $D(f_i)$  cover  $Y$ . Since  $D(f_i) = Z(f_i)^c$ , this is iff  $\bigcap_i Z(f_i) \cap Y = \emptyset$ , equivalently  $Z((f_i)_i) \cap Y$  is empty. But

$$Z((f_i)_i) \cap Y = Z((f_i)_i) \cap Z(\mathcal{I}(Y)) = Z((f_i)_i + \mathcal{I}(Y)),$$

and this is empty iff  $Z((f_i)_i) + \mathcal{I}(Y)$  is all of  $k[\mathbb{A}^n]$  (by Nullstellensatz). Quotienting by  $\mathcal{I}(Y)$  this is equivalent to  $Z((\bar{f}_i)_i) = k[Y]$ . ■

Recall that  $f \in \mathcal{O}_{\text{reg}}(Y)$  iff there is an open cover  $Y = \bigcup_i U_i$  such that  $f|_{U_i}$  is rational for each  $i$ . Since  $Y$  is quasicompact, we can assume this covering is finite, and taking basic open sets, that  $U_i = D_Y(t_i)$  for  $t_i \in k[Y]$ . So  $f = g_i/h_i$  for  $g_i, h_i \in k[\mathbb{A}^n]$  and  $h_i$  nonzero. Equivalently,  $f = \bar{g}_i/\bar{h}_i$  (as  $\bar{g}$  agrees with  $g$  on  $Y$ ).

### 2.2.15 Theorem (Serre)

For every closed  $Y \subseteq \mathbb{A}^n$  there is an equality  $k[Y] = \mathcal{O}_{\text{reg}}(Y)$  (viewing  $k[Y]$  as a subalgebra of  $\text{Fun}(Y, k)$ ).

#### Proof

Since  $\mathcal{O}_{\text{reg}}(Y)$  consists of all polynomial functions, clearly  $k[Y] \subseteq \mathcal{O}_{\text{reg}}(Y)$ . By the above if  $f \in \mathcal{O}_{\text{reg}}(Y)$  then there is a finite set of triples  $\{t_i, g_i, h_i\}$  of polynomials in  $k[Y]$  such that  $D_Y(t_i)$  cover  $Y$  and  $f_i = g_i/h_i$  on  $D_Y(t_i)$  and  $h_i \neq 0$  on  $D_Y(t_i)$ .

By assumption we have that  $D_Y(t_i) \subseteq D_Y(h_i)$ , and so  $D_Y(t_i) = D_Y(t_i) \cap D_Y(h_i) = D_Y(t_i h_i)$ . Since  $g_i/h_i = t_i g_i / t_i h_i$ , we can replace our collection with  $\{t_i h_i, t_i g_i, t_i h_i\}_i$ . In particular we can assume that  $t_i = h_i$ .

Now we want to show that  $f \in k[Y]$ . We first give a short, but tricky, proof and afterward we will give a more insightful one. Notice that for every  $i$  we have

$$f h_i^2 = g_i h_i \in \text{Fun}(Y, k),$$

so  $(f h_i - g_i) h_i = 0$ . This is true in  $D_Y(h_i)$  clearly, and for  $P \in Y - D_Y(h_i)$  we have that  $h_i(P) = 0$  so the equality holds trivially.

Now since  $Y = \bigcup_i D_Y(h_i) = \bigcup_i D_Y(h_i^2)$  we conclude that  $(h_i^2) = k[Y]$  as per the previous lemma. So there are  $r_i \in k[Y]$  such that

$$\sum_i r_i h_i^2 = 1.$$

(The algebraic analogue of a partition of unity.) Thus

$$f = f \sum_i r_i h_i^2 = \sum_i r_i f h_i^2 = \sum_i r_i g_i h_i \in k[Y]$$

as desired. ■

We now give a more insightful proof of Serre's theorem.

### 2.2.16 Lemma

Let  $A$  be a ring and suppose  $f_1, \dots, f_n$  generate the unit ideal. Then

$$0 \rightarrow A \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j},$$

is exact, where  $\varphi(a) = (a/1)_i$  and  $\psi((a_i)_i) = (a_i - a_j)_{i,j}$ .

### Proof

First we show that the sequence is exact at  $A$ , meaning  $\varphi$  is injective. If  $a/1 = 0$  for all  $i$ , then there are  $n_i$  such that  $f_i^{n_i} a = 0$  for each  $i$ . But

$$f_i \in \sqrt{(f_i^{n_i})},$$

and so  $\sqrt{(f_i^{n_i})}$  is equal to the unit ideal. But this means that  $(f_i^{n_i})$  is equal to the unit ideal, and thus  $a = 0$ .

Now we show that  $\text{im } \varphi = \ker \psi$ . Firstly,

$$\psi \circ \varphi(a) = \psi((a/1)_i) = (a/1 - a/1)_{i,j} = 0,$$

so the sequence is a chain complex. We now need only show that  $\ker \psi \subseteq \text{im } \varphi$ . So let  $(a_i/f_i^{n_i})_i$  lie in  $\ker \psi$ . Since  $a_i f_i^{r_i} / f_i^{n_i + r_i} = a_i / f_i^{n_i}$ , we may enlarge the  $n_i$  so that they are equal.

So

$$0 = \psi(a_i/f_i^n) = (a_i/f_i^n - a_j/f_j^n)_{i,j}.$$

This means that  $a_i/f_i^n = a_j/f_j^n$  in  $A_{f_i f_j}$  for all  $i, j$  and thus there is an  $M_{ij}$  such that  $(f_i f_j)^{M_{ij}} (f_j^n a_i - f_i^n a_j) = 0$ . Again by potentially enlarging the  $M_{ij}$  we can assume they are all equal to  $M$ . So we have that for all  $i, j$ ,

$$(f_i f_j)^M (f_j^n a_i - f_i^n a_j) = 0.$$

As before  $f_j^{M+n}$  generate the unit ideal so there are  $b_j \in A$  such that

$$\sum_j b_j f_j^{M+n} = 1,$$

and so if we define  $a = \sum_j b_j f_j^M a_j$ , then

$$f_i^{M+r} a = \sum_j b_j f_j^M f_i^{M+r} a_j = \sum_j b_j f_i^M f_j^{M+r} a_j = f_i^M a_i \sum_j b_j f_j^{M+r} = f_i^M a_i.$$

Thus  $a/1 = a_i/f_i^n$ , and so  $\varphi(a) = (a_i/f_i^n)_i$  as desired. ■

Now in the proof of Serre's theorem, since  $(h_i)_i = k[Y]$  we have the exact sequence

$$0 \rightarrow k[Y] \xrightarrow{\varphi} \prod_i k[Y]_{h_i} \xrightarrow{\psi} \prod_{i,j} k[Y]_{h_i h_j}.$$

Now consider  $\bar{f} = (g_i/h_i) \in \prod_i k[Y]_{h_i}$ , then  $g_i/h_i - g_j/h_j = 0$  in  $k[Y]_{h_i h_j}$  because as we will show,

$$(g_i h_j - g_j h_i) h_i h_j = 0 \text{ in } k[Y].$$

Since  $k[Y] \subseteq \text{Fun}(Y, k)$  we can show that the above expression vanishes for every  $P \in Y$ , and this is sufficient. If  $P \in D_Y(h_i) \cap D_Y(h_j)$  then

$$(g_i/h_i)(P) = (g_j/h_j)(P) = f(P),$$

and the equality follows. For all other  $P$  either  $h_i(P)$  or  $h_j(P)$  is zero, and the equality follows trivially.

Thus by exactness there is a  $\tilde{f} \in k[Y]$  such that  $\varphi(\tilde{f}) = \bar{f}$ , meaning  $\tilde{f}/1 = g_i/h_i \in k[Y]_{h_i}$  for all  $i$ . So for some  $n$ ,  $h_i^n(h_i\tilde{f} - g_i) = 0$ . In particular for  $P \in D_Y(h_i)$  since  $h_i(P) \neq 0$  we have that  $h_i(P)\tilde{f}(P) = g_i(P)$  so

$$\tilde{f}(P) = g_i(P)/h_i(P) = f(P).$$

Thus  $f = \tilde{f} \in k[Y]$  as desired, showing that  $\mathcal{O}_{\text{reg}}(Y) = k[Y]$  as desired.

**Note:**

Abusing terminology, for *any* SWF  $(X, \mathcal{O}_X)$  we call elements of  $\mathcal{O}_X(U)$  **regular functions** on  $U$ .

**2.2.17 Theorem**

Let  $(X, \mathcal{O}_X)$  be a SWF and  $Y \subseteq \mathbb{A}^n$  a subset. Let  $\psi: X \rightarrow Y$  be a map of sets, and set  $\psi_i = \text{proj}_i \circ \psi \in \text{Fun}(X, k)$  to be the projections of  $\psi$ . Then  $\psi$  is a morphism of SWFs  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_{\text{reg}})$  iff  $\psi_i \in \mathcal{O}_X(X)$  for all  $i$ . Equivalently,  $\psi^*: \text{Fun}(Y, k) \rightarrow \text{Fun}(X, k)$  maps  $\mathcal{O}_Y(Y)$  into  $\mathcal{O}_X(X)$ .

**Note:**

Equivalently,  $\psi = (\psi_1, \dots, \psi_n): X \rightarrow Y$  is a morphism iff each  $\psi_i$  is regular.

**Proof**

Notice that  $\text{proj}_i: \mathbb{A}^n \rightarrow k$  is given by a regular function  $x_i$  on  $\mathbb{A}^n$ . So if  $\psi$  is a morphism, then every  $\psi_i = \psi^*(x_i)$  is a regular function on  $X$ .

Conversely assume  $\psi_i \in \mathcal{O}_X(X)$  for all  $i$ . So we want to show that  $\psi$  is continuous, and for every  $f \in \mathcal{O}_Y(U)$  we have  $\psi^*(f) \in \mathcal{O}_X(\psi^{-1}U)$  for all  $U \subseteq Y$  open.

First we show that for every  $f \in k[\mathbb{A}^n]$  we have  $\psi^*(f) \in \mathcal{O}_X(X)$ . Notice that  $\psi^*: \text{Fun}(\mathbb{A}^n, k) \rightarrow \text{Fun}(X, k)$  is a  $k$ -algebra morphism, while  $f$  belongs to the  $k$ -subalgebra generated by the  $x_i$ . Thus  $\psi^*f$  belongs to the  $k$ -algebra generated by the  $\psi^*(x_i) = \psi_i$ . Since  $\psi_i \in \mathcal{O}_X(X)$  by assumption, this means  $\psi^*f \in \mathcal{O}_X(X)$  as desired (since it is a subalgebra).

Now we show that  $\psi$  is continuous. The basic open sets of  $Y$  are  $D_Y(f)$  with  $f \in k[\mathbb{A}^n]$ , and so it is sufficient to show  $\psi^{-1}D_Y(f)$  is open in  $X$ . Notice that

$$\psi^{-1}D_Y(f) = \{x \in X \mid f(\psi x) = 0\} = D_X(\psi^*f).$$

And since  $\psi^*f \in \mathcal{O}_X(X)$  by locality of SWFs we have that  $D_X(\psi^*f)$  is open as desired.

Now we need to show that for every open  $U \subseteq Y$  and  $f \in \mathcal{O}_Y(U)$  we have  $\psi^*f \in \mathcal{O}_X(\psi^{-1}U)$ . Replacing  $Y$  by  $U$  and  $X$  by  $\psi^{-1}U$ , we can assume  $U = Y$ .

First assume that  $f$  is rational, so  $f = g/h$  on  $Y$  with  $g, h \in k[\mathbb{A}^n]$  and  $h$  nonzero. Then  $\psi^*f = \psi^*g/\psi^*h$  and  $\psi^*h(x) \neq 0$  for every  $x \in X$ . By locality  $1/\psi^*h$  is in  $\mathcal{O}_X(X)$  and thus  $\psi^*f$  is in  $\mathcal{O}_X(X)$  as it is a  $k$ -algebra, as desired.

Now in general, locally  $f \in \mathcal{O}_Y(Y)$  has the form  $g/h|_Y$ , the general case follows from the rational case and the sheaf axioms.

And the final equivalence is because if  $\psi^*$  maps  $\mathcal{O}_Y(Y)$  into  $\mathcal{O}_X(X)$  then it maps  $x_i \in \mathcal{O}_Y(Y)$  to  $\psi^*(x_i) = \psi_i$  into  $\mathcal{O}_X(X)$ . ■

**2.2.18 Corollary**



For every SWF  $X = (X, \mathcal{O}_X)$  there is a natural bijection

$$\mathrm{hom}_{\mathrm{SWF}}(X, \mathbb{A}^n) \xrightarrow{\sim} \mathcal{O}_X(X)^n, \quad \psi \mapsto (\mathrm{proj}_1 \circ \psi, \dots, \mathrm{proj}_n \circ \psi).$$

In particular there is a natural identification  $\mathrm{hom}_{\mathrm{SWF}}(X, \mathbb{A}^1) \cong \mathcal{O}_X(X)$ .

Furthermore, for  $X = (X, \mathcal{O}_X)$  and a subset  $Y \subseteq \mathbb{A}^n$ , a set-theoretic map  $\psi: X \rightarrow Y$  is a morphism of SWFs  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_{\mathrm{reg}})$  iff the composition  $X \xrightarrow{\psi} Y \hookrightarrow \mathbb{A}^n$  is a morphism of SWFs.

### Proof

The first point follows from the above theorem with  $Y = \mathbb{A}^n$ . The second follows from the theorem as well, since  $\psi = (\psi_1, \dots, \psi_n)$  being a morphism is equivalent to all  $\psi_i$  being regular which is equivalent to the composition of  $\psi$  with the inclusion being a morphism. ■

### Note:

The second point follows directly from the fact that  $(Y, \mathcal{O}_{\mathrm{reg}})$  is the restriction of  $(\mathbb{A}^n, \mathcal{O}_{\mathrm{reg}})$  to  $Y$ .

Because for an affine scheme  $(X, \mathcal{O}_X)$  by Serre we have an identification  $\mathcal{O}_X(X) \cong k[X]$ , the following is immediate.

### 2.2.19 Corollary

If  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  are affine varieties, then  $\psi: X \rightarrow Y$  is a morphism of SWFs iff it maps  $k[Y]$  to  $k[X]$ .

There is an obvious functor  $\mathrm{SWF}_k^{\mathrm{op}} \rightarrow \mathrm{Alg}_k$  given by mapping  $(X, \mathcal{O}_X)$  to  $\mathcal{O}_X(X)$  and  $\varphi: X \rightarrow Y$  to  $\varphi^*: \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$ . This is called the **global sections** functor, as it takes global sections. It is denoted by  $\Gamma$ , so  $\Gamma(X, \mathcal{O}_X)$  for objects (we may omit one of the two inputs when the other is obvious), and  $\Gamma(\varphi) = \varphi^*$  on morphisms.

### 2.2.20 Lemma

A morphism of SWFs  $\psi: X \rightarrow \mathbb{A}^n$  satisfies  $\varphi(X) \subseteq Y$  for closed  $Y \subseteq \mathbb{A}^n$  iff  $\psi^*: k[\mathbb{A}^n] \rightarrow \mathcal{O}_X(X)$  satisfies  $\mathcal{I}(Y) \subseteq \ker \psi^*$ .

### Proof

We have  $\psi(X) \subseteq Y$  iff  $\psi(x) \in Z(\mathcal{I}(Y))$  for all  $x \in X$  and  $f \in \mathcal{I}(Y)$ , which is iff  $f(\psi(x)) = \psi^*(f)(x) = 0$ . So  $\psi(X) \subseteq Y$  iff  $\psi^*(f) = 0$  for all  $f \in \mathcal{I}(Y)$ , i.e. iff  $\mathcal{I}(Y) \subseteq \ker \psi^*$ . ■

### 2.2.21 Proposition

Let  $(X, \mathcal{O}_X)$  be an SWF and  $(Y, \mathcal{O}_Y)$  be an affine variety. Then  $\Gamma$  is bijective

$$\Gamma: \mathrm{hom}_{\mathrm{SWF}}(X, Y) \rightarrow \mathrm{hom}_{\mathrm{Alg}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)).$$

### Proof

First take  $Y = \mathbb{A}^n$ , so  $\mathcal{O}_Y(Y) = k[\mathbb{A}^n]$ , the free  $k$ -algebra on  $n$  generators. Thus there is a natural bijection

$$\text{nat}: \text{hom}_{\text{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \rightarrow \mathcal{O}_X(X)^n, \quad \varphi \mapsto (\varphi(x_1), \dots, \varphi(x_n)).$$

Now, the composition

$$\text{hom}_{\text{SWF}}(X, \mathbb{A}^n) \xrightarrow{\Gamma} \text{hom}_{\text{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \xrightarrow{\text{nat}} \mathcal{O}_X(X)^n$$

maps  $\varphi$  to  $(\varphi^*(x_1), \dots, \varphi^*(x_n))$ , and as we showed above this is a bijection. Thus  $\Gamma$  must be a bijection, as desired.

Now let  $Y \subseteq \mathbb{A}^n$  be an arbitrary affine variety, so  $\mathcal{O}_Y(Y) = k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y)$ . Recall that  $(Y, \mathcal{O}_Y)$  is induced from  $(\mathbb{A}^n, \mathcal{O}_{\text{reg}})$  so SWF morphisms into  $Y$  are the same as SWF morphisms into  $\mathbb{A}^n$  landing in  $Y$ . Thus by the previous lemma

$$\text{hom}_{\text{SWF}}(X, Y) \cong \{\varphi \in \text{hom}_{\text{SWF}}(X, \mathbb{A}^n) \mid \varphi(X) \subseteq Y\} = \{\varphi \in \text{hom}_{\text{SWF}}(X, \mathbb{A}^n) \mid \mathcal{I}(Y) \subseteq \ker \psi^*\},$$

and as we just showed, this is bijective to

$$\cong \{\varphi^* \in \text{hom}_{\text{Alg}}(k[\mathbb{A}^n], \mathcal{O}_X(X)) \mid \mathcal{I}(Y) \subseteq \ker \psi^*\},$$

and these are precisely the maps which quotient to  $k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y) \rightarrow \mathcal{O}_X(X)$ , i.e. this forms a bijection

$$\cong \text{hom}_{\text{Alg}}(k[Y], \mathcal{O}_X(X)).$$

Now it is easy to see that the composition of these isomorphisms is  $\Gamma$ . ■

### 2.2.22 Corollary

A SWF  $Y$  is an affine variety iff the  $k$ -algebra  $\mathcal{O}_Y(Y)$  is fg and for every SWF  $X$ ,  $\Gamma$  is bijective from  $\text{hom}_{\text{SWF}}(X, Y)$ .

### Proof

We just showed for an affine variety, the property holds. Conversely, if  $\mathcal{O}_Y(Y)$  is fg then it is equal to  $\mathcal{O}_{Y'}(Y')$  for some affine variety  $Y'$  (we showed that reduced fg  $k$ -algebras  $B$  are all of the form  $k[Y']$  for an affine  $Y'$ , and  $\mathcal{O}_Y(Y) \subseteq \text{Fun}(Y, k)$  is reduced always). Thus there is an isomorphism

$$\text{hom}_{\text{SWF}}(X, Y') \cong \text{hom}_{\text{Alg}}(\mathcal{O}_{Y'}(Y'), \mathcal{O}_X(X)) = \text{hom}_{\text{Alg}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \cong \text{hom}_{\text{SWF}}(X, Y),$$

where the final isomorphism is given by  $\Gamma$ . Thus  $\text{hom}(\bullet, Y') \cong \text{hom}(\bullet, Y)$  naturally, so by Yoneda  $Y' \cong Y$ . ■

### 2.2.23 Corollary

$\Gamma$  forms an anti-equivalence of categories from affine varieties to fg reduced  $k$ -algebras:  $\text{AffVar}_k \rightarrow \text{Alg}_k^{\text{fg, red}}$ . In particular  $\Gamma$  reflects isomorphisms, so  $X \cong Y$  iff  $\mathcal{O}_X(X) \cong \mathcal{O}_Y(Y)$  for affine  $X, Y$ .

### Proof

We showed that  $\Gamma$  is fully faithful on affine varieties, and so it is an equivalence to its essential image. Its essential image are the fg reduced  $k$ -algebras.

We can explicitly construct its inverse: for a fg reduced  $k$ -algebra  $A$ , then we map it to  $\text{Specm}(A)$ , and maps  $\varphi: A \rightarrow B$  are mapped to  $|\varphi^*|$ . ■

### 2.2.24 Corollary

Let  $X \subseteq \mathbb{A}^n$  and  $Y \subseteq \mathbb{A}^m$  be affine varieties. A map  $\varphi: X \rightarrow Y$  is a morphism of varieties iff  $\varphi = (f_1, \dots, f_m)$  for some  $f_i \in k[\mathbb{A}^n]$ .

### Proof

$\varphi$  is a morphism iff it is a morphism into  $\mathbb{A}^m$ , which is iff  $\varphi$ 's projections are in  $\mathcal{O}_X(X) = k[X]$ , which are just the restriction of polynomial functions to  $X$ . ■

### 2.2.25 Definition

Let  $X$  be a topological space. A subset  $Y \subseteq X$  is **locally closed** iff  $Y = U \cap F$  for open  $U \subseteq X$  and closed  $F \subseteq X$ .

The following lemma better explains the reasoning for the terminology of “locally closed”.

### 2.2.26 Lemma

For  $Y \subseteq X$  a subspace, the following are equivalent:

- (1)  $Y$  is locally closed,
- (2)  $Y$  is an open subset of a closed subset,
- (3)  $Y$  is a closed subset of an open subset,
- (4) for every  $y \in Y$ , there is a neighborhood  $U$  of  $y$  in  $X$  such that  $Y \cap U$  is closed in  $U$ .

### Proof

Recall that open (resp. closed) subsets of a subspace  $S$  are of the form  $S \cap U$  for  $U$  open (resp.  $S \cap F$  for  $F$  closed). So an open subset of a closed subset  $F$  are the sets of the form  $U \cap F$  for  $U$  open, and similarly for closed subsets of open subsets. Thus we see the equivalence of (1), (2), and (3).

Now if  $Y$  is locally closed, then it is the closed subset of an open subset  $U$ . Thus every  $y \in Y$  has neighborhood  $U$  in which  $Y$  is closed, showing (4). Conversely, for each  $y \in Y$  let  $y \in U_y$  be a neighborhood where  $Y \cap U_y$  is closed in  $U_y$ . Let  $U = \bigcup_y U_y$ , so

$$U - Y = \bigcup_y (U_y - Y),$$

and since  $Y \cap U_y$  is closed in  $U_y$ , we have  $U_y - Y = U_y - Y \cap U_y$  is open, so  $U - Y$  is open and thus  $Y$  is closed in  $U$ . ■

### 2.2.27 Corollary

A locally closed subset of a locally closed subset is locally closed.

**Proof**

So suppose  $Z \subseteq Y \subseteq X$  is a chain of inclusions, where each subset is locally closed in the next. It is sufficient to show that an open/closed subset of  $Y$  is locally closed in  $X$ , since if  $Z = U \cap F$  in  $Y$ , then  $U, F$  are locally closed in  $X$  and then clearly their intersection is too. But clearly by (2) an open subset of  $Y$  is locally closed, and by (3) the same for a closed subset. ■

**2.2.28 Proposition**

Let  $X \subseteq \mathbb{A}^n$  be an affine variety, and let  $f \in k[\mathbb{A}^n]$ . Then  $(D_X(f), \mathcal{O}_{\text{reg}})$  is an affine variety isomorphic to  $(Y, \mathcal{O}_{\text{reg}})$  where  $Y \subseteq \mathbb{A}^{n+1}$  is the closed subset  $Z(\mathcal{I}(X) \cup \{fx_{n+1} - 1\})$ .

**Proof**

We construct explicit inverse morphisms  $\varphi: D_X(f) \rightarrow Y: \psi$ . Define  $\varphi$  by mapping  $(x_1, \dots, x_n) \in X \subseteq \mathbb{A}^n$  to

$$\varphi(x_1, \dots, x_n) = \left( x_1, \dots, x_n, \frac{1}{f(x_1, \dots, x_n)} \right).$$

Its image lies in  $Y$  clearly, and its components are rational functions on  $D_X(f)$  so it forms a morphism into  $\mathbb{A}^n$ , thus into  $Y$ . Similarly define  $\psi$  by mapping  $(x_1, \dots, x_{n+1}) \in Y$  to

$$\psi(x_1, \dots, x_{n+1}) = (x_1, \dots, x_n).$$

Its image lies in  $X$  since  $f(x_1, \dots, x_n)x_{n+1} = 1$  so  $f(x_1, \dots, x_n) \neq 0$ , and it is a morphism into  $\mathbb{A}^n$  and thus into  $X$ . Clearly  $\psi \circ \varphi = \text{id}$  and for  $(x_1, \dots, x_{n+1}) \in Y$  we must have  $x_{n+1} = 1/f(x_1, \dots, x_n)$  by definition so  $\varphi \circ \psi = \text{id}$ . ■

**2.2.29 Example**

The SWF  $(D_{\mathbb{A}^1}(x), \mathcal{O}_{\text{reg}})$  is isomorphic to the hyperbola  $Z(xy - 1) \subseteq \mathbb{A}^2$ .

We can also prove the above proposition via Yoneda. Define  $B = k[X]_f$ . Now we know that  $\Gamma$  defines a bijection  $\text{hom}_{\text{SWF}}(Z, X) \rightarrow \text{hom}_{\text{Alg}}(k[X], \mathcal{O}_Z(Z))$  for any SWF  $Z$ . Furthermore, the universal property of localization says that a morphism  $\zeta: k[x] \rightarrow \mathcal{O}_Z(Z)$  factors through  $B$  iff  $\zeta(f) \in \mathcal{O}_Z(Z)^\times$ , and such a factorization is unique. And by locality,  $\mathcal{O}_Z(Z)^\times$  are just the nowhere vanishing functions in  $\mathcal{O}_Z(Z)$ . Thus  $\Gamma$  restricts to a bijection

$$\text{hom}_{\text{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \{\varphi \in \text{hom}_{\text{SWF}}(Z, X) \mid \varphi^*(f)(z) \neq 0 \text{ for all } z \in Z\}.$$

And we know

$$\varphi^*(f)(z) = f(\varphi(z)),$$

so  $\varphi^*(f)(z) \neq 0$  iff  $\varphi(z) \in D_X(f)$ , thus

$$\text{hom}_{\text{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \{\varphi \in \text{hom}_{\text{SWF}}(Z, X) \mid \varphi(Z) \subseteq D_X(f)\}.$$

And since  $D_X(f) \subseteq X$  has the induced SWF structure (as they are both induced from  $\mathbb{A}^n$ ), this means  $\Gamma$  defines a natural isomorphism

$$\text{hom}_{\text{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \text{hom}_{\text{SWF}}(Z, D_X(f)).$$

Now  $k[X]_f$  is a reduced fg  $k$ -algebra, and thus it is equivalent to some affine variety  $Y$  under  $\Gamma$  (so  $\mathcal{O}_Y(Y) = k[X]_f$ ), so we have natural isomorphisms

$$\text{hom}_{\text{SWF}}(Z, Y) \cong \text{hom}_{\text{Alg}}(k[X]_f, \mathcal{O}_Z(Z)) \cong \text{hom}_{\text{SWF}}(Z, D_X(f)),$$

and by Yoneda  $Y \cong D_X(f)$  as desired.

Notice that this means  $k[D_X(f)] \cong k[X]_f$  (the left-hand side is  $\mathcal{O}_{\text{reg}}(D_X(f))$  which are rational functions on  $D_X(f)$ ). Explicitly this isomorphism is given by the universal property of localization applied to the restriction  $k[X] \rightarrow k[D_X(f)]$ .

This immediately shows the following two corollaries.

### 2.2.30 Corollary

For  $X \subseteq \mathbb{A}^n$  affine and  $f \in k[\mathbb{A}^n]$ , we have

$$\mathcal{O}_{\text{reg}}(D_X(f)) = k[\mathbb{A}^n]_{D_X(f)}[1/f] \subseteq \text{Fun}(D_X(f), k),$$

i.e. regular functions on  $D_X(f)$  are of the form  $p/f$  for  $p \in k[\mathbb{A}^n]$ .

### 2.2.31 Corollary

If  $(X, \mathcal{O}_X)$  is an affine variety, then for every  $f \in \mathcal{O}_X(X)$ , the SWF  $(D_X(f), \mathcal{O}_X|_{D_X(f)})$  is affine. Furthermore, the restriction map  $\mathcal{O}_X(X) \rightarrow \mathcal{O}_X(D_X(f))$  induces an isomorphism of  $k$ -algebras

$$\mathcal{O}_X(X)_f \xrightarrow{\sim} \mathcal{O}_X(D_X(f)).$$

### Note:

Since  $D_X(f) \subseteq X$  is open, the restriction of  $\mathcal{O}_X$  to  $D_X(f)$  is equal to  $\mathcal{O}_X$  but on the subposet of open subsets of  $D_X(f)$ .

### 2.2.32 Corollary

An affine variety has a basis consisting of affine varieties.

### Proof

The  $D_X(f)$  form an open basis for  $X$  (since they do for  $\mathbb{A}^n$ , and  $X$  is a subspace and  $D_X(f) = D_{\mathbb{A}^n}(f) \cap X$ ), and they are affine. ■

### 2.2.33 Definition

Let  $X = (X, \mathcal{O}_X)$  be an algebraic variety. A **closed/open/locally closed subvariety** of  $X$  is an SWF  $(Y, \mathcal{O}_X|_Y)$  where  $Y \subseteq X$  is closed/open/locally closed. A **quasi-affine variety** is an open subvariety of an affine variety.

### 2.2.34 Corollary

A closed/open/locally closed subvariety of an algebraic variety is an algebraic variety.

### Proof

Let  $X_i$  be a finite subcovering of  $X$  of affine varieties. Then  $Y = \bigcup_i (Y \cap X_i)$ , and to show that  $Y$  is an algebraic variety it is sufficient to show that  $Y \cap X_i$  is. So replacing  $X$  by  $X_i$  and  $Y$  by  $Y \cap X_i$ , we assume  $X$  is affine.

We consider the cases when  $Y$  is open and closed, and this gives us the case when  $Y$  is locally closed, since if  $Y$  is open in a closed subset then it is an open subset of an algebraic variety which is then an algebraic variety.

If  $Y$  is closed, then  $(Y, \mathcal{O}_X|_Y)$  is affine since following the isomorphism to a closed subset of  $\mathbb{A}^n$  we see that  $Y$  is a closed subset of  $\mathbb{A}^n$  and thus an affine variety.

Now if  $Y \subseteq X$  is open, then  $\{D_X(f)\}_{f \in \mathcal{O}_X(X)}$  is a basis of  $X$ , and since  $X$  is Noetherian,  $Y$  is quasi-compact, so it is the finite union of  $D_X(f)$ s. Each of these are affine varieties, so  $Y$  is an algebraic variety by definition. ■

### 2.2.35 Example

We have shown some quasi-affine varieties are affine, like  $D_X(f)$ . But not all!

For example let  $X = \mathbb{A}^2 - 0$ , which is quasi affine as it is equal to  $D_{\mathbb{A}^2}(x_1) \cup D_{\mathbb{A}^2}(x_2)$ . The inclusion  $X \hookrightarrow \mathbb{A}^2$  is not an isomorphism, as it is not a homeomorphism, but the induced map  $\text{res}: k[\mathbb{A}^2] \rightarrow \mathcal{O}_X(X)$  is an isomorphism. This cannot happen for affine varieties ( $\Gamma$  reflects isomorphisms).

We see that  $X$  is the pushout of  $D(x_1), D(x_2)$  along their intersection. Thus  $\mathcal{O}_X(X)$  is the pullback of  $k[x_1, x_2]_{x_1}, k[x_1, x_2]_{x_2}$  along  $k[x_1, x_2]_{x_1 x_2}$ . That is,  $\mathcal{O}_X(X)$  consists of pairs  $f_i \in k[x_1, x_2]_{x_i}$  such that  $f_1 = f_2$  in  $k[x_1, x_2]_{x_1 x_2}$ . But a rational function whose denominator is both a power of  $x_1$  and a power of  $x_2$  must be a polynomial, so  $\mathcal{O}_X(X) = k[\mathbb{A}^2]$ .

## 2.3 The category of SWFs

In this section we discuss some categorical constructions and results on  $\text{SWF}_k$ .

There is a clear forgetful functor from  $\text{SWF}_k$  to the category of topological spaces,  $U: \text{SWF}_k \rightarrow \text{Top}$ , which maps  $(X, \mathcal{O}_X)$  to its underlying topological space  $X$  and a morphism  $\varphi: (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  to its underlying continuous map  $\varphi$ . This functor is faithful, but not full (it is easy to construct a continuous map which is not a morphism of SWFs). Though it is a left-adjoint, as we will show.

### 2.3.1 Definition

For a topological space  $X$  and an arbitrary set  $S$ , a function  $f: X \rightarrow S$  is **locally constant** if for every  $p \in X$  there is a neighborhood  $U \subseteq X$  such that  $f|_U$  is constant.

Note that  $f: X \rightarrow S$  being locally constant is the same as it being continuous when  $S$  is endowed with the discrete topology. Indeed, if  $f$  is locally constant then for every  $p \in f^{-1}\{x\}$  there is an open neighborhood of  $p$  contained in  $f^{-1}\{x\}$  by definition, so  $f^{-1}\{x\}$  is open. Thus  $f$  is continuous to discrete  $S$ . Conversely,  $f$  being continuous to discrete  $S$  means that  $f^{-1}\{x\}$  is open and so for every  $p \in X$  we take the neighborhood  $U = f^{-1}(fp)$  to see it is locally constant.

### 2.3.2 Lemma

For an SWF  $(X, \mathcal{O}_X)$ ,  $\mathcal{O}_X(U)$  contains all locally constant functions  $U \rightarrow k$ .

### Proof

Let  $f: U \rightarrow k$  be locally constant, so there is an open cover  $U_i$  of  $U$  on which  $f|_{U_i}$  is constant. Since  $\mathcal{O}_X(U_i)$  is a  $k$ -algebra, it must contain all constant functions (since it contains the unit and all of its multiples), so  $f|_{U_i} \in \mathcal{O}_X(U_i)$ . By gluing, this means  $f \in \mathcal{O}_X(U)$  as desired. ■

### 2.3.3 Proposition

The forgetful functor  $U: \text{SWF}_k \rightarrow \text{Top}$  is a left-adjoint.

#### Proof

Forgetful functors are in general “the most general” object, so in our case in lieu of the previous lemma it seems that the following construction is a good contender. For open  $U \subseteq X$  define  $\mathcal{O}_X(U)$  to be the collection of locally constant  $U \rightarrow k$ . Thus  $\mathcal{O}_X$  is just the sheaf of continuous maps into  $k$ , which is always a sheaf.

For locality, let  $f \in \mathcal{O}_X(U)$  then its preimage on all subsets of  $k$  is open, in particular  $D_U(f)$ . And if  $f$  is locally constant, so too is  $1/f$  ( $f$  and  $1/f$  are constant on the same open subsets). Thus  $(X, \mathcal{O}_X)$  is an SWF which is denoted  $\underline{X}$ .

We claim that  $U \dashv \underline{\cdot}$ . This amounts to constructing a natural isomorphism

$$\text{hom}_{\text{Top}}(U(X, \mathcal{O}_X), Y) \cong \text{hom}_{\text{SWF}}((X, \mathcal{O}_X), \underline{Y}).$$

$U$  defines an injective map from the right to the left, so we just need to show that it is surjective. That is, for  $\varphi: X \rightarrow Y$  continuous,  $\varphi$  lifts to a morphism of SWFs  $(X, \mathcal{O}_X) \rightarrow \underline{Y}$ . So for  $U \subseteq Y$  open we need to show  $\varphi^*$  maps  $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\varphi^{-1}U)$ .

For  $f: U \rightarrow k$  locally constant, then  $f \circ \varphi: \varphi^{-1}U \rightarrow k$  is locally constant as well, as the composition of continuous maps. Thus  $\varphi^*(f) \in \mathcal{O}_X(\varphi^{-1}U)$  by the previous lemma. ■

In particular  $U$  preserves colimits. The following is a technical definition and lemma which we will not prove in full here (see homework).

### 2.3.4 Definition

Let  $\mathcal{B}$  be a basis for a topology, then a **B-SWF** over  $k$  is a presheaf  $\mathcal{O}: \mathcal{B}^{\text{op}} \rightarrow \text{Alg}_k$  satisfying the following axioms:

- (1)  $\mathcal{O}$  is a subpresheaf of the function sheaf,  $\text{Fun}(\bullet, k)$ .
- (2) **B-sheaf axiom:** for any basic set  $B \in \mathcal{B}$  and cover of basic sets  $B = \bigcup_i B_i$ , if  $f: B \rightarrow k$  is a function such that  $f|_{B_i} \in \mathcal{O}(B_i)$  for all  $i$ , then  $f \in \mathcal{O}(B)$ .
- (3) **locality:** for any  $B \in \mathcal{B}$  and  $f \in \mathcal{O}(B)$ ,  $D_B(f)$  is open (i.e. the union of basic sets) and for every basic cover  $B_i$  of  $D_B(f)$ , we have that  $f|_{B_i} \in \mathcal{O}(B_i)$ .

### 2.3.5 Proposition

For every B-SWF  $(\mathcal{B}, \mathcal{O})$  where  $\mathcal{B}$  is a basis of the topology on  $X$ , there is a unique SWF  $(X, \mathcal{O}_X)$  extending it, meaning  $\mathcal{O}_X(B) = \mathcal{O}(B)$  for all  $B \in \mathcal{B}$ .

#### Proof

The construction is as follows: for every  $U \subseteq X$  open and  $\{B_i\} \subseteq \mathcal{B}$  covering  $U$ , define

$$\mathcal{O}_X(U) = \{f: U \rightarrow k \mid f|_{B_i} \in \mathcal{O}(B_i) \text{ for all } i\}.$$

Then one can show without much difficulty that  $\mathcal{O}_X(U)$  is independent on the choice of basic covering. In particular this means that  $\mathcal{O}_X(B) = \mathcal{O}(B)$  for all  $B \in \mathcal{B}$  by choosing the covering  $\{B\}$ .

It is then easy to show that this is a sheaf, and locality follows easily from locality of  $(\mathcal{B}, \mathcal{O})$ . ■

Now we will show that  $\mathbf{SWF}_k$  is cocomplete. Since  $U$  preserves colimits, we already know what the underlying topological spaces of the colimits are.

### 2.3.6 Lemma

Let  $\pi: X \rightarrow Y$  be a quotient map, and suppose there is an SWF structure  $(X, \mathcal{O}_X)$  on  $X$ . Then there is a unique SWF structure  $(Y, \mathcal{O}_Y)$  given by the following universal property:  $\varphi: (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$  is a morphism of SWFs iff  $\varphi \circ \pi: (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$  is a morphism of SWFs.

#### Proof

This is a universal property and thus a solution is unique up to unique isomorphism. Define

$$\mathcal{O}_Y(U) = \{f: U \rightarrow k \mid \pi^*(f) \in \mathcal{O}_X(\pi^{-1}(U))\}.$$

It is not hard to show that this satisfies the universal property. ■

### 2.3.7 Theorem

The category  $\mathbf{SWF}_k$  is cocomplete.

#### Proof

We just need to show that  $\mathbf{SWF}_k$  has coproducts and coequalizers. For coproducts, let  $\{X_i\}_i$  be a collection of SWFs and let  $X = \coprod_i X_i$ . It has a basis of open subsets given by the union of all open subsets of  $X_i$ . Then define  $\mathcal{O}_X$  on  $X$  as a B-SWF where for  $U \subseteq X_i$  open we define

$$\mathcal{O}_X(U) = \mathcal{O}_{X_i}(U).$$

It is easy to see that this is a B-SWF, and thus defines an SWF on  $X$ . It is also not hard to see that this has the universal property of a coproduct.

For coequalizers, let  $\varphi, \psi: X \rightrightarrows Y$  be parallel morphisms of SWFs. The underlying space of their coequalizer will be  $Y/\sim$  where  $\sim$  is the equivalence relation generated by  $\varphi(x) \sim \psi(x)$  for all  $x \in X$ . This is a quotient space, so we just give it the quotient space SWF structure from the previous lemma. This clearly satisfies the universal property of coequalizers as a direct result from the universal property of quotients. ■

#### Note:

While the colimit of SWFs is an SWF, the colimit of algebraic varieties need not be a variety!

### 2.3.8 Corollary

The global sections functor  $\Gamma$  contravariantly preserves colimits.

#### Proof

Since  $\Gamma$  is contravariant, preserving colimits means mapping colimiting cones to limit cones. So it is sufficient to show that

$$\Gamma\left(\coprod_i (X_i, \mathcal{O}_{X_i})\right) \cong \prod_i \Gamma(X_i, \mathcal{O}_{X_i}), \quad \Gamma(\text{coeq}(\varphi, \psi)) \cong \text{eq}(\Gamma(\varphi), \Gamma(\psi)),$$



in a way which maps the colimit cone to the limit cone, which we leave as an exercise.

For coproducts, by the above construction we see that for  $X = \coprod_i X_i$ ,

$$\Gamma\left(\coprod_i (X_i, \mathcal{O}_{X_i})\right) = \{f: X \rightarrow k \mid f|_{X_i} \in \mathcal{O}_{X_i}(X_i)\},$$

so the map  $\Gamma(X) \rightarrow \prod_i \Gamma(X_i)$  given by  $f \mapsto (f|_{X_i})$  is a natural isomorphism.

For a quotient map  $\pi: X \rightarrow Y$ , we have

$$\Gamma(Y) = \{f: Y \rightarrow k \mid f \circ \pi \in \mathcal{O}_X(X)\},$$

meaning  $\pi^*: \Gamma(Y) \rightarrow \Gamma(X)$  is an injection (by  $\pi$ 's surjectivity), meaning  $\Gamma(Y) \cong \text{im}\pi^*$ . So letting  $Z = \text{coeq}(\varphi, \psi)$  we have

$$\Gamma(\text{coeq}(\varphi, \psi)) = \{f: Z \rightarrow k \mid f \circ \pi \in \mathcal{O}_Y(Y)\} \cong \text{im}\pi^*.$$

We know that

$$\text{eq}(\Gamma\varphi, \Gamma\psi) = \ker(\varphi^* - \psi^*).$$

So we want to show that

$$\text{im}\pi^* = \ker(\varphi^* - \psi^*).$$

Firstly, we have  $\pi \circ \varphi = \pi \circ \psi$  by definition so  $\varphi^* \circ \pi^* = \psi^* \circ \pi^*$  and thus  $(\varphi^* - \psi^*) \circ \pi^* = 0$ . So we have left-to-right inclusion. Now suppose that  $f \circ \varphi = f \circ \psi$  for  $f \in \Gamma(Y)$ , then by the equivalence on  $Z$  this means  $f$  lifts to a map  $g: Z \rightarrow k$ , i.e.  $f = g \circ \pi$ . So by definition  $g \in \Gamma(Z)$  since  $f \in \Gamma(Y)$ , and this shows that  $f \in \text{im}\pi^*$  as desired. ■

Let us consider a specific type of colimit. Recall that a pushout is a colimit diagram of the form

$$\begin{array}{ccc} A & \xrightarrow{\quad} & B \\ \downarrow & & \downarrow \\ C & \dashrightarrow & X \end{array}$$

For sets, and by extension topological spaces,  $X$  is the quotient of  $B \amalg C$  obtained by identifying the images of an element from  $A$  in  $B$  and  $C$ . The topology is given by the quotient topology. This kind of colimit is called a **pushout**, it is dual to the **pullback** (also called the **fiber product**). It is the colimit of a functor on the diagram  $\bullet \leftarrow \bullet \rightarrow \bullet$ .

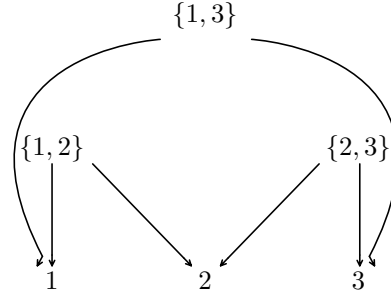
If our category is **Top**, then the pushout can be viewed geometrically as gluing together  $B$  and  $C$  along their “intersection”  $A$ .

When we are dealing with topological spaces though (and in our case SWFs), often times we would like to glue together a family of spaces and not just two. We can generalize the pushout to account for this.

### 2.3.9 Definition

Let  $I$  be an indexing set, then the **gluing category** over  $I$  is the category  $\text{Glue}_I$  whose objects are  $I \sqcup \binom{I}{2}$  (i.e.  $I$  and the set of all two-element subsets of  $I$ ). For every  $\{i, j\} \in \binom{I}{2}$ ,  $\text{Glue}_I$  has unique morphisms from  $\{i, j\}$  to  $i$  and to  $j$ .

For example, if  $I = 1, 2, 3$  then  $\text{Glue}_I$  looks like



And when  $I$  has two elements then  $\text{Glue}_I$  is just the pushout category  $\bullet \leftarrow \bullet \rightarrow \bullet$ .

A **gluing datum** is a functor whose domain is a gluing category, i.e. a functor  $X: \text{Glue}_I \rightarrow \mathcal{C}$ . Equivalently it is given by a collection of objects  $X_i \in \mathcal{C}$  and  $X_{ij} \in \mathcal{C}$  for  $i, j \in \mathcal{C}$  ( $X_{ij}$  is unordered), as well as morphisms  $f_{ij}: X_{ij} \rightarrow X_i$  and  $f_{ji}: X_{ij} \rightarrow X_j$ . A **gluing** is the colimit of a gluing datum. Geometrically, for a gluing datum  $X: \text{Glue}_I \rightarrow \mathcal{C}$ , we interpret  $X_i$  as the spaces we want to glue, and  $X_{ij}$  as the common subspace over which we want to glue.

We know that  $\text{SWF}_k$  is cocomplete so it has all gluings. Our main concern regarding gluings is that we can view an SWF as the gluing of an open cover of induced SWFs.

Denote the inclusion (canonical) map  $X_i \rightarrow \coprod_i X_i$  by  $\text{inc}_i$ .

### 2.3.10 Lemma

For a gluing datum  $X: \text{Glue}_I \rightarrow \mathcal{C}$ , its gluing is the coequalizer of the diagram (where  $X_{ij} = X_{ji} = X_{\{i,j\}}$ )

$$\coprod_{i \neq j} X_{ij} \begin{matrix} \xrightarrow{\varphi} \\ \xrightarrow{\psi} \end{matrix} \coprod_i X_i \rightarrow \text{colim } X.$$

Where  $\varphi$  is the map whose component on  $X_{ij}$  is  $f_{ij}$  composed with inclusion  $X_i \rightarrow \coprod_i X_i$ . Similarly  $\psi$  is the map whose component on  $X_{ij}$  is  $f_{ji}$  composed with inclusion  $X_j \rightarrow \coprod_i X_i$ .

### Proof

Cones under  $X$  are equivalent (classically) to cones under the coequalizer diagram

$$\coprod_{\{i,j\} \rightarrow i} X_{ij} \begin{matrix} \xrightarrow{\alpha} \\ \xrightarrow{\beta} \end{matrix} \coprod_i X_i \amalg \coprod_{\{i,j\}} X_{ij},$$

where  $\alpha$ 's component on  $X_{ij}$  for  $\{i,j\} \rightarrow i$  is  $f_{ij}$  composed with inclusion into the large coproduct (on the right). And  $\beta$ 's component on  $X_{ij}$  is just the inclusion into the large coproduct.

Explicitly this means that

$$\alpha \circ \text{inc}_{ij \rightarrow i} = \text{inc}_i \circ f_{ij}, \quad \beta \circ \text{inc}_{ij \rightarrow i} = \text{inc}_{\{i,j\}}, \quad \varphi \circ \text{inc}_{ij} = \text{inc}_i \circ f_{ij}, \quad \psi \circ \text{inc}_{ij} = \text{inc}_j \circ f_{ji}.$$

Cones under this are maps  $f$  from the large coproduct to an object  $Y$  such that  $f \circ \alpha = f \circ \beta$ . We claim that restricting  $f$  to  $\coprod_i X_i$  gives a cone under the diagram in the lemma. We have that

$$f \circ \alpha \circ \text{inc}_{ij \rightarrow i} = f \circ \beta \circ \text{inc}_{ij \rightarrow i}$$

which is equivalent to

$$f \circ \text{inc}_i \circ f_{ij} = f \circ \text{inc}_{\{i,j\}}.$$

Thus we see that this is also equal to  $f \circ \text{inc}_j \circ f_{ji}$ .

A cone under the diagram in the lemma is a map  $f$  such that

$$f \circ \varphi \circ \text{inc}_{ij} = f \circ \psi \circ \text{inc}_{ij} \iff f \circ \text{inc}_i \circ f_{ij} = f \circ \text{inc}_j \circ f_{ji}$$

for all  $i, j$ . So we have shown that a cone under the classical coequalizer diagram restricts to one of the one in the lemma.

Conversely if we have  $f \circ \text{inc}_i \circ f_{ij} = f \circ \text{inc}_j \circ f_{ji}$  for  $f: \coprod_i X_i \rightarrow Y$ , then this uniquely extends to a cone over the classical diagram. This is because we must have

$$f \circ \text{inc}_i \circ f_{ij} = f \circ \text{inc}_{\{i,j\}}.$$

Thus we can just define  $f$ 's component on  $X_{ij}$  to be  $f \circ \text{inc}_i \circ f_{ij}$ , and by the identity for  $f \circ \varphi = f \circ \psi$  this is well-defined (symmetric in  $i$  and  $j$ ).

So there is an isomorphism between the cones under the classical and the diagram in the lemma, so they define the same coequalizers.  $\blacksquare$

### 2.3.11 Lemma

Let  $X: \text{Glue}_I \rightarrow \text{Top}$  be a gluing datum. Then its gluing is

$$\text{colim } X = \coprod_{i \in I} X_i / \sim,$$

where  $\sim$  is the equivalence induced by  $f_{ij}(x) \sim f_{ji}(x)$  for all  $x \in X_{ij}$ .

### Proof

Consider the coequalizer diagram in the previous diagram

$$\coprod_{i \neq j} X_{ij} \xrightarrow[\psi]{\varphi} \coprod_i X_i \rightarrow \text{colim } X.$$

The coequalizer is just the quotient of  $\coprod_i X_i$  by identifying the images of  $\varphi$  and  $\psi$  of the same element.  $\varphi$  on  $X_{ij}$  is just  $f_{ij}$  and  $\psi$  is  $f_{ji}$ , so this is just the equivalence  $f_{ij}(x) \sim f_{ji}(x)$  for  $x \in X_{ij}$  as desired.  $\blacksquare$

Since we know what the coproduct of SWFs and the quotient of an SWF is, we can then determine the SWF obtained by gluing.

Let  $X: \text{Glue}_I \rightarrow \text{SWF}_k$  be a gluing datum. Write  $X_i = (X_i, \mathcal{O}_i)$ ,  $X_{ij} = (X_{ij}, \mathcal{O}_{ij})$ , and  $\varphi_{ij} = X(\{i, j\} \rightarrow i)$  for the data. Let us write  $\hat{X} = \coprod_i X_i$ . Then its gluing is the SWF  $(X, \mathcal{O}_X)$  where  $X = \hat{X} / \sim$  where  $\varphi_{ij}(x_{ij}) \sim \varphi_{ji}(x_{ij})$ . Let us denote by  $\pi: \hat{X} \rightarrow X$  the projection.

For  $U \subseteq X$  open, then  $\pi^{-1}U \subseteq \hat{X}$  is open and thus has the form  $\pi^{-1}U = \bigcup_i U_i$  where  $U_i = X_i \cap \pi^{-1}U$  are open. Then by our construction of quotient and coproduct SWFs,

$$\mathcal{O}_X(U) = \{f: U \rightarrow k \mid f \circ \pi \in \mathcal{O}_{\hat{X}}(\pi^{-1}U)\} = \{f: U \rightarrow k \mid (f \circ \pi)|_{U_i} \in \mathcal{O}_i(U_i) \text{ for all } i\}.$$

We can now prove our goal regarding gluing.

### 2.3.12 Theorem

Let  $(X, \mathcal{O}_X)$  be an SWF and  $X = \bigcup_i U_i$  an open cover. Then  $(X, \mathcal{O}_X)$  is isomorphic to the gluing of  $(U_i, \mathcal{O}_X|_{U_i})$  along their intersections.

### Proof

This is a rather lengthy proof. It is entirely technical, and just ensures that everything behaves as it should. Let us write  $\mathcal{O}_i$  for  $\mathcal{O}_X|_{U_i}$  and  $\mathcal{O}_{ij}$  for  $\mathcal{O}_X|_{U_i \cap U_j}$ . Define  $(Y, \mathcal{O}_Y)$  to be the gluing of the  $U_i$  along their intersections. Let  $\hat{Y} = \coprod_i U_i$ , so as a topological space  $Y = \hat{Y} / \sim$  where we identify elements of  $U_i, U_j$  if they lie in the intersection. For an element  $x \in U_i$  let  $x_i \in \hat{Y}$  denote its copy in  $\hat{Y}$ . Then our equivalence is

induced by  $x_i \sim x_j$  for  $x \in U_i \cap U_j$ . Furthermore for  $x \in \hat{Y}$  let  $\pi(x) = [x] \in Y$  denote its equivalence class, and for  $x \in U_i \subseteq X$  we define  $[x] = [x_i]$ .

There is a clear map  $\hat{\varphi}: \hat{Y} \rightarrow X$  which maps an element of  $\hat{Y}$  to its corresponding element in  $X$ , i.e.  $x_i \mapsto x$ . This is a continuous map, since for  $U \subseteq X$  open we have  $\hat{\varphi}^{-1}(U) = \coprod_i (U_i \cap U)$  open. Furthermore it respects the equivalence  $\sim$ , since if  $x \in U_i \cap U_j$  then  $x_i$  and  $x_j$  are both mapped to  $x$ . So by the universal property of quotients, this defines a map  $\varphi: Y \rightarrow X$  given by  $\varphi[x] = x$  (so we have shown that such a map is well-defined and continuous). Clearly  $\varphi$  has an inverse given by  $x \mapsto [x]$ , so it is bijective.

Now we claim that  $\hat{\varphi}$  is an open map. This is clear since it maps basic open sets  $U \subseteq U_i$  to  $U \subseteq X$  which are open (since  $U_i$  is open in  $X$ ). So we have a commutative diagram

$$\begin{array}{ccc} \hat{Y} & \xrightarrow{\hat{\varphi}} & X \\ & \searrow \pi & \nearrow \varphi \\ & Y & \end{array}$$

Every open set of  $Y$  is of the form  $\pi(\hat{U})$  for open  $\hat{U} \subseteq \hat{Y}$  (since  $U \subseteq Y$  is open iff  $\pi^{-1}U \subseteq \hat{Y}$  is open, and  $U = \pi(\pi^{-1}U)$  by surjectivity). And we see  $\varphi(\pi(\hat{U})) = \hat{\varphi}(\hat{U})$  is open. So  $\varphi$  is a bijective open continuous map, thus a homeomorphism.

Now we need to show that for all  $V \subseteq X$  open,  $\varphi^*$  defines an isomorphism  $\mathcal{O}_X(V) \rightarrow \mathcal{O}_Y(\varphi^{-1}V)$ . As we saw with B-SWFs, it is sufficient to restrict our attention to basic open sets. We take as a basis for  $X$  the open subsets of  $U_i$ , as  $i$  ranges. These define a basis of open subsets of  $\hat{Y}$ , and thus projecting them via  $\pi$  defines a basis of  $Y$ .

For  $V \subseteq U_i$  we have that restricting  $\hat{\varphi}$  to  $V \subseteq \hat{Y}$  gives the identity  $V \rightarrow V \subseteq X$ . That is

$$V = \hat{\varphi}(V) = \varphi \circ \pi(V) \implies \varphi^{-1}(V) = \pi(V).$$

So if we show that  $\varphi^*$  is an isomorphism  $\mathcal{O}_X(V) \rightarrow \mathcal{O}_Y(\varphi^{-1}V) = \mathcal{O}_Y(\pi(V))$  then we will have finished, as we have exhausted both the basic open sets of  $X$  as well as  $Y$ 's.

By above we know for  $U \subseteq Y$  open we have

$$\mathcal{O}_Y(U) = \left\{ f: U \rightarrow k \mid (f \circ \pi)|_{\hat{U}_i} \in \mathcal{O}_i(\hat{U}_i) \text{ for all } i \right\},$$

where  $\hat{U}_i = \pi^{-1}(U) \cap U_i$ . In our case, for  $V \subseteq U_i$  basic we are concerned with  $U = \pi(V)$ . Then  $\hat{U}_j = \pi^{-1}(\pi(V)) \cap U_j$  which are all  $x \in U_j$  such that  $x$  is identified with an element of  $V \subseteq U_i$ . So we see  $\hat{U}_j = V \cap U_j$  and so

$$\mathcal{O}_Y(\pi(V)) = \left\{ f: \pi(V) \rightarrow k \mid (f \circ \pi)|_{V \cap U_j} \in \mathcal{O}_j(V \cap U_j) \text{ for all } j \right\}.$$

Since  $V \cap U_j$  form an open cover of  $V$ , this is just equal to

$$\mathcal{O}_Y(\pi(V)) = \{ f: \pi(V) \rightarrow k \mid (f \circ \pi)|_V \in \mathcal{O}_X(V) \}.$$

We know that on  $V$ ,  $\hat{\varphi}$  is just the identity so  $\varphi \circ \pi = \text{id}$  and since  $\varphi$  is bijective,  $\varphi$  and  $\pi$  are inverses. So for  $f \in \mathcal{O}_X(V)$  we have  $\varphi^*(f) \circ \pi = f \circ \varphi \circ \pi = f$  thus  $\varphi^*(f) \in \mathcal{O}_Y(\pi(V))$ . So  $\varphi$  is indeed a morphism of SWFs.

And  $\varphi^*$  is bijective, as we can map  $f \in \mathcal{O}_Y(\pi(V))$  to  $f \circ \pi \in \mathcal{O}_X(V)$ . Then  $\varphi^*(f \circ \pi) = f \circ \pi \circ \varphi = f$  as they are inverses on  $V$ , and  $\varphi^*(f) \circ \pi = f \circ \varphi \circ \pi = f$ . Thus  $\varphi$  is an isomorphism as desired. ■

Our main interest in the above theorem is due to algebraic varieties. By definition, algebraic varieties are SWFs  $X$  with an affine cover  $U_i$ . This theorem says we can view  $X$  as a certain colimit of the  $U_i$  with  $U_i \cap U_j$ . In particular we can use the fact that  $\Gamma$  preserves colimits to compute the global sections of  $X$  as a certain limit of  $\Gamma(U_i)$  and  $\Gamma(U_i \cap U_j)$ .

And if  $Y \subseteq X$  is a subspace, then by the above theorem  $Y$  is the gluing of  $Y \cap U_i$  along their intersections  $Y \cap U_i \cap U_j$ . Thus global sections of  $Y$ , which are just sections of  $Y$ , are a certain limit of  $\Gamma(Y \cap U_i)$  and  $\Gamma(Y \cap U_i \cap U_j)$ . In particular for  $V \subseteq X$  open,  $\mathcal{O}_X(V)$  is a certain limit of  $\mathcal{O}_X(U \cap U_i)$  and  $\mathcal{O}_X(U \cap U_i \cap U_j)$ . So if we know  $U_i$  and the intersections  $U_i \cap U_j$  well, we can deduce properties of  $X$ .

We now consider another functorial construction on  $\text{SWF}_k$ .

### 2.3.13 Definition

Let  $(X, \mathcal{O}_X)$  be an SWF and  $p \in X$  a point. Define its **stalk** at  $p$  to be

$$\mathcal{O}_{X,p} = \text{colim}_{x \in U} \mathcal{O}_X(U),$$

where the colimit is taken of the functor  $\mathcal{O}_X$  restricted to the subposet  $\mathcal{N}(p) \subseteq \text{Op}(X)$  of open neighborhoods of  $p$ . Explicitly, this is

$$\mathcal{O}_{X,p} = \coprod_{x \in U} \mathcal{O}_X(U) / \sim,$$

where  $\sim$  is the equivalence identifying  $f \in \mathcal{O}_X(U)$  and  $g \in \mathcal{O}_X(V)$  if there is an open subset  $x \in W \subseteq U, V$  such that  $f|_W = g|_W$  (i.e.  $f$  and  $g$  are equal locally around  $p$ ). Such equivalence classes are called **germs**.

Since the colimit of  $k$ -algebras is a  $k$ -algebra, the stalk of an SWF is a  $k$ -algebra.

We sometimes write  $(f, U)$  to mean  $f \in \mathcal{O}_X(U)$  or to mean its germ in  $\mathcal{O}_{X,p}$ . Notice that for a germ  $(f, U)$  of  $p$ , the value  $f(p)$  makes sense, as all functions in the same germ must have the same value at  $p$ .

Consider the category of **pointed spaces of functions**:  $\text{SWF}_k^*$ , whose objects are SWFs with a distinguished point:  $(X, \mathcal{O}_X, p)$  for  $p \in X$ , and a morphism of pointed SWFs is a morphism of SWFs which preserve the distinguished points:  $\varphi: (X, \mathcal{O}_X, p) \rightarrow (Y, \mathcal{O}_Y, q)$  is a morphism of SWFs such that  $\varphi(p) = q$ .

Then taking stalks (**stalkification**) is functorial  $(\text{SWF}_k^*)^{\text{op}} \rightarrow \text{Alg}_k$ . This is because a morphism of pointed SWFs  $\varphi: (X, \mathcal{O}_X, p) \rightarrow (Y, \mathcal{O}_Y, q)$  induces a morphism of functors  $\varphi^*: \mathcal{O}_Y \rightarrow \mathcal{O}_X$  as well as a functor  $\varphi^{-1}: \mathcal{N}(q) \rightarrow \mathcal{N}(p)$  via mapping  $q \in V$  to  $\varphi^{-1}V$ . Thus it forms a morphism of functors  $\mathcal{O}_Y|_{\mathcal{N}(q)} \rightarrow \mathcal{O}_X|_{\mathcal{N}(p)}$  and since taking colimits is itself functorial, we get a morphism  $\mathcal{O}_{Y,q} \rightarrow \mathcal{O}_{X,p}$ .

Explicitly  $\varphi$  induces the map  $\varphi^*: \mathcal{O}_{Y,q} \rightarrow \mathcal{O}_{X,p}$  on germs by mapping  $(f, U) \in \mathcal{O}_{Y,q}$  to  $(\varphi^*(f), \varphi^{-1}U) \in \mathcal{O}_{X,p}$ .

Further notice that if  $p \in U \subseteq X$  is a neighborhood of  $p$  then the stalks of  $\mathcal{O}_X|_U$  and  $\mathcal{O}_X$  at  $p$  are isomorphic. This can be shown by the germ definition, or by noting that the stalk is a filtered colimit, and  $\mathcal{N}(p, U)^{\text{op}}$  is a cofinal subset of  $\mathcal{N}(p, X)^{\text{op}}$ , and filtered colimits over the restriction of a functor to a cofinal subcategory is isomorphic to the colimit over the whole subcategory.

### 2.3.14 Lemma

Let  $Y$  be an affine variety and  $P \in Y$ . Then under the isomorphism  $Y \cong \text{Specm}(k[Y])$ ,  $P$  corresponds to the maximal ideal  $\mathfrak{m}_P \leq k[Y]$  given by

$$\mathfrak{m}_P = \{f \in k[Y] \mid f(P) = 0\}.$$

#### Proof

As  $Y$  is affine, it is of the form  $Z_{\mathbb{A}^n}(I)$  for some ideal  $I \leq k[\mathbb{A}^n]$ ,  $\mathcal{I}(Y) = I$ . The isomorphism  $Z_{\mathbb{A}^n}(I) \cong \text{Specm}(k[Y])$  is given by mapping  $P = (a_1, \dots, a_n)$  to  $\mathfrak{m}_P = (x_1 - a_1, \dots, x_n - a_n)$ , quotiented by  $I$ . Clearly  $\mathfrak{m}_P \subseteq \{f \in k[Y] \mid f(P) = 0\}$ , and the right-hand side is a proper ideal so there is equality. ■

### 2.3.15 Corollary

Let  $(X, \mathcal{O}_X)$  be an algebraic variety,  $P \in X$ , and  $P \in U$  an affine neighborhood. Then  $\mathcal{O}_{X,P} \cong k[U]_{\mathfrak{m}_P}$  where  $\mathfrak{m}_P \leq k[U]$  is the ideal corresponding to  $P$  as in the above lemma.

#### Proof

As explained, taking the filtered colimit over a cofinal subsets doesn't affect the result, and so we can assume  $X = U$  is affine. Furthermore,  $\{D(f)\}_{f \in \mathcal{O}_X(X)}$  is a cofinal subset because it is a basis. Thus we can take the colimit over  $D(f)$  where  $f(P) \neq 0$ , i.e.

$$\mathcal{O}_{X,P} \cong \operatorname{colim}_{f \in k[X], f(P) \neq 0} \mathcal{O}_X(D(f)) \cong \operatorname{colim}_{f(P) \neq 0} k[X]_f.$$

For  $f(P) \neq 0$  we have  $f \notin \mathfrak{m}_P$  and so it is in the localizing set of  $k[X]_{\mathfrak{m}_P}$ . In particular, we have natural maps  $k[X]_f \rightarrow k[X]_{\mathfrak{m}_P}$  for all  $f$ . We now show that  $k[X]_{\mathfrak{m}_P}$  satisfies the universal property of the colimit.

Suppose that we have maps  $k[X]_f \rightarrow A$  for all  $f$  commuting with restrictions. In particular these each give maps  $k[X] \rightarrow A$  which map  $f$  to a unit. Since the maps  $k[X]_f \rightarrow A$  commute with restrictions, we see that this is a single map  $k[X] \rightarrow A$  mapping every  $f(P) \neq 0$ , so  $f \notin \mathfrak{m}_P$ , to a unit. By the universal property of localization this induces a map  $k[X]_{\mathfrak{m}_P} \rightarrow A$ . Thus  $k[X]_{\mathfrak{m}_P}$  is the colimit, as desired. ■

#### Note:

Notice that this shows that  $\mathcal{O}_{X,P}$  is a local ring for an algebraic variety  $X$ . This is true in general for an SWF: let  $p \in X$  arbitrary then let  $\mathfrak{m} \leq \mathcal{O}_{X,p}$  be the set of all germs  $(U, f)$  such that  $f(p) = 0$ . This is clearly an ideal of  $\mathcal{O}_{X,p}$ , and it is proper as it doesn't contain the unit. Now for  $(U, f) \notin \mathfrak{m}$  we know that  $p \in D_U(f)$  and thus  $(D_U(f), 1/f)$  is a germ in  $\mathcal{O}_{X,p}$  and a multiplicative inverse for  $(U, f)$ . So  $\mathcal{O}_{X,p} - \mathfrak{m} = \mathcal{O}_{X,p}^\times$  so  $(\mathcal{O}_{X,p}, \mathfrak{m})$  is a local ring.

This lends to a generalization of SWFs. As discussed in the definition of SWFs, the sheaf condition generalizes naturally to give a structure called a sheaf. These are presheaves valued in the category of rings (or  $R$ -algebras) satisfying the sheaf condition (a certain diagram is an equalizer diagram).

But this does not explain how to generalize the locality condition. Utilizing stalks, we can. Call a pair  $(X, \mathcal{O}_X)$  where  $X$  is a topological space and  $\mathcal{O}_X$  a sheaf valued in the category of rings (resp.  $\mathbf{Alg}_R$ ) a **ringed space** (resp.  **$R$ -ringed space**). This generalizes the sheaf condition.

Now a  $(R)$ -ringed space  $(X, \mathcal{O}_X)$  is called a **locally ringed space** if all of its stalks  $\mathcal{O}_{X,p}$  are local rings (resp. local  $R$ -algebras). This generalizes the locality condition.

## 2.4 Quasi-projective varieties

### 2.4.1 Definition

A **graded ring** is a ring  $S$  together with a decomposition  $S = \bigoplus_{i \geq 0} S_i$ , where each  $S_i \subseteq S$  is an Abelian group and  $S_i \cdot S_j \subseteq S_{i+j}$ . In particular  $S_0 \subseteq S$  is a subring, and  $S_i \subseteq S$  is an  $S_0$ -submodule, and the direct sum  $S_+ = \bigoplus_{i > 0} S_i$  is an ideal of  $S$ .

Elements of  $S_i$  are called **homogeneous of degree  $i$** , and an element is just **homogeneous** if it is homogeneous of some degree. Denote the set of homogeneous elements of  $S$  by  $S_{\text{hom}}$ .

An ideal  $I \leq S$  is **homogeneous** if it respects the decomposition:  $I = \bigoplus_{i \geq 0} (I \cap S_i)$ .

### 2.4.2 Example

Let  $S = k[x_1, \dots, x_n]$ , then  $S$  is graded by  $S_d = \operatorname{span}_k \{x_1^{i_1} \cdots x_n^{i_n} \mid i_1 + \cdots + i_n = d\}$ , called the space of homogeneous polynomials of degree  $d$ .

For  $S$  graded and  $a \in S$ , we can uniquely write  $a$  as  $a = \sum_d a_d$  for  $a_d \in S_d$ , all but finitely many being zero. The element  $a_d$  is called the **homogeneous part of  $a$  of degree  $d$** . Clearly  $(a + b)_d = a_d + b_d$  so the map  $S \rightarrow S_d$  mapping an element to its homogeneous part of degree  $d$  is an Abelian group morphism. Furthermore,

$$\left( \sum_d a_d \right) \left( \sum_d b_d \right) = \sum_d \sum_k a_k b_{d-k}, \quad \sum_k a_k b_{d-k} \in S_d,$$

so that  $(ab)_d = \sum_k a_k b_{d-k}$ .

We let  $\deg(a)$  denote the maximal  $d$  such that  $a_d \neq 0$ . So we see that if  $\deg(a) \leq n$  and  $\deg(b) \leq m$ , then  $\deg(ab) \leq n + m$  and  $(ab)_{n+m} = a_n b_m$ .

And clearly an ideal  $I$  is homogeneous iff for all  $a \in I$  we have  $a_d \in I$ .

### 2.4.3 Lemma

- (1) An ideal  $I \leq S$  is homogeneous iff it is generated by homogeneous elements, so  $I = (I \cap S_{\text{hom}})$ .
- (2) Homogeneous ideals are closed under sums, products, intersections, and radicals.
- (3) A non-unit homogeneous ideal  $I \leq S$  is prime iff for all  $a, b \in S_{\text{hom}}$  such that  $a, b \notin I$  we have  $ab \notin I$ .
- (4) A finitely generated homogeneous ideal is generated by finitely many homogeneous elements.

### Proof

- (1) If  $I \leq S$  is homogeneous, then

$$I = \sum_i (I \cap S_i) = \left( \bigcup_i I \cap S_i \right) = (I \cap S_{\text{hom}}).$$

Conversely suppose  $(b_i)$  generate  $I$ , where  $b_i \in S_{d_i}$ . Then elements of  $I$  are of the form  $a = \sum_i a_i b_i$  with  $a_i \in S$ . Then

$$a_d = \sum_i (a_i b_i)_d = \sum_i (a_i)_{d-d_i} b_i \in I,$$

so  $I$  is homogeneous.

- (2) This is clear from (1) for sums and products. And if  $a \in I = \bigcap_i I_i$  then  $a_d \in I_i$  for all  $i$  so  $a_d \in I$  and thus  $I$  is homogeneous.

Now if  $I$  is homogeneous, we induct on  $d$  to show that for every  $f \in \sqrt{I}$  such that  $f = \sum_{i=0}^d f_i$ ,  $f_i \in \sqrt{I}$ . For  $d = 0$  we have  $f = f_0$  so this is clear. Now inductively, it is sufficient to show that  $f_d \in \sqrt{I}$  since then  $f - f_d = \sum_{i=0}^{d-1} f_i \in \sqrt{I}$  and we proceed by induction.

By definition there is some  $n$  such that  $f^n \in I$ . Now, we know  $\deg(f) \leq d$  and thus inductively  $\deg(f^n) \leq dn$  so  $(f^n)_{dn} = f_d^n$ . Since  $f^n \in I$  we have  $f_d^n = (f^n)_{dn} \in I$  and thus  $f_d \in \sqrt{I}$  as desired.

- (3) One direction is clear. Now suppose we have that  $ab \notin I$  when  $a, b \in S_{\text{hom}} - I$ . For  $a, b \in S - I$  we have  $a = \sum_i a_i$  and  $b = \sum_j b_j$ . Let us write  $a_I = \sum_{i, a_i \in I} a_i$  and  $a^I = \sum_{i, a_i \notin I} a_i$  so  $a = a_I + a^I$  and similarly  $b = b_I + b^I$ . So  $a_I, b_I \in I$  and

$$ab = a_I b_I + a_I b^I + a^I b_I + a^I b^I,$$

so it is sufficient to show that  $a^I b^I \notin I$ .

That is, it is sufficient to consider the case when  $a_i \notin I$  and  $b_i \notin I$  to show  $ab \notin I$ . Let  $n = \deg(a)$  and  $m = \deg(b)$  so  $(ab)_{n+m} = a_n b_m$ , and since  $a_n, b_m \in S_{\text{hom}} - I$  we have  $a_n b_m \notin I$ , and by homogeneity  $ab \notin I$ .

- (4) Let  $T \subseteq I$  be finite and generate  $I$ . Then let

$$T_{\text{hom}} = \{a_i \mid a \in T, i \geq 0\}.$$

Since  $I$  is homogeneous,  $T_{\text{hom}} \subseteq I$ . And since elements of  $T$  are finite sums of elements from  $T_{\text{hom}}$ , we have  $I = \langle T \rangle \subseteq \langle T_{\text{hom}} \rangle \subseteq I$  so  $\langle T_{\text{hom}} \rangle = I$ . Clearly  $T_{\text{hom}}$  is a finite subset of  $S_{\text{hom}}$ . ■

For an algebraically closed field  $k$ , the group  $k^\times$  acts on  $\mathbb{A}^n = k^n$  via multiplication: for  $P = (a_1, \dots, a_n) \in k^n$  and  $\lambda \in k^\times$  we have the action  $\lambda P = (\lambda a_1, \dots, \lambda a_n)$ .

For the graded ring  $S = k[\mathbb{A}^n]$  with grading  $S_d = \text{span}_k \{x^I \mid \sum I = d\}$ , we see that for  $f \in S_d$  we have  $f(\lambda P) = \lambda^d f(P)$ . This is clear for monomials  $x^I$ , and follows in general by linearity.

#### 2.4.4 Lemma

A closed subset  $Z \subseteq \mathbb{A}^n$  is invariant under  $k^\times$ 's action iff  $\mathcal{I}(Z) \subseteq S$  is homogeneous. For a general subset  $X \subseteq \mathbb{A}^n$ , if  $X$  is  $k^\times$ -invariant then  $\mathcal{I}(X) \subseteq S$  is homogeneous (only one direction of the previous equivalence holds).

#### Proof

Suppose  $\mathcal{I}(Z) \subseteq S$  is homogeneous, and let  $P \in Z$  and  $\lambda \in k^\times$ . Since  $Z$  is closed,  $Z = Z(\mathcal{I}(Z))$  it is sufficient to show that for  $f \in \mathcal{I}(Z)$  we have  $f(\lambda P) = 0$ . If  $f$  is homogeneous, this is clear as  $f(\lambda P) = \lambda^d f(P) = 0$ . Then in general we have  $f = \sum_d f_d$  and  $f_d \in \mathcal{I}(Z)$  so  $f_d(\lambda P) = 0$  and thus  $f(\lambda P) = 0$ .

Now if  $X$  is  $k^\times$ -invariant, we want to show that if  $f \in \mathcal{I}(X)$  then  $f_d \in \mathcal{I}(X)$ . Fix  $P \in X$ , so we want to show  $f_d(P) = 0$  for all  $d$ . For every  $\lambda \in k^\times$ , by invariance we have  $\lambda P \in X$  so  $f(\lambda P) = 0$ . Thus

$$0 = f(\lambda P) = \sum_d f_d(\lambda P) = \sum_d \lambda^d f_d(P).$$

That is, the polynomial  $Q(t) = \sum_d f_d(P)t^d$  satisfies  $Q(\lambda) = 0$  for all  $\lambda \in k^\times$ . So  $Q$  has infinitely many roots, and thus must be the zero polynomial, so  $f_d(P) = 0$  for all  $d$ , as desired. ■

Notice that since  $\overline{X} = Z(\mathcal{I}(X))$  and thus  $\mathcal{I}(\overline{X}) = \mathcal{I}(X)$ , we see that if  $X$  is  $k^\times$ -invariant so too is  $\overline{X}$ .

We view  $S = k[\mathbb{A}^{n+1}]$  as the graded  $k$ -algebra  $k[x_0, \dots, x_n]$ , so we begin indexing at 0 here. Since  $\mathbb{A}^{n+1} - 0$  is  $k^\times$ -invariant, we can quotient by  $k^\times$  to get  $\mathbb{P}^n = \mathbb{P}^n$ . That is, we define  $\mathbb{P}^n$  to be the set of all  $k^\times$ -orbits of  $\mathbb{A}^{n+1} - 0$ . We denote the orbit of  $(a_0, \dots, a_n) \in \mathbb{A}^{n+1} - 0$  by  $[a_0 : \dots : a_n] \in \mathbb{P}^n$ . Note that equivalently,  $\mathbb{P}^n$  can be viewed as the set of all lines in  $\mathbb{A}^{n+1}$  passing through the origin, and then  $[a_0 : \dots : a_n]$  corresponds to the line  $t(a_0, \dots, a_n)$ .

Let  $\pi: \mathbb{A}^{n+1} - 0 \rightarrow \mathbb{P}^n$  be the canonical projection map:  $\pi(a_0, \dots, a_n) = [a_0 : \dots : a_n]$ . This is surjective, and thus we can endow  $\mathbb{P}^n$  with the quotient topology. In particular this makes  $\mathbb{P}^n$  into an SWF  $(\mathbb{P}^n, \mathcal{O}_{\text{reg}})$  where

$$\mathcal{O}_{\text{reg}}(U) = \{f: U \rightarrow k \mid f \circ \pi \in \mathcal{O}_{\text{reg}}(\pi^{-1}U)\}.$$

That is,  $f: U \rightarrow k$  is regular iff the pullback  $\pi^*(f): \pi^{-1}U \rightarrow k$  is regular.

#### 2.4.5 Definition

For a homogeneous polynomial  $f \in S_{\text{hom}}$  and  $P \in \mathbb{P}^n$  we write  $f(P) = 0$  to mean that  $f(\tilde{P}) = 0$  for some representative (equivalently every representative)  $\tilde{P} \in \pi^{-1}(P)$ . For  $f \in S_{\text{hom}}$  let

$$Z_{\mathbb{P}^n}(f) = \{P \in \mathbb{P}^n \mid f(P) = 0\},$$

and in general for  $T \subseteq S_{\text{hom}}$  let

$$Z_{\mathbb{P}^n}(T) = \bigcap_{f \in T} Z_{\mathbb{P}^n}(f).$$

For an homogeneous ideal  $I \leq S$  we define  $Z_{\mathbb{P}^n}(I) = Z_{\mathbb{P}^n}(I \cap S_{\text{hom}})$ . If  $T$  is any of the above (an homogeneous polynomial/set/ideal) then we define  $D_{\mathbb{P}^n}(T) = \mathbb{P}^n - Z_{\mathbb{P}^n}(T)$ .

Finally for any  $Y \subseteq \mathbb{P}^n$  we define the homogeneous ideal  $\mathcal{I}_{\text{hom}}(Y) \leq S$  by

$$\mathcal{I}_{\text{hom}}(Y) = \langle f \in S_{\text{hom}} \mid f(P) = 0 \text{ for all } P \in Y \rangle.$$

As it is generated by homogeneous elements, it is an homogeneous ideal.



### 2.4.6 Lemma

- (1)  $Z_{\mathbb{P}^n}$  is order reversing,
- (2)  $Z_{\mathbb{P}^n}(0) = \mathbb{P}^n$  and  $Z_{\mathbb{P}^n}(1) = \emptyset$ ,
- (3)  $Z_{\mathbb{P}^n}(T_1 \cdot T_2) = Z_{\mathbb{P}^n}(T_1) \cup Z_{\mathbb{P}^n}(T_2)$  and  $Z_{\mathbb{P}^n}(\bigcup_i T_i) = \bigcap_i Z_{\mathbb{P}^n}(T_i)$ ,
- (4) For every  $T \subseteq S_{\text{hom}}$ , there is an equality  $Z_{\mathbb{P}^n}(T) = Z_{\mathbb{P}^n}((T))$ .

### Proof

Points (1) and (2) are clear. Point (3) is shown in essentially the same way as in the affine case. For point (4) we have

$$Z_{\mathbb{P}^n}((T)) = Z_{\mathbb{P}^n}((T) \cap S_{\text{hom}}),$$

and  $T \subseteq (T) \cap S_{\text{hom}}$  so we have an inclusion  $Z_{\mathbb{P}^n}((T)) \subseteq Z_{\mathbb{P}^n}(T)$ . Conversely let  $P \in Z_{\mathbb{P}^n}(T)$  and  $f \in (T) \cap S_{\text{hom}}$  then  $f = \sum_i a_i t_i$  for  $a_i \in S$  and  $t_i \in T$ . Suppose  $\deg(f) = d$  then  $\deg(a_i t_i) = d$  for all  $i$  which means that each  $a_i$  must be homogeneous as well and  $\deg(a_i) = d - \deg(t_i)$ . Since everything is homogeneous, since  $t_i(P) = 0$  for all  $i$  we have  $f(P) = 0$  so we have the other inclusion as well. ■

### 2.4.7 Proposition

- (1) For  $T \subseteq S_{\text{hom}}$  we have the equality

$$\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n+1}-0}(T).$$

- (2)  $Z \subseteq \mathbb{P}^n$  is closed iff  $Z = Z_{\mathbb{P}^n}(T)$  for some  $T \subseteq S_{\text{hom}}$ .
- (3) The collection  $\{D_{\mathbb{P}^n}(f)\}_{f \in S_{\text{hom}}}$  forms a basis of open sets of  $\mathbb{P}^n$ .

### Proof

We know that for  $f \in S_{\text{hom}}$  and  $\tilde{P} \in \mathbb{A}^{n+1}-0$ ,  $f(\tilde{P}) = 0$  iff  $f(\pi(\tilde{P})) = 0$ . Thus  $\pi(\tilde{P}) \in Z_{\mathbb{P}^n}(T)$  iff  $f(\pi(\tilde{P})) = 0$  for all  $f \in T$  iff  $f(\tilde{P}) = 0$  for all  $f \in T$  iff  $\tilde{P} \in Z_{\mathbb{A}^{n+1}-0}(T)$  as desired.

The preimage  $\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n+1}-0}(T)$  is closed in  $\mathbb{A}^{n+1}-0$ . Since  $\mathbb{P}^n$  has the quotient topology, this means  $Z_{\mathbb{P}^n}(T)$  is closed in  $\mathbb{P}^n$ . Conversely if  $Z \subseteq \mathbb{P}^n$  is closed, then  $\pi^{-1} Z \subseteq \mathbb{A}^{n+1}-0$  is closed and  $k^\times$ -invariant. Since  $\pi^{-1} \overline{Z} = Z_{\mathbb{A}^{n+1}}(\mathcal{I}(\pi^{-1} Z))$ , we see that  $I = \mathcal{I}(\pi^{-1} Z)$  is homogeneous. So  $I = (I \cap S_{\text{hom}})$  and thus  $\pi^{-1} Z = Z_{\mathbb{A}^n}(I) = Z_{\mathbb{A}^n}(I \cap S_{\text{hom}})$  so we have equality

$$\pi^{-1} Z = \pi^{-1} \overline{Z} \cap (\mathbb{A}^{n+1}-0) = Z_{\mathbb{A}^{n+1}-0}(I \cap S_{\text{hom}}) = \pi^{-1} Z_{\mathbb{P}^n}(I \cap S_{\text{hom}}).$$

Thus we see  $Z = Z_{\mathbb{P}^n}(I \cap S_{\text{hom}})$  as desired. The reverse direction is clear since  $\pi^{-1} Z_{\mathbb{P}^n}(T) = Z_{\mathbb{A}^{n+1}-0}(T)$  is closed.

Every closed set in  $\mathbb{P}^n$  is of the form  $Z_{\mathbb{P}^n}(T) = \bigcap_{f \in T} Z_{\mathbb{P}^n}(f)$  so all open sets are unions of  $D_{\mathbb{P}^n}(f)$ . ■

For homogeneous polynomials of the same degree  $g, h \in S_d$  and  $\tilde{P} \in D_{\mathbb{A}^{n+1}}(h)$  and  $\lambda \in k^\times$  we see

$$\frac{g}{h}(\lambda \tilde{P}) = \frac{g(\lambda \tilde{P})}{h(\lambda \tilde{P})} = \frac{\lambda^d g(\tilde{P})}{\lambda^d h(\tilde{P})} = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

So homogeneous rational functions  $g/h$  define maps  $D_{\mathbb{P}^n}(h) \rightarrow k$ .

This allows us to better describe the SWF  $(\mathbb{P}^n, \mathcal{O}_{\text{reg}})$ .

**2.4.8 Lemma**

For  $U \subseteq \mathbb{P}^n$  open, a function  $f: U \rightarrow k$  is regular iff there are homogeneous polynomials  $g, h \in S_d$  such that  $U \subseteq D_{\mathbb{P}^n}(h)$  and  $f = g/h|_U$ .

**Proof**

We know that  $f$  is regular iff  $\pi^*(f): \pi^{-1}U \rightarrow k$  is regular. This is iff it is rational, so there are polynomials  $g, h \in S$  such that  $\pi^{-1}U \subseteq D(h)$  and for every  $\tilde{P} \in \pi^{-1}U$  we have

$$\pi^*(f)(\tilde{P}) = f(\pi(\tilde{P})) = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

In particular, for  $g, h \in S_d$  the function  $f = g/h$  is regular.

Conversely we want to show that two polynomials satisfying the above equality must be homogeneous of the same degree. First we claim that  $\pi^{-1}U \subseteq D(h)$  implies  $h$  homogeneity. Suppose not, so there are  $i_1 \neq i_2$  such that  $h_{i_1}, h_{i_2}$  are nonzero. We have open non-empty subsets  $D(h_{i_1}), D(h_{i_2}), \pi^{-1}U \subseteq \mathbb{A}^{n+1}$ , and since  $\mathbb{A}^{n+1}$  is irreducible this means that their intersection is nonempty. So there is  $\tilde{P} \in \pi^{-1}U$  such that  $h_{i_1}(\tilde{P}), h_{i_2}(\tilde{P}) \neq 0$ .

Then for every  $\lambda \in k^\times$  we have  $\lambda\tilde{P} \in \pi^{-1}U \subseteq D(h)$  so  $h(\lambda\tilde{P}) \neq 0$ . Hence

$$0 \neq h(\lambda\tilde{P}) = \sum_i h_i(\lambda\tilde{P}) = \sum_i \lambda^i h_i(\tilde{P}),$$

then as before this means  $Q(t) = \sum_i h_i(\tilde{P})t^i$  has no zeroes in  $k^\times$ . Since  $k$  is algebraically closed, this means its only zero is 0 and thus  $Q(t) = at^d$  for some  $a \in k$ . This means that  $h_i(\tilde{P}) = 0$  for all  $i \neq d$ , so there is at most a single index where  $h_i(\tilde{P}) \neq 0$ , a contradiction.

Now we claim that  $g$  is homogeneous. For every  $\tilde{P} \in \pi^{-1}U$  and  $\lambda \in k^\times$  we have

$$\frac{g(\lambda\tilde{P})}{\lambda^d h(\tilde{P})} = \frac{g(\lambda\tilde{P})}{h(\lambda\tilde{P})} = f(\pi(\lambda\tilde{P})) = f(\pi(\tilde{P})) = \frac{g(\tilde{P})}{h(\tilde{P})}.$$

Thus we see  $g(\lambda\tilde{P}) = \lambda^d g(\tilde{P})$ . Thus for all  $\lambda \in k^\times$ ,

$$\sum_i \lambda^i g_i(\tilde{P}) = \lambda^d \sum_i g_i(\tilde{P}).$$

Defining  $Q(t) = t^d \sum_i g_i(\tilde{P}) - \sum_i t^i g_i(\tilde{P})$  we see that this has infinitely many zeroes and is thus identically zero, so  $g_i(\tilde{P}) = 0$  for all  $i \neq d$ . So  $g(\tilde{P}) = g_d(\tilde{P})$  for all  $\tilde{P} \in \pi^{-1}U$ . Since  $\mathbb{A}^{n+1}$  is Noetherian,  $\pi^{-1}U$  is dense and so  $g = g_d$  on all  $\mathbb{A}^{n+1}$  as desired.  $\blacksquare$

**Note:**

The above results, namely that  $Z \subseteq \mathbb{P}^n$  is closed iff it is of the form  $Z_{\mathbb{P}^n}(T)$  and  $f: U \rightarrow k$  is regular iff it is homogeneous rational, are many times taken to be definitions of projective space. We had to work a little harder to show that the more obvious definition, induced from affine space, agrees with this.

**2.4.9 Corollary**

Let  $Y \subseteq \mathbb{P}^n$  be an arbitrary subspace, so  $(\mathbb{P}^n, \mathcal{O}_{\text{reg}})$  induces an SWF  $(Y, \mathcal{O}_{\text{reg}})$ .

- (1) The closed sets of  $Y$  are those of the form  $Z_Y(T) = \{P \in Y \mid f(P) = 0 \text{ for all } f \in T\}$ .
- (2) The collection  $D_Y(f) = \{P \in Y \mid f(P) \neq 0\}$  forms a basis of open sets of  $Y$ .

- (3) A function  $f: Y \rightarrow k$  is regular iff it is locally given by homogeneous rational functions. That is, for every  $P \in Y$  there is an open subset  $U \subseteq \mathbb{P}^n$  such that  $f|_U$  is homogeneous rational.

### Proof

Points (1) and (2) are clear from the definition of the subspace topology, the obvious equalities

$$Z_Y(T) = Z_{\mathbb{P}^n}(T) \cap Y, \quad D_Y(T) = D_{\mathbb{P}^n}(T) \cap Y,$$

and the results we proved above. Point (3) is immediate from the definition of induced SWFs and the previous result. ■

#### 2.4.10 Lemma

The SWF  $(\mathbb{P}^n, \mathcal{O}_{reg})$  is an algebraic variety.

### Proof

Cover  $\mathbb{A}^{n+1} - 0$  by  $\{D(x_i)\}_{i=0}^n$ , so through  $\pi$  we cover  $\mathbb{P}^n$  by  $\{D_{\mathbb{P}^n}(x_i)\}_{i=0}^n$ . And clearly we have

$$U_i = D_{\mathbb{P}^n}(x_i) = \{[a_0 : \dots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}.$$

We claim that these  $U_i$  are affine.

Define  $\varphi: U_i \rightarrow \mathbb{A}^n$  by

$$\varphi[a_0 : \dots : a_n] = \left( \frac{a_0}{a_i}, \dots, \frac{\hat{a}_i}{a_i}, \dots, \frac{a_n}{a_i} \right).$$

Each map  $[a_0 : \dots : a_n] \mapsto a_j/a_i$  is regular, as it is homogeneous rational (given by  $x_j/x_i$ ), so this is a morphism into  $\mathbb{A}^n$ .  $\varphi$  is bijective with inverse

$$\psi(a_1, \dots, a_n) = [a_1 : \dots : a_i : 1 : a_{i+1} : \dots : a_n].$$

To show that  $\psi$  is a morphism of SWFs, we note that it is equal to the composition

$$\mathbb{A}^n \xrightarrow{\tilde{\psi}} \mathbb{A}^{n+1} \xrightarrow{\pi} \mathbb{P}^n,$$

where  $\tilde{\psi}$  maps  $(a_1, \dots, a_n)$  to  $(a_1, \dots, a_i, 1, a_{i+1}, \dots, a_n)$ , and this is clearly a morphism of SWFs as its components are regular (polynomial even). Thus  $\psi$  is the composition of morphisms of SWFs and thus a morphism itself. ■

#### 2.4.11 Definition

A **projective variety** is an SWF  $X$  which is isomorphic to an SWF  $(Y, \mathcal{O}_{reg})$  where  $Y \subseteq \mathbb{P}^n$  is closed.

A **quasi-projective variety** is an SWF  $X$  which is isomorphic to an open subspace of an open subvariety of a projective variety.

Clearly projective varieties are quasi-projective. Notice that quasi-projective varieties are just SWFs which are isomorphic to locally closed subvarieties of  $\mathbb{P}^n$ .

**2.4.12 Corollary**

- (1) Quasi-projective varieties are algebraic varieties.
- (2) Every locally closed subvariety of a quasi-projective variety is quasi-projective.
- (3) Every quasi-affine variety is quasi-projective.

**Proof**

The first point follows from the fact that locally closed subvarieties of an algebraic variety are algebraic. A locally closed subvariety of a quasi-projective variety is isomorphic to a locally closed subvariety of a locally closed subvariety of  $\mathbb{P}^n$ . Local closure is a transitive property, so this is just a locally closed subvariety of  $\mathbb{P}^n$  and thus quasi-projective, showing (2).

Since  $\mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ , we see that affine space  $\mathbb{A}^n$  is projective. Quasi-affine varieties are locally closed subvarieties of  $\mathbb{A}^n$ , and thus locally closed subvarieties of a projective variety, and thus quasi-affine by (2). ■

**2.4.13 Proposition**

The global sections of projective space are just scalars:  $\mathcal{O}_{\text{reg}}(\mathbb{P}^n) \cong k$ .

**Proof**

We have an affine cover of  $\mathbb{P}^n$  by  $U_i$  above. There is an isomorphism  $U_i \cong \mathbb{A}^n$  and through this isomorphism we see that we identify  $\Gamma(U_i)$  with  $k[x_1, \dots, \hat{x}_i, \dots, x_n]$ , which we denote by  $k[\bar{x}_{-i}]$ . Then  $U_i \cap U_j = D_{\mathbb{P}^n}(x_i x_j)$  is identified with  $D(x_j) \subseteq \mathbb{A}^n$ , and thus its global sections are  $\Gamma(U_i \cap U_j) = k[\bar{x}_{-i}]_{x_j}$ .

As discussed previously,  $\mathbb{P}^n$  is then isomorphic to the gluing of  $U_i$  along their intersections. Since  $\Gamma$  contravariantly preserves colimits, we have an equalizer diagram

$$\Gamma(\mathbb{P}^n) \rightarrow \prod_i \Gamma(U_i) \begin{matrix} \xrightarrow{\varphi} \\ \xrightarrow{\psi} \end{matrix} \prod_{i \neq j} \Gamma(U_i \cap U_j),$$

where  $\varphi$  maps  $(s_i)_i$  to  $(f_{ij}^*(s_i))_{i,j}$  and  $\psi$  maps it to  $(f_{ji}^*(s_j))_{i,j}$ . So the equalizer is the collection of  $(s_i)_i$  such that  $f_{ij}^*(s_i) = f_{ji}^*(s_j)$  for all  $i, j$ .

Since  $f_{ij}: U_i \cap U_j \rightarrow U_i$  is just the inclusion  $f_{ij}^*$  is just the restriction to  $U_i \cap U_j$ . So we need  $s_i = s_j$  on  $U_i \cap U_j$  for all  $i, j$ . But  $s_i \in \Gamma(U_i)$  is a polynomial in all variables other than  $x_i$ , but it is equal to  $s_j$  a polynomial function not in  $x_j$ . Thus  $s_i$  is constant in  $x_j$  as well, but this is true for all  $j$  so  $s_i$  is constant.

So the equalizer is just a tuple of constants which agree on  $U_i \cap U_j$ , so tuples of the form  $(\lambda, \dots, \lambda)$ . Clearly this is just isomorphic to  $k$ .

Note that here we used the fact  $U_i \cap U_j$  is nonempty and thus dense to justify the fact that  $s_i$  being constant in  $x_j$  in  $U_i \cap U_j$  implies it is constant in  $x_j$  on the entire space. ■

**2.4.14 Corollary**

$\mathbb{P}^n$  is not affine.

**Proof**

For affine  $Y \subseteq \mathbb{A}^n$  we have that  $\mathcal{O}_{\text{reg}}(Y) = k[Y] = k[\mathbb{A}^n]/\mathcal{I}(Y)$ . This is isomorphic to  $k$  iff  $\mathcal{I}(Y)$  is maximal,

so of the form  $\mathcal{I}(Y) = (x_1 - a_1, \dots, x_n - a_n)$ . But then  $Y = Z(\mathcal{I}(Y))$  is just the singleton  $\{(a_1, \dots, a_n)\}$ , and  $\mathbb{P}^n$  is not a singleton. ■

## 3 Dimension Theory

### 3.1 Affine and finite morphisms

**Note:**

For any algebraic variety  $X$ , we will usually write  $k[X]$  for  $\mathcal{O}_X(X)$ .

#### 3.1.1 Definition

A morphism  $f: X \rightarrow Y$  of algebraic varieties is

- (1) **affine** if there is an open affine covering  $Y = \bigcup_i U_i$  such that  $f^{-1}(U_i) \subseteq X$  is affine for all  $i$ ,
- (2) **finite** if there is an open affine covering  $Y = \bigcup_i U_i$  such that  $f^{-1}(U_i) \subseteq X$  is affine, and  $k[f^{-1}U_i]$  is a finite  $k[U_i]$ -algebra for all  $i$ .

Note that  $f: X \rightarrow Y$  induces a morphism  $f^*: k[U] \rightarrow k[f^{-1}U]$  for all open  $U \subseteq Y$ . So in the definition of a finite morphism, we mean that  $f^*$  is finite for each  $i$ .

#### 3.1.2 Lemma

$f: X \rightarrow Y$  is affine iff  $f^{-1}U \subseteq X$  is affine for every open affine  $U \subseteq Y$ .

Similarly  $f$  is finite iff  $f^{-1}U \subseteq X$  is affine and  $k[f^{-1}U]$  is a finite  $k[U]$ -algebra for every open affine  $U \subseteq Y$ .

From this lemma it is clear that the composition of affine/finite morphisms is affine/finite.

#### 3.1.3 Definition

A property  $P$  of morphisms is **local on the target** if for any morphism  $f: X \rightarrow Y$  and an open cover  $\{U_i\}$  of  $Y$ ,  $f$  satisfies  $P$  iff restricting  $f$  to  $f^{-1}U_i \rightarrow U_i$  satisfies  $P$ .

We then see that being affine/finite is local on the target.

We state a few properties of morphisms of affine varieties which are proven in homework (they can be proven directly or by translating to the case of affine schemes).

#### 3.1.4 Proposition

Let  $f: X \rightarrow Y$  be a morphism of affine varieties, with induced morphism  $f^*: k[Y] \rightarrow k[X]$ .

- (1) A closed subset  $Z \subseteq X$  is irreducible iff  $\mathcal{I}(Z) \subseteq k[X]$  is prime, and it is a point iff  $\mathcal{I}(Z)$  is a point.
- (2) For every  $T \subseteq X$ ,  $(f^*)^{-1}(\mathcal{I}_X(T)) = \mathcal{I}_Y(f(T))$ .
- (3) For every closed  $S \subseteq Y$ ,  $\mathcal{I}_X(f^{-1}S) = \langle f^*\mathcal{I}_Y(S) \rangle$ .
- (4)  $f^*$  is injective iff  $f$  is **dominant**, i.e.  $f(X) \subseteq Y$  is dense.

- (5)  $f^*$  is surjective iff  $f$  is a closed embedding, i.e. its image is closed and it forms an isomorphism  $X \cong f(X)$ .

### 3.1.5 Corollary

Let  $f: X \rightarrow Y$  be a morphism of affine varieties which induces  $f^*: k[Y] \rightarrow k[X]$ .

- (1) If a prime ideal  $\mathfrak{p} \leq k[X]$  corresponds to the closed irreducible subset  $Z \subseteq X$ , then  $(f^*)^{-1}\mathfrak{p} \subseteq k[Y]$  corresponds to the closure  $\overline{f(Z)} \subseteq Y$ .
- (2) If a closed subset  $Z \subseteq Y$  corresponds to the radical ideal  $I \leq k[Y]$ , then the irreducible components  $Z_i$  of  $f^{-1}Z$  correspond to minimal prime ideals  $\mathfrak{p}_i \leq k[X]$  such that  $(f^*)^{-1}\mathfrak{p}_i \supseteq I$ .

### Proof

- (1) From the above proposition,

$$(f^*)^{-1}\mathcal{I}_X(Z) = \mathcal{I}_Y(f(Z)) = \mathcal{I}_Y(\overline{f(Z)}),$$

so it corresponds to  $\overline{f(Z)}$ .

- (2)  $Z_i$  is the maximal closed irreducible subset of  $X$  such that  $f(Z_i) \subseteq Z$ , equivalently  $\overline{f(Z_i)} \subseteq Z$ . From (1) this corresponds to  $(f^*)^{-1}\mathfrak{p}_i$  for prime  $\mathfrak{p}_i$  containing  $\mathcal{I}_Y(Z) = I$ . ■

### 3.1.6 Corollary

- (1) For an affine variety  $X$ , the correspondence  $x \mapsto \mathcal{I}_X(x)$  is a bijection between points of  $X$  and maximal ideals of  $k[X]$ .
- (2) For a morphism  $f: X \rightarrow Y$  inducing  $f^*: k[Y] \rightarrow k[X]$ , then for every maximal  $\mathfrak{m} \leq k[X]$ ,  $(f^*)^{-1}\mathfrak{m} \leq k[Y]$  is maximal. Furthermore, under the correspondence of (1), the map  $x \mapsto f(x)$  corresponds to  $\mathfrak{m} \mapsto (f^*)^{-1}\mathfrak{m}$ .

### Proof

Immediate. ■

### 3.1.7 Lemma

For algebraic varieties, the following hold.

- (1) If  $\iota: Z \hookrightarrow X$  is a closed embedding, it is finite.
- (2) For every  $f \in k[X]$ , the inclusion  $j: D_X(f) \hookrightarrow X$  is affine.

### Proof

For (1) we may assume  $Z \subseteq X$  is a closed subset. Then for every affine  $U \subseteq X$ ,  $\iota^{-1}U = U \cap Z \subseteq U$  is a closed subspace of an affine variety, thus it itself is affine. So  $\iota: \iota^{-1}U \rightarrow U$  is a closed embedding and thus  $\iota^*: k[U] \rightarrow k[\iota^{-1}U]$  is surjective, and in particular finite (since a surjective map of rings is finite).

For (2), for  $U \subseteq X$  affine,  $j^{-1}U = D_U(f)$  is affine. ■

### 3.1.8 Lemma

Let  $f: X \rightarrow Y$  be a finite morphism, then for every closed  $Z \subseteq X$  and every closed  $f(Z) \subseteq W$ ,  $f|_Z: Z \rightarrow W$  is finite.

### Proof

If  $X, Y$  are affine, then  $k[X]$  is a finite  $k[Y]$ -algebra. For a closed  $Z \subseteq X$ , by the above lemma  $k[Z]$  is a finite  $k[X]$ -algebra and thus a finite  $k[Y]$ -algebra. Since  $Z \rightarrow Y$  factors as  $Z \rightarrow W \rightarrow Y$ ,  $k[Y] \rightarrow k[Z]$  factors as  $k[Y] \rightarrow k[W] \rightarrow k[Z]$ , and since  $k[Y] \rightarrow k[W]$  is finite as well, we see that  $k[W] \rightarrow k[Z]$  must be finite.

In general, let  $Y = \bigcup_i U_i$  be an affine covering such that  $f^{-1}U_i \subseteq X$  is affine and  $k[f^{-1}U_i]$  is a finite  $k[U_i]$ -algebra. Then  $W = \bigcup_i W \cap U_i$  is an open affine covering of  $W$ , and  $f|_Z^{-1}(U_i \cap W) = f^{-1}(U_i) \cap Z$  is closed inside  $f^{-1}U_i$ , thus affine. So from the previous paragraph,  $k[f|_Z^{-1}(U_i \cap W)]$  is a finite  $k[U_i \cap W]$ -algebra, showing that  $f|_Z$  is finite. ■

We recall the following results, proven in our section on integral and finite algebras.

### 3.1.9 Lemma

Let  $\varphi: A \rightarrow B$  be an integral (in particular finite) morphism of rings.

- (1) A prime ideal  $\mathfrak{q} \leq B$  is maximal iff  $\varphi^{-1}\mathfrak{q} \leq A$  is maximal.
- (2) If  $\mathfrak{q}_1 \subsetneq \mathfrak{q}_2 \leq B$  are prime, then  $\varphi^{-1}\mathfrak{q}_1 \subsetneq \varphi^{-1}\mathfrak{q}_2$ .
- (3) If  $\varphi$  is injective, then  $\varphi^*: \text{Spec}(B) \rightarrow \text{Spec}(A)$  is surjective: for every prime  $\mathfrak{p} \leq A$  there is a prime  $\mathfrak{q} \leq B$  such that  $\varphi^{-1}\mathfrak{q} = \mathfrak{p}$ .

### 3.1.10 Proposition

Let  $f: X \rightarrow Y$  be a finite morphism, then

- (1)  $f$  is a closed map, meaning it maps closed subsets of  $X$  to closed subsets of  $Y$ .
- (2) if  $Z_1 \subsetneq Z_2 \subseteq X$  are closed irreducible, then  $f(Z_1) \subsetneq f(Z_2)$ .

### Proof

First we show (1) and (2) in the case that  $X, Y$  are affine.

The induced morphism  $f|_Z: Z \rightarrow \overline{f(Z)}$  is finite as well, from an above proposition, and dominant by definition. So replacing  $f$  with  $f|_Z$ , it is sufficient to show that dominant finite morphisms are surjective.

The map  $x \mapsto f(x)$  corresponds to  $\mathfrak{m} \mapsto (f^*)^{-1}\mathfrak{m}$ , so we show the second is surjective. Let  $\mathfrak{m} \leq k[Y]$  be maximal, then since  $f$  is dominant  $f^*$  is injective and finite, and by the above lemma this means there is a prime  $\mathfrak{p} \leq k[X]$  such that  $(f^*)^{-1}\mathfrak{p} = \mathfrak{m}$ . And by the same lemma, this means  $\mathfrak{p}$  is maximal, so  $(f^*)^{-1}$  is indeed surjective.

Since  $Z_1 \subsetneq Z_2$  are closed irreducible,  $\mathcal{I}_X(Z_1) \supsetneq \mathcal{I}_X(Z_2)$  are prime (the inequality is due to  $\mathcal{I}_X$  being a

bijection on closed subsets). Since  $k[X]$  is a finite  $k[Y]$ -algebra, by the above lemma

$$\mathcal{I}_Y(f(Z_1)) = (f^*)^{-1}\mathcal{I}_X(Z_1) \supsetneq (f^*)^{-1}\mathcal{I}_X(Z_2) = \mathcal{I}_Y(f(Z_2)),$$

and thus  $f(Z_1) \subsetneq f(Z_2)$ .

Now in general take  $Y = \bigcup_i U_i$  an affine covering such that  $f^{-1}U_i \subseteq X$  is affine and  $k[f^{-1}U_i]$  is a finite  $k[U_i]$ -algebra. Since  $Z \subseteq X$  is closed,  $Z \cap f^{-1}U_i \subseteq f^{-1}U_i$  is closed. Then as in the affine case,  $f(Z \cap f^{-1}U_i) \subseteq f(U_i)$  is closed. Since  $f(Z \cap f^{-1}U_i) = f(Z) \cap U_i$ , we see that  $f(Z)$  is closed as desired.

This is because being closed is “local”: if  $Z \cap U_i$  is closed for all  $i$  in an open covering then  $Z$  is closed. And this is because if we let  $W = Y - Z$ , we see  $W \cap U_i$  is open for all  $i$  and thus  $W$  is open and so  $Z$  is closed.

For (2), let  $i$  be such that  $f(Z_2) \cap U_i \neq \emptyset$ . Then  $Z_1 \cap f^{-1}U_i \subseteq Z_2 \cap f^{-1}U_i$  are closed in  $f^{-1}U_i$ . And  $Z_2 \cap f^{-1}U_i \subseteq Z_2$  is open and nonempty, thus dense. Since  $Z_1 \cap f^{-1}U_i \subseteq Z_1 \subsetneq Z_2$ , it is not dense in  $Z_2$ , so  $Z_1 \cap f^{-1}U_i \subsetneq Z_2 \cap f^{-1}U_i$ . These are closed irreducible subsets of an affine variety, so  $f(Z_1 \cap f^{-1}U_i) \subsetneq f(Z_2 \cap f^{-1}U_i)$ , and thus we must have  $f(Z_1) \subsetneq f(Z_2)$ . ■

### 3.1.11 Corollary

If  $X$  is an algebraic variety, its points are all closed.

#### Proof

For  $x \in X$ ,  $x$  is a point of an affine variety and thus itself an affine variety. Its ideal is maximal, which means  $k[\{x\}] = k$  and so  $\{x\} \hookrightarrow X$  is finite as  $k$  is a finite  $k[X]$ -algebra. So by the above proposition the inclusion is a closed map, and thus  $x$  is closed in  $X$ .

Alternatively if we choose an affine cover  $U_i$  of  $X$ , then  $\{x\} \cap U_i$  is either empty or  $\{x\} \subseteq U_i$ . In each case it is closed, and thus  $\{x\}$  is closed. ■

### 3.1.12 Definition

A morphism of varieties  $f: X \rightarrow Y$  is **quasi-finite** if it has finite fibers:  $f^{-1}y$  is a finite set for all  $y \in Y$ .

### 3.1.13 Corollary

Every finite morphism of varieties is quasi-finite.

#### Proof

Let  $f: X \rightarrow Y$  be finite, and  $y \in Y$ . It is sufficient to show that every irreducible component of  $f^{-1}y$  is a point, as varieties have finitely many irreducible components. Let  $Z$  be an irreducible component, and let  $x \in Z$ , then  $\{x\} \subseteq Z$  are closed irreducible sets and  $f(Z) = f(\{x\}) = y$ . By the proposition, we must have  $\{x\} = Z$ . ■

### 3.1.14 Corollary

- (1) For every affine variety  $X$  there is a finite surjective morphism  $X \rightarrow \mathbb{A}^n$  such that  $k[X]$  is a finite  $k[\mathbb{A}^n]$ -algebra.
- (2) For every non-constant function  $f \in k[\mathbb{A}^n] - k$ , there is a finite surjective morphism  $Z(f) \rightarrow \mathbb{A}^{n-1}$



such that  $k[Z(f)]$  is a finite  $k[\mathbb{A}^{n-1}]$ -algebra.

### Proof

By the Noether normalization lemma, since  $k[X]$  is fg there is a finite embedding  $k[\mathbb{A}^n] \hookrightarrow k[X]$  for some  $n$ . Thus  $X \rightarrow \mathbb{A}^n$  is finite and dominant. It is then closed, and since it is dominant it is surjective.

For (2), we have  $k[Z(f)] = k[\mathbb{A}^n]/(f)$ . Then in Noether normalization our injective embedding was  $k[\mathbb{A}^{n-1}] \rightarrow k[b\mathbb{A}^n]/(f)$ . ■

## 3.2 Dimension theory

### 3.2.1 Definition

Let  $X$  be a topological space. A **chain of irreducible subsets** is a sequence  $\emptyset \neq Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subseteq X$  where each  $Z_i$  is irreducible. The length of such a chain is  $n$ .

Then we define the **(Krull) dimension** of  $X$  to be the supremum of the lengths of all chains of irreducible subsets. It is denoted  $\dim X$ . For  $X = \emptyset$  we set  $\dim X = -1$ .

### Note:

For Hausdorff  $X$ , the only irreducible subsets are the points, and so  $\dim X = 0$ . We will thus consider only Noetherian topological spaces.

The Krull dimension of a Noetherian topological space may be infinite. While each chain is finite (and can only be extended finitely), there need not be a bound on the length of all chains. We will nevertheless show that for algebraic varieties the dimension is finite.

### 3.2.2 Lemma

If  $Y$  is irreducible and  $X \subseteq Y$  is closed with  $\dim X = \dim Y < \infty$ , then  $X = Y$ .

### Proof

We show that if  $\dim X = n$  and  $X \subsetneq Y$  then  $\dim Y \geq n + 1$ . By definition, there is a chain of irreducible subsets of  $X$  of length  $n$ :  $\emptyset \neq X_0 \subsetneq \cdots \subsetneq X_n$ . These are then closed irreducible in  $Y$ , and we can append  $Y$  to this chain to obtain a chain of length  $n + 1$ . ■

### 3.2.3 Lemma

Let  $X$  be an arbitrary topological space.

- (1) If  $Y \subseteq X$  is a subspace,  $\dim Y \leq \dim X$ .
- (2) If  $X = \bigcup_i X_i$  is a finite union of closed subsets, then  $\dim X = \max_i \dim X_i$ .
- (3) If  $X = \bigcup_i U_i$  is an open covering, then  $\dim X = \sup_i \dim U_i$ .

### Proof

For (1), we note that if  $Y_n \subseteq Y$  is closed irreducible then  $Y_n = Y \cap X_n$  for  $X_n \subseteq X$  irreducible. Indeed,  $Y_n = Y \cap X_n$  for  $X_n$  closed. Then  $X_n$  is the union of its irreducible components, and then  $Y_n$  is the union of the intersection of  $Y$  with these irreducible components. There are all closed in  $Y_n$ , and thus this union cannot be proper, so  $Y_n$  is equal to the intersection of  $Y$  with one of these irreducible components.

So every chain in  $Y$  gives rise to a chain in  $X$ , showing  $\dim Y \leq \dim X$ .

For (2) and (3), by (1) we have that  $\max_i \dim X_i, \sup_i \dim U_i \leq \dim X$ . Conversely let  $\emptyset \neq Z_0 \subsetneq \cdots \subsetneq Z_n$  be a chain of irreducible subsets of  $X$ . Then for each  $j$ ,

$$Z_j = \bigcup_i Z_j \cap X_i.$$

Since  $Z_j$  is irreducible and this is a finite union of closed subsets, this means there is an  $i$  where  $Z_j = Z_j \cap X_i$  and so  $Z_j \subseteq X_i$ . For  $j = n$  this shows that this chain is contained in some  $X_i$  so  $\dim X \leq \max_i \dim X_i$ .

For (3) we note that open subsets of irreducible subsets are irreducible (since irreducibility is equivalent to open subsets having nonempty intersection). Since  $Z_0$  is nonempty there is some  $U_i$  where  $Z_0 \cap U_i \subseteq U_i$  is nonempty. In such a case we also have  $Z_j \cap U_i \subsetneq Z_{j+1} \cap U_i$  since  $Z_{j+1} \cap U_i$  is dense in  $Z_{j+1}$  while  $Z_j$  is not. So we get an irreducible chain  $\emptyset \neq Z_0 \cap U_i \subsetneq \cdots \subsetneq Z_n \cap U_i$  of  $U_i$ , so  $\sup_i \dim U_i \geq \dim X$ . ■

### 3.2.4 Lemma

If  $f: X \rightarrow Y$  is a finite surjective morphism of algebraic varieties. Then  $\dim X = \dim Y$ .

#### Proof

If  $\emptyset \neq X_0 \subsetneq \cdots \subsetneq X_n$  is a chain of closed irreducible subsets of  $X$ , then by our previous results  $\emptyset \neq f(X_0) \subsetneq \cdots \subsetneq f(X_n)$  is a chain of closed irreducible subsets of  $Y$ , so  $\dim Y \geq \dim X$ .

Conversely let  $\emptyset \neq Y_0 \subsetneq \cdots \subsetneq Y_n$  be a chain of closed irreducible subsets of  $Y$ . We claim that there is a chain  $\emptyset \neq X_0 \subsetneq \cdots \subsetneq X_n$  of closed irreducible subsets of  $X$  such that  $f(X_i) = Y_i$  for all  $i$ .

Write  $f^{-1}Y_n$  as a union of its irreducible components,  $f^{-1}Y_n = \bigcup_j K_j$ . Since  $f$  is surjective,  $Y_n = \bigcup f(K_j)$  and  $f$  is a closed map so  $f(K_j) \subseteq Y$  is closed. Thus  $f(K_j) = Y_n$  for some  $j$ ; let  $X_n = K_j$ .

The restriction  $f|_{X_n}: X_n \rightarrow Y_n$  is finite surjective, so inducting on  $n$  we construct  $\emptyset \neq X_0 \subsetneq \cdots \subsetneq X_n$  as desired. ■

#### Note:

Later we will be able to show that for a morphism of varieties  $f: X \rightarrow Y$ ,

- (1) if  $f$  is dominant, then  $\dim X \geq \dim Y$ ,
- (2) if  $f$  is quasi-finite, then  $\dim X \leq \dim Y$ .

So if  $f$  is dominant and quasi-finite (e.g. surjective and finite) then  $\dim X = \dim Y$ .

### 3.2.5 Theorem

- (1) For  $n \geq 0$ ,  $\dim \mathbb{A}^n = n$ .
- (2) For  $n > 0$  and  $f \in k[\mathbb{A}^n] - k$ ,  $\dim Z(f) = n - 1$ .

#### Proof

We prove both points by induction on  $n$ . For  $n = 0$  point (2) holds vacuously and point (1) holds trivially since  $\mathbb{A}^0$  is a single point. So assume  $n > 0$ .

For point (2), there is a finite surjective morphism  $Z(f) \rightarrow \mathbb{A}^{n-1}$  so by the above lemma,  $\dim Z(f) = \dim \mathbb{A}^{n-1}$  which is  $n - 1$  by induction. For (1), consider the chain  $X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n$  of  $\mathbb{A}^n$  where

$$X_i = Z(x_{i+1}, \dots, x_n).$$

These are all irreducible as  $(x_{i+1}, \dots, x_n)$  is prime and the strict inclusion is clear. So  $\dim \mathbb{A}^n \geq n$ .

Conversely let  $X_0 \subsetneq \cdots \subsetneq X_m$  be a chain of closed irreducible subsets of  $\mathbb{A}^n$ . Now we have that  $\dim X_{m-1} \geq m - 1$ , since this chain gives a chain of length  $m - 1$  of  $X_{m-1}$ , and  $X_{m-1} \subsetneq \mathbb{A}^n$ . Thus there is a  $f \in k[\mathbb{A}^n] - 0$  such that  $X_{m-1} \subseteq Z(f)$  (because  $\mathcal{I}(X_{m-1}) \neq 0$ ), and thus we have  $f \notin k$ . So inductively  $\dim Z(f) = n - 1$  and thus

$$m - 1 \leq \dim X_{m-1} \leq \dim Z(f) = n - 1 \implies m \leq n$$

as desired. ■

### 3.2.6 Corollary

The dimension of any algebraic variety is finite.

### Proof

The dimension of any affine variety is finite, as a subspace of  $\mathbb{A}^n$ . If  $X$  is an algebraic variety, it has a finite cover by open affine subvarieties. Thus  $X$ 's dimension is the maximum of these dimensions, which is finite. ■

### 3.2.7 Definition

Let  $L/K$  be a field extension.

- (1) A subset  $S \subseteq L$  is **algebraically independent** over  $K$  if for every  $s_1, \dots, s_n \in S$  and  $0 \neq f \in K[x_1, \dots, x_n]$  we have  $f(s_1, \dots, s_n) \neq 0$ .
- (2) A subset  $S \subseteq L$  is a **transcendence basis** of  $L/K$  if it is algebraically independent over  $K$  and  $L$  is an algebraic extension of  $K(S)$ .
- (3) It can be shown that every field extension has a transcendence basis, and they all have the same cardinality. The cardinality of a transcendence basis is called the **transcendence degree** of the extension, and is denoted  $\text{trdeg}(L/K)$  or  $\text{trdeg}_K(L)$ .

### Note:

To prove the existence of a transcendence basis, and the uniqueness of its cardinality, we can use the concept of a **pregeometry**. A **pregeometry** is a pair  $(S, \text{cl})$  where  $\text{cl}: \mathcal{P}(S) \rightarrow \mathcal{P}(S)$  is a **closure operation** satisfying the following axioms:

- (1) **monotonicity**:  $\text{cl}$  is monotone increasing and dominates: if  $A \subseteq B$  then  $A \subseteq \text{cl}(A) \subseteq \text{cl}(B)$ .
- (2) **idempotent**:  $\text{cl}$  is idempotent:  $\text{cl}(\text{cl}(A)) = \text{cl}(A)$ .
- (3) **finite character**: for each  $a \in \text{cl}(A)$  there is some finite  $F \subseteq A$  with  $a \in \text{cl}(F)$ .
- (4) **exchange**: if  $b \in \text{cl}(A \cup \{c\}) - \text{cl}(A)$ , then  $c \in \text{cl}(A \cup \{b\})$ .

For a pregeometry  $(S, \text{cl})$ , a subset  $D \subseteq S$  is **independent** if  $d \notin \text{cl}(D - d)$  for all  $d \in D$ . A **basis** for  $S$  is an independent set  $D$  such that  $\text{cl}(D) = S$ . Equivalently this is a maximally independent set; and thus it can be shown easily via Zorn that a basis must exist.

Then we can show that if  $A$  is independent and  $B$  generates  $S$  (meaning  $\text{cl}(B) = S$ ), then  $|A| \leq |B|$ . Clearly this implies two bases have the same cardinality. We omit the details of the proofs here, they can be found in most standard texts on model theory for example (or my final assignment in model theory).

Pregeometries are also called **matroids** outside of model theory. They generalize the concept of independence found in linear algebra and are used to prove important results in model theory. Our main concern here is their use for algebraic independence.

Given a field extension  $L/K$ , we define a pregeometry on  $L$  by

$$\text{cl}(S) = \{a \in L \mid a \text{ is algebraic over } K(S)\}.$$

This is clearly monotonic and dominates, idempotence is easily seen as well. Finite character is easy to see as well: if  $a$  is algebraic over  $S$  then it is the root of some polynomial in  $K(S)[x]$ . Any such polynomial must use finitely many elements from  $S$ . If  $a$  is algebraic over  $K(S \cup b)$  but not  $K(S)$ , then there is a polynomial in  $K(S \cup b)[x]$  with root  $a$ , and it must have some coefficient with  $b$ . Then viewing  $b$  as the variable and not  $a$ , we see  $b \in \text{cl}(A \cup a)$ , so this satisfies exchange.

If  $S$  generates  $L$  then every  $a \in L = \text{cl}(S)$  is algebraic over  $K(S)$ , so  $L/K(S)$  is algebraic. And if  $S$  is independent, then let  $s_1, \dots, s_n \in S$  and suppose  $f(s_1, \dots, s_n) = 0$  for  $0 \neq f \in K[x_1, \dots, x_n]$ . Take  $n$  minimal, so each  $x_i$  has nonzero coefficient. This shows that  $s_1$  is algebraic over  $K(s_2, \dots, s_n)$ , which contradicts independence. So if  $S$  is independent in the pregeometry, then it is algebraically independent over  $K$  (the converse is readily shown as well).

Thus a basis of the pregeometry is the same as a transcendence basis of the field extension, and we have proven the result.

If we have field extensions  $M/L/K$ , then let  $S_0$  be a transcendence basis of  $M/L$ , and  $S_1$  a transcendence basis of  $L/K$ . These are disjoint as  $S_0 \subseteq M - L$ , and their union forms a transcendence basis of  $M/K$ . So

$$\text{trdeg}(M/K) = \text{trdeg}(M/L) + \text{trdeg}(L/K).$$

If  $L/K$  is finitely generated, so  $L = K(y_1, \dots, y_n)$ , then any maximal algebraically independent subset of  $y_1, \dots, y_n$  over  $K$  is a transcendence basis. Thus  $\text{trdeg}(L/K) \leq n$ .

### 3.2.8 Definition

Let  $X$  be an algebraic variety. We define  $k(X)$  to be the colimit of sections of dense open subsets:

$$k(X) = \text{colim}_{U \subseteq X \text{ dense}} \mathcal{O}_X(U).$$

Explicitly, we can write  $k(X)$  in terms of equivalence classes. We identify two pairs  $(U, f)$  and  $(V, g)$  where  $f \in \mathcal{O}_X(U), g \in \mathcal{O}_X(V)$  if there is a dense open subset  $W \subseteq U \cap V$  such that  $f|_W = g|_W$ . Then  $k(X)$  is the quotient

$$k(X) = \coprod_{U \subseteq X \text{ dense}} \mathcal{O}_X(U) / \sim.$$

A **rational function** is an element of  $k(X)$ , so an equivalence class.

### 3.2.9 Lemma

- (1)  $k(X)$  has a natural  $k$ -algebra structure.
- (2) For any dense open subset  $U \subseteq X$ , restriction induces a  $k$ -algebra isomorphism  $k(X) \cong k(U)$ .

- (3) If  $X = \bigcup_{i=1}^n Z_i$  is the decomposition of  $X$  into irreducible components, then  $k(X) \cong \prod_i k(Z_i)$  where the components of the map are restrictions.
- (4) If  $X$  is irreducible,  $k(X)$  is a field.
- (5) If  $X$  is affine and irreducible, then  $k(X)$  is the fraction field of  $k[X]$ . More precisely the map  $k[X] \rightarrow k(X)$  given by  $f \mapsto (X, f)$  factors through an isomorphism  $\text{Frac}(k[X]) \cong k(X)$ .

### Proof

Point (1) is clear as a colimit of  $k$ -algebras is a  $k$ -algebra. Point (2) is clear since dense open subsets of  $U$  are dense in  $X$ , and form a cofinal subcategory of dense open subsets of  $X$ . This is because  $U$  is dense, so for every  $W \subseteq X$  open dense we have  $W \cap U \subseteq U$  is open dense.

For (3), define  $U_i = Z_i - \bigcup_{j \neq i} Z_j$ . Since  $Z_i$  are irreducible,  $U_i \neq \emptyset$ , and since  $U_i = X - \bigcup_{j \neq i} Z_j$ ,  $U_i$  is open. And the  $U_i$  are all disjoint. Define  $U = \bigcup_i U_i$ , then we claim  $U$  is dense in  $X$ .

Indeed, let  $V \subseteq X$  be nonempty open. Then  $V \cap Z_i \neq \emptyset$  for some  $i$ , so  $V \cap Z_i$  is nonempty open in  $Z_i$  and thus dense. So  $V \cap Z_i \cap U_i \neq \emptyset$ , and so  $V \cap U \neq \emptyset$  meaning  $U$  is indeed dense.

But  $U = \bigsqcup_i U_i$  is a disjoint union of irreducible  $U_i$ , so a dense open subset of  $U$  is the same as a disjoint union  $W = \bigsqcup_i W_i$  for  $W_i \subseteq U_i$  dense open. By the sheaf condition,  $k[W] = \prod_i k[W_i]$ . This says that the functor  $W \mapsto \mathcal{O}_X(W)$  is the same as the product functor  $(W_i) \mapsto \prod_i k[W_i]$ . Colimits and limits commute, so we see

$$k(U) \cong \prod_i k(U_i).$$

Now,  $U \subseteq X$  is dense and  $U_i \subseteq Z_i$  is dense, so by (2) this forms an isomorphism

$$k(X) \cong \prod_i k(Z_i).$$

For (4), note that all nonempty open  $U \subseteq X$  are dense. For nonzero  $f \in k[U]$ ,  $D_U(f) \subseteq X$  is nonempty open and thus dense. So  $(D_U(f), 1/f) \in k(X)$  is a multiplicative inverse for  $(U, f)$ .

For (5), since  $k[X] \subseteq k(X)$  and  $k(X)$  is a field, it remains to show that every element of  $k(X)$  can be written as  $g/h$  for  $g/h \in k[X]$ . Indeed, the basic open subsets  $D_X(h)$  form a basis of  $X$ , so every  $f \in k(X)$  has representative in  $k[D_X(h)] = k[X]_h$ . Such an  $f$  is of the form  $g/h^k$  for  $g \in k[X]$  as desired. ■

### 3.2.10 Corollary

If  $X$  is a nonempty irreducible affine variety,  $\dim X = \text{trdeg}_k(k(X))$ .

### Proof

There is a finite surjective morphism  $X \rightarrow \mathbb{A}^n$ , so  $\dim X = \dim \mathbb{A}^n = n$ . Since  $k[\mathbb{A}^n] \subseteq k[X]$  and  $k[X]$  is a finite  $k[\mathbb{A}^n]$ -algebra, we have that  $k(X)$  is a finite extension of  $k(\mathbb{A}^n) = k(x_1, \dots, x_n)$  (as it is the fraction field of  $k[x_1, \dots, x_n]$ ). Since  $x_1, \dots, x_n$  are algebraically independent over  $k$  and  $k(X)/k(x_1, \dots, x_n)$  is finite and thus algebraic, we see that  $\text{trdeg}_k(k(X)) = n$ . ■

### 3.2.11 Corollary

If  $X$  is an algebraic variety and  $U \subseteq X$  is open dense, then  $\dim U = \dim X$ .

**Proof**

If  $X$  is affine and irreducible, then there is a  $f \in k[X]$  such that  $D_X(f) \subseteq U \subseteq X$ . Thus  $\dim D_X(f) \leq \dim U \leq \dim X$ . On the other hand  $k[D_X(f)] = k[X]_f$ , and since  $X$  is irreducible so is  $D_X(f)$  and thus  $k(D_X(f))$  is the fraction field of  $k[X]_f$ , which is just  $k(X)$ . Thus  $\dim D_X(f) = \dim X = \text{trdeg}_k(k(X))$  so  $\dim U = \dim X$ .

Now suppose  $X$  is just affine, and let  $X_1, \dots, X_n$  be the irreducible components of  $X$ . Since  $U \subseteq X$  is open dense,  $U \cap X_i \subseteq X_i$  is open dense and  $X_i$  is affine so  $\dim(U \cap X_i) = \dim X_i$ . Then  $U$  is the finite union of the closed  $U \cap X_i$ , so  $\dim U = \max_i \dim X_i = \dim X$  (the last equality is because  $X_i$  form a finite closed cover of  $X$ ).

In the general case, choose an open affine covering  $X = \bigcup_i X_i$ , then  $U \cap X_i \subseteq X_i$  is open dense so  $\dim(U \cap X_i) = \dim X_i$ . So

$$\dim U = \max_i \dim(U \cap X_i) = \max_i \dim X_i = \dim X,$$

as desired. ■

**3.2.12 Corollary**

If  $X$  is a nonempty irreducible variety, then  $\dim X = \text{trdeg}_k(k(X))$ .

**Proof**

Choose an open affine neighborhood  $U \subseteq X$ . Since  $X$  is irreducible,  $U$  is open dense. Then  $k(X) \cong k(U)$  and  $\dim U = \dim X$ . Since  $\dim U = \text{trdeg}_k(k(U))$  we are finished. ■

If we define  $\text{trdeg}_k(0) = -1$  for the zero ring, then the previous corollary holds for the empty variety.

**3.2.13 Proposition**

- (1) For  $g \in k[\mathbb{A}^n] - k$ , every irreducible component of  $Z(g)$  has dimension  $n - 1$ .
- (2) Conversely if  $X \subseteq \mathbb{A}^n$  is a closed subvariety such that every irreducible component of  $X$  has dimension  $n - 1$ , then there is a  $g \in k[\mathbb{A}^n] - k$  such that  $X = Z(g)$ . If  $X$  is irreducible, then  $g$  can be chosen to be irreducible.

**Proof**

- (1) If  $g \in k[\mathbb{A}^n]$  is irreducible, then  $Z(g)$  is irreducible as  $(g)$  is prime, and we have  $\dim Z(g) = n - 1$  as before. If  $g$  decomposes as  $g = \prod_i g_i$  where  $g_i$  are irreducible, then  $Z(g) = \bigcup_i Z(g_i)$  so  $Z(g_i)$  are the irreducible components of  $Z(g)$ . Furthermore  $\dim Z(g_i) = n - 1$  as claimed.
- (2) We may assume  $X$  is irreducible, as if  $X = \bigcup_i X_i$  for  $X_i = Z(g_i)$  then  $X = Z(g_1 \cdots g_n)$ . So  $X$  is closed irreducible and  $\dim X = n - 1$ , in particular  $X \subsetneq \mathbb{A}^n$ . Thus there is a nonzero  $g \in k[\mathbb{A}^n]$  such that  $X \subseteq Z(g)$ . If we decompose  $g$  to irreducible components  $g_1 \cdots g_n$ , then  $Z(g) = \bigcup_i Z(g_i)$ . Since  $X$  is irreducible,  $X \subseteq Z(g_i)$  for some  $i$ ; so we can assume  $g$  is irreducible.  
But then  $\dim Z(g) = n - 1$ , and so  $\dim X = \dim Z(g)$  meaning  $X = Z(g)$ . ■

**3.2.14 Corollary**

For nonzero  $0 \neq f \in k[\mathbb{A}^n]$ , if  $Z(f) \neq \emptyset$  then  $\dim Z(f) = n - 1$ .

**Proof**

$f$  cannot be a scalar as nonzero scalars have empty zero sets. So  $f \in k[\mathbb{A}^n] - k$ , so  $\dim Z(f) = n - 1$  either by the above proposition or from our general result previously on  $\dim Z(f)$ . ■

**3.2.15 Lemma**

Let  $X$  be an irreducible affine variety of dimension  $n$ . Let  $0 \neq f \in k[X]$  be such that  $Z_X(f) \neq \emptyset$ . Then  $\dim Z_X(f) = n - 1$ .

**Proof**

There is a finite surjective  $\varphi: X \rightarrow \mathbb{A}^m$  such that  $k[X]$  is a finite  $k[\mathbb{A}^m]$ -algebra. Since this implies  $\dim X = \dim \mathbb{A}^m = m$  we have  $m = n$ .

We wish to find a function  $g \in k[\mathbb{A}^n]$  such  $\varphi$  induces a surjective morphism  $Z_X(f) \rightarrow Z_{\mathbb{A}^n}(g)$ . This would then be finite as the domain and codomain are closed, which would imply  $\dim Z_X(f) = \dim Z_{\mathbb{A}^n}(g) = n - 1$ .

So consider  $\tilde{\Phi}: X \rightarrow \mathbb{A}^{n+1}$  given by  $\tilde{\Phi}(x) = (\varphi(x), f(x))$ . Since  $\mathbb{A}^{n+1}$  is the categorical product of  $\mathbb{A}^n$  and  $\mathbb{A}^1$ , this is a continuous map. Since  $\varphi = \text{proj}_1 \circ \tilde{\Phi}$ , we have that  $k[X]$  is a finite  $k[\mathbb{A}^{n+1}]$ -algebra and  $\tilde{\Phi}$  is a finite morphism.

As  $\tilde{\Phi}$  is finite, it is closed so  $Y = \tilde{\Phi}(X) \subseteq \mathbb{A}^{n+1}$  is closed. Since  $X$  is irreducible, so too must  $Y$  be (if  $Y = Z_0 \cup Z_1$  then taking preimages we must have that  $X = \tilde{\Phi}^{-1}Z_0$  or  $\tilde{\Phi}^{-1}Z_1$  which then implies  $Y = Z_0$  or  $Z_1$ ). As a finite morphism, its restriction to closed sets is finite, so  $\Phi: X \rightarrow Y$ , the restriction of  $\tilde{\Phi}$ , is surjective finite. Thus we have  $\dim Y = \dim X = n$ , as this is  $(n+1) - 1$  and  $Y$  is irreducible by the previous lemma there is an irreducible  $h \in k[\mathbb{A}^{n+1}]$  such that  $Y = Z(h)$ .

Since  $f = \Phi^*(x_{n+1})$  we have  $Z_X(f) = \Phi^{-1}(Z_Y(x_{n+1}))$  and so  $\Phi$  induces a surjective morphism  $Z_X(f) \rightarrow Z_Y(x_{n+1})$ . Write  $h = \sum_{i=0}^m a_i x_{n+1}^i$  with  $a_i \in k[x_1, \dots, x_n]$  so we have an equality of ideals  $(h, x_{n+1}) = (a_0, x_{n+1})$ . Thus

$$Z_Y(x_{n+1}) = Z_{\mathbb{A}^{n+1}}(x_{n+1}) \cap Y = Z(x_{n+1}) \cap Z(h) = Z(x_{n+1}, h) = Z(a_0, x_{n+1}) = Z_{\mathbb{A}^n}(a_0) \times \{0\}.$$

Consider the composition

$$Z_X(f) \rightarrow Z_Y(x_{n+1}) \cong Z_{\mathbb{A}^n}(a_0) \times \{0\} \xrightarrow{\text{proj}} Z_{\mathbb{A}^n}(a_0).$$

This is a composition of finite morphisms, thus finite surjective.

Since  $Z_X(f) \neq \emptyset$  we have  $Z_{\mathbb{A}^n}(a_0) \neq \emptyset$  and thus if we have  $a_0 \neq 0$  then  $\dim Z_{\mathbb{A}^n}(a_0) = n - 1$ . So  $\dim Z_X(f) = \dim Z_{\mathbb{A}^n}(a_0) = n - 1$  as desired. And indeed if  $a_0 = 0$  then that would imply  $(h, x_{n+1}) = (x_{n+1})$  so  $x_{n+1}$  divides  $h$ , but  $h$  is irreducible so  $h = x_{n+1}$ . But then  $x_{n+1}|_Y = 0$  since  $Y = Z(h)$ , and since  $f = \Phi^*(x_{n+1}|_Y)$  this would imply  $f = 0$ , a contradiction. ■

**3.2.16 Theorem**

Let  $X$  be an irreducible variety, then for every nonzero  $f \in k[X]$  every irreducible component of  $Z_X(f)$  is of dimension  $\dim X - 1$ .

**Proof**

Let  $Y$  be an irreducible component of  $Z_X(f)$ , and define  $Z$  to be the union of all irreducible components of  $Z_X(f)$  other than  $Y$ , and set  $U = X - Z$ . Then  $Z \subseteq Z_X(f)$  is closed as a finite union of closed sets and  $Z \cup Y = Z_X(f)$  and  $Y$  is not a subset of  $Z$ . And

$$Z_U(f) = Z_X(f) \cap U = (Z \cup Y) \cap U = Y \cap U.$$

This is nonempty since  $Y$  is not a subset of  $Z$ . Since  $X, Y$  are irreducible,  $Y \cap U \subseteq Y$  and  $U \subseteq X$  are open dense. Thus  $\dim Y = \dim(Y \cap U) = \dim Z_U(f)$  and  $\dim X = \dim U$ .

Replacing  $X$  by  $U$  and  $f$  by  $f|_U$ , it is sufficient to show that if  $Z_X(f)$  is irreducible then  $\dim Z_X(f) = \dim X - 1$ .

For any  $x \in Z_X(f)$ , take an affine neighborhood  $U \subseteq X$  of  $x$ . Then  $x \in Z_X(f) \cap U = Z_U(f)$ , and we can replace  $X$  with  $U$  and  $f$  with  $f|_U$  (as  $\dim X = \dim U$ ) we can assume  $X$  is affine. And in such a case  $\dim Z_X(f) = \dim X - 1$  by the previous lemma. ■

### 3.2.17 Lemma

Let  $Y$  be a topological space with finitely many irreducible components  $Y_1, \dots, Y_m$ . Let  $Z \subseteq Y$  be a closed subset then any irreducible component  $Z' \subseteq Z$  is an irreducible component of some  $Z \cap Y_i$ .

#### Proof

Since  $Z' \subseteq Z \subseteq \bigcup_i Y_i$  and  $Z'$  is irreducible,  $Z' \subseteq Y_i$  for some  $i$ . So  $Z' \subseteq Z \cap Y_i$ . Since  $Z'$  is maximally closed irreducible subset of  $Z$ , it is a maximally closed irreducible subset of  $Z \cap Y_i$ , thus an irreducible component. ■

### 3.2.18 Corollary

Assume that all irreducible components of a variety  $Y$  have dimension  $\geq d$ . Then for every  $f_1, \dots, f_n \in k[Y]$ , every irreducible component  $Z'$  of  $Z = Z_Y(f_1, \dots, f_n)$  has dimension at least  $d - n$ .

#### Proof

Let  $Y_1, \dots, Y_m$  be the irreducible components of  $Y$ , and let  $Z'$  be an irreducible component of  $Z_Y(f)$ . Then  $Z'$  is an irreducible component of some  $Z_Y(f) \cap Y_i = Z_{Y_i}(f)$ . By the previous theorem, this has dimension  $\dim Y_i - 1 \geq d - 1$  if  $f \neq 0$  and  $\dim Y_i \geq d$  if  $f = 0$ . This proves the result for  $n = 1$ , and then we proceed by induction. ■

#### Note:

This corollary would be false if we considered just  $Y$  and  $Z$  instead of their irreducible components.

For example take  $Y$  to be the disjoint union of  $\mathbb{A}^n$  and  $\text{pt}$  (the one-point space). Then consider  $f \in k[Y]$  given by  $f|_{\mathbb{A}^n} = 1$  and  $f|_{\text{pt}} = 0$ , so  $Z_Y(f) = \text{pt}$ . So  $\dim Y = m$  while  $\dim Z_Y(f) = 0$ .

The issue here is  $Y$  has an irreducible component of dimension zero.

### 3.2.19 Lemma

Let  $X$  be an irreducible algebraic variety and  $Z \subsetneq X$  be closed irreducible with  $\dim Z \leq \dim X - 2$ . Then there is a closed irreducible  $Z \subsetneq Y \subsetneq X$  with  $\dim Y = \dim X - 1$ .

#### Proof

If  $X$  is affine, then since  $Z \subsetneq X$  there is a function  $f \in k[X]$  such that  $Z \subseteq Z_X(f)$ . Since  $Z$  is irreducible, there must be an irreducible component  $Y$  of  $Z_X(f)$  such that  $Z \subseteq Y \subseteq X$ . The irreducible components of  $Z_X(f)$  are of dimension  $\dim X - 1$ , so  $\dim Y = \dim X - 1$ , meaning both inclusions are strict and we are



finished.

In the general case, choose an open affine subset  $U \subseteq X$  such that  $U \cap Z \neq \emptyset$ . Then  $U \subseteq X, U \cap Z \subseteq Z$  are open dense, and then

$$\dim(Z \cap U) = \dim Z \leq \dim X - 2 = \dim(X \cap U) - 2.$$

So by the affine case, there is a closed irreducible  $Z \cap U \subsetneq Y_U \subsetneq X \cap U$  with  $\dim Y_U = \dim X - 1$ . Let  $Y \subseteq X$  be the closure of  $Y_U$ , which is irreducible since  $Y_U$  is. We claim  $\dim Y_U = \dim Y$ . Since  $Y_U = Y \cap U$  is open dense in  $Y$ , this is clear. ■

### 3.2.20 Corollary

Let  $X$  be an algebraic variety, and  $\emptyset \neq X_0 \subsetneq \cdots \subsetneq X_m$  a maximal chain of closed irreducible sets. Then  $X_m$  is an irreducible component of  $X$ , and  $\dim X_i = i$  for all  $i$ . In particular if  $X$  is irreducible then  $\dim X = m$ , meaning all maximal chains of closed irreducible sets are of length  $\dim X$ .

### Proof

Let  $p \in X_0$ , if  $X_0 \neq \{p\}$  then we can extend the chain by  $\{p\} \subsetneq X_0$ , contradicting maximality. Thus  $\dim X_0 = \dim \{p\} = 0$ .

And  $X_m$  as it is irreducible is contained in an irreducible component  $X'$  of  $X$ . If  $X_m \neq X'$  then we can extend the chain by  $X_m \subsetneq X'$ , contradicting maximality. Thus  $X_m$  is irreducible.

Now we know that  $\dim X_{i+1} > \dim X_i$ , and if  $\dim X_{i+1} = \dim X_i + 1$  then we are finished. Otherwise  $\dim X_{i+1} \geq \dim X_i + 2$ , which by the previous lemma means there is a closed irreducible  $X_i \subsetneq X' \subsetneq X_{i+1}$ , contradicting maximality. ■

### 3.2.21 Lemma

Suppose  $Y$  is affine and  $X \subseteq Y$  is closed such that all irreducible components of  $Y$  (resp.  $X$ ) are of dimension  $\dim Y$  (resp.  $\dim X$ ). Then there are  $f_1, \dots, f_{\dim Y - \dim X} \in k[Y]$  such that  $X \subseteq Z_Y(f_1, \dots, f_n)$  and  $\dim Z_Y(f_1, \dots, f_n) = \dim X$ . Moreover,  $X$  is a union of the irreducible components of  $Z_Y(f_1, \dots, f_n)$ .

### Proof

Let  $Y_1, \dots, Y_m$  be the irreducible components of  $Y$ . If  $\dim X < \dim Y$ , we have that  $Y_i$  is not contained in  $X$  (considering dimensions).

Now we claim there is a  $f \in k[Y]$  such that  $f|_X = 0$  and  $f|_{Y_i} \neq 0$  for all  $i$ . For each  $i$ , let  $y_i \in Y_i - X$ , and let  $X' = X \cup \{y_1, \dots, y_m\} \subseteq Y$ . As a finite union of closed sets,  $X' \subseteq Y$  is closed. Thus the map  $k[Y] \rightarrow k[X']$  is surjective, and so there is a  $f \in k[Y]$  where  $f|_X = 0$  and  $f(y_i) = 1$ .

Every irreducible component of  $Z_Y(f)$  is an irreducible component of some  $Z_{Y_i}(f) = Y_i \cap Z_Y(f)$  (as showed previously), and thus have dimension  $\dim Y_i - 1 = \dim Y - 1$ . So we simply induct on  $n$ .

Now we need to consider the case  $\dim X = \dim Y$ , then every irreducible component  $X'$  of  $X$  is an irreducible component of  $Y$ . Indeed,  $X' \subseteq \bigcup Y_i$  and  $X'$  is irreducible so  $X' \subseteq Y_i$ . Since  $\dim X' = \dim X = \dim Y = \dim Y_i$ , we have  $X' = Y_i$ , and we see it is indeed an irreducible component of  $Y$ . ■

### 3.2.22 Corollary

Let  $Y$  be an irreducible affine variety, then for every  $y \in Y$  there are  $f_1, \dots, f_{\dim Y} \in k[Y]$  such that  $Z_Y(f_1, \dots, f_{\dim Y})$  is a finite set containing  $y$ . Thus  $\{y\} \subseteq Z_Y(f_1, \dots, f_{\dim Y})$  is a connected component, so there is an open neighborhood  $y \in U$  such that  $Z_U(f_1, \dots, f_{\dim Y}) = \{y\}$ .

**Proof**

By the above lemma with  $X = \{y\}$ , we have  $f_1, \dots, f_{\dim Y} \in k[Y]$  such that  $Z_Y(f_1, \dots, f_{\dim Y})$  contains  $y$  and has dimension zero. Thus it is a finite set containing  $Y$ , so we can separate  $y$  from the other points in  $Z$  by means of an open set. Meaning  $y \in U$  and  $Z \cap U = \{y\}$ . ■

**3.2.23 Corollary**

Let  $\varphi: X \rightarrow Y$  be a morphism of irreducible varieties. Then for every  $y \in Y$ , each irreducible component of  $\varphi^{-1}y$  is of dimension at least  $\dim X - \dim Y$ .

**Proof**

Let  $y \in U$  be an open affine neighborhood of  $y$ , so  $\dim Y = \dim U$ . Similarly we can replace  $X$  by  $\varphi^{-1}U$  without affecting dimension. Thus we can assume  $Y$  is affine.

Then there are regular  $f_1, \dots, f_{\dim Y} \in k[Y]$  such that  $y \in Z_Y(f_1, \dots, f_{\dim Y})$ . We can further replace  $Y$  by an open subset, so we can assume  $Z_Y(f_1, \dots, f_{\dim Y}) = \{y\}$ . Then

$$\varphi^{-1}y = \varphi^{-1}Z_Y(f_1, \dots, f_{\dim Y}) = Z_X(\varphi^*(f_1), \dots, \varphi^*(f_{\dim Y}))$$

whose irreducible components have dimension at least  $\dim X - \dim Y$  (as shown). ■

## 4 Types of Varieties

### 4.1 Products of varieties

The goal of this section is to construct the categorical product of varieties. First we will construct the product of SWFs.

Recall that when constructing the colimits of SWFs we knew that the underlying topological space was the colimit of the spaces. This is not true for limits; but we will see that the underlying set is the limit, but endowed with a different topology.

Let  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  be SWFs, then we construct a topology on  $Z = X \times Y$  as follows. For  $U \subseteq X$  and  $V \subseteq Y$  open, and  $f: U \times V \rightarrow k$  of the form  $f(x, y) = \sum_i g_i(x)h_i(y)$  for  $g_i \in k[U]$  and  $h_i \in k[V]$ , define

$$D_{U \times V}(f) = \{(x, y) \in U \times V \mid f(x, y) \neq 0\}.$$

This collection of  $D_{U \times V}(f)$  forms a basis for our topology on  $Z$ . Call such  $f$  **basic regular** relative to  $U, V$ .

This allows us to define a structure sheaf on  $Z$ . For  $W \subseteq Z$  open, let  $\mathcal{O}_Z(W)$  be the set of functions  $\varphi: W \rightarrow k$  such that there is a basic open cover  $W \subseteq \bigcup_i D_{U_i \times V_i}(f_i)$  and restricting  $\varphi$  to  $W \cap D_{U_i \times V_i}(f_i)$  is of the form  $\varphi_i/f_i$  where  $\varphi_i$  is basic regular relative to  $U, V$ .

Notice that the projections  $\pi_X: Z \rightarrow X$  and  $\pi_Y: Z \rightarrow Y$  are morphisms of SWFs. Indeed,  $\pi_X^{-1}(D_U(f)) = D_U(f) \times Y = D_{U \times Y}(f \cdot 1)$  is open. Since  $D_U(f)$  forms a basis of open sets of  $X$ , we see that  $\pi_X$  is continuous. And for  $f: U \rightarrow k$  regular,  $\pi_X^*(f)(x, y) = f(x)$  is regular as well since  $1 \in k[Y]$ .

Let us denote  $(Z, \mathcal{O}_Z)$  by  $(X, \mathcal{O}_X) \times (Y, \mathcal{O}_Y)$ , or just  $X \times Y$ .

**4.1.1 Lemma**

$(Z, \mathcal{O}_Z)$  is the categorical product of  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  in  $\text{SWF}_k$ .

**Proof**

For  $\varphi_X: T \rightarrow X$  and  $\varphi_Y: T \rightarrow Y$ , in order for this to factor through  $\varphi: T \rightarrow Z$  we must have that  $\varphi = (\varphi_X, \varphi_Y)$  (since  $Z$  is the product of sets). It remains to show that  $\varphi$  is a morphism.

For every open  $U \subseteq X, V \subseteq Y$ , we have  $\varphi^{-1}(U \times V) = \varphi_X^{-1}(U) \cap \varphi_Y^{-1}(V)$  is open. And for every  $f: U \times V \rightarrow k$  basic regular of the form  $f(x, y) = \sum_i g_i(x)h_i(y)$ , we have

$$\varphi^*(f) = \sum_i \varphi_X^*(g_i) \varphi_Y^*(h_i): \varphi^{-1}(U \times V) \rightarrow k$$

which is regular as  $\varphi_X, \varphi_Y$  are morphisms and  $k[\varphi^{-1}(U \times V)]$  is an algebra.

Thus we have that

$$\varphi^{-1}(D_{U \times V}(f)) = D_{\varphi^{-1}(U \times V)}(\varphi^*(f)) \subseteq T$$

is open, so  $\varphi$  is continuous.

Now we need to show that  $\varphi^*$  induces a map of  $k$ -algebras. But this is immediate, as it maps basic regular maps to regular maps, and thus quotients of basic regular to regular maps. And the rest follows by the sheaf condition. ■

Since limits are functorial, if we have morphisms  $f_i: X_i \rightarrow Y_i$  for  $i = 1, 2$ , the product  $f_1 \times f_2: X_1 \times X_2 \rightarrow Y_1 \times Y_2$  is a morphism as well.

#### 4.1.2 Lemma

Let  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  be SWFs and  $X' \subseteq X, Y' \subseteq Y$  be subsets. Then the following two SWFs on  $X' \times Y'$  are the same

$$(X', \mathcal{O}_X|_{X'}) \times (Y', \mathcal{O}_Y|_{Y'}) = ((X, \mathcal{O}_X) \times (Y, \mathcal{O}_Y))|_{X' \times Y'}.$$

#### Proof

Let  $Z_1, Z_2$  denote the SWFs; we want to show  $Z_1 = Z_2$ . We do this by showing that the identity of  $X' \times Y'$  gives a morphism in both directions.

In one direction,  $\text{id}: Z_1 \rightarrow Z_2$  is a morphism since it is the restriction of the embedding  $X' \times Y' \hookrightarrow X \times Y$ . This is just the product of the inclusions  $(X', \mathcal{O}_X|_{X'}) \rightarrow (X, \mathcal{O}_X)$  with the same one for  $Y$ , which is a morphism.

Conversely we show by the universal property that  $(X \times Y)|_{X' \times Y'} \rightarrow X$  obtained by projecting is a morphism. But this is just the restriction of the projection  $X \times Y \rightarrow X$ , which is a morphism. ■

#### 4.1.3 Lemma

For every two sets  $X, Y$ , there is a natural injective homomorphism

$$\mu: \text{Fun}(X, k) \otimes_k \text{Fun}(Y, k) \rightarrow \text{Fun}(X \times Y, k), \quad \mu(f \otimes g): (x, y) \mapsto f(x)g(y).$$

#### Proof

$\mu$  is clearly bilinear, so it is well-defined. Since  $\text{Fun}(X, k), \text{Fun}(Y, k)$  are  $k$ -vector spaces, to show injectivity it is sufficient to show that  $\mu$  maps a basis to a linearly independent set. Take a basis  $\{f_i\}_{i \in I}$  of  $\text{Fun}(X, k)$  and a basis  $\{g_j\}_{j \in J}$  of  $\text{Fun}(Y, k)$ . Then  $\{f_i \otimes g_j\}_{i \in I, j \in J}$  is a basis for their tensor product, and  $\mu$  maps this to  $f_i g_j$ . Suppose we have a finite zeroing sum

$$\sum_{i,j} \alpha_{ij} f_i g_j = 0.$$

Then for every  $y \in Y$  we have the zeroing sum

$$\sum_i \sum_j \alpha_{ij} g_j(y) f_i = 0,$$

which means that for all  $i$ ,  $\sum_j \alpha_{ij} g_j(y) = 0$  by linear independence. Since this is true for all  $y$  this means  $\sum_j \alpha_{ij} g_j = 0$ , and thus  $\alpha_{ij} = 0$ . ■

#### 4.1.4 Theorem

If  $X, Y$  are affine varieties, then  $X \times Y$  is an affine variety. Furthermore, the natural map  $k[X] \otimes_k k[Y] \rightarrow k[X \times Y]$ ,  $f \otimes g \mapsto \pi_X^*(f) \cdot \pi_Y^*(g)$  is an isomorphism.

If  $X, Y$  are algebraic varieties, then so is  $X \times Y$ .

#### Proof

We first prove the first point, where  $X, Y$  are affine. Since  $k[X], k[Y]$  are fg  $k$ -algebras, so is  $k[X] \otimes_k k[Y]$ . Furthermore,  $k[X] \subseteq \text{Fun}(X, k)$  and  $k[Y] \subseteq \text{Fun}(Y, k)$ , and as  $k$  is a field all  $k$ -modules are free and thus flat so tensoring preserves subobjects, thus  $k[X] \otimes_k k[Y] \subseteq \text{Fun}(X, k) \otimes_k \text{Fun}(Y, k)$ . By the previous lemma this is a subalgebra of  $\text{Fun}(X \times Y, k)$  which is reduced. Thus  $k[X] \otimes_k k[Y]$  is a reduced fg  $k$ -algebra.

This means there is an affine  $k$ -variety  $Z$  such that  $k[Z] = k[X] \otimes_k k[Y]$ . Given any other SWF, we have natural bijections

$$\begin{aligned} \text{hom}_{\text{SWF}}(W, Z) &\cong \text{hom}_{\text{Alg}}(k[Z], \mathcal{O}_W(W)) \\ &\cong \text{hom}_{\text{Alg}}(k[X] \otimes_k k[Y], \mathcal{O}_W(W)) \\ &\cong \text{hom}_{\text{Alg}}(k[X], \mathcal{O}_W(W)) \times \text{hom}_{\text{Alg}}(k[Y], \mathcal{O}_W(W)) \\ &\cong \text{hom}_{\text{SWF}}(W, X) \times \text{hom}_{\text{SWF}}(W, Y) \end{aligned}$$

where the first is due to a result we showed in general: the global sections functor  $\Gamma$  is a natural isomorphism for  $Z$  affine. The third is because tensoring over  $k$  is the coproduct of  $k$ -algebras. Thus  $Z$  satisfies the universal property of products, so  $Z \cong X \times Y$  is affine.

Alternatively, we could note that  $\mathbb{A}^n \times \mathbb{A}^m \cong \mathbb{A}^{n+m}$  via the identity. Then for  $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m, X \times Y \subseteq \mathbb{A}^{n+m}$  is closed as it is equal to  $Z(\mathcal{I}(X) \cup \mathcal{I}(Y))$ . Thus it is affine.

Now in general, let  $X = \bigcup_i X_i$  and  $Y = \bigcup_j Y_j$  be open affine coverings. Then  $X \times Y = \bigcup_{i,j} X_i \times Y_j$  is an open affine cover by the first point, so  $X \times Y$  is algebraic. ■

Note that since  $A \otimes_k k[x_1, \dots, x_n] \cong A[x_1, \dots, x_n]$  naturally, we see that

$$k[X \times \mathbb{A}^n] \cong k[X] \otimes_k k[\mathbb{A}^n] \cong k[X][x_1, \dots, x_n]$$

for  $X$  affine.

#### 4.1.5 Corollary

Let  $X, Y$  be algebraic varieties,  $X' \subseteq X, Y' \subseteq Y$  be open/closed subvarieties. Then  $X' \times Y' \subseteq X \times Y$  is an open/closed subvariety.

#### Proof

$X' \times Y'$  is open/closed in  $X \times Y$  and  $X' \times Y'$  is a variety. Furthermore we showed that the induced SWF structure on  $X' \times Y'$  from  $X \times Y$  is the same as the product of induced structures from  $X, Y$ . Thus  $X' \times Y'$  is a subvariety of  $X \times Y$ . ■

#### 4.1.6 Proposition

Let  $X, Y$  be algebraic varieties, then

- (1)  $\dim(X \times Y) = \dim X + \dim Y$ ,
- (2) If  $X$  and  $Y$  are irreducible, so is  $X \times Y$ .

#### Proof

First we show (2): if  $X, Y$  are irreducible varieties, then so is  $X \times Y$ . Suppose  $X \times Y = Z_1 \cup Z_2$  for  $Z_i$  closed, then define

$$X_i = \{x \in X \mid x \times Y \subseteq Z_i\} = \bigcap_{y \in Y} \{x \in X \mid (x, y) \in Z_i\}.$$

Notice that  $X = X_1 \cup X_2$ , and  $X_i$  are closed as each  $\{x \in X \mid (x, y) \in Z_i\}$  is closed as the preimage of  $Z_i$  under  $x \mapsto (x, y)$ . Thus  $X = X_i$  for some  $i$ , thus  $X \times Y \subseteq Z_i$  and we have the desired result.

Note that the map  $x \mapsto (x, y)$  is continuous as its components are the identity and the constant map, both of which are continuous. So it is even a morphism of SWFs.

Now we claim that if  $X, Y$  are affine then  $\dim(X \times Y) = \dim X + \dim Y$ . Let  $\dim X = n, \dim Y = m$ , then there are finite surjective maps  $X \rightarrow \mathbb{A}^n, Y \rightarrow \mathbb{A}^m$ . The product map  $X \times Y \rightarrow \mathbb{A}^{n+m}$  is then finite surjective, and thus  $\dim(X \times Y) = \dim(\mathbb{A}^{n+m}) = n + m$  as desired.

Then in general for  $X = \bigcup_i X_i, Y = \bigcup_j Y_j$  open affine covers,  $X \times Y = \bigcup_{i,j} X_i \times Y_j$ . Then we have

$$\dim(X \times Y) = \max_{i,j} \dim(X_i \times Y_j) = \max_{i,j} (\dim X_i + \dim Y_j) = \max_i \dim X_i + \max_j \dim Y_j = \dim X + \dim Y$$

as desired. ■

#### 4.1.7 Theorem

The product  $\mathbb{P}^n \times \mathbb{P}^m$  is projective.

#### Proof

Define

$$S: \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{(n+1)(m+1)-1}, \quad (\ell_1, \ell_2) \mapsto \ell_1 \otimes \ell_2.$$

Explicitly, this maps lines  $\ell_1 \subseteq k^{n+1}$  and  $\ell_2 \subseteq k^{m+1}$  to  $\ell_1 \otimes \ell_2 \subseteq k^{(n+1)(m+1)}$  which we then project down to  $\mathbb{P}^{(n+1)(m+1)-1}$ . Alternatively  $S$  maps  $[x_0 : \dots : x_n]$  and  $[y_0 : \dots : y_m]$  to

$$[z_{ij}]_{0 \leq i \leq n, 0 \leq j \leq m}, \quad z_{ij} = x_i \cdot y_j.$$

We claim that  $S$  is a closed embedding, so  $\text{im} S$  is closed and  $S$  induces an isomorphism  $S: \mathbb{P}^n \times \mathbb{P}^m \xrightarrow{\sim} \text{im} S$ . Being a morphism, and even a closed embedding, is local on the target.

So take the standard affine open covering  $D(Z_{ij}) \subseteq \mathbb{P}^{(n+1)(m+1)-1}$ , then

$$S^{-1}D(Z_{ij}) = \{(X, Y) \mid X_i Y_j \neq 0\} = D(X_i) \times D(Y_j) \subseteq \mathbb{P}^n \times \mathbb{P}^m.$$

So we need to show that  $D(X_i) \times D(Y_j) \rightarrow D(Z_{ij})$  is a closed embedding.

Both the source and target are affine varieties, so this is a closed embedding iff  $S^*$  is a surjective morphism of coordinate rings. Identifying  $D(X_i) \cong \mathbb{A}^n$  by  $[X_0 : \dots : X_n] \mapsto (X_0/X_i, \dots, X_n/X_i)$ . Let  $z_{rs} = Z_{rs}/Z_{ij}$ ,  $x_r = X_r/X_i$ , and  $y_s = Y_s/y_j$ , so  $k[D_{ij}] \cong k[z_{rs}]_{r,s}$ . Then

$$S^*: k[\mathbb{A}^{(n+1)(m+1)-1}] \rightarrow k[\mathbb{A}^n] \otimes_k k[\mathbb{A}^m] \cong k[x_0, \dots, x_n, y_0, \dots, y_m]$$

is given by mapping  $z_{rs} = Z_{rs}/Z_{ij}$  to  $X_r Y_s / X_i Y_j = x_r y_s$ . In particular  $S^*(z_{rj}) = x_r y_j = x_r$ , and similar for  $y$ , so that  $S^*$  is indeed surjective. ■

**4.1.8 Definition**

The closed embedding  $S$  in the above proof is called the **Segre embedding**.

**4.1.9 Corollary**

The product of projective varieties is projective, and the product of quasi-projective/quasi-affine varieties is quasi-projective/quasi-affine.

**Proof**

For  $X \subseteq \mathbb{P}^n, Y \subseteq \mathbb{P}^m$  closed, then  $X \times Y$  is a closed subvariety of  $\mathbb{P}^n \times \mathbb{P}^m$ , and thus projective. The rest follows. ■

**4.2 Applications to dimension theory****4.2.1 Definition**

For a variety  $X$ , consider the morphism  $\Delta_X: X \rightarrow X \times X$  given by  $\Delta_X(x) = (x, x)$ . This is a morphism since its components are the identity; it is called the **diagonal morphism**.

Note that the image  $\Delta_X(X) = \{(x, x) \mid x \in X\}$  is isomorphic to  $X$  via  $\Delta_X$ . Indeed, its inverse is the composition

$$\Delta_X(X) \hookrightarrow X \times X \xrightarrow{\text{proj}_1} X.$$

Thus for every  $Y_1, Y_2 \subseteq X$ , the diagonal morphism forms an isomorphism

$$Y_1 \cap Y_2 \xrightarrow{\sim} \Delta_X(Y_1 \cap Y_2) = \Delta_X(X) \cap (Y_1 \times Y_2).$$

The left-hand side has structure induced from  $X$ , while the right-hand side's structure is induced from  $X \times X$ .

**4.2.2 Example**

Let  $X = \mathbb{A}^n$ , then  $X \times X = \mathbb{A}^{2n}$  and  $\Delta(\mathbb{A}^n) \subseteq \mathbb{A}^{2n}$  is given by  $Z(x_1 - y_1, \dots, x_n - y_n)$ .

**4.2.3 Definition**

A subset  $C \subseteq \mathbb{A}^{n+1}$  is called a **cone** if it is  $k^\times$ -invariant and  $0 \in C$ .

If we let  $\pi: \mathbb{A}^{n+1} - 0 \rightarrow \mathbb{P}^n$  be the projection, we see that  $C$  is a cone iff  $C = \pi^{-1}(\pi(C - 0)) \cup 0$ . In particular the maps  $C \mapsto [C] = \pi(C - 0)$  and  $X \mapsto C(X) = \pi^{-1}(X) \cup 0$  define a bijection between cones  $C \subseteq \mathbb{A}^{n+1}$  and subsets of  $\mathbb{P}^n$ .

**4.2.4 Lemma**

- (1) For every closed  $X \subseteq \mathbb{P}^n$ , its cone  $C(X) \subseteq \mathbb{A}^{n+1}$  is closed.

(2) If  $X$  is irreducible, then so is  $C(X)$ .

(3)  $\dim C(X) = \dim X + 1$ .

### Proof

For  $X \subseteq \mathbb{P}^n$  closed,  $X = Z_{\mathbb{P}^n}(f_1, \dots, f_m)$  for homogeneous  $f_i$ . Then  $C(X) = Z_{\mathbb{A}^{n+1}}(f_1, \dots, f_m)$  is closed.

For (2), if  $X = Z_{\mathbb{P}^n}(f_1, \dots, f_m)$  is irreducible, then  $(f_1, \dots, f_m)$  is prime and thus  $C(X) = Z_{\mathbb{A}^{n+1}}(f_1, \dots, f_m)$  is irreducible.

If  $X = \bigcup_i X_i$  is a union of closed irreducible subsets, then  $C(X) = \bigcup_i C(X_i)$  is a union of closed irreducible subsets. Then  $\dim C(X) = \max \dim C(X_i)$ , so we can assume that  $X$  is closed irreducible. Since  $\pi^{-1}X = C(X) - 0$  is open dense in  $C(X)$ , it is sufficient to show that  $\dim \pi^{-1}X = \dim X + 1$ .

Consider the standard open affine covering  $\mathbb{P}^n = \bigcup_i U_i$ , then  $X = \bigcup_i X \cap U_i$  and  $\pi^{-1}X = \bigcup \pi^{-1}(X \cap U_i)$ . So it is sufficient to show

$$\dim \pi^{-1}(X \cap U_i) = \dim(X \cap U_i) + 1.$$

Notice that  $\pi^{-1}U_i = D(x_i)$ , and the map

$$(\pi, \text{proj}_i): \pi^{-1}U_i \rightarrow U_i \times (\mathbb{A}^1 - 0)$$

Is an isomorphism with inverse

$$s: ([x_0 : \dots : x_n], t) \mapsto \left( \frac{x_0}{x_i}t, \dots, \frac{x_n}{x_i}t \right).$$

These are both clearly inverses, and morphisms.

Thus

$$\pi^{-1}U_i \cong U_i \times (\mathbb{A}^1 - 0),$$

and this induces an isomorphism

$$\pi^{-1}(U_i \cap X) \cong (U_i \cap X) \times (\mathbb{A}^1 - 0).$$

This has dimension  $\dim(U_i \cap X) + 1$  as desired. ■

### 4.2.5 Theorem

- (1) Let  $X, Y \subseteq \mathbb{A}^n$  be closed irreducible subvarieties. Then every irreducible component of  $X \cap Y$  has dimension at least  $\dim X + \dim Y - n$ . In particular if  $X \cap Y$  is nonempty, then  $\dim(X \cap Y) \geq \dim X + \dim Y - n$ .
- (2) Let  $X, Y \subseteq \mathbb{P}^n$  be closed irreducible subvarieties such that  $\dim X + \dim Y \geq n$ . Then  $X \cap Y$  is nonempty and every irreducible component of  $X \cap Y$  is of dimension at least  $\dim X + \dim Y - n$ .

### Proof

For (1), there is an isomorphism

$$X \cap Y \xrightarrow{\sim} (X \times Y) \cap \Delta(\mathbb{A}^n) \subseteq \mathbb{A}^{2n}.$$

Since  $X$  and  $Y$  are irreducible, so is  $X \times Y$  and  $\dim(X \times Y) = \dim X + \dim Y$ . Furthermore,

$$(X \times Y) \cap \Delta(\mathbb{A}^n) = Z_{X \times Y}(x_1 - y_1, \dots, x_n - y_n) \subseteq X \times Y$$

is the zeroset of  $n$  regular functions and thus the dimension of its irreducible components are each at least  $\dim(X \times Y) - n = \dim X + \dim Y - n$ .

For (2), we know that  $C(X), C(Y)$  are closed irreducible subsets of  $\mathbb{A}^{n+1}$ . Moreover,  $0 \in C(X) \cap C(Y)$  so they have nonempty intersection. Thus from (1)

$$\dim C(X) \cap C(Y) \geq \dim C(X) + \dim C(Y) - (n+1) = (\dim X + 1) + (\dim Y + 1) - (n+1) = \dim X + \dim Y - n + 1 \geq 1.$$

In particular  $C(X) \cap C(Y)$  is not finite, so there is an  $x \in C(X) \cap C(Y) - 0$  so  $\pi(x) \in X \cap Y$ , meaning  $X \cap Y$  is nonempty.

Let  $Z \subseteq X \cap Y$  be an irreducible component, then choose  $U_i$  such that  $Z \cap U_i \neq \emptyset$ . Then  $X \cap U_i, Y \cap U_i$  are closed irreducible subsets of  $U_i \cong \mathbb{A}^n$  and  $Z \cap U_i$  is an irreducible component of their intersection. The result then follows from (1),

$$\dim Z = \dim(Z \cap U_i) \geq \dim(X \cap U_i) + \dim(Y \cap U_i) - n = \dim X + \dim Y - n. \quad \blacksquare$$

#### 4.2.6 Corollary

Let  $f_1, \dots, f_m$  be homogeneous non-scalar polynomials in  $k[x_0, \dots, x_n]$ , with  $m \leq n$ . Then the closed set  $Z = Z_{\mathbb{P}^n}(f_1, \dots, f_m)$  is nonempty, and the dimension of each irreducible component of  $Z$  is at least  $n - m$ .

#### Proof

We induct on  $m$ , for  $m = 1$  this is clear. Let  $Z_m = Z_{\mathbb{P}^n}(f_1, \dots, f_m)$  then inductively we have

$$Z_{m+1} = Z_m \cap Z_{\mathbb{P}^n}(f_{m+1})$$

and  $\dim Z_m + \dim Z_{\mathbb{P}^n}(f_{m+1}) \geq n - m + (n - 1) = n + (n - (m + 1)) \geq n$  since  $m + 1 \leq n$ . Thus  $Z_{m+1}$  is nonempty by the above theorem. And an irreducible component of  $Z_{m+1}$  has dimension at least  $\dim Z_m + \dim Z_{\mathbb{P}^n}(f_{m+1}) - n \geq n - (m + 1)$  as desired.  $\blacksquare$

#### 4.2.7 Lemma

Let  $\varphi: X \rightarrow Y$  be a dominant morphism of affine irreducible varieties. Then there exists an open dense subset  $U \subseteq Y$  such that  $\varphi|_{\varphi^{-1}U}: \varphi^{-1}U \rightarrow U$  decomposes as

$$\varphi = \varphi^{-1}U \xrightarrow{g} U \times \mathbb{A}^n \xrightarrow{\text{proj}_U} U$$

where  $g$  is a finite surjective morphism.

#### Proof

Since  $\varphi$  is dominant,  $\varphi^*: k[Y] \rightarrow k[X]$  is injective. Since  $k[X]$  is a fg  $k$ -algebra, it is a fg  $k[Y]$ -algebra. Both  $X, Y$  are irreducible and thus  $k[X], k[Y]$  are domains so we can consider their fraction fields. Let  $\mathfrak{p} = 0 \subseteq k[Y]$ , then the localization  $k[X]_{\mathfrak{p}}$  is a fg  $k[Y]_{\mathfrak{p}}$ -algebra.

By Noether normalization, there are  $y_1, \dots, y_n \in k[X]_{\mathfrak{p}}$  which are algebraically independent over  $k(Y)$  and  $k[X]_{\mathfrak{p}}$  is finite over  $k(Y)[y_1, \dots, y_n]$ .

We claim there is an  $f \in k[Y]$  such that  $y_1, \dots, y_n \in k[X]_f$  and  $k[X]_f$  is a finite  $k[Y]_f[y_1, \dots, y_n]$ -algebra.

Since  $y_1, \dots, y_n \in k[X]_{\mathfrak{p}}$  there is an  $f' \in k[Y]$  such that  $y_1, \dots, y_n \in k[X]_{f'}$  (take the product of the denominators of the  $y_i$ s). Let  $a_1, \dots, a_r$  generate  $k[X]$  as a  $k[Y]$ -algebra. Then  $a_1, \dots, a_r$  are integral over



$k(Y)[y_1, \dots, y_n]$  as  $k[X]$  is finite over it, and thus integral over  $k[Y]_{f''}[y_1, \dots, y_n]$  for some  $f'' \in k[Y]$  as we can take common denominators. Then  $f = f'f''$  does the job.

With this in mind, the embeddings

$$k[Y]_f \hookrightarrow k[Y]_f[y_1, \dots, y_n] \hookrightarrow k[X]_f$$

define morphisms  $D_X(f) \xrightarrow{g} D_Y(f) \times \mathbb{A}^n \xrightarrow{\text{proj}} D_Y(f)$  of affine varieties. This is because these are all affine and

$$k[D_Y(f) \times \mathbb{A}^n] \cong k[D_Y(f)] \otimes_k k[\mathbb{A}^n] \cong k[Y]_f \otimes_k k[\mathbb{A}^n] \cong k[Y]_f[y_1, \dots, y_n].$$

Since  $k[Y]_f[y_1, \dots, y_n] \hookrightarrow k[X]_f$  is injective and finite,  $g$  is surjective and finite. ■

#### 4.2.8 Theorem

Let  $\varphi: X \rightarrow Y$  be a dominant morphism of irreducible varieties. Then  $f(X)$  contains a dense open subset  $U \subseteq Y$  such that for every  $y \in U$ ,  $\varphi^{-1}y$  is nonempty and each irreducible component of  $\varphi^{-1}y \subseteq X$  is of dimension  $\dim X - \dim Y$ .

#### Note:

Previously we showed that for any morphism of irreducible varieties, the fibers have dimension  $\geq \dim X - \dim Y$ . This theorem strengthens this for dominant morphisms.

#### Proof

For  $X$  and  $Y$  affine, then by the previous lemma there is a dense open  $U \subseteq Y$  such that  $\varphi$  factors as

$$\varphi^{-1}U \xrightarrow{g} U \times \mathbb{A}^n \xrightarrow{\text{proj}} U$$

where  $g$  is finite surjective. Since  $U \subseteq Y$  and  $\varphi^{-1}U \subseteq X$  are open dense,

$$\dim X = \dim \varphi^{-1}U = \dim U \times \mathbb{A}^n = \dim U + n = \dim Y + n.$$

So  $n = \dim X - \dim Y$ .

And for  $y \in U$ ,  $g: \varphi^{-1}U \rightarrow U \times \mathbb{A}^n$  induces a finite surjective morphism  $g: \varphi^{-1}y \rightarrow \mathbb{A}^n$ . Thus  $\dim \varphi^{-1}y = n$ , so  $\varphi^{-1}y \neq \emptyset$  and each irreducible component has dimension  $\leq n = \dim X - \dim Y$ . We know in general the dimension is at least  $\dim X - \dim Y$  so we have equality.

In general, we can replace  $Y$  with a dense open affine subset  $U \subseteq Y$  and  $X$  by  $\varphi^{-1}U$ . So we can assume that  $Y$  is affine. Then choose an affine covering  $X = \bigcup_i X_i$ , then  $\varphi|_{X_i}: X_i \rightarrow Y$  is a dominant morphism of affine varieties (since  $X_i$  is dense,  $\varphi(X_i)$  is dense in  $\varphi(X)$  which is then dense in  $Y$ ). Thus each  $\varphi(X_i)$  contains an open dense  $U_i$  such that for  $y \in U_i$ , the irreducible components of  $\varphi|_{X_i}^{-1}y = \varphi^{-1}y \cap X_i$  all have dimension  $\dim X_i - \dim Y = \dim X - \dim Y$ .

Let  $U = \bigcap_i U_i$ , then for each  $y \in U$ ,  $\varphi^{-1}y = \bigcup_i \varphi^{-1}y \cap X_i$  is an open cover. Since the irreducible components of  $\varphi^{-1}y \cap X_i$  have dimension  $\dim X - \dim Y$ , the same is true for  $\varphi^{-1}y$ . ■

What kinds of sets can be the images of morphisms of varieties? If  $Y$  is a variety and  $X \subseteq Y$  is locally closed, then it itself is a variety and thus the image of the inclusion  $X \hookrightarrow Y$ . Furthermore, if we have a finite collection of morphisms  $\varphi_i: X_i \rightarrow Y$ , then they define a morphism  $\varphi: \coprod_i X_i \rightarrow Y$  whose image is the union of images of  $\varphi_i$ . Thus the possible images of morphisms are closed under finite unions, in particular it contains finite unions of locally closed sets.

#### 4.2.9 Definition

A **constructible** subset of a topological space is a finite union of locally closed subsets.

Note that the collection of constructible subsets of a space  $X$  is the minimal collection containing all open subsets of  $X$  that is also closed under finite unions and complements. This is because it must contain closed sets, as the union of open sets, and be closed under intersections. Thus it must contain all locally closed sets, as the intersection of closed and open sets, and thus all constructible sets.

In other words, the collection of constructible subsets of a topological space is the algebra generated by its open subsets.

Chevalley's theorem implies that the image of a morphism of varieties is always constructible.

#### 4.2.10 Theorem (Chevalley)

Let  $f: X \rightarrow Y$  be a morphism of varieties. Then the image under  $f$  of any constructible subset of  $X$  is a constructible subset of  $Y$ .

#### Proof

Since direct images commute with unions, it is sufficient to show that the image of any locally closed subset  $S \subseteq X$  is constructible. Since locally closed subsets of a variety are varieties themselves, we can assume that  $S = X$  and show that  $f(X)$  is constructible.

We claim that for every closed  $Z \subseteq X$ ,  $f(Z) \subseteq Y$  is constructible. Consider the set of all closed  $Z \subseteq X$  such that  $f(Z)$  is not constructible. If it is non-empty, because  $X$  is Noetherian it contains a minimal subset  $Z_0$ . Replacing  $X$  by  $Z_0$  and  $f$  by  $f|_{Z_0}$ , we show that  $f(X)$  is constructible if  $f(Z)$  is constructible for all closed  $Z \subsetneq X$ .

Assume not, then  $X$  must be irreducible as otherwise there exists a proper closed union  $X = X_1 \cup X_2$ . Then  $f(X) = f(X_1) \cup f(X_2)$  is a union of constructible sets and thus constructible.

Because  $X$  is irreducible, so too is  $f(X)$  and thus  $\overline{f(X)} \subseteq Y$ . Because being locally closed is transitive, it is sufficient to show that  $f(X)$  is constructible in  $\overline{f(X)}$ . So we may assume  $\overline{f(X)} = Y$ ; that is, we may assume  $Y$  is irreducible and  $f$  is dominant.

Thus  $f(X)$  contains an open dense subset  $U \subseteq Y$  such that for every  $y \in U$ , the fiber  $f^{-1}y$  is nonempty and all of its irreducible components are of dimension  $\dim X - \dim Y$ . In particular it contains an open subset  $U$ . Since  $f(X) = U \cup (f(X) - U)$ , it is sufficient to show that  $f(X) - U$  is constructible. We have that

$$f(X) - U = f(X - f^{-1}U) \subseteq Y$$

and  $X - f^{-1}U$  is closed in  $X$  and thus by assumption its image is constructible. ■

#### 4.2.11 Lemma

Let  $X$  be a Noetherian topological space. Then  $C \subseteq X$  is closed iff for every irreducible closed subset  $Z \subseteq X$  such that  $C \cap Z$  is dense in  $Z$ , one has  $Z \subseteq C$ .

#### Proof

If  $C$  is closed and  $Z \cap C$  is dense in  $Z$ , then it is closed and dense in  $Z$  and thus equal to  $Z$ . So  $Z \subseteq C$  as desired.

Conversely let  $\overline{C} = \bigcup_i Z_i$  be the irreducible components of  $\overline{C}$ . As  $C$  is dense in  $\overline{C}$ ,  $C \cap Z_i$  is dense in  $Z_i$  for all  $i$ . Thus  $Z_i \subseteq C$  for all  $i$  by assumption so  $\overline{C} = C$ . ■

Now consider the concept of dimension, which is very much a global one. For example if  $X$  is the union of a line and a plane, its dimension is 2. But for points lying on the line and not the plane, locally  $X$  looks one-dimensional. This is a bit unfortunate as many properties are local. So we remedy this.

**4.2.12 Definition**

Let  $X$  be a topological space, and  $x \in X$ . Then  $X$ 's **Krull dimension at  $x$**  is

$$\dim_x X = \min \{ \dim U \mid x \in U \text{ open} \}.$$

Note that if  $x \in Z$  is irreducible, then  $\dim Z = \dim(Z \cap U)$  for all  $U$  open with nonempty intersection with  $Z$ . Conversely if  $X$  is Noetherian and thus has finitely many irreducible components,  $\dim U = \max_Z \dim(Z \cap U)$  where the maximum runs over the collection of irreducible subsets of  $X$ . Thus if  $X$  is Noetherian

$$\dim_x X = \max \{ \dim Z \mid x \in Z \text{ is an irreducible component of } X \}.$$

In particular this is true for varieties.

**4.2.13 Theorem**

Let  $\varphi: X \rightarrow Y$  be a morphism of varieties. Then for all  $n$ , the set

$$\{x \in X \mid \dim_x f^{-1}(f(x)) \geq n\} \subseteq X$$

is closed.

**Proof**

We use the previous lemma. So let  $Z \subseteq X$  be irreducible closed such that  $Z^{(n)} = \{z \in Z \mid \dim_z f^{-1}f(z) \geq n\}$  is dense in  $Z$ . We want to show that  $Z = Z^{(n)}$ .

We can replace  $(X, Y)$  by  $(f^{-1}(\overline{f(Z)}), \overline{f(Z)})$  and thus assume that  $f$  maps  $Z$  dominantly into  $Y$  and thus  $Y$  is irreducible. Let  $X_1, \dots, X_m$  be the irreducible components of  $X$ , and let  $f_i: X_i \rightarrow Y$  be the restriction of  $f$  to  $X_i$ . Then define

$$Z_i^{(n)} = \{z \in Z \cap X_i \mid \dim_z f_i^{-1}f_i(z) \geq n\}$$

so that  $Z^{(n)} = \bigcup_i Z_i^{(n)}$ .

As  $Z$  is irreducible and  $Z^{(n)}$  is dense in  $Z$ , it follows that  $Z_i^{(n)}$  is dense in  $Z$  for some  $i$ . Thus  $Z \cap X_i$  is dense in  $Z$  for that  $i$ , and thus  $Z \subseteq X_i$ . So we may replace  $X$  with  $X_i$  and assume that  $X$  is irreducible, and thus  $f$  is dominant.

There is a dense open  $U \subseteq Y$  such that for all  $y \in U$ ,  $\dim f^{-1}y = \dim X - \dim Y$ . Because  $f$  maps  $Z$  dominantly into  $Y$ ,  $Z^{(n)} \cap f^{-1}U \neq \emptyset$ . For  $z$  in this intersection

$$\dim X - \dim Y = \dim(f^{-1}f(z)) \geq \dim_z f^{-1}f(z) \geq n,$$

and thus  $\dim X - \dim Y \geq n$ .

Conversely, for every  $x \in X$  every irreducible component of  $f^{-1}f(x)$  has dimension  $\geq \dim X - \dim Y \geq n$ . In particular for  $x \in Z$ , so  $x \in Z^{(n)}$  and thus we have equality as desired. ■

**Note:**

A map  $f: X \rightarrow \mathbb{R}$  is **upper semicontinuous** if  $f^{-1}(-\infty, c)$  is open for all  $c \in \mathbb{R}$ . Equivalently  $f^{-1}[c, \infty)$  is closed.

Thus this theorem can be reformulated as  $x \mapsto \dim_x f^{-1}f(x)$  is upper semicontinuous for any morphism  $f$  of varieties.

**4.3 Separated and complete varieties**

Recall that for a topological space  $X$ ,  $X$  is Hausdorff iff its diagonal  $\Delta(X) \subseteq X \times X$  is closed. For varieties, the product  $X \times X$  is not the topological product and thus this does not imply Hausdorff-ness. In fact non-trivial affine, and thus all non-trivial varieties, are not Hausdorff.

Nevertheless this property is still important.

#### 4.3.1 Definition

A variety  $X$  is **separated** if its diagonal  $\Delta(X)$  is closed in the product variety  $X \times X$ .

For a morphism  $\varphi: X \rightarrow Y$  of varieties, let  $\Gamma_\varphi: X \rightarrow X \times Y$  be the morphism corresponding to  $(\text{id}, \varphi)$ , i.e.  $\Gamma_\varphi(x) = (x, \varphi(x))$ . Define  $\Gamma(\varphi)$  to be the image of  $\Gamma_\varphi$  in  $X \times Y$ , which then has the induced structure of a SWF.  $\Gamma(\varphi)$  is isomorphic to  $X$  by the inverse morphisms

$$X \xrightarrow{\Gamma_\varphi} \Gamma(\varphi) \hookrightarrow X \times Y \xrightarrow{\text{proj}_X} X.$$

For example,  $\Gamma_{\text{id}}$  is the diagonal morphism  $\Delta_X$  and its image is the diagonal  $\Gamma(\text{id}) = \Delta(X)$ .

#### 4.3.2 Lemma

- (1) Let  $\varphi: X \rightarrow Y$  be a morphism of varieties with  $Y$  separated. Then the graph  $\Gamma(\varphi) \subseteq X \times Y$  is closed.
- (2) If  $X$  is separated and  $Y \subseteq X$  is a subvariety, then  $Y$  is separated.
- (3) The product of separated varieties is separated.
- (4) Quasi-affine and quasi-projective varieties are separated.

#### Proof

- (1) Notice that

$$\Gamma(\varphi) = (\varphi \times \text{id}_Y)^{-1} \Delta(Y) \subseteq X \times Y$$

and it is therefore closed.

- (2) Since  $X$  is separated,  $\Delta(X) \subseteq X \times X$  is closed. Thus  $\Delta(Y) = \Delta(X) \cap (Y \times Y) \subseteq Y \times Y$  is closed.
- (3) Consider the projections  $\text{proj}_X: (X \times Y)^2 \rightarrow X^2$  and  $\text{proj}_Y: (X \times Y)^2 \rightarrow Y^2$ . Then

$$\Delta(X \times Y) = \text{proj}_X^{-1}(\Delta(X)) \cap \text{proj}_Y^{-1}(\Delta(Y))$$

is therefore closed as  $X$  and  $Y$  are separated.

- (4) As quasi-affine and quasi-projective varieties are subvarieties of  $\mathbb{A}^n$  and  $\mathbb{P}^n$ , it is sufficient to show that these are separated by (2). For  $\mathbb{A}^n$  this is clear because  $\Delta(\mathbb{A}^n) = Z(x_1 - y_1, \dots, x_n - y_n) \subseteq \mathbb{A}^n \times \mathbb{A}^n$ .

Recall the Segre embedding, which is closed,

$$S: \mathbb{P}^n \times \mathbb{P}^n \rightarrow \mathbb{P}^{n^2+2n}, \quad ([x_0 : \dots : x_n], [y_0 : \dots : y_n]) \mapsto [z_{i,j}],$$

where  $z_{i,j} = x_i y_j$ . We claim that

$$\Delta(\mathbb{P}^n) = S^{-1}(Z(z_{ij} - z_{ji}))$$

and this is clearly sufficient.

Indeed,  $S([x_0 : \dots : x_n], [y_0 : \dots : y_n]) \in Z(z_{ij} - z_{ji})$  iff  $x_i y_j = x_j y_i$  for all  $i, j$ . This is then iff  $[x_0 : \dots : x_n] = [y_0 : \dots : y_n]$  as desired. ■

**4.3.3 Lemma**

For every morphism  $\varphi: X \rightarrow Y$ , then graph  $\Gamma(\varphi) \subseteq X \times Y$  is locally closed.

**Proof**

Since  $\Gamma(\varphi) = (\varphi \times \text{id})^{-1}\Delta(Y)$ , it is sufficient to show that  $\Delta(Y) \subseteq Y \times Y$  is locally closed. Choose an affine covering  $Y = \bigcup_i Y_i$  and define  $W = \bigcup_i (Y_i \times Y_i)$ . We show that  $\Delta(Y) \subseteq W$  is closed.

Since being closed is a local property, it is sufficient to show that  $\Delta(Y) \cap (Y_i \times Y_i)$  is closed. Indeed,

$$\Delta(Y) \cap (Y_i \times Y_i) = \Delta(Y_i) \subseteq Y_i \times Y_i$$

is closed since  $Y_i$  is affine and thus separated. ■

**4.3.4 Proposition**

Let  $X$  be a variety, then the following are equivalent.

- (1)  $X$  is separated.
- (2) For every open affine  $U, V \subseteq X$ , their intersection is affine, and the natural map  $k[U] \otimes k[V] \rightarrow k[U \cap V]$  is surjective.
- (3) There exists an open affine covering  $X = \bigcup_i X_i$  such that  $X_i \cap X_j$  is affine for all  $i, j$  and  $k[X_i] \otimes k[X_j] \rightarrow k[X_i \cap X_j]$  is surjective.

**Proof**

Assume (1), then since  $X$  is separated

$$U \cap V \cong \Delta(X) \cap (U \times V) \subseteq U \times V$$

is closed. As  $U \times V$  is affine, so too is  $U \cap V$ . Furthermore the restriction map

$$k[U] \otimes k[V] \xrightarrow{\sim} k[U \times V] \rightarrow k[U \cap V]$$

is surjective. Notice that the composition

$$k[U] \rightarrow k[U] \otimes k[V] \xrightarrow{\sim} k[U \times V] \rightarrow k[U \cap V]$$

is the restriction map, and similarly for  $k[V]$ . Thus the surjection is the natural map as desired, showing (2).

(2) implies (1) is clear.

Now suppose that we have an affine covering as in (3). Then  $X \times X = \bigcup_{i,j} X_i \times X_j$  is an open covering. It is sufficient to show that  $\Delta(X) \cap (X_i \times X_j)$  is closed in  $X_i \times X_j$  for all  $i, j$ . That is,  $X_i \cap X_j$  is closed in  $X_i \times X_j$ . Since  $X_i \cap X_j$  is affine and  $k[X_i \times X_j] \rightarrow k[X_i \cap X_j]$  is surjective, we have the desired result. ■

**4.3.5 Definition**

A variety  $X$  is **universally closed** if for every other variety  $Y$ , the projection  $X \times Y \rightarrow Y$  is a closed

map.

A separated universally closed variety is called **complete**.

#### 4.3.6 Lemma

- (1) A closed subvariety of a complete variety is complete.
- (2) If  $\varphi: X \rightarrow Y$  is a morphism of varieties, with  $X$  complete and  $Y$  separated, then  $\varphi(X) \subseteq Y$  is closed and  $\varphi(X)$  is a complete variety.

#### Proof

- (1) Suppose  $Y \subseteq X$  is a closed subvariety of a complete variety. Then  $Y$  is separated as a subvariety of a separated variety. And for every  $Z$ , projection factorizes as

$$Y \times Z \hookrightarrow X \times Z \rightarrow Z$$

where both maps are closed (the first because  $Y$  is closed and the second because  $X$  is complete).

- (2)  $\varphi$  factors as

$$X \xrightarrow{\Gamma_\varphi} X \times Y \xrightarrow{\pi_Y} Y$$

and so  $\varphi(X) = \pi_Y(\Gamma(\varphi))$ . Since  $Y$  is separated  $\Gamma(\varphi)$  is closed and since  $X$  is complete we see that  $\varphi(X)$  is closed.

And  $\varphi(X)$  is separated because  $Y$  is. And for any  $Z$ , consider the surjection  $\varphi \times \text{id}: X \times Z \rightarrow \varphi(X) \times Z$ . Then for any closed  $C \subseteq \varphi(X) \times Z$ , the preimage  $C' = (\varphi \times \text{id})^{-1}C$  is closed and thus

$$\text{proj}_Z^{\varphi(X)}(C) = \text{proj}_Z^{\varphi(X)} \circ (\varphi \times \text{id}) \circ (\varphi \times \text{id})^{-1}(C) = \text{proj}_Z^{\varphi(X)} \circ (\varphi \times \text{id})(C') = \text{proj}_Z^X(C')$$

is closed. ■

#### 4.3.7 Example

$\mathbb{A}^1$  is not complete, since the projection  $\pi: \mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$  maps

$$Z(xy - 1) = \{(x, y) \mid xy = 1\} \mapsto \{y \neq 0\} = D(y) \subseteq \mathbb{A}^1$$

which is not closed.

Alternatively,  $\mathbb{P}^1$  is separated but the image of the embedding  $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$  is not closed. And thus by the previous lemma,  $\mathbb{A}^1$  is not complete.

#### 4.3.8 Corollary

If  $X$  is complete, then every  $f \in k[X]$  is locally constant.

#### Proof

A function  $f \in k[X]$  defines a morphism  $f: X \rightarrow \mathbb{A}^1$  whose image is then closed and complete. Thus  $f(X)$  is either finite or all of  $\mathbb{A}^1$ . Since  $\mathbb{A}^1$  is not complete,  $f(X)$  is finite. Since finite subsets of  $\mathbb{A}^1$  are disconnected, for every connected component  $X'$  of  $X$ ,  $f(X')$  is connected and then must be a point. ■

#### 4.3.9 Corollary

Quasi-affine complete varieties are finite.

#### Proof

Since  $X$  is quasi-affine, every two points in  $X$  are separated by a global section. That is, for all  $a \neq b \in X$  there is an  $f \in k[X]$  such that  $f(a) \neq f(b)$ . Indeed, we may assume  $X \subseteq \mathbb{A}^n$  and then one of the coordinate functions does the job.

Since  $X$  is complete,  $f$  must be locally constant, and thus  $a$  and  $b$  must belong to different connected components. That is, the singletons of  $X$  are the connected components. But  $X$  has finitely many connected components, as it is Noetherian, so finitely many points. ■

#### 4.3.10 Theorem

Projective varieties are complete.

#### Proof

It is sufficient to show this for  $\mathbb{P}^n$  as closed subvarieties of complete varieties are complete. Projective  $n$ -space is separated, so we must just show universal closure. This is tedious (recitation).

**TODO: This**

#### Note:

For a complex variety  $X$ , we can endow it with the **complex analytic topology** as follows. For  $X = \mathbb{A}^n$ , the underlying set is  $\mathbb{C}^n$ , so we equip it with the usual Euclidean topology on  $\mathbb{C}^n$ . If  $X$  is affine, then choose a closed embedding  $X \hookrightarrow \mathbb{A}^n$  and equip it with the subspace topology. This is independent on choice of closed embedding as they are all isomorphic and thus homeomorphic.

Then for a general variety, choose an open affine cover  $X = \bigcup_i X_i$ , and give it the final topology with respect to the inclusions  $X_i \hookrightarrow X$ . That is,  $U \subseteq X$  is open iff  $U \cap X_i$  is open for all  $i$ . This is again independent on choice of affine cover.

Then it can be shown that  $X$  is separated/complete iff it is Hausdorff/compact Hausdorff in the complex analytic topology.

## 4.4 Rational maps

#### 4.4.1 Lemma

Let  $\varphi, \psi: X \rightarrow Y$  be morphisms of algebraic varieties with  $Y$  separated. Let  $U \subseteq X$  be dense open. If  $\varphi|_U = \psi|_U$  then  $\varphi = \psi$ .

**Proof**

Consider  $F = (\varphi, \psi): X \rightarrow Y \times Y$ , and since  $Y$  is separated  $\Delta(Y) \subseteq Y \times Y$  is closed. Furthermore  $F(U) \subseteq \Delta(Y)$ , and so  $U \subseteq F^{-1}\Delta(Y)$ . Since  $F^{-1}\Delta(Y)$  is closed, this means  $F^{-1}\Delta(Y) = X$ , and since

$$F^{-1}\Delta(Y) = \{x \in X \mid \varphi(x) = \psi(x)\},$$

we are finished. ■

**Note:**

This is not true when  $Y$  is not separated. For example let  $Y = (\mathbb{A}^1 \times \{0\}) \sqcup (\mathbb{A}^1 \times \{1\}) / \sim$  where we identify  $(x, 0) \sim (x, 1)$  for  $x \neq 0$ . Let  $\varphi, \psi: \mathbb{A}^1 \rightarrow Y$  be given by  $\varphi(x) = [(x, 0)]$  and  $\psi(x) = [(x, 1)]$ . These agree on the open dense  $\mathbb{A}^1 - 0$  but are not equal at  $x = 0$ .

Notice that since  $\mathbb{A}^1$  is separated, this means that if  $f, g \in k[X]$  agree on a dense open subset, they agree on all of  $X$ . This can also be shown directly, since the set that  $f$  and  $g$  agree on is  $Z(f - g)$  which is closed; if it contains a dense subset it must be all of  $X$ .

**4.4.2 Definition**

Let  $X, Y$  be varieties with  $Y$  separated. A **rational map** is an equivalence class of pairs  $(U, \varphi_U)$  where  $U \subseteq X$  is open dense and  $\varphi_U: U \rightarrow Y$  is a morphism. We identify pairs  $(U, \varphi_U) \sim (V, \varphi_V)$  iff  $\varphi_U$  and  $\varphi_V$  agree on an open dense  $W \subseteq U \cap V$ . Note that this is equivalent (by the above lemma) to agreeing on  $U \cap V$ .

We write  $\varphi: X \dashrightarrow Y$  to mean that  $\varphi$  is a rational map from  $X$  to  $Y$  (i.e. an equivalence class). Denote the set of rational maps  $X \dashrightarrow Y$  by  $\text{rat}(X, Y)$ .

**4.4.3 Lemma**

Every rational map  $\varphi: X \dashrightarrow Y$  has a largest dense open subset  $U_\varphi \subseteq X$  such that  $\varphi$  has a representative  $U_\varphi \rightarrow Y$ .

This  $U_\varphi$  is called the **domain of definition** of  $\varphi$ .

**Proof**

Note that we can glue morphisms together ( $\text{hom}(\bullet, Y)$  is a sheaf on  $X$ ). So let  $U_\varphi$  be the union of all  $U$  where  $(U, \varphi_U)$  is a representative of  $\varphi$ . Since all the  $\varphi_U$  agree on their intersections this is well-defined and maximal. ■

Notice that for every open dense  $U \subseteq X$  there is a natural bijection

$$\text{rat}(X, Y) \cong \text{rat}(U, Y) \quad U \subseteq X \text{ open dense}$$

obtained by mapping  $(V, \varphi_V) \in \text{rat}(X, Y)$  to  $(V \cap U, \varphi_{V \cap U})$  and  $(V, \varphi_V) \in \text{rat}(U, Y)$  to  $(V, \varphi_V) \in \text{rat}(X, Y)$ .

Note that rational maps  $X \dashrightarrow \mathbb{A}^1$  are precisely the rational *functions* we defined to be  $k(X)$ . In general, we have that

$$\text{rat}(X, Y) = \text{colim}_{U \subseteq X} \text{hom}(U, Y)$$

where  $U$  range over open dense subsets of  $X$ . Thus  $\text{rat}(X, Y)$  is immediately seen to be a  $k$ -algebra.

Let  $X_1, \dots, X_n$  be the irreducible components of  $X$ . Then for  $U \subseteq X$  open dense,  $U \cap X_i \subseteq X_i$  is open dense if nonempty. Thus we have natural maps

$$\text{res}_{X_i}: \text{rat}(X, Y) \rightarrow \text{rat}(X_i, Y), \quad (\varphi: U \rightarrow Y) \mapsto (\varphi|_{X_i \cap U}: X_i \cap U \rightarrow Y).$$



**4.4.4 Lemma**

The product of the natural maps

$$\prod_i \text{res}_{X_i}: \text{res}(X, Y) \rightarrow \prod_i \text{res}(X_i, Y)$$

is an isomorphism.

**Proof**

We construct an explicit inverse. For each  $i$  consider  $X'_i = X_i - \bigcup_{j \neq i} X_j \subseteq X_i$ . This is disjoint from all other  $X_j$  and is open in  $X_i$ . As it is nonempty and  $X_i$  is irreducible, it is open dense in  $X_i$ . Thus  $X' = \bigsqcup_i X'_i \subseteq X$  is open dense.

For an  $n$ -tuple  $\varphi_i: X_i \dashrightarrow Y$  with representatives  $\varphi_{U_i}: U_i \rightarrow Y$  for  $U_i \subseteq X_i$  open dense, let  $U'_i = X'_i \cap U$  and  $U = \bigsqcup_i U'_i$ . Then  $U \subseteq X' \subseteq X$  is open dense, and since the  $U'_i$  are disjoint there is a unique  $\varphi: U \rightarrow Y$  such that  $\varphi|_{U'_i} = \varphi_i|_{U'_i}$  for all  $i$ .

Mapping  $(\varphi_i)_i \mapsto \varphi$  is the desired inverse.  $\blacksquare$

In lieu of this lemma, we may assume that  $X$  is irreducible.

**4.4.5 Definition**

A rational map  $\varphi: X \dashrightarrow Y$  is **dominant** if  $\varphi_U: U \rightarrow Y$  is dominant for some representative of  $\varphi$ . This implies every representative of  $\varphi$  is dominant.

If  $\varphi: X \dashrightarrow Y$  and  $\psi: Y \dashrightarrow Z$  are rational with  $\varphi$  dominant, we form their composition  $\psi \circ \varphi: X \dashrightarrow Z$  as follows. For any representatives  $\varphi_U: U \rightarrow Y, \psi_V: V \rightarrow Z$ , let  $W = \varphi_U^{-1}V$  which is nonempty since  $\varphi_U$  is dominant. Then define  $\psi \circ \varphi$  to be the class of  $\psi_V \circ (\varphi_U|_W)$ . This is independent on choice of representatives.

We can thus form a category whose objects are separated varieties over  $k$  and whose morphisms are the dominant rational maps between them. We will primarily focus on the subcategory whose objects are furthermore irreducible. Call these categories  $\text{RatVar}_k$  and  $\text{RatVar}_k^{\text{irr}}$  respectively. We let their hom-sets be denoted by  $\text{rat}_{\text{dom}}(X, Y)$ .

Furthermore, a dominant rational map  $\varphi: X \dashrightarrow Y$  defines the pullback  $\varphi^*: k(Y) \rightarrow k(X)$  by  $\varphi^*(f) = f \circ \varphi$ . We can compose these maps and we see that  $(\varphi_1 \circ \varphi_2)^* = \varphi_2^* \circ \varphi_1^*$ . This defines a contravariant functor from the aforementioned category to the category of  $k$ -algebras. As we consider the subcategory consisting of irreducible separated varieties, this functor maps into field extensions of  $k$ .

Recall that if  $X$  is affine (and irreducible) then  $k(X)$  is the field of fractions of  $k[X]$ , and then  $\varphi^*: k(X) \rightarrow k(Y)$  is the unique extension of  $\varphi^*: k[X] \rightarrow k[Y]$ .

**4.4.6 Definition**

Let  $X, Y$  be irreducible separable varieties. A **birational map** is a dominant rational map  $X \dashrightarrow Y$  which is an isomorphism in the category  $\text{RatVar}^{\text{irr}}$ .

$X$  and  $Y$  are **birationally equivalent** if they are isomorphic in  $\text{RatVar}^{\text{irr}}$ .

Since there is a natural embedding  $\text{hom}_{\text{Var}}(X, Y) \hookrightarrow \text{rat}(X, Y)$ , we can also call a dominant morphism of (separable, irreducible) varieties a birational map if the associated rational map is.

**4.4.7 Example**

If  $U \subseteq X$  is open dense, then the inclusion  $\iota: U \hookrightarrow X$  is a birational equivalence. Its inverse is the rational

map  $X \dashrightarrow U$  represented by  $(U, \text{id}_U)$ .

Note that for every birational equivalence  $\alpha: X' \dashrightarrow X$ , the induced map

$$\alpha^*: \text{rat}(X, Y) \rightarrow \text{rat}(X', Y), \quad \varphi \mapsto \varphi \circ \alpha$$

is a bijection. Its inverse is given by precomposition with  $\alpha$ 's inverse.

Postcomposition is only defined for dominant rational maps, in which case it is still a bijection. That is, for  $\alpha: Y \dashrightarrow Y'$  postcomposition is a bijection:

$$\alpha_*: \text{rat}_{\text{dom}}(X, Y) \xrightarrow{\sim} \text{rat}_{\text{dom}}(X, Y'), \quad \varphi \mapsto \alpha \circ \varphi.$$

#### 4.4.8 Theorem

There is an antiequivalence of categories  $\text{RatVar}_k^{\text{irr}}$  and finite field extensions of  $k$ . The equivalence is given by mapping  $X$  to  $k(X)$  and  $\varphi \mapsto \varphi^*$ .

#### Proof

We need to show that the functor is fully faithful and essentially surjective. We reduce full faithfulness to the fact that global sections is fully faithful on affine varieties.

Because  $X, Y$  are birationally equivalent to their nonempty open subsets, we can replace them with any affine subset. So we may assume  $X, Y$  are affine. We first show that  $\varphi \mapsto \varphi^*$  is injective. If  $\varphi_1, \varphi_2: X \dashrightarrow Y$  are rational maps, we can replace  $X$  with an affine subset of the intersection of the domains of  $\varphi_1, \varphi_2$ . So  $\varphi_1^*, \varphi_2^*: k(Y) \rightarrow k(X)$  are equal, and thus their restrictions  $\varphi_1^*, \varphi_2^*: k[Y] \rightarrow k[X]$  are equal. Since global sections is fully faithful on affine varieties,  $\varphi_1 = \varphi_2$ .

Now let  $\psi: k(Y) \rightarrow k(X)$  be a  $k$ -algebra morphism. Since  $k[Y] \subseteq k(Y)$  is a finitely generated  $k$ -subalgebra, there is a  $f \in k[X]$  such that  $\psi(k[Y]) \subseteq k[X]_f = k[D_X(f)]$ . Thus  $\psi|_{k[Y]}$  is a morphism  $k[Y] \rightarrow k[D_X(f)]$  and thus comes from a morphism  $\varphi: D_X(f) \rightarrow Y$  of affine varieties. Then  $\varphi$  defines a rational map  $D_X(f) \dashrightarrow Y$  whose pullback is  $\psi$ .

For essential surjectivity, let  $L/k$  be a finite field extension. Suppose  $a_1, \dots, a_n \in L$  generate the field extension. Let  $A = k[a_1, \dots, a_n]$  so that  $L = \text{Frac}(A)$  and  $A$  is a domain, finitely generated over  $k$ . Thus there is an affine variety  $X$  such that  $k[X] \cong A$  and thus

$$k(X) \cong \text{Frac}(k[X]) \cong \text{Frac}(A) \cong L$$

as desired. ■

The following two corollaries are immediate.

#### 4.4.9 Corollary

Let  $\varphi: X \dashrightarrow Y$  be a dominant rational map between irreducible separated varieties. The following are equivalent.

- (1)  $\varphi$  is birational.
- (2) There exist dense open subsets  $U \subseteq X, V \subseteq Y$  such that  $\varphi$  induces an isomorphism  $U \xrightarrow{\sim} V$ .
- (3) The pullback  $\varphi^*: k(Y) \rightarrow k(X)$  is an isomorphism.

#### 4.4.10 Corollary

Let  $X, Y$  be two irreducible separated varieties. The following are equivalent.

- (1)  $X$  and  $Y$  are birationally equivalent.
- (2) There exist dense open subsets  $U \subseteq X, V \subseteq Y$  such that  $U$  and  $V$  are isomorphic varieties.
- (3)  $k(X) \cong k(Y)$ .

### 4.5 Curves

#### 4.5.1 Definition

Let  $X$  be a variety and  $x \in X$ . The **cotangent space** of  $X$  at  $x$  is the  $k$ -vector space  $T_x^*X = \mathfrak{m}_x/\mathfrak{m}_x^2$  where  $\mathfrak{m}_x$  is the maximal ideal of the stalk  $\mathcal{O}_{X,x}$ . The **tangent space** of  $X$  at  $x$  is the dual space  $T_xX = (\mathfrak{m}_x/\mathfrak{m}_x^2)^*$ .

The motivation for this definition is due to the following definition and result.

#### 4.5.2 Definition

Assume  $X = Z_{\mathbb{A}^n}(f_1, \dots, f_m)$  is an affine variety, and let  $x \in X$ . The **Jacobian** of  $X$  at  $x$  is the linear map  $J_X(x): k^m \rightarrow k^n$  given by

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) & \cdots & \frac{\partial f_k}{\partial x_1}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_k}(x) & \cdots & \frac{\partial f_k}{\partial x_k}(x) \end{pmatrix}.$$

#### 4.5.3 Proposition

Let  $x \in X$  be a point in an algebraic variety. Then  $T_x^*X \cong \text{coker}(J_X(x))$ .

#### 4.5.4 Definition

A point  $x \in X$  is a **regular** (or **nonsingular/smooth**) point of  $X$  if  $\dim_k(T_x^*X) = \dim_x X$ . A variety is **regular** (or **nonsingular/smooth**) if all of its points are so.

The following two corollaries are then immediate.

#### 4.5.5 Corollary

A point  $x \in X$  is regular iff  $\text{corank}(J_X(x)) = n - \text{rank}(J_X(x)) = \dim_x X$ .

#### 4.5.6 Corollary

Let  $X = Z_{\mathbb{A}^n}(f)$ , with  $f$  nonconstant. Then by Krull's principal ideal theorem  $\dim_x X = n - 1$ . The

Jacobian of  $X$  at  $x$  is just the gradient  $\nabla f(x)$ , and  $X$  is smooth at  $x$  iff that gradient is nonzero.

#### 4.5.7 Definition

A variety  $X$  is **equidimensional** if all of its irreducible components have the same dimension. Equivalently  $\dim_x X$  is constant for all  $x \in X$ .

#### 4.5.8 Definition

A **curve** is an equidimensional variety of dimension 1.

Now we make a few remarks. If  $A$  is a Noetherian ring, and  $S \subseteq A$  is a multiplicative subset, then ideals of  $S^{-1}A$  are of the form  $S^{-1}I$ . Thus  $S^{-1}A$  is Noetherian: if we have a collection of ideals  $\Sigma$  of  $S^{-1}A$  let  $\Sigma_A$  be the collection of ideals  $I \leq A$  such that  $S^{-1}I \in \Sigma$ . Then  $\Sigma_A$  has a maximal ideal whose localization is then a maximal ideal of  $\Sigma$ .

In particular if  $Y$  is a variety, then  $\mathcal{O}_{Y,y}$  is Noetherian. Indeed, if  $y \in U$  is an affine neighborhood then  $\mathcal{O}_{Y,y}$  is isomorphic to a localization of  $k[U]$  which is Noetherian (by Hilbert's basis theorem).

Furthermore, for any  $y \in Y$ ,  $\dim_y Y = \dim(\mathcal{O}_{Y,y})$  where the right-hand side is the Krull dimension of the ring, i.e. the supremum on the length of chains of prime ideals. Indeed, because this claim is local we can assume  $Y$  is affine. Then  $\mathcal{O}_{Y,y} = k[Y]_{\mathfrak{m}_y}$ , so prime ideals of the stalk are in one-to-one correspondence with prime ideals of  $k[Y]$  contained in  $\mathfrak{m}_y$ . These correspond, in an order-reversing manner, to closed irreducible subsets of  $Y$  via  $Z(\bullet)$  which contain  $Z(\mathfrak{m}_y) = \{y\}$ . So any chain of prime ideals of  $k[Y]_{\mathfrak{m}_y}$  gives rise to a chain of closed irreducible subsets of  $Y$  containing  $y$ , which gives the desired result.

Also if  $Y$  is irreducible, then  $\mathcal{O}_{Y,y}$  can be viewed as a subring of  $k(Y)$ . This is obvious considering their definitions via colimits or even via germs. We can then view  $k(Y)$  as  $\text{Frac}(\mathcal{O}_{Y,y})$ .

Thus if  $X$  is a curve, so  $\dim_x X = 1$  for all  $x \in X$  and thus its stalks  $\mathcal{O}_{X,x}$  are all Noetherian rings of dimension 1. Furthermore if  $X$  is smooth at  $x \in X$  then  $1 = \dim_x X = \dim_k \mathfrak{m}_x / \mathfrak{m}_x^2$ . If  $\pi \in \mathfrak{m}_x$  generates the 1-dimensional space  $\mathfrak{m}_x / \mathfrak{m}_x^2$ , by Nakayama  $\pi$  generates the ideal  $\mathfrak{m}_x$ , so it is principal. This leads to the following definition.

#### 4.5.9 Definition

A **discrete valuation ring (DVR)** is a one-dimensional Noetherian local ring  $(A, \mathfrak{m})$  for which  $\mathfrak{m}$  is principal. A generator of  $\mathfrak{m}$  is called a **uniformizer** of  $A$ .

So stalks of curves at smooth points are DVRs.

By Nakayama,  $\pi \in \mathfrak{m}$  is a uniformizer iff  $\bar{\pi} \in \mathfrak{m}/\mathfrak{m}^2$  generates the module over  $A/\mathfrak{m} \cong k$ . Because  $\mathfrak{m} \neq \mathfrak{m}^2$ , as otherwise  $0 \in \mathfrak{m}$  would generate it by Nakayama and thus  $\mathfrak{m} = 0$ , we see that  $\pi \in \mathfrak{m}$  is a uniformizer iff  $\pi \in \mathfrak{m} - \mathfrak{m}^2$ . This is because such a uniformizer must exist, so  $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 1$  and so any non-zero element generates it.

#### 4.5.10 Lemma

Let  $(A, \mathfrak{m})$  be a DVR and  $\pi \in A$  be a uniformizer. Then  $A$  is a domain and every nonzero element of  $A$  can be written as  $\pi^n u$  for a unique  $n \in \mathbb{Z}_{\geq 0}$  and  $u \in A^\times$ .

#### Proof

Let  $0 \neq a \in A$ . If  $a$  is not a unit, then  $a \in \mathfrak{m}$  so  $a = \pi a_1$  for some  $0 \neq a_1 \in A$ . If  $a_1$  is not a unit, we can continue this. Thus we get an increasing chain of ideals  $(a) \subseteq (a_1) \subseteq \dots$ . Note that this chain is strictly

increasing: if  $a_{m+1} \in (a_m)$  then  $a_{m+1} = a_m b$  for  $b \in A$ , so  $a_m = \pi a_{m+1} = \pi b a_m$ . Thus  $(1 - \pi b)a_m = 0$ , but  $1 - \pi b \notin \mathfrak{m}$  and is thus invertible so  $a_m = 0$ , and thus  $a = 0$  a contradiction.

So this construction must terminate; at some point  $a_n = u$  is a unit. And  $a = \pi^n u$  as desired.

To show that  $A$  is a domain, because its elements can be written as  $\pi^n u$ , it is sufficient to show that  $\pi$  is not nilpotent. If it were, then  $\mathfrak{m}^r = 0$  for some  $r > 0$ . Thus  $\mathfrak{m} \subseteq \text{nil}(A)$  and so  $\dim A = \dim(A/\text{nil}(A)) = \dim(A/\mathfrak{m}) = 0$  since  $A/\mathfrak{m}$  is a field. But by assumption  $A$  has dimension one, so it is a domain.

Now for uniqueness of representation, it is sufficient to show that  $\pi^n$  cannot be written as  $\pi^m u$  for  $n \neq m$ ,  $u \in A^\times$ . So suppose  $\pi^n = \pi^m u$ , and we can assume  $n > m$ . Then  $\pi^m(u - \pi^{n-m}) = 0$  so because  $A$  is a domain,  $u = \pi^{n-m}$  which means  $\pi^{n-m}$  is a unit, and thus  $\pi$  is a unit. But this is impossible. ■

#### 4.5.11 Proposition

Let  $\varphi: X \rightarrow Y$  be a rational map from a curve  $X$  to a projective (and thus separated) variety  $Y$ . Then every smooth point of  $X$  lies in the domain of definition of  $\varphi$ .

#### Proof

First assume that  $Y = \mathbb{P}^n$ , and let  $x \in X$  be a smooth point. Then  $\mathcal{O}_{X,x}$  is a DVR and let  $\pi \in \mathcal{O}_{X,x}$  be a uniformizer. Thus  $\pi \in \mathcal{O}_X(U)$  for some open  $x \in U$ . Then  $Z_U(\pi)$  is a finite set containing  $x$ , and by potentially shrinking  $U$  we may assume  $Z_U(\pi) = \{x\}$ . We can shrink  $U$  further still so that  $\varphi$  defines a morphism  $U - x \rightarrow Y$  (since  $\varphi$ 's domain of definition is dense, it intersects  $U$ ). By restricting  $\varphi$  to the preimage of some  $V_i = D_{\mathbb{P}^n}(X_i) \cong \mathbb{A}^n$ , we may assume that  $\varphi$  lands in  $\mathbb{A}^n$  (as the  $V_i$  cover  $\mathbb{P}^n$ , one of the preimages must intersect  $U$ ).

Thus

$$\varphi|_{U-x} = \pi \circ (f_0, \dots, f_i = 1, \dots, f_n)$$

for  $f_j \in \mathcal{O}_X(U - x)$  and  $\pi: \mathbb{A}^{n+1} - 0 \rightarrow \mathbb{P}^n$  is the projection. Then each  $f_j$  lies in  $k(X) = \text{Frac}(\mathcal{O}_{X,x})$ , and so  $f_j$  can be written uniquely as  $f_j = \pi^{m_j} u_j$  for  $m_j \in \mathbb{Z}$  and  $u_j \in \mathcal{O}_{X,x}^\times$  (note here because it is in the fraction field,  $m_j$  may be nonnegative). Thus, by potentially shrinking  $U$  further, each  $u_j$  defines a non-vanishing regular function.

Let  $m = \min_j m_j$  so that  $\pi^{-m} f_j$  is regular on  $U$ , since  $\pi^{-m} f_j$  is a nonnegative power of  $\pi$  times  $u_j$ . Thus  $\pi \circ (\pi^{-m} f_0, \dots, \pi^{-m} f_n)$  is a regular function on all of  $U$ . Furthermore it agrees with  $\varphi$  on  $U - x$  (since  $\pi^{-m} f_i$  is a nonzero multiple of  $f_i$  and thus projecting it gives the same value as  $f_i$ ). Thus we can extend  $\varphi$  to include  $x$  in its domain, as desired.

For an arbitrary projective variety  $Y$ , it is closed in some  $\mathbb{P}^n$ . Let  $\iota: Y \hookrightarrow \mathbb{P}^n$  be the inclusion, so as before  $\iota \circ \varphi$  extends over some open neighborhood  $V$  of  $x$ . Let us denote this extended morphism by  $\psi$ , so  $\psi^{-1}Y \subseteq V$ . Thus  $\psi^{-1}Y$  is closed in  $V$  and also contains a dense open subset (the subset on which  $\varphi$  is defined) and thus must be equal to  $V$ . So  $\psi$  is a morphism  $V \rightarrow Y$ , so  $\varphi$  can be extended to a neighborhood of  $x$ ,  $V$ . ■

#### Note:

Both assumptions in the above result, that  $Y$  is projective and that  $x \in X$  is smooth, are critical. For example, the rational map  $\mathbb{P}^1 \rightarrow \mathbb{A}^1$  defined to be the identity on a copy of  $\mathbb{A}^1 \subseteq \mathbb{P}^1$  does not extend to  $\infty \in \mathbb{P}^1$  (which is the point whose first/second coordinate is zero). This is because it would define a non-constant morphism  $\mathbb{P}^1 \rightarrow \mathbb{A}^1$ , which projective spaces do not possess ( $k[\mathbb{P}^n] = k$ ).

And we need smoothness, as if we consider  $X = Z_{\mathbb{A}^2}(y^2 - x^2 - x^3)$  and the map  $X - 0 \rightarrow \mathbb{P}^1$  given by  $(x, y) \mapsto [y : x]$ , we cannot extend this  $(0, 0)$ .

Recall our antiequivalence of categories

$$\text{RatVar}_k^{\text{irr}} \cong k/\text{Field}^{\text{fin}}, \quad X \mapsto k(X), \quad \varphi \mapsto \varphi^*.$$

Note here that  $k/\text{Field}^{\text{fin}}$  is the category of finite field extensions of  $k$ . This is denoted by  $k/\text{Field}$  because it corresponds to a general categorical construction, the undercategory, whose objects are morphisms  $k \rightarrow L$ . This

conflicts with the notation  $L/k$  for a field extension, which is unfortunate. In any case, we will not use this notation again.

We claim that this equivalence of categories restricts to an equivalence of categories

$$\{\text{irreducible smooth projective curves}\} \cong \{\text{finite field extensions } L/k \text{ of transcendence degree } 1\}.$$

Firstly, if  $X$  is an irreducible smooth projective curve, then we need to show that  $k(X)/k$  has transcendence degree 1. If we let  $x \in X$  be arbitrary, then  $k(X) = \text{Frac}(\mathcal{O}_{X,x})$  and  $\mathcal{O}_{X,x}$  is a DVR and thus has a uniformizer  $\pi \in \mathcal{O}_{X,x}$ . Elements of  $k(X)$  are of the form  $\pi^m u$  for  $u \in \mathcal{O}_{X,x}^\times$  and  $m \in \mathbb{Z}$ , and so  $\pi$  generates the extension. Since  $\dim \mathcal{O}_{X,x} = 1$ ,  $k(X)$  is a proper extension of  $k$  and thus  $\pi$  is a transcendence basis.

Thus the functor  $X \mapsto k(X), \varphi \mapsto \varphi^*$  is well-defined and remains fully faithful.

To show essential surjectivity we must work a little harder. The idea is that given  $L$ , there is an irreducible affine variety  $X$  with  $k(X) \cong L$  (as before). Furthermore it is a curve since  $\dim_k X = \dim_k L = 1$ . It is then closed in some  $\mathbb{A}^n$  which we can embed in  $\mathbb{P}^n$  in the usual manner. Taking the Zariski closure of  $X \subseteq \mathbb{P}^n$ , we obtain an irreducible projective curve  $\bar{X}$  with the same function field  $L$ . The issue is that the curve may not be smooth, which we remedy by “smoothing it out”.

#### 4.5.12 Definition

A variety  $X$  is **normal** when the local ring  $\mathcal{O}_{X,x}$  is a normal integral domain for all  $x \in X$ . Recall that an integral domain being normal means being integrally closed in its field of fractions.

#### 4.5.13 Lemma

Suppose  $x \in X$  lies in more than one irreducible component of  $X$ . Then  $\mathcal{O}_{X,x}$  is not a domain; thus normal varieties are disjoint unions of irreducible varieties.

#### Proof

Recall that the minimal prime ideals of  $\mathcal{O}_{X,x}$  are in bijection with the irreducible components of  $X$  containing  $x$ . A domain has a single minimal prime ideal: the zero ideal. Thus if  $x$  lies in more than a single irreducible component,  $\mathcal{O}_{X,x}$  cannot be a domain. ■

If  $X$  is normal, then by this lemma  $X$  is the disjoint union of its irreducible components:  $X = \bigsqcup_i X_i$ . Thus each irreducible component is open and closed. Since being normal at a point is clearly local, this means that all the  $X_i$  are normal subvarieties as well.

#### 4.5.14 Definition

For a variety  $X$ , its **normalization** is a normal variety  $\bar{X}$  along with a dominant morphism  $\nu: \bar{X} \rightarrow X$  which is universal. That is, for any other dominant morphism from a normal variety  $\varphi: Y \rightarrow X$ , there is a unique  $\psi: Y \rightarrow \bar{X}$  such that  $\varphi = \nu \circ \psi$ .

As this is a universal property, normalization is unique up to unique isomorphism. We will show that every variety has a normalization.

#### 4.5.15 Lemma

- (1) If  $\nu: \bar{X} \rightarrow X$  is dominant from a normal variety, it is a normalization iff for every *irreducible* and normal  $Y$  and dominant  $\varphi: Y \rightarrow X$ , there is a unique  $\psi: Y \rightarrow \bar{X}$  such that  $\varphi = \nu \circ \psi$ .
- (2) Let  $X$  be a variety with irreducible components  $X_1, \dots, X_n$ . Suppose that the normalization  $\bar{X}_i$  exists for each  $i$ . Then  $\bar{X} = \bigsqcup_i \bar{X}_i$  is the normalization of  $X$ .

Thus we can restrict the construction of normalizations to irreducible varieties.

### Proof

For (1), one direction is clear. In the other, suppose  $\varphi: Y \rightarrow X$  is dominant and  $Y$  is normal. Then as  $Y$  is normal, it is the disjoint union of its irreducible components,  $Y = \bigsqcup_i Y_i$  which are all clopen subsets and thus normal. By assumption, we have unique morphisms  $\psi_i: Y_i \rightarrow \overline{X}$  such that  $\varphi|_{Y_i} = \nu \circ \psi_i$ . Then letting  $\psi = \bigsqcup_i \psi_i: Y \rightarrow \overline{X}$  we see that  $\varphi = \nu \circ \psi$ , and uniqueness of  $\psi$  follows from the uniqueness of the  $\psi_i$ s.

For (2) let  $\nu_i: \overline{X}_i \rightarrow X_i$  be the normalization maps so  $\nu = \bigsqcup_i \nu_i: \overline{X} \rightarrow X$ . As being normal is local,  $\overline{X}$  is normal, and since  $\nu(\overline{X}) \cap X_i$  is dense in  $X_i$  for each  $i$  and  $X_i \subseteq X$  is closed,  $\nu(\overline{X})$  is dense in  $X$ . So  $\nu$  is dominant.

Now suppose  $\varphi: Y \rightarrow X$  is dominant from irreducible normal  $Y$ . Then  $\varphi(Y) \subseteq X$  is irreducible and thus contained in some  $X_i$ . Thus there is some  $\psi: Y \rightarrow \overline{X}_i \subseteq \overline{X}$  such that  $\varphi = \nu_i \circ \psi$ ; thus  $\varphi = \nu \circ \psi$  as desired. By (1) this means that  $\nu: \overline{X} \rightarrow X$  is the normalization. ■

### 4.5.16 Lemma

Let  $A$  be an integral domain. Then  $A$  is normal iff  $A_{\mathfrak{m}}$  is normal for all maximal  $\mathfrak{m} \leq A$ .

### Proof

Let  $B$  be the integral closure of  $A$  in  $\text{Frac}(A)$ . Since forming the integral closure commutes with localization,  $B_{\mathfrak{m}}$  is the integral closure of  $A_{\mathfrak{m}}$  in  $\text{Frac}(A)_{\mathfrak{m}} = \text{Frac}(A)$ . Thus  $A_{\mathfrak{m}}$  is normal for every  $\mathfrak{m}$  iff  $0 = B_{\mathfrak{m}}/A_{\mathfrak{m}} = (B/A)_{\mathfrak{m}}$  for all  $\mathfrak{m}$ . This holds iff  $B/A = 0$ , i.e. iff  $A$  is normal. ■

Thus if  $A$  is a fg  $k$ -algebra which is a domain, then  $A$  is normal then iff  $X = \text{Specm}(A)$  is. Indeed, we know that  $\mathcal{O}_{X,x}$  is the localization of  $k[X] = A$  by the maximal ideal corresponding to  $x \in \text{Specm}(A)$ . By the above lemma,  $A$  is normal iff all of these are normal, i.e. iff  $X$  is normal.

We now state the following result without proof.

### 4.5.17 Theorem

Let  $k$  be a field and  $A$  a fg  $k$ -algebra that is a domain. Then  $\overline{A}$ , the integral closure of  $A$  in its fraction field  $L$ , is a finite  $A$ -algebra.

### Note:

For a proof, see *Matsumura, Commutative Algebra* theorem 72 part 31.

Thus if we have an affine variety  $X = \text{Specm}(A)$  with  $A$  an fg domain over  $k$ , then we also obtain an affine variety  $\text{Specm}(\overline{A})$ .

### 4.5.18 Proposition

Let  $A$  be a fg  $k$ -algebra that is a domain, and let  $X = \text{Specm}(A)$  be its corresponding irreducible affine variety. Let  $\overline{A}$  be  $A$ 's integral closure in its fraction field, and let  $\overline{X} = \text{Specm}(\overline{A})$ . Then the natural map  $\nu: \overline{X} \rightarrow X$  is the normalization of  $X$ .

### Proof

First suppose that  $Y = \operatorname{Specm}(B)$  is an irreducible normal affine variety, and we have a dominant morphism  $\varphi: Y \rightarrow X$ . This corresponds to an injective  $k$ -algebra morphism  $\varphi^*: A \hookrightarrow B$ . This extends to a morphism of their field of fractions, which we denote by  $\varphi^*$  as well. Then  $\varphi^*(\overline{A})$  is contained inside the integral closure of  $B$  in  $\operatorname{Frac}(B)$ , which is just  $B$  as  $Y$  is normal. So  $\varphi^*$  extends to a map  $\varphi^*: \overline{A} \rightarrow B$  through which  $\nu$  factors.

This proves the universal property in the affine case.

Now suppose that  $Y$  is an arbitrary irreducible normal variety and there is a dominant map  $\varphi: Y \rightarrow X$ . Let  $U_i$  be an affine open cover of  $Y$ . Then each  $U_i$  is irreducible, normal, and affine, and  $\varphi|_{U_i}: U_i \rightarrow X$  is dominant. Thus there are unique  $\psi_i: U_i \rightarrow \overline{X}$  such that  $\varphi|_{U_i} = \nu \circ \psi_i$ .

These  $\psi_i$  must agree on the intersections  $U_i \cap U_j$ , which can be seen by considering an affine cover of  $U_i \cap U_j$ . Thus we glue  $\psi_i$  together to get  $\psi: Y \rightarrow \overline{X}$  which restricts to  $\psi_i$  on each  $U_i$ . Thus  $\varphi = \nu \circ \psi$ , and uniqueness of  $\psi$  is due to  $\psi_i$ 's uniqueness. ■

#### 4.5.19 Proposition

The normalization  $\nu: \overline{X} \rightarrow X$  exists for every variety  $X$ , and is a finite surjective birational morphism.

#### Proof

We may assume that  $X$  is irreducible. If we take an open affine cover  $U_i$  of  $X$ , these are all irreducible affine varieties and thus possess a normalization  $\overline{U}_i \rightarrow U_i$ . By universality, we can uniquely identify  $\overline{U}_i$  and  $\overline{U}_j$  over  $U_i \cap U_j$ . Then by gluing the  $\overline{U}_i$  together, we get the desired normalization.

To show that the normalization is finite surjective birational, we may work locally on  $X$  and thus assume that  $X$  is irreducible affine. So let  $X = \operatorname{Specm}(A)$  and then  $\overline{X} = \operatorname{Specm}(\overline{A})$ , where  $A$  is a fg  $k$ -algebra which is a domain and  $\overline{A}$  is its integral closure in  $\operatorname{Frac}(A)$ . Because  $\overline{A}$  is also a domain,  $\overline{X}$  is irreducible as well.

So

$$k(X) \cong \operatorname{Frac}(A) \cong \operatorname{Frac}(\overline{A}) \cong k(\overline{X}),$$

which means  $X$  and  $\overline{X}$  are birationally equivalent. And  $\overline{A}$  is a finite  $A$ -algebra, so  $\nu$  is finite. Because finite morphisms are closed, and  $\nu$  is dominant,  $\nu$  is surjective. ■

We return now to curves; the relevance of normalization is due to the following result.

#### 4.5.20 Proposition

A 1-dimensional Noetherian local domain  $(A, \mathfrak{m})$  is a DVR iff  $A$  is normal.

#### Proof

Because every element of a DVR is of the form  $\pi^n u$  for unique  $n \in \mathbb{Z}_{\geq 0}$  and  $u \in A^\times$ , it follows immediately that DVRs are UFDs. UFDs are normal (the proof is the same as the case of  $\mathbb{Z}$ ), so one direction is clear.

Conversely if  $A$  is normal, we will show that  $\mathfrak{m}$  is principal. Notice that since  $A$  is 1-dimensional, the only prime ideals are  $0 \subsetneq \mathfrak{m}$ . Furthermore,  $\mathfrak{m} \neq \mathfrak{m}^2$  as otherwise by Nakayama we would have that  $\mathfrak{m} = 0$  so that  $A$  is a field and then not one-dimensional. So let  $x \in \mathfrak{m} - \mathfrak{m}^2$ , and let  $0 \neq y \in \mathfrak{m}$ .

Then  $\mathfrak{m}$  is no longer a proper ideal in the localization  $A_y$ , and thus  $A_y$  is a zero-dimensional domain (as its only prime ideal is 0). Thus it is a field, so  $1/x \in A_y$ . Thus  $y^m \in (x)$  for some  $m > 0$ . Since  $\mathfrak{m}$  is fg and  $y \in \mathfrak{m}$  was arbitrary,  $\mathfrak{m}^n \subseteq (x)$  for some  $n > 0$  which we choose to be minimal; we claim  $n = 1$  so that  $\mathfrak{m} = (x)$ .

Now let  $z \in \mathfrak{m}^{n-1} - (x)$ , and then  $z\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (x)$ . Define  $\alpha = z/x$  so  $\alpha \in K = \operatorname{Frac}(A)$ . But  $\alpha \notin A$ , and  $\alpha\mathfrak{m} \subseteq A$  so it is an ideal of  $A$ .

If  $\alpha\mathfrak{m} \leq \mathfrak{m}$  then  $\mathfrak{m}$  is a faithful fg  $A[\alpha]$ -module over  $A$ . This would then imply that  $\alpha$  is integral over  $A$ . But  $A$  is normal, which would mean  $\alpha \in A$ , a contradiction.



Thus  $\alpha\mathfrak{m}$  is not contained in  $\mathfrak{m}$  and thus it is not a proper ideal,  $\alpha\mathfrak{m} = A$ . In particular  $x/z \in \mathfrak{m}$  and so  $z\mathfrak{m} = (x)$ . But then  $x \in z\mathfrak{m} \subseteq \mathfrak{m}^n$ , and so if  $n \geq 2$  this would mean  $x \in \mathfrak{m}^2$  a contradiction. Thus  $n = 1$  as desired. ■

#### 4.5.21 Corollary

A curve  $X$  is smooth at  $x \in X$  iff it is normal at  $\mathcal{O}_{X,x}$ . Thus an irreducible curve  $X$  is smooth iff it is normal.

We may now complete the proof of essential surjectivity of our functor  $X \mapsto k(X)$  and  $\varphi \mapsto \varphi^*$ . So let  $L/k$  be a finite field extension of transcendence degree 1, which determines an irreducible affine variety  $X$  such that  $k(X) \cong L$ . There is then a closed embedding  $X \subseteq \mathbb{A}^r$ , which we can then embed as open dense in  $\mathbb{P}^r$ . Let  $\bar{X} \subseteq \mathbb{P}^r$  denote the Zariski closure, which is then an irreducible projective variety with a dense open subvariety  $X$ , so  $k(\bar{X}) \cong k(X)$ . Let  $\nu: Y \rightarrow \bar{X}$  be its normalization, so  $Y$  is an irreducible normal curve and thus a smooth curve by the above corollary. Furthermore since  $\nu$  is finite surjective,

$$k(Y) \cong k(\bar{X}) \cong k(X) \cong L.$$

It remains to show that  $Y$  is projective.

#### 4.5.22 Lemma

Let  $\varphi: X \rightarrow Y$  be a finite morphism such that  $Y$  is complete. Then  $X$  is also complete.

#### Proof

Universal closure is clear, since for a variety  $Z$ , projection  $\text{proj}_Z^X: X \times Z \rightarrow Z$  factors as

$$X \times Z \xrightarrow{\varphi \times \text{id}} Y \times Z \xrightarrow{\text{proj}_Z^Y} Z$$

which is a composition of closed maps.

We now need to show that  $X$  is separated. Now since  $\varphi$  is finite, it is closed and thus so is  $\varphi \times \varphi$ . So  $\varphi(X \times X) \subseteq Y \times Y$  and  $\varphi(\Delta X) \subseteq \Delta Y$  are closed. Thus  $\varphi(\Delta X) \subseteq \varphi(X \times X)$  is closed, and thus taking preimages  $\Delta X \subseteq X \times_Y X = \{(x_0, x_1) \in X \times X \mid \varphi x_0 = \varphi x_1\} \subseteq X \times X$  is closed.

Now we claim that  $\Delta X \subseteq X \times_Y X$  is closed, and this is sufficient. Because  $\varphi$  is affine, we may assume that  $X$  and  $Y$  are. Indeed, let  $U_i$  be an affine cover of  $Y$ , then  $V_i = \varphi^{-1}U_i$  is an affine cover of  $X$ . Then

$$\Delta X = \bigcup_i \Delta(V_i), \quad X \times_Y X = \bigcup_i V_i \times_{U_i} V_i,$$

and so if  $\Delta(V_i) \subseteq V_i \times_{U_i} V_i$  is closed for all  $i$ , so is  $\Delta X \subseteq X \times_Y X$ .

But affine varieties are separated, so  $\Delta(X) \subseteq X^2$  is closed. Thus  $\Delta(X) \subseteq X \times_Y X$  is closed as well, as desired. ■

#### 4.5.23 Lemma

Let  $X$  be projective and  $x_1, \dots, x_n \in X$ . Then there is an affine open neighborhood of  $X$  containing all of the  $x_i$ .

#### Proof

If  $X = \mathbb{P}^m$ , we note that  $\mathrm{GL}_{m+1}(k)$  acts continuously and transitively on  $\mathbb{P}^m$ . Thus every matrix in  $\mathrm{GL}_{m+1}(k)$  forms an automorphism of  $\mathbb{P}^m$ . We claim that there is a  $g \in \mathrm{GL}_{m+1}(k)$  such that the first rows of  $g(x_1), \dots, g(x_n)$  are all non-zero and thus lie in  $D_{\mathbb{P}^m}(X_1)$ . We construct such a  $g$  as follows.

First, there is a linear functional  $\ell \in (k^{n+1})^*$  such that  $\ell(x_i) \neq 0$  for all  $i$ . Indeed, define the subspace  $H_i = \{\ell \in (k^{n+1})^* \mid \ell(x_i) = 0\}$ ; it is proper since  $x_i \neq 0$ . Then  $\bigcup_i H_i$  is not the whole dual space, since the union of subspaces is only a subspace if they are all contained in one of the subspaces. Thus there is an  $\ell$  not in any  $H_i$ , as desired. Now extend  $\ell$  to a basis  $\ell, \ell_1, \dots, \ell_n$  and define  $g$  to be  $g(x) = (\ell(x), \ell_1(x), \dots, \ell_n(x))$ ; this has the desired properties.

Thus  $g(x_i)$  all lie in  $D_{\mathbb{P}^m}(X_1) \cong \mathbb{A}^m$ . Then  $x_i$  all lie in  $g^{-1}D_{\mathbb{P}^m}(X_1)$ , which is still affine since  $g$  is an automorphism. ■

Now we can show that  $Y$  is projective. The map  $\nu: Y \rightarrow \overline{X}$  is a birational equivalence of curves. The domain of definition of  $\nu^{-1}$  has dimension 1 (it is dense), and thus its complement has dimension 0 since if it were 1 then as a closed set it would equal all of  $\overline{X}$ . Thus the complement of  $\nu^{-1}$ 's domain of definition is a finite subset  $S \subseteq \overline{X}$ .

So  $\nu$  defines an isomorphism between  $Y - \nu^{-1}S \rightarrow \overline{X} - S$ ; let  $y_1, \dots, y_n = \nu^{-1}S$  (because  $\nu$  is finite, it is quasi-finite). By the previous lemma, there is an affine neighborhood  $V \subseteq \overline{X}$  containing  $S$ . Let  $W = \nu^{-1}V \subseteq Y$  which is affine as well since  $\nu$  is finite. We can then embed  $W$  into  $\mathbb{P}^m$ , and because  $W$  is normal as an open subset of  $Y$ , it extends to a map  $\psi: Y \rightarrow \mathbb{P}^m$ .

Let  $\varphi$  be the composition  $Y \xrightarrow{\nu} \overline{X} \hookrightarrow \mathbb{P}^r$ . Then consider the map  $\Phi = (\varphi, \psi): Y \rightarrow \mathbb{P}^r \times \mathbb{P}^n$ . We claim that this is a closed embedding.

Now  $Y$  is complete because  $\nu: Y \rightarrow \overline{X}$  is finite and  $\overline{X}$  is projective, and thus complete,  $Y$  is complete by the previous lemma. So  $\Phi$  is a morphism from a complete to a separated variety, and thus has closed image  $Z = \Phi(Y)$ . Further,  $\varphi$  separates any two points of  $Y$  unless they are the  $y_i$ , in which case  $\psi$  does. So  $\Phi$  is an injection.

Projection onto the first component yields  $\varphi$ , which is an isomorphism excluding the  $y_i$ . And projection onto the second component yields  $\psi$ , which is an isomorphism on  $W$ . Thus there is an inverse  $Z \rightarrow Y$  whose domain of definition contains all of  $Z$ . Thus  $\Phi$  has an isomorphism, meaning  $Y \cong Z$  which is projective.

This concludes the proof of essential surjectivity and the following theorem.

#### 4.5.24 Theorem

There is an antiequivalence of categories

$$\{\text{irreducible smooth projective curves}\} \xrightarrow{\sim} \{\text{finite field extensions } L/g \text{ of transcendence degree } 1\}$$

given by  $X \mapsto k(X)$  and  $\varphi \mapsto \varphi^*$ .